

UNITED STATES TARIFF COMMISSION

SYNTHETIC RESINS AND THEIR RAW MATERIALS

A SURVEY OF THE TYPES AND USES OF SYNTHETIC
RESINS, THE ORGANIZATION OF THE INDUSTRY,
AND THE TRADE IN RESINS AND RAW
MATERIALS, WITH PARTICULAR
REFERENCE TO FACTORS
ESSENTIAL TO TARIFF
CONSIDERATION

UNDER THE GENERAL PROVISIONS OF SECTION 332, TITLE III,
PART II, TARIFF ACT OF 1930

REPORT No. 131
SECOND SERIES



UNITED STATES
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ERRATA

Since publication of the report on Synthetic Resins the Commission's attention has been called to certain necessary corrections.

[Page 37](#)—2d line under heading “Production in the United States”

Strike out “The Resinous Products and Chemical Co., Inc.,” and insert “Rohm and Haas,”

[Page 154](#)—Last item under “Vinyl Resins”

Transfer the name of E. I. du Pont de Nemours and Co., Wilmington, Del. to line below so that it will not be opposite a trade name. This company manufactures Vinyl Resins but not “Koroseal”.

December 1938

Transcriber's Note: The errata have been corrected for this e-text, together with a number of sundry typos.

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1. INTRODUCTION

This survey deals with the several commercially important types of synthetic resins covered by paragraphs 2, 11, and 28 of the Tariff Act of 1930 and with the raw materials necessary for their production. It is made under the general investigatory powers of the Tariff Commission as provided in section 332 of that act.

The field of synthetic resins is a comparatively new one, most of its commercial development having occurred within the past 10 years. In 1937 the domestic output was more than 160 million pounds as compared with slightly more than 10 million pounds in 1927.

The first important patents on synthetic resins were granted about 25 years ago. These patents covered phenolic resins probably intended for use as substitutes for certain natural resins. It was soon found that these synthetics offered possibilities of application vastly greater than the natural materials. At first progress in their application was slow as is usually the case with new products. During the World War the shortage of phenol promoted interest in the use of the other tar acids as raw materials for synthetic resins and intensive research developed resins from the cresols and higher boiling tar acids. These resins possessed properties sufficiently different from those made from phenol to establish them permanently.

In the meantime research on other types of resins was carried on in the United States and in Europe. The tar-acid resins for molding were the only commercially important ones on the market until about 1929. About that time, however, new commercial products began to appear rapidly. Cast phenolic resins became available as material for novelties of unusual brilliancy and beauty, the urea resins to meet the requirements for light colored thermosetting resins in molded articles, and the alkyd resins for use in new surface coatings which replaced conventional paint materials.

Later there followed a number of thermoplastic materials offering new and unusual properties. Vinyl resins found application in molded products and in safety glass. The acrylate resins became the nearest approach to organic glass yet developed. The polystyrene resins, long in the research stage, made their commercial appearance in 1937. Resins from petroleum, from furfural, from adipic acid, and from aniline are on the market. Many others are under investigation and some of them will undoubtedly become important.

The versatility of synthetic resins is most unusual. In various uses they have successfully displaced glass, wood, metal, hard rubber, bone, glue, cellulose plastics, protein plastics, and conventional paint materials. They compete with glass in shades and reflectors and offer properties which will increase their use for this purpose. Cases

for scales, radios, and clocks, formerly of wood and metal, are now made of these synthetic resins.

Scope and purpose.

This survey deals with the synthetic resins, the nature and trade in the raw materials necessary for their production, the processes by which they are made, trade in them in the United States and between nations, and the nature of the competition which they meet. It does not go into the details of manufacture of and trade in the multitude of articles made of synthetic resins but stops at the point where these materials are turned over to the resin fabricator. The synthetic resins are but one of four broad groups of organic plastics. The others—natural resins, cellulose ethers and esters, and protein plastics—are discussed herein only as they relate to or compete with the synthetic resins.

The purpose of the survey is to bring together in one publication the available information on synthetic resins so as to provide a basis for consideration of future tariff problems. Because the industries involved are comparatively young and are expanding rapidly, their present day importance is not generally realized. The rapidity with which the synthetic resin industry is developing causes any comprehensive report on the subject to be practically out of date before it can be published. Notwithstanding the progress made each year in the quantity of production, new applications, and new commercial products, the industry may be said to be still in the industrial nursery. This circumstance necessarily limits the period during which any treatment of the subject will be representative.

Fundamental definitions.

The scope of this report has been stated to include synthetic resins up to the point where they are further manufactured, and the raw materials used in producing them. It was also stated that natural resins and synthetic plastics other than resins, such as the cellulose compounds and modified rubber compounds, are excluded. The boundaries of these categories are therefore important.^[1]

The term “resin” was formerly applied exclusively to a group of natural products, principally of vegetable origin, although at least one important resin, shellac, is of animal origin.^[2] These natural resins are widely used in paints, varnishes, and lacquers for decorative and protective surface coatings. They also have extensive use in textile impregnation, adhesives, soap, paper, and in cold-molded articles. In recent years the natural resins have had to compete with synthetic products, and each gravitates toward uses which demand the quality or combination of qualities which it can most completely supply.

A resin may be defined as a semisolid or solid, complex, amorphous mixture of organic compounds with no definite melting point and no tendency to crystallize. The resins are characterized by a typical luster and a conchoidal fracture rather than by definite chemical composition. The term includes natural resins, such as colophony (ordinary rosin), copal, damar, lac, mastic, sandarac, shellac, etc., sometimes called gums or gum resins although none of them are true gums.

A synthetic resin is a resin made by synthesis from nonresinous organic compounds. The term includes materials ranging from viscous liquids to hard, infusible, amorphous solids. As a rule synthetic resins possess properties distinct from those of natural resins. The term "plastics," sometimes applied to synthetic resins, also includes many materials which are not resins.

A plastic is anything possessing plasticity; that is, anything which can be deformed under mechanical stress without losing its coherence or its ability to keep its new form. According to this definition the term includes such materials as putty, cement, clay, glass, and metals, as well as certain modified natural or semisynthetic products, such as cellulose acetate, cellulose nitrate, and casein more commonly so designated. To speak of the plastics industries is almost meaningless because of their enormous scope, including as they do those producing cement, ceramics, confectionery and rubber, as well as those producing the semisynthetic products mentioned.

The resin industry embraces two main types of materials, thermoplastic and thermosetting. Thermoplastic materials are those which, although rigid at normal temperatures, may be deformed and molded under heat and pressure. Among such materials are the cellulose esters, acrylate resins, vinyl resins, polystyrene resins, etc. The recent development of injection molding has given thermoplastics a new significance.

Thermosetting substances are thermoplastic at some stage of their existence, but become hard, rigid, and permanently infusible upon the application of the proper heat and pressure. They are then irreversible whereas the thermoplastics are reversible. Outstanding among the thermosetting resins are tar-acid resins, urea resins, and the alkyd resins.

Tariff history.

The earliest mention of synthetic resins in the tariff laws of the United States was the provision in group III of the Emergency Tariff Act of 1916 for a duty of 30 percent ad valorem and 5 cents per pound on synthetic phenolic resins. None of the non-coal-tar synthetic resins were specifically mentioned prior to the Tariff Act of 1930.

The Tariff Act of 1922 (par. 28) provided for synthetic phenolic resin and all resinlike products, solid, semisolid or liquid, prepared from phenol, cresol, phthalic anhydride, coumarone, indene, or from any other article or material provided for in

paragraph 27 or 1549. The rate of duty was 60 percent ad valorem based on American selling price or United States value and 7 cents per pound, with a provision that the ad valorem rate should be reduced to 45 percent 2 years after the passage of the act.

The Tariff Commission made two investigations of synthetic resins under section 316 of the act of 1922. The first was undertaken April 16, 1926, upon complaints of several domestic manufacturers, of unfair methods of competition and unfair acts in the importation and sale of synthetic phenolic resin, Form C, and articles made wholly or in part therefrom, in infringement of the patent rights of the Bakelite Corporation. Following the investigation, the Commission recommended on May 25, 1927, that this material (as described under United States Patents No. 942,809 and 1,424,738) be excluded from entry into the United States. Importers appealed from the findings of the Commission to the Court of Customs Appeals, and the judicial proceedings were ended on October 13, 1930, by denial of a writ of certiorari for the Supreme Court of the United States to review the judgment of the Court of Customs and Patent Appeals. The latter court had held, among other things, that there was substantial evidence in support of each finding of the Commission. On November 26, 1930, the Treasury Department issued an order prohibiting the importation of synthetic phenolic resin, Form C, with certain exceptions. (T. D. 44411.)

The second investigation by the Tariff Commission was instituted on December 23, 1927, also under section 316 of the act of 1922. It concerned unfair methods of competition and unfair acts in the importation into the United States of laminated products of paper or other materials and insoluble, infusible condensation products of phenols and formaldehyde. The Commission recommended to the President that, until March 4, 1929, inclusive, certain products covered by United States Letters Patent Nos. 1,018,385, 1,019,406, and 1,037,719 be excluded from entry into the United States. These products were laminated cloth, paper or the like, combined with insoluble, infusible condensation products of phenols and formaldehyde. The order of the President prohibiting the importation was contained in T. D. 42801 issued June 11, 1928.

Under the Tariff Act of 1930, practically no changes were made in the provisions of paragraph 28 that concern coal-tar synthetic resins. Paragraph 2 was extended to include, among other things, the resins (polymers) of certain organic compounds. The only commercial products covered by this provision are the vinyl resins. The rate of duty was 30 percent ad valorem on foreign value and 6 cents per pound. Under the trade agreement with Canada, the duty on vinyl acetate, polymerized or unpolymerized, and on synthetic resins made in chief value therefrom was reduced to 15 percent ad valorem and 3 cents per pound (effective Jan. 1, 1936).

The Tariff Act of 1930 contains a provision, in paragraph 11, for synthetic gums and resins not specially provided for, 4 cents per pound and 30 percent ad valorem on foreign value.

Broadening use of synthetic resins.

The application of synthetic resins has extended into practically every branch of industry. This marked expansion is not surprising when the adaptability of these products is considered. Their uses range from jewelry and bottle closures to building materials; from adhesives and new types of surface coatings to light reflectors and shades. They are being substituted for natural materials, such as wood, metal, and glass at an increasing rate. They have provided new uses for raw materials formerly used in antiseptics, disinfectants, explosives, embalming fluids, fertilizers, moth repellants, and as solvents. The speed of expansion of their use in resin manufacture has been such as to create a serious shortage of some of these raw materials.

New applications for synthetic resins appear almost daily. They are used in furniture, wall panels, builders' hardware, electrical fixtures, and in thousands of small appliances. The automobile industry is probably the largest single user. An interesting application here is in silent gears and shaft bearings where the use of synthetic resins makes water lubrication possible. Other automotive uses are in distributor heads, horn buttons, gear shift knobs, dome light reflectors, control knobs and the finishing lacquers. Additional uses contemplated for the near future are in accelerator pedals and instrument panels. A new type of safety glass in which vinyl resins are used was introduced in 1936.

In decorative uses remarkable progress has been made. Panels of laminated resins are widely used in store fronts, lobbies of office buildings, and hotels; doors faced with this material are in use. The liner *Queen Mary* is paneled, in part, with laminated resins, as is the annex to the Library of Congress. Lamp shades of urea resin are used in many Pullman cars and are available for home and office use.

Other things being equal, the cheaper a synthetic resin, the more widely it may be applied as a substitute for other materials. As a result many an apparently useless byproduct, such as oat hulls which yield furfural, is either already used or being tested as a source of raw material. Other materials which have already found a place or may do so, are soybean meal, sugar, and certain petroleum distillates.

Each of the important groups of synthetic resins has been sponsored by one or more manufacturers of established reputation and large capital resources. When a product reaches the commercial stage, after heavy research cost, its future importance is therefore usually assured.

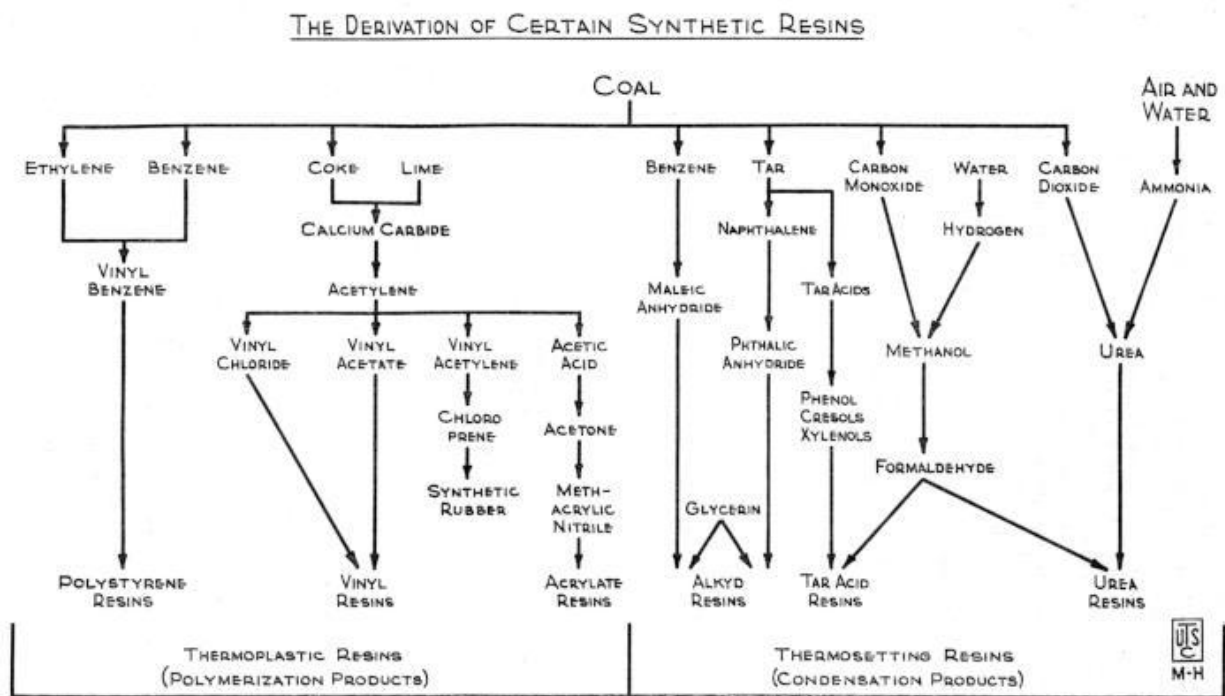
Relation of synthetic resins to their raw materials.

Most of the commercially important synthetic resins are derived directly or indirectly from coal. The chart ([p. 6](#)) shows the derivation of certain synthetic resins

from the principal raw materials used in their manufacture and the intermediate products back to the original source of the material.

The polystyrene resins, for example, are made by polymerizing styrene or vinyl benzene. Although basically from ethylene and benzene, vinyl benzene may be formed in several ways. Ethylene is found in the gases from the destructive distillation of coal but is obtained commercially by cracking natural gas or petroleum. Styrene, found already formed in the light oil fractions from coal tar, causes gum-forming in motor benzol and certain industrial gases.

When coke and lime are mixed and heated in an electric furnace to 2,000° C., calcium carbide is formed. This compound with water yields acetylene, the starting point for a long list of important products, including several types of synthetic resins. When acetylene gas is passed through acetic acid (itself obtained from acetylene) vinyl acetate is obtained. If hydrochloric acid is used instead of acetic acid, vinyl chloride is obtained. These compounds, when polymerized, yield the vinyl resins. The acrylate resins may be obtained from the same basic raw material by an entirely different procedure. Synthetic rubber is also derived from acetylene, as are acetic anhydride and acetic acid (used in making cellulose acetate plastics) and many other chemicals of commercial importance.



Derivation of certain synthetic resins.

When naphthalene (from coal tar) is treated with air at elevated temperatures, phthalic anhydride is formed. Substituting benzene for naphthalene yields maleic anhydride. Both of these substances when condensed with glycerin, a byproduct of the soap industry, yield alkyd resins.

The tar acids from coal tar, either separated or mixed, when condensed with formaldehyde give the highly important tar-acid resins. Or if formaldehyde is condensed with urea, obtained from carbon dioxide and ammonia, the urea resins are formed.

The chart indicates the synthetic resins which are thermoplastic, that is, which become plastic again upon reheating, and those which are thermosetting, that is, pass into an infusible stage at a certain critical temperature and pressure and do not again become plastic upon subsequent reheating.

Sources of information.

The data used in this report were obtained from a great variety of sources. The several American and British trade journals were freely consulted as were the various text books on this subject. Much of the information on the domestic industry was obtained by personal contact with producers and by correspondence. Field work included visits to most of the domestic producers of resins and a representative group of fabricators. Information of this type which was nonconfidential or which could be combined so as not to reveal individual operations was invaluable. Even where it was such that it could not be published it became part of the general background.

The data pertaining to the industry in foreign countries were, for the most part, furnished the Tariff Commission by Department of Commerce representatives stationed abroad, in response to inquiries by the Commission.

2. SUMMARY

Growth of the industry.

The coal-tar synthetic resin industry in the United States began on a small scale some years before the World War. The output then was confined to a few types of tar-acid resins and the applications were quite limited until 1927, when certain of the basic patents expired. The output of about 1.5 million pounds in 1921 had increased to more than 13 million pounds in 1927 and the average unit value of sales had dropped from 81 cents per pound to 47 cents. Production continued to increase and the unit value to decrease annually until 1932 when general economic conditions forced a slight curtailment for 1 year. Since then the annual increase in volume and variety has been rapid. Production of non-coal-tar synthetic resins was started on a small scale in 1929 when both urea and vinyl resins entered the picture. Commercial production of the petroleum resins began in 1936 and of the acrylate resins in 1937. Table 1 shows the production and sales of coal-tar resins and of non-coal-tar resins, from 1921 through 1937.

TABLE 1.—*Synthetic resins: United States production and sales, 1921-37*

Year	Production	Sales		
		Quantity	Value	Unit value
	<i>Pounds</i>	<i>Pounds</i>		
Coal-tar resins: ¹				
1921	1,643,796	1,674,456	\$1,352,166	\$0.81
1922	5,944,133	6,415,931	4,315,196	.67
1923-26	(²)			
1927	13,452,230	13,084,313	6,094,656	.47
1928	20,411,465	20,778,856	7,211,958	.35
1929	33,036,490	30,660,513	10,393,397	.33
1930	30,867,752	24,014,093	7,323,656	.30

1931	34,179,000	29,343,000	7,862,000	.27
1932	29,039,000	23,891,000	5,001,000	.21
1933	41,628,485	31,657,653	7,238,560	.23
1934	56,059,489	43,350,876	10,126,849	.23
1935	90,913,162	65,923,334	12,777,195	.19
1936	117,301,780	86,213,735	17,056,099	.20
1937	141,098,844	108,284,175	20,165,064	.19
Non-coal-tar resins:				
1932	1,898,000	1,787,000	796,000	.45
1933	3,571,717	3,256,411	1,745,102	.54
1934	(²)	3,500,829	1,491,145	.43
1935	(²)	(²)	(²)	
1936	15,611,041	14,766,640	3,591,467	.24
1937	21,005,869	18,891,277	5,680,600	.30

¹ Does not include resins from adipic acid, coumarone and indene, hydrocarbon, polystyrene, succinic acid and sulfonamides. With the exception of coumarone and indene resins in recent years production of the resins not included was small.

² Not publishable. Figures would reveal operations of individual producers.

Source: Compiled from annual reports of the Tariff Commission on dyes and other synthetic organic chemicals in the United States.

Many factors have contributed to the growth of the synthetic resin industry. Among these are the intensive research and development work carried on by many individuals and firms; their widespread application in many fields competing with wood, metal, and glass; and the development of processes for raw materials which have greatly reduced their cost and made their wider use possible.

Raw materials.—Although the chief raw materials consumed in the synthetic resin industry are coal-tar derivatives and formaldehyde, many others are utilized. The rapid expansion of the industry has created new demands for materials in increasing quantities and has not only increased the markets for well-known materials but has resulted in the production on a huge scale of materials entirely new to commerce. Practically all the raw materials now used can be derived from a few natural substances, such as air, water, coal, petroleum crudes, salt, sulphur, and limestone. The air

yields nitrogen which may be converted to ammonia, a raw material for urea, one of the components of the urea resins. Coal, as is well known, yields a great variety of substances, many of which are essential to synthetic resin manufacture. Benzene is the starting point for synthetic phenol; naphthalene is used to make phthalic anhydride and maleic anhydride; coke is converted to calcium carbide, which in turn yields acetylene, acetic acid, and many other synthetics; carbon monoxide which is converted to methanol and formaldehyde; and the natural tar acids such as phenol, the cresols, and the xylenols. Limestone is a component of calcium carbide, and salt yields needed alkalies and acids. Water is broken down, and the hydrogen is converted to ammonia, methanol, formaldehyde, and ethylene.

Some idea of the expansion in production of these raw materials whose principal use is in synthetic resins may be had by comparing the output in 1923 of tar acids, formaldehyde, phthalic anhydride, maleic anhydride, urea, vinyl acetate, and vinyl chloride, which amounted to 35 million pounds, with the output of 270 million pounds in 1936. The manufacture of these materials is largely by coal-tar distilling companies and makers of chemicals.

Resins.—The coal-tar resins are the most important in quantity, value, and variety of application. This class includes four groups: (a) tar acid, (b) alkyd, (c) coumarone and indene, and (d) polystyrene. Of these, resins from tar acids (phenol, cresols, and xylenols) are produced in the largest quantity, the output having increased from about 15 million pounds in 1932 to about 80 million pounds in 1937. In the latter year about 40 percent of the consumption of tar acid resins was in molded articles, 25 percent in paint and varnishes, 20 percent in laminated products, and 15 percent in miscellaneous uses.

The alkyd resins have shown a remarkable increase in output. Production totaled slightly less than 10 million pounds in 1933; in 1937 it amounted to about 61 million pounds. Practically all of the alkyds have been consumed in paints and varnishes.

The coumarone and indene resins have increased steadily over a number of years and are now one of the most important groups.

The polystyrene resins have been in an experimental stage for a long time, with the volume of production small. In 1937, however, commercial

production of a water-white product was announced, and it is believed that the output of these resins will increase sharply in the near future.

The non-coal-tar resins were of little importance prior to 1930 and production amounted to less than 2 million pounds in 1932. Since then, however, progress has been rapid, both in types and output. Resins from urea constitute an important part of this class and the output has increased practically every year since 1929 when production was started. Most of the output is used in molded articles where light and pastel shades are required. In 1936, for the first time, appreciable quantities were consumed in laminating and in surface coatings.

The vinyl resins have been produced in increasing quantities for the past 8 years. Production reached a new high in 1937, and with the acceptance of this type of resin for safety glass laminations it is expected that the output will increase materially in the near future. In 1937 the application in surface coatings, molded articles, and laminations were of approximately equal importance.

The acrylate resins are among the newest commercial developments in this industry. Of the several types now manufactured, one appears valuable in surface coatings and adhesives and another, in the form of its cast or molded polymer, in airplane windows, machined articles, and lenses.

Petroleum resins were first produced in commercial quantities in 1936, but the output in that year was appreciable. These low-priced synthetics are used in surface coatings, laminations, and miscellaneous uses.

The industry abroad.

World production of synthetic resins at this time is estimated at 300 million pounds annually, of which the United States accounts for 45 percent. Germany produces about 27 percent and Great Britain about 20 percent of the total and a number of countries including France, Italy, Czechoslovakia, Canada, and Japan produce the remainder. Practically all types are made in Germany and Great Britain although in lesser quantities than here. The urea resins originated abroad, as did the acrylates and polystyrenes.

Commercial development of the synthetic resins abroad has been somewhat behind that in the United States, although in recent years the increase there has been so rapid as to seriously affect the international raw material market. Germany, formerly one of our principal sources of crude naphthalene, for a time restricted exports of that commodity in order to conserve the available supply for home consumption, presumably in alkyd resins. Great Britain, formerly the principal exporter of phenol, has found it necessary to supplement production of natural phenol with synthetic phenol. It is possible that in the future similar conditions may arise in world markets for cresylic acid.

International trade.

International trade in the synthetic resins has been small. Germany has been the principal exporting country. There are a number of reasons for the negligible movement of these materials in international trade, the chief of which are active home markets in the principal producing countries; the existence of patents of a basic nature which limited trade to the owners and licencees under them; affiliation of producing companies in different countries with allocation of the world market; and high tariff barriers in many countries.

The principal domestic producer of tar-acid resins is affiliated with firms in Germany, the United Kingdom, France, Italy, Canada, and Japan. The two principal American makers of urea resins have or have had agreements as to patents, exchange of technical information, and probably markets, with producers in Great Britain. Similar conditions exist with other types of resins.

In 1937 production of all synthetic resins in the United States amounted to 162 million pounds and imports to less than 674,000 pounds (see [table 13](#), p. [58](#)). Production of tar-acid resins in that year amounted to 79.8 million pounds; alkyd resins to 61.2 million pounds and all coal-tar resins to 141 million pounds. Imports of all coal-tar synthetic resins (which would include both tar acid and alkyd as well as others) amounted to only 19,000 pounds. Coal-tar resins are dutiable at 7 cents per pound and 45 percent ad valorem based on American selling price. On the small imports in 1937 the duty collected averaged 54 percent ad valorem on American selling price

and would have averaged much higher had it been calculated upon foreign value as are most duties.

In 1937 the production of non-coal-tar resins totaled about 21 million pounds. In that year imports of non-coal-tar resins totaled 65,000 pounds. Imports of non-coal-tar resins, other than vinyl resins, amounted to less than 2,000 pounds. These were dutiable at 4 cents per pound and 30 percent ad valorem on foreign value, equivalent on the average to 48 percent ad valorem. The vinyl resins have been imported into the United States in increasing quantities in recent years. The principal foreign producer, in Canada, developed markets in the United States, but is a joint owner of a plant now under construction in this country. Imports of vinyl resins in 1937 were 653,000 pounds. These were dutiable at 3 cents per pound and 15 percent ad valorem on foreign value, equivalent to 25 percent ad valorem.

It is apparent that foreign competition with United States producers in the home market has been and is likely to continue insignificant under existing duties. With a large home market and generally favorable conditions with respect to the necessary raw materials and the technical skills, this situation would probably continue even under lower duties. Moreover, as international trade develops in these materials, this country is more likely to be a net exporter than a net importer.

3. TAR-ACID RESINS

The tar-acid resins were the first true synthetic resins to appear in commerce, but they were preceded by two plastics, celluloid and casein. Probably the first successful attempt to make a semisynthetic or modified natural product as a substitute for natural materials was the discovery of celluloid in 1868 by John Wesley Hyatt. By treating cotton with nitric acid he obtained a material which could be substituted for ivory in billiard balls. The Celluloid Corporation grew out of this discovery and the product was widely used to replace amber, ivory, mother-of-pearl, tortoise shell and other materials.

The discovery of casein plastic took place in 1890. Adolph Spitteler of Hamburg, Germany, in trying to make a white blackboard, found that casein (from milk) could be hardened by treating it with formaldehyde. Casein plastics are now widely used in buttons, buckles, and other ornaments.

As early as 1872 the reactions between coal-tar acids and aldehydes were being studied, and by 1900 many research workers were investigating phenol-formaldehyde condensation products. During the period 1900-1910, the study of these products increased greatly, both with regard to process of production and to applications, such as its substitution for shellac and other natural resins. United States Patents Nos. 942,699 and 942,809 issued December 7, 1909, to Dr. L. H. Baekeland and commonly known as the heat and pressure patents were probably the basic patents on phenol-formaldehyde resins. Baekeland so modified these resins by methods of hardening under heat and pressure that rigid molded articles could be made. The range of uses of tar-acid-formaldehyde molding compositions has steadily widened. Molded articles such as pencil and pen barrels, ash trays, bottle closures, parts for automobiles, cameras, precision instruments, dynamos, motors, and other electrical equipment, cafeteria trays, table and counter tops are well known to the public.

During the life of these and other basic patents issued about 1909 the domestic production of phenol-formaldehyde molding compositions was practically restricted to one company. Since the expiration of these patents in 1926 a number of other producers have been established. In 1937 there were

36 domestic makers of tar-acid-formaldehyde resins for molding, laminating, and surface coating applications.

The early work done on phenol-formaldehyde resins gave dark-colored products which were too hard and brittle to be machined or worked on a lathe. Investigations by F. Pollak and A. Ostersetzer, in Vienna, resulted in a process for the manufacture of cast phenolic resin with a range of color possibilities from water-white transparency through all shades and degrees of translucency and opaqueness. This product is cast into sheets, rods, tubes, and special castings, all of which may be turned or milled on automatic machines. United States Patent No. 1,854,600, issued April 19, 1932, to F. Pollak and A. Ostersetzer and assigned to Pollopas, Ltd., London, is considered the basic patent for cast phenolic resins. American rights under this and related patents are owned by the Catalin Corporation of America who have licensed other domestic makers. The German equivalent of rights under this patent is owned by a subsidiary of I. G. Farbenindustrie Aktiengesellschaft and rights under the French equivalent by Établissements Kuhlmann.

In the early days of the phenol-formaldehyde resin industry (1909-16) there was considerable uneasiness about the supply of phenol. World production was not large and Germany and England controlled most of it. The output of the United States was almost entirely for medicinal use, although our potential production was large (see p. [111](#)). This situation caused many research workers to study the resins made from other tar acids, principally meta and para cresols and the xylenols. The investigations resulted in many new types of resins and in modifications of the phenol-formaldehyde type. The World War changed conditions materially. Imports of phenol were shut off and prices soared. Production of synthetic phenol was begun, and, although the wartime production went into explosives, its development had an important bearing on the synthetic resin industry. Unusual demand for phenol, toluene, and other coal-tar crudes resulted in a great expansion of production. With the cessation of hostilities there was an ample supply of cheap phenol and the expansion of the coal-tar industry continued so that the supply of tar acids kept pace with the new demand for use in the production of synthetic resin.

In 1926, the early patents on resins from tar acids began to expire and the second era of the industry began. Since that year most of the research work has been for materials that would give different properties to the resultant resins. The past 10 years have seen a greater diversification in the manufacture of resins from tar acids and substantial reductions in their prices. Tar-acid resins

averaged \$1.29 per pound in 1920, 23 cents per pound in 1934, and 19 cents per pound in 1937. The production of certain resins of this class which are soluble in drying oils has been an important achievement. They yield varnishes of improved type that are quick-drying.

The three stages of a tar-acid resin.

About 28 years ago the Journal of Industrial and Engineering Chemistry published the original paper of Dr. Leo H. Baekeland on the Synthesis, Constitution, and Uses of Bakelite. According to Baekeland's theory, the reaction between phenol and formaldehyde consists of condensation and polymerization taking place in three stages. The first product formed, called "initial condensation product A" is usually a liquid or semisolid which on continued heating is converted to "intermediate condensation product B." B is an insoluble solid which can be softened by heat, and is the material used by molders, laminators, and other fabricators.

The final stage, known as "final condensation product C," is probably the result of polymerization of B, by heat and pressure. C product is infusible, indifferent to all solvents, and cannot be distilled or melted; hence the tar-acid resins belong to the thermosetting group. The conversion to C takes place in the presses of the molder or final fabricator of the resin. This theory is generally accepted and the designations of the several stages are in universal use in the trade.

Classification of tar-acid resins.

All the synthetic resins obtained by the condensation of a tar acid, or a mixture of tar acids, with an aldehyde are popularly called phenolic resins, regardless of whether they are made from phenol, the isomeric cresols, xylenols, other high boiling tar acids, or any mixture of these materials. A more accurate designation and that used in this survey is tar-acid resins, reserving the term phenolic resins for those made from pure phenol.

The tar-acid resins might be classified in a number of ways; for example, by composition, physical form, or general application. Each of these has its shortcomings. To classify them by composition, that is, by the kind of tar acid used, is not satisfactory because of the vast number of types made from mixed tar acids. For the purpose of this discussion it seems best to classify the tar-

acid resins by their general application into six groups: for molding, for casting, for laminating, for surface coating (paints, varnishes, and lacquers), for adhesives, and for miscellaneous uses.

In 1937 approximately 66 percent of the United States production of tar-acid resins was made from phenol; 18 percent from phenol-cresol mixtures; 13 percent from cresol-cresylic acid mixtures; and 3 percent from cresol-xylene mixtures. Table 2 shows for recent years production and sales of tar-acid resins by type of raw material. Pure phenol is used for cast resins. Molding resins are usually made from pure phenol or from tar-acid mixtures, chiefly phenol. Laminating and coating resins are usually made from mixtures containing substantial amounts of the cresols and xylenols (frequently spoken of by the trade as cresylic acid).

TABLE 2.—*Tar-acid resins: United States production and sales, by type of raw material, 1933-37*

Year	Phenol			Tar-acid mixtures ¹		
	Production (net resin content)	Sales		Production (net resin content)	Sales	
		Quantity (net resin content)	Value		Quantity (net resin content)	Value
	<i>1,000 pounds</i>	<i>1,000 pounds</i>	<i>1,000 dollars</i>	<i>1,000 pounds</i>	<i>1,000 pounds</i>	<i>1,000 dollars</i>
1933	25,163	21,851	5,383	6,535	6,152	1,182
1934	29,777	27,995	7,332	10,887	8,091	1,705
1935	36,323	34,597	6,568	16,654	12,371	2,200
1936	51,603	49,053	9,419	18,747	12,908	2,325
1937	52,472	50,209	8,616	27,373	23,337	4,685
	Phenol-cresol mixtures		Cresol-cresylic acid mixtures	Cresol-xylene mixtures		
	Production (net resin content)	Sales (net resin content)	Production (net resin content)	Sales (net resin content)	Production (net resin content)	Sales (net resin content)
	<i>1,000 pounds</i>	<i>1,000 pounds</i>	<i>1,000 pounds</i>	<i>1,000 pounds</i>	<i>1,000 pounds</i>	<i>1,000 pounds</i>
1937	14,046	13,238	10,702	8,467	2,625	1,632

¹ Includes phenol-cresol mixtures, cresol-cresylic acid mixtures, and cresol-xylenol mixtures. For 1937, where it is possible, the totals of tar-acid mixtures are broken down into these three groups.

Source: Dyes and Other Synthetic Organic Chemicals in the United States, U. S. Tariff Commission.

Processes of resin manufacture.

The processes of and patents for the manufacture of tar-acid-formaldehyde resins are numerous. No attempt is made here to describe in detail the several processes of manufacture or the endless number of variations and modifications. In general the processes in operation may be designated (a) one stage wet, (b) two stage wet, and (c) dry.

The one-stage wet process consists in heating molecular proportions of tar acid and formaldehyde (40-percent solution) in the presence of an acid or alkaline catalyst. The formaldehyde is added all at once and the reaction proceeds with the elimination of water. The difficulty with this process is that of obtaining uniform batches because it cannot be controlled exactly.

The two-stage process is probably the one most widely used today and consists in introducing formaldehyde in two or more stages as the reaction progresses. Much better process control and more uniform results are so obtained. A soluble, fusible resin is formed from which the water is easily removed. Fillers and pigments may be added during the latter part of the operation.

The dry process is the least important and is used only where cast resins are being made. Light-colored, transparent resins are obtained and the operation is carried on to the final stage (C resin). In this process the aldehyde used is solid paraformaldehyde or hexamethylenetetramine. These materials are more costly than formaldehyde solution.

Proportions of raw materials used vary widely—Baekeland suggested 7 mols of formaldehyde and 6 mols of phenol (210 parts of 100-percent formaldehyde to 564 parts of phenol), with a yield of resin equivalent to 118 percent of the phenol. Larger proportions of formaldehyde are said to increase the yield to as much as 140 percent of the phenol.

Catalysts used to aid in the condensation of the reacting bodies may be acids or bases. Certain properties of the resins may be varied by the kind and quantity of catalyst used. Large proportions of basic or acidic catalysts may affect the filler or metal inserts. Basic catalysts used include caustic soda,

caustic potash, ammonia, carbonates, and alkali sulphites. Acid catalysts are usually one of the mineral acids such as hydrochloric acid or sulphuric acid.

While formaldehyde in the form of a 40-percent solution is the principal aldehyde used with the tar acids, certain other aldehydes are used in small amounts. Among these are acetaldehyde, butyraldehyde, benzaldehyde, and others. Resins from furfural and phenol are discussed as “Furfural Resins,” page [51](#).

Production in the United States.

The production of tar-acid resins in the United States has increased markedly in the last 10 years. Table [3](#) shows the production and sales of all coal-tar resins in 1927 and 1928 (when there was no further break-down available but when this classification was made up chiefly of tar-acid resins) and of tar-acid resins from 1929 to 1937. The figures given are in net resin content and do not include fillers, modifiers, or pigments. From 1929 to 1937 production increased from 26 million pounds to 80 million pounds; sales from 25 million pounds valued at 9.9 million dollars to 74 million pounds valued at 13.3 million dollars; the value per pound dropped from 39 cents to 19 cents.

In 1937 the production of tar-acid resins for molding accounted for about 40 percent of the total; those for surface coatings, about 25 percent; those for lamination, about 20 percent; and those for miscellaneous uses, about 15 percent.

TABLE 3. *Tar-acid resins: United States production and sales 1927-37*

Year	Production (net resin content)	Sales		
		Quantity (net resin content)	Value	Unit value
	<i>Pounds</i>	<i>Pounds</i>		
1927 ¹	13,452,230	13,084,313	\$6,094,656	\$0.47
1928 ¹	20,411,465	20,778,856	7,211,958	.35
1929 ²	26,235,792	25,129,701	9,869,274	.39
1930 ²	18,338,389	17,428,687	6,576,023	.38

1931 ²	22,647,000	21,496,000	6,646,000	.31
1932 ²	17,163,000	15,042,000	3,946,000	.26
1933 ²	31,697,780	28,002,799	6,564,670	.23
1934 ²	40,663,565	36,086,008	9,037,861	.25
1935 ²	52,731,728	46,733,378	8,730,438	.19
1936 ²	70,349,328	61,961,200	11,743,978	.19
1937 ²	79,844,825	73,545,880	13,300,870	.19

¹ All coal-tar resins.

² Resins from tar acids only.

Source: Compiled from annual reports of the Tariff Commission on dyes and other synthetic organic chemicals in the United States.

Imports into the United States.

Imports of tar-acid resins into the United States are dutiable under paragraph 28 at 7 cents per pound and 45 percent ad valorem based upon American selling price. Discussion of this rate, other restrictions upon imports in the earlier years, and of the rates upon articles made of these resins will be found on pages [59 to 61](#).

Imports of tar-acid resins are not shown separately in official statistics; the classification under which such imports are entered includes all synthetic resins of coal-tar origin. Table [4](#) shows the quantity and value of imports of all coal-tar resins since 1918, and table [5](#) shows the principal sources of imports for certain years.

Invoice analyses of imports in the last 3 years show only very small quantities of phenolic resins being imported. In 1934 there was an importation of 950 pounds of Bakelite molding compound; in 1935 imports of 100 pounds of molding compound and 22 pounds of aminophenol resin are recorded, and in 1936 imports of Bakelite filament compound totaled 250 pounds and other resins 8,851 pounds.

Even if in the years up to 1933 all of the imports of resins of coal-tar origin were tar-acid resins, imports of tar-acid resins have been negligible when compared with production. The smallness of imports may be accounted for by a combination of factors, (1) the prohibition of imports of certain types, which conflicted with patent rights; (2) the rate of duty upon imports; (3) the fact that

the manufacture of tar-acid resins developed more rapidly in the United States than in most foreign countries; and (4) the allocation of markets through agreements between affiliated producers in different countries. (See p. [58](#).)

TABLE 4.—*Synthetic resins of coal-tar origin: United States imports for consumption, 1919-37*

Year	Quantity	Dutiable value	Value per pound	Computed ad valorem rate	Computed specific rate
	<i>Pounds</i>			<i>Percent</i>	<i>Per pound</i>
1919	1,114	\$2,860	\$2.57	32.0	\$0.82
1920	2,479	2,681	1.08	34.6	.37
1921	1,420	2,366	1.67	33.0	.55
1922	2,518	3,498	1.39	52.3	.73
1923	3,183	10,512	3.30	62.1	.20
1924	8,756	4,183	.48	68.9	.33
1925	1,537	889	.58	57.1	.33
1926	1,649	1,298	.79	53.9	.42
1927	11,359	4,266	.38	63.6	.24
1928	60,547	10,984	.18	83.6	.15
1929	67,529	17,503	.26	72.0	.19
1930	46,464	10,417	.22	76.2	.17
1931	6,074	6,180	1.02	51.9	.53
1932	6,403	3,905	.61	56.5	.34
1933	3,776	2,508	.66	55.5	.37
1934	15,711	8,680	.55	57.7	.32
1935	18,015	6,075	.34	65.8	.22
1936	18,598	13,643	.73	54.5	.40
1937 ¹	18,977	14,278	.75	54.3	.41

¹ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

TABLE 5.—*Synthetic resins of coal-tar origin: United States imports for consumption, by principal sources, in specified years, 1929-37*

Imported from—	1929	1931	1933	1934	1935	1936	1937 ¹
	Quantity (pounds)						
Germany	50,770	3,166	2,724	9,801	2,220	10,750	13,950
France	20	2,331	740		297	168	
United Kingdom	336			1,065	13,242	1,979	2,215
Switzerland	3,473	1,781	4,384	1,716			
Canada	1,372	135	1,266	594			
All other countries	16,403	577	312		340	51	502
Total	67,529	6,074	3,776	15,711	18,015	18,598	18,977
	Value						
Germany	\$11,771	\$4,053	\$1,913	\$5,303	\$1,959	\$9,700	\$11,960
France	21	1,760	465		236	177	
United Kingdom	2,235			255	2,476	1,090	659
Switzerland				2,621	1,308	2,154	1,197
Canada				501	46	486	214
All other countries	3,476	367	130		50	36	248
Total	17,503	6,180	2,508	8,680	6,075	13,643	14,278
	Unit value						
Germany	\$0.23	\$1.28	\$0.70	\$0.54	\$0.88	\$0.90	\$0.86
France	1.05	.76	.63		.79	1.05	
United Kingdom	6.65			.24	.19	.55	.30
Switzerland				.75	.73	.49	.70
Canada				.37	.34	.38	.36
All other countries	.21	.64	.42		.15	.71	.49
Average	.26	1.02	.66	.55	.34	.73	.75
	Percent of total quantity						
Germany	75.2	52.1	72.1	62.4	12.3	57.8	73.5
France	.1	38.4	19.6		1.6	.9	
United Kingdom	.5			6.8	73.5	10.6	11.7
Switzerland				22.1	9.9	23.6	9.1
Canada				8.7	.8	6.8	3.1
All other countries	24.2	9.5	8.3		1.9	.3	2.6

¹ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

Exports from the United States.

Appreciable quantities of phenolic resins are exported annually in the form of molding compounds and as finished articles of wide variety. Statistics of these exports are not compiled separately by the Department of Commerce.

Exportation is limited by a number of factors, such as licensing agreements, patents, allocation of markets, and high tariffs or embargoes in certain countries. The largest domestic maker is affiliated with producers in Great Britain, Germany, France, Italy, Canada, and Japan. Other domestic firms have agreements as to patents and markets with producers in England, Germany, and other countries.

TAR-ACID RESINS FOR MOLDING

The tar-acid resins were first developed for molding and they are still used in large volume in this way. An article produced in large quantity is more likely to be made of molded resin. The cost of the mold, which may amount to several thousand dollars, then becomes very small per unit produced. If the article is of such a shape that it would require a great deal of labor to produce in metal or wood, it may be produced in quantity much more cheaply from resin, since it will come from the mold almost in finished form.

A few of the large molders find it economical to make their own resins when they use one type in large volume or desire some special modification. Most of the molders buy resins for molding in the form of either powder or pre-formed pellets ready for use.

Molding powders and pellets.

Molding powder is made from B-stage resin (see p. [13](#)), a filler, a pigment, a lubricant, and a plasticizer. These materials are mixed and put through rolls at a moderate heat and pressure. The resin softens and amalgamates with the other materials. It hardens upon cooling and is ground to powder. A pre-formed pellet may be made from the powder by pressure; use in this form saves the

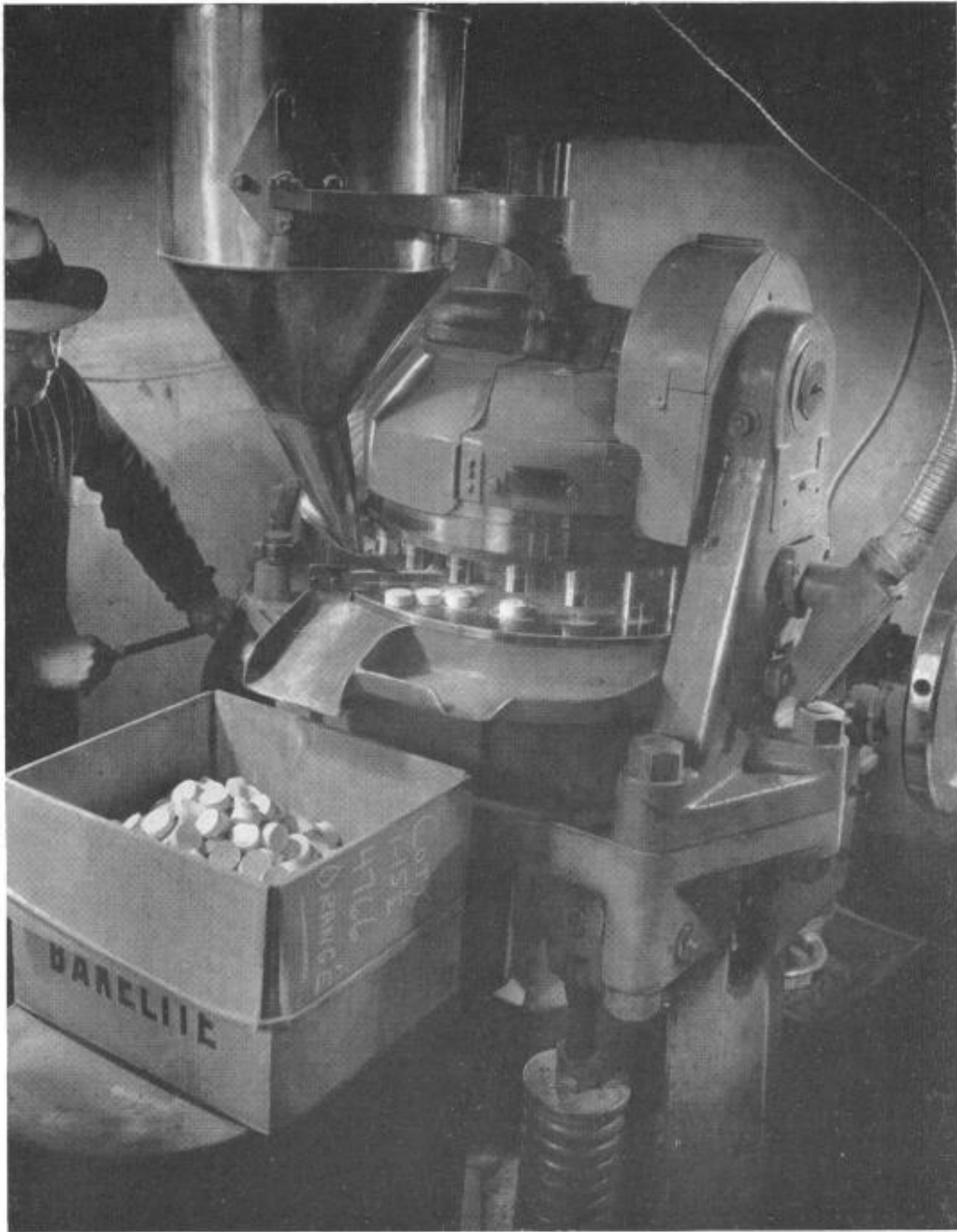
time of the molder when filling the mold, since he is not required to measure the powder.

The proper selection of the filler in a molding powder is important in influencing the quality of the molded article. Fibrous fillers improve the mechanical strength and shock resistance of the finished article. Wood flour is the most widely used filler in tar-acid resins as well as in other thermosetting resins. Pine, spruce, and fir are the principal kinds used, and consideration must be given to the bulk, gum content, color, and the size and shape of the wood particles. Color is the least important since most of the tar-acid resins give brown or black moldings. When the molding must withstand high temperatures, asbestos fiber is used as a filler. In articles requiring high shock resistance, such as golf club heads, a filler of paper pulp is used. Where high electrical insulation and dielectric properties are required, ground mica is used as the filler. Certain inorganic fillers such as powdered slate, gypsum, barium sulphate, calcium sulphate, china clay, zinc oxide, and infusorial earths, are sometimes used. Large proportions of these may be used where hardness is more important than strength, as in phonograph records. Other materials used include rubber, graphite, horn, bone, starch, pumice, and cork.

Coloring matter used may be coal-tar dyes or pigments such as bone black, carbon black, and iron oxides. Pigments are usually more satisfactory, although dyes are sometimes preferred in articles for insulation.

A lubricant is added to the molding mixture to overcome the tendency to stick in the mold. Metallic soaps, stearates, and stearic acid are those most commonly used.

Sometimes a plasticizer is included, its function being to act as a solvent for the resin, thus increasing the flow of the material in the mold. The plasticizer should be one which will become infusible or at least remain solid in the molded article.



PREFORM PRESS MAKING PELLETS FOR USE IN MOLDING.
Source: Bakelite Corporation, 247 Park Avenue, New York, N. Y.



VACUUM CLEANER PARTS OF TAR-ACID RESIN ILLUSTRATING THE INTRICATE MOLDED SHAPES POSSIBLE.

Source: Bakelite Corporation, 247 Park Avenue, New York, N. Y.



RADIO CABINET AND TELEPHONE SET OF MOLDED TAR-ACID RESIN.

Source: Bakelite Corporation, 217 Park Avenue, New York, N. Y.

A typical molding powder or pre-form pellet will contain by weight:

Resin	40 to 50 percent
Filler	35 to 50 percent
Plasticizer	5 percent
Lubricant	1 percent
Pigment	1 percent

The molding of tar-acid resins.

Ordinarily the molds used are made of hardened steel, highly polished. They must stand working pressures of several thousand pounds per square inch. The mold is placed in a hydraulic press, heated by steam, electricity, or gas, and the molding material is placed in the mold. The press is closed and heat and pressure are applied. The temperatures used range between 250° F. and 365° F., and the pressures between 1,000 and 8,000 pounds per square inch. The molding time depends on the shape and size of the article and on the composition of the molding material. As little as one-half minute is required for small objects and as long as 10 minutes for large objects. Average molding time is about 3 minutes. The article is removed from the mold, allowed to cool, and is then trimmed, sanded, filed, or polished. Since the mold is highly polished, the finishing operation is usually needed only to remove the flash. Inserts, such as metal parts (binding posts, electrical contacts, etc.), or inlays of polished metal in name plates, and signs, are often molded in; gear shift knobs are molded over a hollow metal core; rubber inserts are used in castors, electrical plugs, and similar objects.

The molding operation is an art, and has made remarkable progress in recent years. Many articles molded of tar-acid resins are well-known to the public. The automotive industry is the best customer, using such molded parts as gear shift knobs, horn buttons, accelerator pedals, light switches, ignition parts, and distributor heads. Other well-known applications are builders' hardware, electrical switch plates, switches and fixtures, fountain pens, radio parts, telephone parts, handles for stoves, vacuum cleaners, and other appliances, buttons, buckles, costume jewelry, camera cases, radio cabinets, small containers, and hundreds of others.

The importance of tar-acid resins in molded articles is shown by the fact that more than 75 percent of all synthetic resin molded articles made in 1937 used this type of resin as a binder.

Production of tar-acid molding resins.

Domestic production of tar-acid molding powders and pellets was reported to the Tariff Commission by 15 makers in 1937. Most of these firms have specialized in resin development and manufacture. Among the well-known

brands are Bakelite, Durez, Durite, Resinox, Indur, and others (see p. [153](#) for list of trade names).

Statistics of production and sales of tar-acid resins used in molding were collected separately for the first time in 1935. They show a net resin output of about 21,000,000 pounds, with sales of 18,000,000 pounds or about 40 percent of the total tar-acid resins. The average unit value was 17 cents per pound. In 1937 the production of tar-acid resins for molding exceeded 32,000,000 pounds, again about 40 percent of the total. These statistics are based on net resin and do not include fillers, modifiers, pigments, or inert material of any kind.

CAST PHENOLIC RESINS

Process of manufacture.

The production of cast phenolic resins requires pure materials, expensive equipment, and extreme care in the control of the operation. A mixture of phenol and formaldehyde and a catalyst (usually sodium or potassium hydroxide) is charged into a nickel-lined reaction kettle and heated until the water separates and is removed. The reaction is then allowed to proceed to the desired point. Glycerin is added to aid in forming a transparent product. All equipment, including pipe lines, valves, and pumps, is nickel or nickel lined except that used for formaldehyde, which is made of aluminum.

The resin is usually made in 1,000 pound batches, and the reaction cycle ranges from 6 to 18 hours. It is colored with soluble coal-tar dyes and cast into lead molds. These are placed in a heated room and allowed to cure for 3 to 6 days. The resin is removed from the mold with air hammers, and the lead molds are melted.

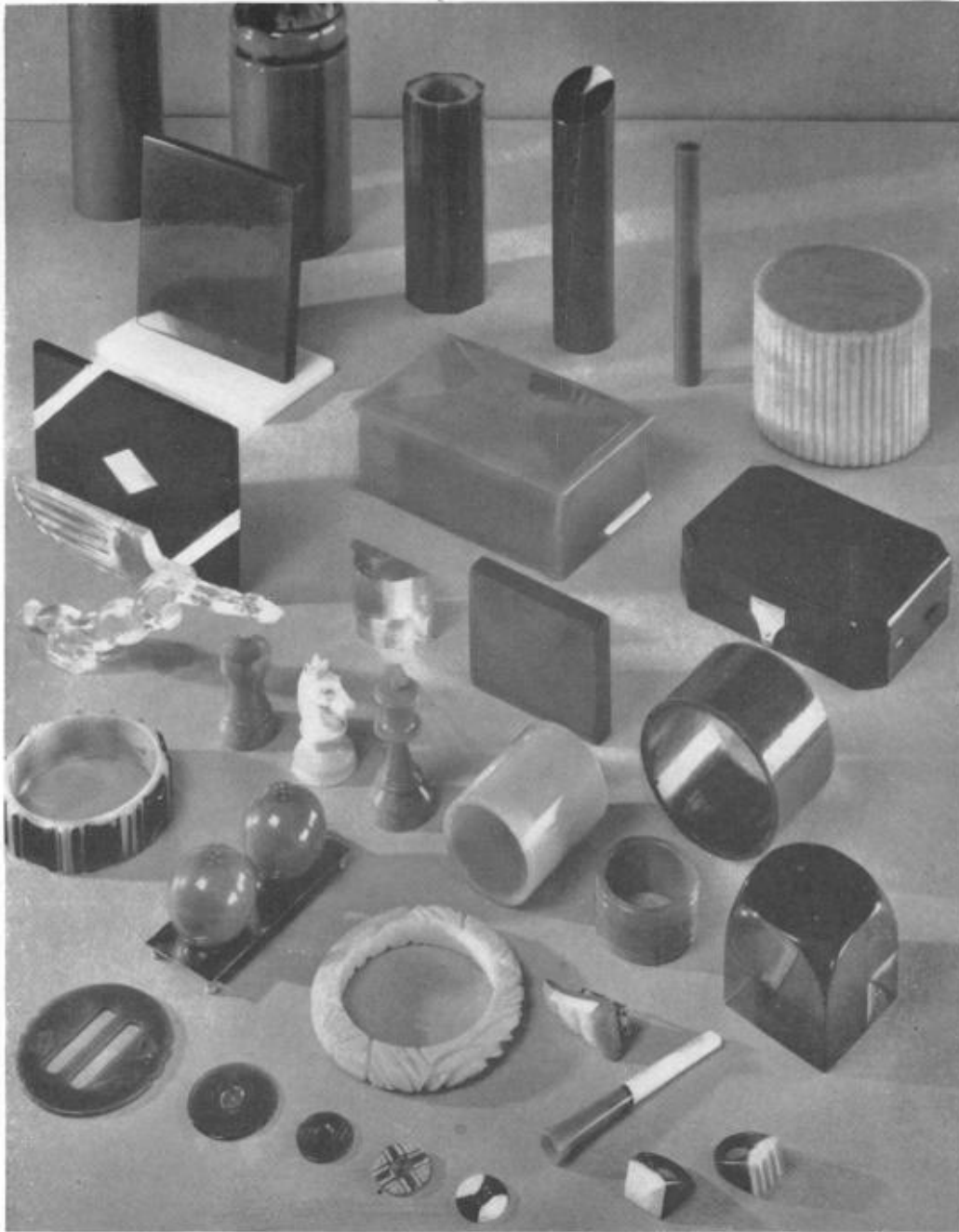
The appearance of the resin may be changed by varying its water content, by the addition of dyes and fillers, and by the addition of other substances to produce some desired effect, such as imitation ivory or marble. The clarity of the resin depends upon its water content—the greater the degree of dehydration the clearer the product. Range of colors is complete, from crystal clear to the darker shades, with any degree of transparency, translucency, or opaqueness.

Casting is in the form of sheets, rods, tubes, or special forms suitable for the production of buckles, jewelry, and other small products. Molds of complicated shape cannot be used, which means that most articles if produced of cast resin must be produced from standard shapes by subsequent working. Recently small radio cabinets have been cast.

Uses.

Cast phenolic resin can be machined in the same manner as hard wood. It must be polished after machining, usually by tumbling with shoe pegs and pumice or with muslin wheels. The smooth finish and low degree of heat conduction give the material a pleasant feel, not cold to the touch as is metal. The coloring is not superficial and therefore does not chip or wear off. Electrical properties are excellent. A slow polymerization continues for some time after fabrication, resulting in slight shrinkage.

Cast phenolic resins are marketed by the producers as rods, sheets, cylinders, and special castings. Standard round rods range from $\frac{3}{8}$ inch to more than 5 inches in diameter. Special rods are available in such forms as square, hexagon, octagon, and fluted. Standard sheets are in sizes from 12 by 24 inches to 36 by 72 inches, and from $\frac{1}{8}$ to 1 inch thick. Stock cylinders are available in a wide range of inside and outside diameters.



CAST PHENOLIC RESINS, STANDARD SHAPES AND SMALL ARTICLES FABRICATED FROM THEM.

Source: Bakelite Corporation, 247 Park Avenue, New York, N. Y.

Stock material is fabricated by a number of firms into an endless variety of articles. Among these are toilet articles such as combs, backs for brushes, cosmetic containers, and trinkets; fittings for automobiles, electrical appliances, furniture, and display fixtures; jewelry, dress ornaments, clock

cases, handbag frames, vanity cases, smokers' articles, signs and advertising specialties, picture frames, handles for cutlery, chessmen, pens, desk penholders, pencils, and many others. Probably the largest consumption is in the making of buttons and buckles.

The cast phenolic resins are odorless, tasteless, nonflammable, resistant to oils and greases, and practically nonbreakable.

Patents and licensing.

The basic patent covering the manufacture of cast phenolic resins is United States Patent No. 1,854,600, issued April 19, 1932, to F. Poliak and A. Ostersetzer, of Vienna, and assigned to Pollopas, Ltd., of London. Many other patents have been granted on variations and modifications of this one. The basic process is also patented in England, France, Germany, and other countries.

United States and Canadian patent rights were purchased by the American Catalin Corporation; German rights by the Interessen Gemeinschaft Industrie A. G. (German I. G.); French rights by Kuhlmann Co., and British rights by the Imperial Chemical Industries. These licensing arrangements limited the licensee to sales in his own and, in some instances, nearby countries.

The American Catalin Corporation has successfully defended the validity of this patent and has licensed a number of domestic manufacturers to produce cast phenolic resins on a royalty basis.

In 1937 there were seven domestic makers of cast phenolic resins located in New Jersey, New York, Massachusetts, and Pennsylvania. These firms produce and market resins under the following trade names: Catalin, Prystal, Joanite, Fiberlon, Phenolin, and Marblette.

Production of cast phenolic resins.

Production was initiated about 1929 by the American Catalin Corporation. The output increased substantially every year from that year through 1933. Statistics of production and sales are not publishable for the years prior to 1934 because they would reveal the operations of individual firms; they are given in table [6](#) for subsequent years.

TABLE 6.—*Cast phenolic resins: United States production and sales, 1934-37*

Year	Production	Sales		
		Quantity	Value	Unit value
	<i>Pounds</i>	<i>Pounds</i>		
1934	4,968,445	4,793,658	\$2,099,035	\$0.44
1935	5,566,621	5,454,490	2,205,879	.40
1936	6,111,632	6,013,855	2,476,619	.41
1937	5,459,654	5,335,746	2,180,620	.41

Source: Dyes and Other Synthetic Organic Chemicals in the United States, U. S. Tariff Commission.

Imports and exports.

The licensing agreements, as outlined above, provide for the allocation of markets for cast phenolic resins. Because of this arrangement there are little or no imports and exports of this material.

TAR-ACID RESINS FOR LAMINATING

By laminating is meant the impregnation of sheets of paper, fiber, or cloth with a solution of synthetic resin and the building up of these layers into sheets of reinforced synthetic resin of various thicknesses. When a tar-acid resin is used the paper or cloth is immersed in or coated with a solution of the B-stage resin, dried, and layers of the material are compressed and consolidated, under heat and pressure to form sheets, rods, tubes, blocks, and other forms, in the infusible C-stage.

The coating of sheets of paper with solutions of natural resin and the compacting of these sheets by heat and pressure is an old practice, especially for electrical uses. Shellac and copal have been widely used and yield a laminated board of good electrical and mechanical properties when used at temperatures under 70° C. Above 70° C. the resin softens and the desirable properties are lost. Since temperatures above 70° C. are not uncommon in electrical equipment, the limitations of these natural resins in this use can readily be seen. The use of tar-acid resins to impregnate insulation material removes the temperature limitation and otherwise improves the product;

insulators so made are widely used in all sorts of electrical and radio equipment.

Uses of tar-acid resin laminated products.

Laminated sheets of tar-acid resin are made with paper, canvas, duck, linen, pulpboard, vulcanized fiber, plywood, and other materials. Paper is the material generally used for electrical insulation, although cloth is sometimes used when greater strength is needed. Canvas is used where maximum strength is required, as in gears for automobiles and industrial machinery. Impregnated linen is adapted to punched parts and small gears.

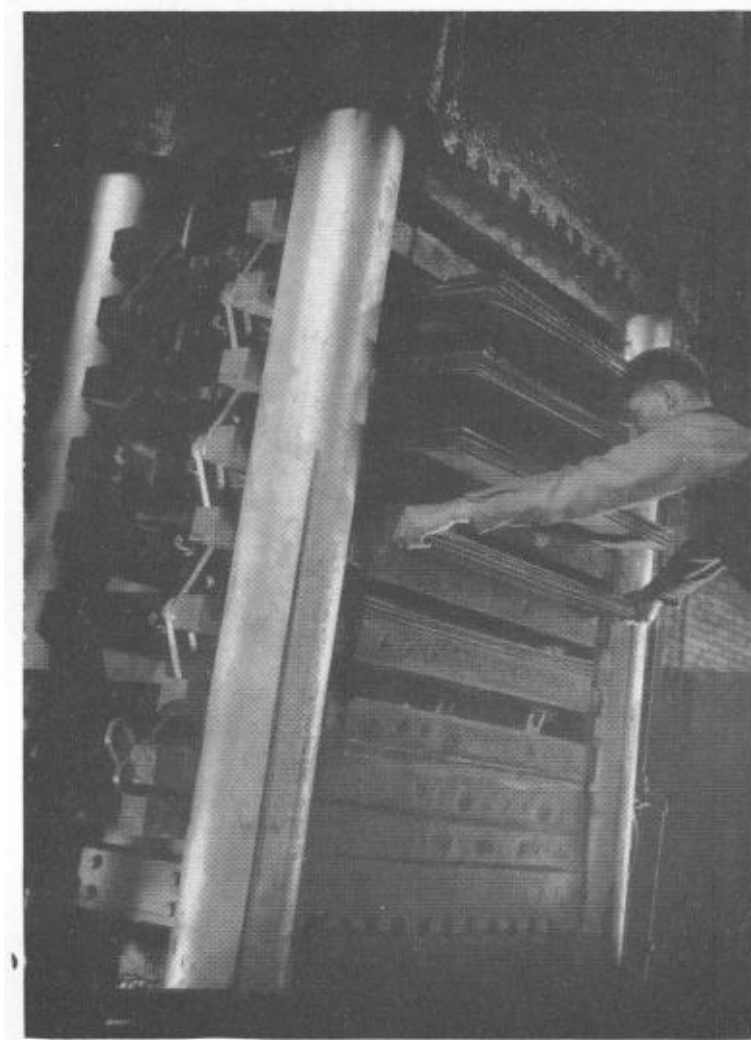
These laminated materials are uniformly dense, tough, resilient and light in weight. They are nonabsorptive, have low thermal conductivity, and a low coefficient of expansion. Their dielectric strength is excellent and chemically they are inert to oils, brine, most acids, weak alkalies, and many solvents. Structurally they are strong under tension, compression, flexion, or impact; they are easy to machine and are sound absorbing.

Gears made of laminated canvas are widely used; they are silent and outwear those made of metal. The development of such gears was brought about by the demand for a positive drive without the clash and clatter resulting from metal to metal contact. The laminated gear absorbs vibrations, eliminates noise, and reduces wear. The laminated material is one-seventh the weight of brass, one-sixth the weight of steel, one-fifth the weight of cast iron and one-half the weight of aluminum. Laminated gear blanks may be cut on automatic machines into helical, spur, bevel, or worm gears.

Timing gears in automobiles are frequently of this type; they require no adjustment and seldom need replacement during the life of the motor. The light weight of the material reduces to a minimum flywheel effect on the camshaft. Where lubrication is difficult a graphite impregnated blank may be used.

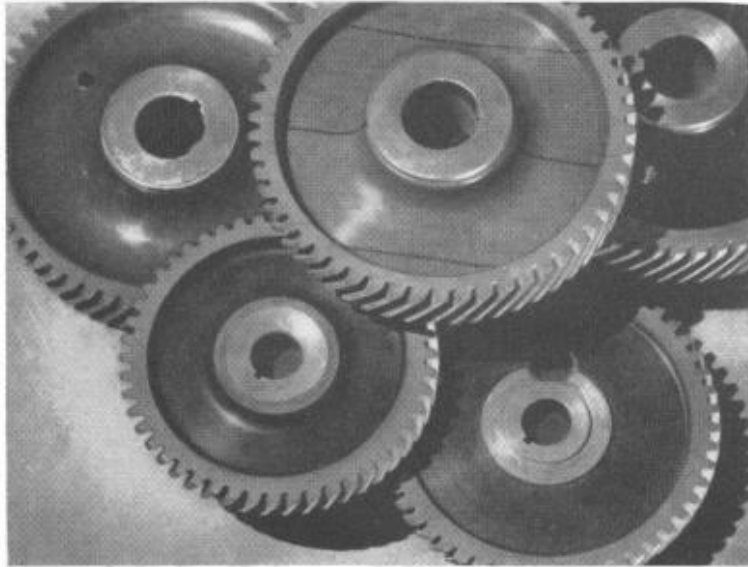
Bearings made from laminated fabric are successfully used in heavy rolling mills where they reduce replacement costs and decrease power consumption. The laminated material possesses strength, smooth surface, density, good load carrying capacity, high impact resistance, nonscoring properties, and is practically frictionless. Power consumption is said to be reduced as much as 40 to 60 percent of that of metal bearings and the life of the laminated bearing has

been as much as 10 times that of the metal ones. It replaces Babbitt metal, brass, bronze, white metal, gun metal, or lignum vitae in this application.



LAMINATING SHEET PRESS.

Source: Bakelite Corporation, 247 Park Avenue, New York, N. Y.



GEARS MADE OF LAMINATED TAR-ACID RESIN.

Source: Bakelite Corporation, 247 Park Avenue, New York, N. Y.



COCKTAIL LOUNGE USING TAR-ACID LAMINATED DECORATIVE MATERIAL.

Source: Bakelite Corporation, 247 Park Avenue, New York, N. Y.

In decorative uses, laminated materials have made remarkable progress in recent years. In this application the material made from laminated paper is veneered on wood or fiber board, and the surface is so durable that refinishing is probably not necessary during the life of the equipment. Table tops for

public rooms such as restaurants, cafeterias, and bars are widely used because of the beautiful designs obtainable and because the material is not discolored by lighted cigarettes, alcohol or other liquids, and does not chip or crack. Laminated sheets are used for bathroom and kitchen walls, doors, window sills, store and theater fronts, lobby walls in hotel and office buildings, and counter tops in banks and post offices. The liner *Queen Mary* is equipped with panels of this material as is also the new Library of Congress Annex. Most of the leading hotels have installed bar and cocktail lounges of laminated materials because of the range of color and the ease with which novel designs may be carried out.

Almost any solid color, design, or imitation of another material may be given the laminated sheet simply by printing it upon the top sheet of paper used in the impregnated assembly. Thus a beautiful piece of walnut or mahogany may be photographed, inexpensively reproduced upon paper, and the finished laminated sheet will closely imitate the polished wood. The combination of beauty with long life should permit the widespread use of this type of material in all sorts of building and equipment. It has been suggested as a possibility in automobile body construction.

Other important uses are in trim and door strips for mechanical refrigerators, in cafeteria trays, buckets and special containers, tires for factory trucks, textile spools, miners' safety helmets, gaskets, valve discs and rings for pumps, pulleys, besides many others.

Production of tar-acid resins for laminating.

Statistics of production and sales of synthetic resins for laminating were not separately compiled prior to 1935. Since that year the resins made from cresylic acid have been used to the greatest extent in laminating, followed by those made from phenol. Tar-acid resins reported as "used in paints, varnishes, and lacquers" may include appreciable quantities of resin varnishes used for laminating. The total production and sale in 1937 of tar-acid resins used in laminating, therefore, would be the sum of the 20 percent of the total (see table [3](#)) reported for laminating plus some part of the 25 percent reported for surface coatings.

Domestic producers of tar-acid resins for laminating are located in Delaware, New Jersey, New York, Illinois, Massachusetts, and Pennsylvania. The makers of the laminated materials are located in Delaware, New Jersey,

New York, Ohio, Illinois, Pennsylvania, Indiana, and Connecticut. Their products are marketed under a number of trade names, including Micarta, Dilecto, Celoron, Formica, Textolite, Phenolite, Insurok, Spauldite, Synthane and Phenol Fibre.

Imports into the United States.

There has been practically no importation of synthetic resins for laminating. Imports of laminated products (rods, tubes, blocks, strips, blanks, or other forms) of which synthetic resin is the chief binding agent totaled only 215 pounds, valued at \$612 in 1931 (principally from the United Kingdom); 13 pounds, valued at \$71 in 1932; none in 1933 and 1934; 609 pounds, valued at \$579 in 1935 from Canada, Germany, and the Netherlands; and 3,260 pounds, valued at \$9,468 in 1936 from Austria, Germany, and the United Kingdom.

Exports from the United States.

Exports of phenolic or other synthetic resins for laminating and of laminated articles are not separately recorded in official statistics. It is known that appreciable quantities of laminated articles are exported to Canada, England, and other countries.

TAR-ACID RESINS FOR SURFACE COATINGS

Synthetic resins are widely used for surface coatings, chiefly because of the ease with which new types can be produced to meet special requirements and because of their uniformity. Tar-acid resin coatings may be varied in composition and properties to meet a particular purpose. Possible variations depend on the type or mixture of tar acid used (phenol, cresols, xylenols, tertiary amyl phenol, tertiary butyl phenol, phenyl phenol), whether the condensation takes place in the presence of an acid or an alkali, and on the proportion of formaldehyde used. The resin so formed may be modified with natural resins, synthetic resins of the alkyd type, fatty acids, or other materials. The almost endless opportunities for different types can, therefore, readily be appreciated.

Types of resin used and the resultant coatings.

The tar-acid resins used in varnishes and other surface coatings are usually oil-soluble types. They may be divided into three general classes: (1) Phenol-formaldehyde condensation products rendered oil-soluble by chemical combination or physical dispersion in other materials, such as rosin and copal; (2) condensation products made from tar acids other than simple phenol, which are themselves soluble in drying oils and thinners; and (3) products from the condensation of the substituted phenols and formaldehyde. These three classes of oil-soluble tar-acid resins differ widely in their chemical and physical properties and in their functions. The first group are usually called modified phenolic resins, the second group are referred to as unmodified or 100-percent soluble, and the third group are known as substituted phenolic resins.

The unmodified resins are extensively used in long-oil tung varnishes, to which they impart greater drying speed, durability, and resistance to alkalis and gases. The modified types impart the same properties to tung oil varnishes but to a lesser extent. In addition the modified types possess considerable hardness so that greater gloss and fullness are obtained. Modifiers are either drying oils or natural resins; tung oil is the most widely used oil and rosin the principal natural resin. Substituted phenols such as para tertiary amyl phenol and para tertiary butyl phenol may be used in place of simple phenol; while these are relatively high priced components, the resins made therefrom have increased in recent years to an appreciable volume because of their improved properties.

Other synthetic resins, such as those of the alkyd, petroleum, urea, and vinyl types, are sometimes incorporated with the phenolics in the same surface coating to obtain some desired property. The addition of a plasticizer, such as tricresyl phosphate or dibutyl phthalate, improves the flexibility of the film.

Spirit varnishes, in which the synthetic resin is dissolved in a solvent, are also available. In this type the soluble fusible resin (form A) is dissolved in an organic solvent such as acetone or the various alcohols, and conversion of the resin to the insoluble, infusible state (form C) is effected by baking the film.

Coatings made from tar-acid resins are widely used in so-called 4-hour enamels and varnishes, for both interior and exterior application. They are also used in the manufacture of linoleum, artificial leather, adhesives, and printing inks. When incorporated with nitrocellulose or cellulose acetate lacquers they improve the adhesion, luster, and resistance to alkalis.

Production in the United States.

In 1937 the output of tar-acid resins for surface coatings exceeded 20 million pounds (net resin). Those from phenol and the substituted phenols accounted for a very large part of the total. They were followed by resins from cresylic acids and the xylenols in that order.

In 1937 there were about 20 domestic makers of this type of synthetic resin, with factories located in California, Connecticut, Illinois, Indiana, New Jersey, New York, Massachusetts, Michigan, Missouri, Ohio, Pennsylvania, and Rhode Island.

Imports into and exports from the United States.

Imports of oil-soluble phenolic resins have been negligible. This is due, in part, to licenses and agreements between certain domestic and foreign makers, to the remarkable advancement and pioneering work done in this country, to the holding of many basic patents by Americans, and to the relatively high duty on imports.

Exports of these products, usually in the form of enamels, varnishes, and lacquers, have been appreciable and are probably increasing each year. Official statistics are not reported separately.

TAR-ACID RESINS IN ADHESIVES

A comparatively new use for tar-acid resins is in the manufacture of wood adhesives. Ordinary vegetable and animal glues have long been used, although their deficiencies in certain characteristics are well known. These include (a) their inability to produce uniform products, (b) the tendency of most alkaline glues to stain wood, (c) the bad effects of moisture on them, and of bacteria and fungi in the case of animal glues. The tar-acid resins have none of these objectionable qualities. Being chemically inert they are free from attack by fungi and bacteria. Moisture does not affect them, and they do not stain wood.

Three types of resins are used as wood adhesives, principally in bonding plywoods and veneers: (1) Hot press liquid, (2) cold press liquid, and (3) resin film. Furniture, radio cabinets, games, and building products constructed from plywoods bonded with resins can be shipped to tropical countries, the bond not being affected by extreme climatic conditions.

These resin adhesives are more expensive than the usual animal and vegetable glues, a factor which has limited their application. Their advantages may, however, open up to resin bonded plywoods uses in which the more ordinary types are not satisfactory.

TAR-ACID RESINS FOR OTHER USES

The application of tar-acid resins in casting, molding, laminating, surface coatings, and adhesives has been described. There are many other uses, but most of them approach the types of application dealt with.

Impregnation of all sorts of materials with tar-acid resins is an increasing use; such applications are in fabrics for aircraft, crease resistant textiles, wood, asbestos, concrete, and electrical coils. Wood with resin forced into the fiber under pressure is used for furniture, flooring, heads for golf clubs, and handles for utensils. Resin is used as a binder in the manufacture of brake linings for automobiles, as well as in the manufacture of abrasive and grinding wheels.

An interesting application is in the construction of corrosion-resistant chemical plant equipment. In 1922 the German firm of Saureschutz Gesellschaft was incorporated to fabricate equipment composed of a special acid-resisting type of phenolic resin and asbestos. Sometime later its manufacture was started in the United States. All sorts of industrial plant equipment is now available, including cylindrical and rectangular tanks up to 9 feet in diameter and 12 feet high, piping for corrosive liquids and gases, valves, pumps, fans and ventilators, filter press plates and frames, buckets, dippers, etc.

Another new use is for making matrices in which to mold rubber printing plates. Such plates are used at present chiefly in printing cotton and paper bags but extensive experimentation promises to broaden their use. The matrix is made of fiber board of very open structure impregnated with tar-acid resin in the process of manufacture.

4. ALKYD RESINS

Description and uses.

The alkyd resins, used principally in paints, varnishes, and lacquers, are a group of condensation products synthesized by reacting polyhydric alcohols, such as glycerin and the glycols, with dibasic organic acids, such as phthalic, maleic, succinic, and sebacic. The condensation product is almost always modified to give properties to the resin desirable or essential to the specific application contemplated. The modifying agent may be a drying, semidrying, or nondrying oil; the fatty acid of an oil; a natural resin, such as rosin; a synthetic resin of the tar-acid group or of the urea-formaldehyde type; or other substance. Up to the present time unmodified alkyd resins have not been commercially important.

A wide variety of types is obtained by the use of different materials and different modifiers. The variations begin with the dibasic acid used, and with the polyhydric alcohol used. The modifications possible are practically endless, and almost any fixed oil or the corresponding fatty acid, and most of the natural or synthetic resins may be used. The importance of the modifier is shown by the proportion used in most alkyd resins. On the average, approximately 50 percent of the total weight of the drying and semidrying alkyd resin products is modifier, 30 percent dibasic acid, and 20 percent polyhydric alcohol. The proportions will, of course, vary with individual types. Certain types on the market contain only 25 percent modifier while others have as much as 75 percent.

In a new industry such as this, rapid changes in types and applications must be expected. Extensive research is being carried on by various groups. The raw material makers are seeking cheaper products or those with special properties; the resin makers are investigating an endless number of modifications, and the makers of surface coatings are testing most of the new types offered.

Development and patents.

Probably the earliest record of research leading to the development of the alkyds was that of van Bemmelen, who reported in a German technical journal in 1856 the sirupy products obtained by heating together succinic acid and glycerin or citric acid and glycerin. The first investigation of the phthalic anhydride-glycerin resins was recorded in 1901.^[3] Watson Smith, while engaged in research on phthalein dyes, obtained a transparent, highly refractive resinlike substance when glycerin and phthalic anhydride were heated together. Smith recommended the product as a cement for ceramic wares.

During the period 1910-16 the research laboratories of the General Electric Co., engaged in research on a synthetic resin from glycerin and phthalic anhydride. As a result of these studies numerous patents were granted for this type of resin to which the trade name Glyptal was applied. Intensive research was carried on by several firms, many variations were developed, and literally hundreds of patents were granted.

The paint and varnish industry has been undergoing radical readjustment. Methods and natural products, which for decades or centuries had changed very little, are giving way to synthetic creations of our laboratories. The first important departure from the traditional practices was the development of nitrocellulose lacquers. The commercial application of the alkyd resins followed, and their use is increasing rapidly. Because this development is still comparatively young, the large number of modifications offered has confused the coating manufacturer. It is probable that many of the synthetic products now being marketed have no special technical or economic justification and that they will in time lose out in competition with better products known at present, or still to be developed.

United States Patent No. 1,893,873, dated January 10, 1933, granted to R. H. Kienle and assigned to the General Electric Co., was considered one of the basic patents in this field. Early in 1936 it was declared invalid in a suit claiming infringement brought against the Paramet Chemical Co. of Brooklyn, N. Y. The decision in this case seems to have opened the glycerin-phthalic anhydride resins to a large number of manufacturers.

Among the principal brands of alkyd resins now on the domestic market are Beckosol, Dulux, Esterol, Glyptal, Rezyl, and Teglac. Each of these trade names identifies a series of products.

Classification of alkyd resins.

A number of classifications of the alkyd resins are possible and practical. Since by far the most important applications are in surface coatings, and their use in molding compositions is relatively unimportant, it seems advisable at this time to emphasize the more important use. For the purpose of this survey the following classification is used:

- (1) Drying alkyd resins.
 - (a) Unmodified.
 - (b) Modified with natural materials.
 - (c) Modified with other synthetic resins.
 - (d) Modified with other synthetic resins and oil extended.
- (2) Semidrying alkyd resins.
- (3) Nondrying alkyd resins.
- (4) Miscellaneous modified alkyd resins.
- (5) Alkyd resins in water dispersion.
- (6) Alkyd resins in molding compositions.

At least 75 percent of the alkyd resin finishes used at present are of the drying type and about 15 percent of the nondrying type.

Unmodified drying alkyd resins.—This class of alkyd resins consists of a series of compounds made from polyhydric alcohols, polybasic acids, and fatty acids in chemical combination. The alcohol is usually glycerin, and the polybasic acid largely phthalic anhydride or acid, although others, such as maleic anhydride (acid) are increasing rapidly in importance. The fatty acid or oil used may be linseed, tung, perilla, hempseed, soybean, sunflower, safflower, or other drying oil. It is believed that tung oil and perilla oil are the most important at this time.

Unmodified drying alkyd resins are characterized by excellent durability but limited resistance to water in air-dried finishes. Both in air-dried and in baked finishes they are outstanding as to flexibility, quick drying, long

luster life, and permanent adhesion. Their principal uses are in finishes for interior walls and woodwork, automobiles, coatings on steel such as for refrigerators, railway equipment, bridges, advertising signs, and lithographed containers. In these applications the products of this type compete with nitrocellulose lacquers and the older types of varnishes and paints. While the initial cost is higher, greater durability is obtained together with faster drying, flexibility, and hardness.

Probably the largest field for surface coatings is outdoor wood finishes. Several attempts have been made to adapt pure alkyd finishes to this use but with limited success because the hard and non-porous finish does not permit the escape of moisture contained in the wood and the pressure developed from vaporization of the moisture by the sun's rays tends to lift the coating from the wood surface. Recently it has been found practicable to incorporate from 15 to 20 percent alkyd resins in conventional types of outdoor paints for wood. Here the use of alkyds has contributed greater durability and retention of fresh appearance over a longer period. Paints of this type are now on the retail market.

Drying alkyd resins modified with natural materials.—This type of alkyd resin is modified principally with natural resins, such as rosin, damar, mastic, shellac, or copal. The use of these natural resins imparts hardness to the resin but shortens its durability. They make the product less expensive, permit easier incorporation of the drying oil, and in some instances increase the water resistance.

Their principal application is to modify nitrocellulose lacquers and lacquer sealers, in order to impart gloss, hardness, and easy sanding. It has been said that the commercial production of drying alkyds modified with natural resins was as important a development in the surface coating industry as the discovery of the alkyds themselves.

Drying alkyd resins modified with other synthetic resins.—Drying alkyd resins may be modified with tar-acid formaldehyde resins, tar-acid furfural resins, urea-formaldehyde resins, petroleum resins, and the coumarone and indene resins.

Modification with tar-acid resins gives a quicker setting, harder drying finish with a higher gloss. Alkyd resins so modified are adapted to both air-drying and baked undercoats and finishes; they have good durability and

adhesion and good resistance to grease, oils, alcohol and abrasion. For some uses the tar-acid resin modification gives better qualities than either component possesses alone, but in light colored finishes it has a tendency to cause the finish to yellow. Coatings made of drying alkyd resins modified with tar acid resins are widely used on automobile chassis, fenders, and bodies, machinery coatings, steel fixtures and toys; they are especially suitable for primers, undercoats, and finishes on metal.

Modification with urea resins produces baked-finish coatings. As much as 40 percent of the urea resin is incorporated. It makes possible coatings with a full range of permanent colors and improves their hardness and mar-proofness, whereas without the ureas the combination of color range with hardness had been difficult to obtain. The urea resin modified alkyds find use on metal surfaces of articles which must stand rough handling, such as toys, furniture, and motors.

Modification with petroleum resins produces air-dried finishes. For industrial use on metal they give coatings with better adhesion, dispersion of pigments, and resistance to acids, alkalies, and moisture at a lower cost than is obtained by ester gum or tar-acid resin modification. The petroleum resin modification minimizes skinning and improves the luster and the flow.

Drying alkyd resins modified with other synthetic resins and oil extended.—Excellent water resistance and versatility are the characteristics of finishes made of alkyd resins modified with other synthetic resins (usually tar-acid) and oil extended. The incorporation of drying oils gives a low cost finish with better compatibility and brushing and with the combined properties of a quick-setting varnish and an alkyd resin. Although not so durable or quick setting as the unmodified finishes, they have better water resistance. These finishes may be brushed or sprayed, air-dried or baked. They have wide industrial and architectural uses.

Semidrying alkyd resins.—Cottonseed oil is the principal modifier in semidrying alkyd resins. Alkyd resins of this type are used in finishes requiring maximum gloss and color retention. When baked on metal at high temperatures they show no tendency to wrinkle. They are used as reinforcing agents to increase flexibility and durability, and to plasticize other finishes.

Nondrying alkyd resins.—The nondrying or nonoxidizing alkyd resins are those containing a nondrying oil, such as castor oil or coconut oil, or the fatty acid of a nondrying oil, such as stearic, palmitic, or oleic acid. Nondrying oils make the resin less sensitive to heat hardening and impart greater flexibility. These resins are used principally as plasticizers in nitrocellulose lacquers. In this use they have the advantage of better retention of plasticizing efficiency than other plasticizers, many of which are lost by evaporation, migration, absorption, or oxidation. These modified nitrocellulose lacquers, either clear or pigmented, are used for coating wood, composition board, cloth, paper, rubber, leather, and similar surfaces.

Miscellaneous modified alkyd resins.—This group includes alkyd resins modified with materials other than those already discussed. To date (1938) there has been little, if any, commercial production of such resins. There are many modifiers which have been suggested and which might be used but for the fact that they are too expensive. Among these are butyl alcohol and benzoic acid.

Alkyd resins in water dispersion.—Emulsions of alkyd resins in water are now available for use in clear and pigmented coatings. These are sold in the form of paste containing 40 to 50 percent solids and are diluted with water at the time of application. They are especially suitable for coating porous surfaces, such as brick, concrete, plaster, stucco, and masonry of all kinds. They are applied by brushing or spraying and they combine the ease of application of water paints with the durability, washability, and hardness of oil paints. They dry quickly, and the dried film cannot again be dissolved or suspended in water; the coating can therefore be washed or, after several weeks, scrubbed with cleansers. Compared with oil paints, they give better coverage, are easier to apply, and cost appreciably less. Compared with other types of water paints, such as kalsomine, they give a glossier coating of greater durability and superior appearance; they seal porous surfaces better; their covering capacity is greater; and their applied cost is slightly less per square yard of surface.

Coatings of this type may be applied directly over fresh plaster without a sizing coat, since they allow the curing of the plaster to continue. The usual paint pigments may be incorporated.

A special use of the water dispersed alkyds is on asphalt or tar since they are nonbleeding in the solvents of these materials. This quality permits their use for traffic and zone markers on streets.

Alkyd resins in molding compositions and other uses.—The alkyd resins are much less important as binders in molded articles than in coatings and finishes. Conversion of the resin to the insoluble infusible form is extremely slow, requiring days as compared with minutes for the tar-acid and urea resins.

The alkyds are used as binders for flake, powder, and split mica to produce insulation material of high electrical strength. Other uses are in the production of linoleums; gaskets; brake linings; laminated fabric, paper, and cardboard sheets; printing inks; and coated paper, textiles, and leathers.

Pigments and solvents in alkyd finishes.

Since the alkyd resins are largely used in surface coatings and finishes and since this application in this field is producing great changes in the industry, it is appropriate to consider the effect of their use on other materials.

The average alkyd resin consists of 50 percent glycerol phthalate modified with 50 percent oil, fatty acid, natural resin, or synthetic resin. The alkyd and modifier are dissolved in a solvent, usually a coal-tar light oil such as toluol, or xylol, or a petroleum solvent, and pigmented with titanium dioxide or other pigment. Highly basic pigments such as zinc oxide, carbonate white lead, whiting and aluminum hydrate (all important pigments in the conventional types of finishes) are not used in alkyd finishes.

Production in the United States.

Prior to 1929, the domestic production of resins from phthalic anhydride was confined largely to one maker. The quantities produced were relatively small. In 1929 there were three producers, the volume of whose production exceeded one million pounds for the first time. Beginning with 1933 the Tariff Commission collected and compiled production and sales statistics for these resins. They are shown in table [7](#).

TABLE 7.—*Alkyd resins from phthalic and maleic anhydride: United States production and sales, 1933-37*

Year	Number of makers	Production	Sales		
			Quantity	Value	Unit value
		<i>Pounds</i>	<i>Pounds</i>		
1933	6	9,930,705	3,654,854	\$673,890	\$0.18
1934	10	15,219,247	7,084,602	1,022,436	.14
1935	15	34,312,713	15,836,942	3,482,078	.22
1936 ¹	31	46,952,452	24,252,535	5,312,121	.22
1937 ¹	39	61,254,019	34,738,295	6,864,194	.20

¹ Includes resins from maleic anhydride.

Source: Dyes and Other Synthetic Organic Chemicals in the United States, U. S. Tariff Commission.

In 1933 there were 6 makers of resins from phthalic anhydride, in 1935 there were 15, and in 1937 there were 35. The 1937 output of alkyd resins from phthalic anhydride was 58,450,032 pounds net resin, with sales of 32,583,307 pounds valued at \$6,446,011. Producing plants are well scattered through northern and eastern United States. In 1936 fewer than one-third of the makers accounted for about 90 percent of the output.

The domestic production of resins from maleic anhydride was reported for the first time in 1933. The output in that year consisted of experimental quantities produced by two firms. A small increase in production occurred in 1934 when another maker began operation. In 1936 there were eight producers and the output was many times that of 1934. In 1937 there were 12 makers of these resins with an output of 2,803,987 pounds and sales of 2,154,988 pounds, valued at \$418,183. It is the opinion of some persons in the industry that in volume of production and sales the resins from maleic anhydride will in the near future approach that obtained from phthalic anhydride.

Imports into and exports from the United States.

No imports of alkyd resins have been recorded in official statistics.

Exports of alkyd resin coatings and finishes are not separately shown, but data collected from the several producers show that appreciable quantities were exported in recent years, principally to Central and South American countries.

5. UREA RESINS

One of the most important series of thermosetting resins is the group made by condensing urea and formaldehyde. As early as 1897 it was discovered that an amorphous condensation product was obtained from the reaction of urea and formaldehyde. The clear glass-like mass obtained led to considerable research work toward the development of a substitute for glass. It was found, however, that the resin obtained absorbed moisture, resulting in a dimming of its luster, and that on standing for a time, the condensation continued producing cracks, fissures, and disfigurements in the molded article. In 1926 a successful commercial product was developed in England by the use of thiourea. Cost of production, however, was high. The addition of thiourea gave the product greater strength and water resistance than that obtained with urea alone but retarded the rate of cure. Also the sulphur present attacked steel molds, which necessitated the use of expensive chromium plated or stainless steel molds.

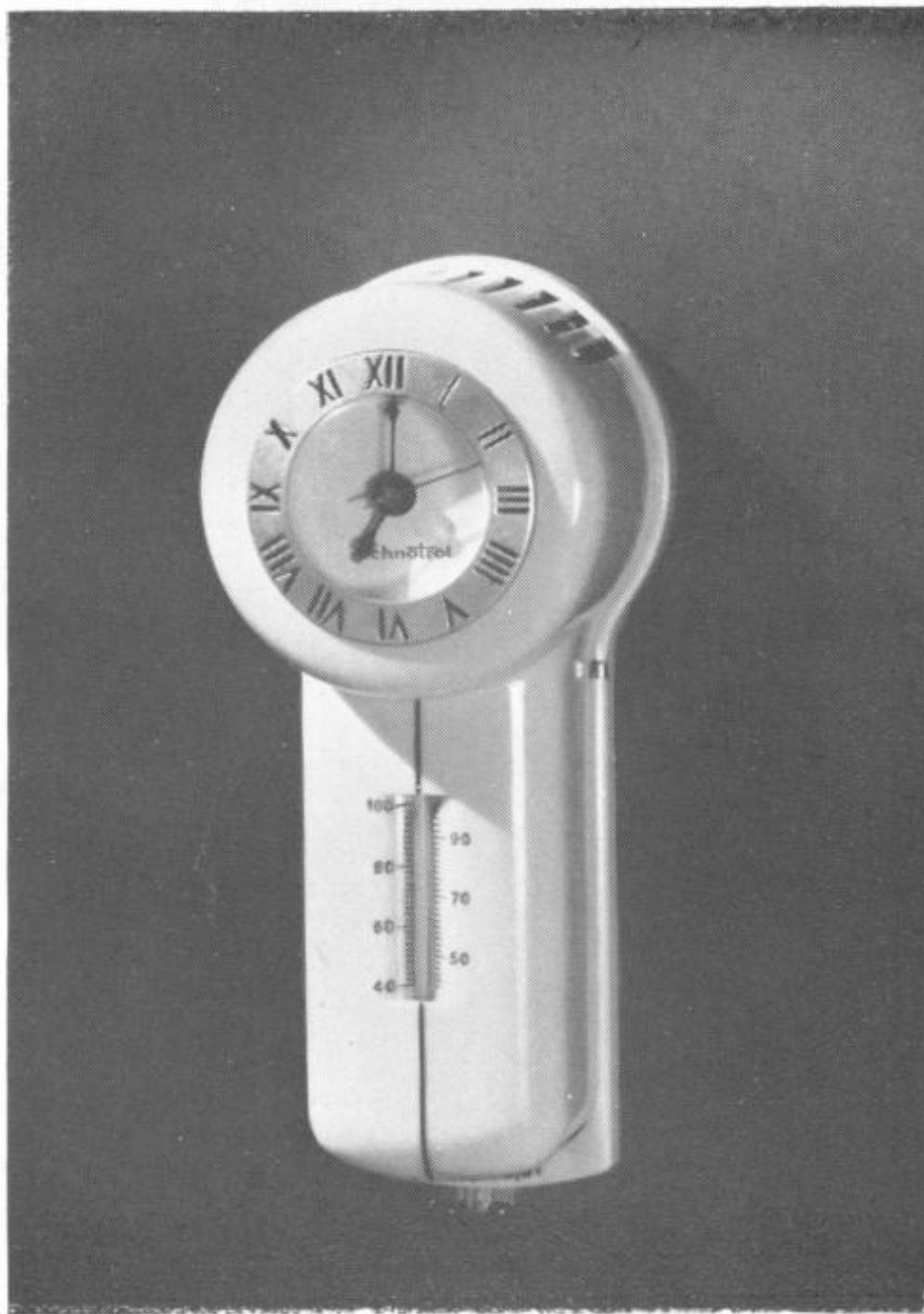
About 1929 the first successful straight urea product was perfected in the United States. It was found that a filler, such as highly refined alpha cellulose, minimized the stresses. The filler (as much as 30 to 40 percent is usually incorporated), destroys the transparency but permits the manufacture of translucent articles in a wide range of color. Many of the colors possible with the urea resins, particularly the light shades, cannot at present be obtained in molded tar-acid resins.

An interesting fact concerning these resins is that they are produced indirectly from four gases: Ammonia, carbon dioxide, hydrogen, and carbon monoxide. Ammonia and carbon dioxide react to form urea, and hydrogen and carbon monoxide yield methyl alcohol which is converted to formaldehyde.

Description and uses.

The urea resins are outstanding largely because of their brilliancy and depth of color, properties not readily obtained in other thermosetting resins. Being odorless and tasteless and completely resistant to oils and greases, they are adapted to use in the manufacture of cosmetic containers. Concentrated acids and alkalies attack the resin. The electrical properties of the urea resins compare favorably with those of the tar-acid resins. They have a lower power factor at

high-frequencies than the tar-acid resins, and are replacing, to some extent, established materials in heavy duty electrical equipment where “tracking” causes trouble. Molded articles made from urea resins are resilient but not unbreakable.



THERMOSTAT CASE OF MOLDED UREA RESIN.

Source: Plaskon Company, Inc., 2112 Sylvan Avenue, Toledo, Ohio.



SCALES CASE OF MOLDED UREA RESIN.

Source: Plaskon Company, Inc., 2112 Sylvan Avenue, Toledo, Ohio.

The important uses of the urea resins are dictated by their pleasing color and appearance. In 1935 the largest outlets were in buttons and buckles, in bottle closures, and in such premium items as biscuit cutters and cereal bowls distributed by a large food manufacturer. Tableware, bathroom fixtures, all sorts of containers and closures, housings for radios, clocks, scales, and other

machines for retail stores, and light-colored wall plates and switches, knobs, handles, and trim on dash panels of automobiles, and handles and trimming on gas and electric ranges were among the widespread applications of the urea resins. In 1938 probably the fastest growing outlet for urea resins is in lighting equipment. Use in packaging, in closures, and in housings, is also increasing. Tableware, the principal outlet for a number of years, is declining markedly.

A comparatively new use is in shades and reflectors, replacing opal glass. The unpigmented resin is highly translucent and gives high light transmission and an exceptional degree of light diffusion. These properties, together with low unit manufacturing costs, reduced shipping costs, and resistance to breakage make the urea resins an ideal material for all sorts of shades and reflectors for direct and indirect lighting fixtures. Many of the shades used in railway cars are of this material. The resin is available in degrees of denseness and opacity to give particular ratios of reflection and transmission. Reflectors as large as 28 inches in diameter are on the market.

Although molded articles are the large outlets for the urea resins, other applications are of increasing importance. Sirups used to impregnate paper and cloth are used in laminating and the resulting materials have unusual decorative possibilities. The surface is hard and durable and the wide range of colors possible permits very attractive applications. The urea resins are used both as the principal binding material for laminated sheets or on the surface laminae of sheets where tar-acid resins are used as the chief binder. The latter practice permits a wide color range in decorative materials without loss of strength or other characteristics of the tar-acid resins. In 1937 there were seven makers, and their production of urea resins for laminating accounted for slightly less than 10 percent of the total of all urea resins.

Another application of urea resins which has grown rapidly in the past 2 years is in combination with alkyd resins in surface coatings. In 1937 there were three makers, and their output of urea resins for coatings amounted to more than 10 percent of the total production of urea. Until recently the use of urea resins in paints and varnishes was discouraged by their insolubility in organic solvents and their instability. On the other hand, their lack of color, their high transparency, their hardness, and their freedom from after-yellowing were desirable characteristics. The development of methods for preparing condensates, which overcome the undesirable properties, has made available resins for this use. They are marketed as water-white viscous solutions in a mixture of organic solvents and are intended for use in baking finishes. They cannot be used alone because the cured resin is extremely hard and brittle and

lacks adhesion. When combined with more elastic film-forming materials such as drying or nondrying oil alkyd resins, they produce coatings that are mar-proof, resistant to alcohol, grease, oil, and fruit acids, and available in a full range of colors. Applications are in metal furniture finishes, toys, refrigerators, can, and drum coatings.

The value of urea resins as adhesives has been known for many years and one of the first patents issued for such use was United States Patent No. 1,355,834 granted in 1920. Commercial development and application, however, did not take place until the last 2 years. Several brands of urea adhesives are now on the market. These meet the need for a hot-press adhesive which is applied in liquid form, cures rapidly at moderate temperatures, and is economical. For greater economy, the urea adhesive may be mixed with various proportions of flour (up to 50 percent) without affecting its water resistance. Diluted thus it comes within the cost range of animal and vegetable glues and is more durable. At present, it sells for 18 to 20 cents per pound; mixing it with 50 percent flour gives an adhesive for plywood, costing about 10 cents per pound. In 1937 three producers made urea resins for this use.

Other uses are in the treatment of textiles to obtain crease-proof properties and in the impregnation of wood. United States Patent No. 1,951,994 issued on March 20, 1934, reports the preparation of artificial silk from urea resins.

Production in the United States.

Commercial production of urea resins in the United States was reported for the first time in 1929. Early in that year the American Cyanamid Co. concluded an arrangement with the British Cyanides Co. of England for the American rights to manufacture and sell in the United States a resin made from urea, thiourea, and formaldehyde and marketed as Beetle molding powder. A manufacturing unit was built at Bound Brook, N. J., and in 1930 the output was substantial.

In 1931 another producer, the Toledo Synthetic Products Co., began manufacture of urea resins. Several years prior to that time the Toledo Scale Co. started a search for a material light in weight to replace the heavy porcelain-on-steel used in cases for scales. The search led to the urea resins and to commercial production by their subsidiary. In 1935 the Toledo Synthetic Products Company reached an agreement with the Imperial Chemical Industries of England for the interchange of technical and commercial information and of

free patent licenses on urea molding and laminating resins. The name of the domestic firm was later changed to the Plaskon Co.

In 1932 the Unyte Corporation started commercial production of urea resins at Grasselli, N. J. This firm was affiliated with the American I. G. Corporation. Late in 1936 the Plaskon Co. took over the Unyte Corporation.

The output of urea resins increased markedly in 1936 and 1937. Statistics for those years cannot be published without disclosing operations of individual firms. It may be stated, however, that the increase in both years over the previous year was considerably greater than for any earlier period. Most of the production was used in molded articles although appreciable quantities were consumed in laminated articles, in surface coatings, in the impregnation of fabric, and in adhesives.

There were 10 domestic makers of these resins in 1937.

Domestic production and sales of urea resins are shown in table [8](#).

TABLE 8.—*Urea resins: United States production and sales, 1933-37*

Year	Production	Sales		
		Quantity	Value	Unit value
	<i>Pounds</i>	<i>Pounds</i>		
1933	3,234,356	2,977,791	\$1,422,671	\$0.48
1934	3,470,916	3,115,608	1,290,802	.41
1935	4,202,536	4,005,083	1,828,565	.46
1936-37	(¹)	(¹)	(¹)	

¹ Not publishable; figures would reveal operations of individual firms.

Source: Dyes and Other Synthetic Organic Chemicals in the United States, U. S. Tariff Commission.

United States imports and exports.

Resins obtained from urea and thiourea, if imported, would probably be classified under paragraph 11 of the Tariff Act of 1930. The present rate of duty under this classification is 4 cents per pound and 30 percent ad valorem.

There has been no importation of these resins. This is due principally to the international licensing arrangements which usually include the allocation of

markets.

Exports are not shown separately in official statistics.

6. ACRYLATE RESINS

A new development of widespread importance in the synthetic resin industry is the commercial production of the polymers of certain derivatives of acrylic acid. The commercial exploitation of the acrylates is another example of the belated realization of the value of substances known for many years. Acrylic acid has been known for about a hundred years, and the polymer of methyl acrylate was first described in 1880. It was not until 1927, however, that a suitable method for their commercial production was developed. The study of the many derivatives of acrylic and methacrylic acids leads to the conclusion that those of greatest practical application in the resin field are the lower esters, such as methyl and ethyl, polymerized separately or together.

Colorless transparency, stability against aging, thermoplasticity, and chemical resistance to many reagents are the general characteristics of the acrylate resins. In consistency they range from soft, sticky, semiliquids to hard, tough, thermoplastic solids. Since these widely varying properties are obtained by control of manufacturing conditions, rather than by the use of plasticizers, the resins retain their initial properties indefinitely. Aging and weathering have no effect as they are stable under exposure to heat, light, and oxidizing agents. The methacrylates are harder and tougher but less elastic than the acrylates.

Properties and uses.

The acrylate resins are marketed in a number of forms, such as solutions in organic solvents, dispersions in water, solid cast sheets, rods and tubes, and molding powders. All of these are distinguishable from many other resins by their colorless transparency, adhesive qualities, great elasticity, and chemical resistance. The brilliant water-white color makes it possible to secure masses having a high degree of light transmission and great optical clarity.

The earliest commercial use of the acrylate resins was in laminated safety glass marketed as Plexigum in the United States and as Luglas and Sigla in Europe. The extensibility and elasticity of the resin film gives the laminated glass a flexible or yielding type of break when subjected to a hard impact. Having excellent adhesion to glass there is no need of an auxiliary cement to bond the resin to the glass, nor is it necessary to seal the edges since the resin has good resistance to moisture. The acrylate resin used for this purpose is in the form of a viscous solution in an organic solvent. A film is applied to each sheet of glass, the solvent removed by drying, and the sheets are pressed together.

The harder acrylic resins are used in the form of solid thermoplastics. Methyl methacrylate is of special interest. As the monomer is a mobile liquid it can be cast-polymerized to a solid of any desired shape in predesigned molds or produced in finely divided form for use as molding powder. The cast resin is marketed in this country as Crystalite, Plexiglas and Lucite, and in England as Diakon.

The solid acrylate resins are clearer than cast phenolic resins, not as brittle as the polystyrene resins, and not as tough as cellulose acetate or nitrocellulose plastics. Their transparency and resistance to aging and weather permit their use in applications not previously considered for synthetic resins. Sheets of this resin may be formed or molded into many useful shapes. The aircraft industry has found them suitable for windshields and cockpit enclosures to effect streamlining and thus greatly reduce wind resistance.

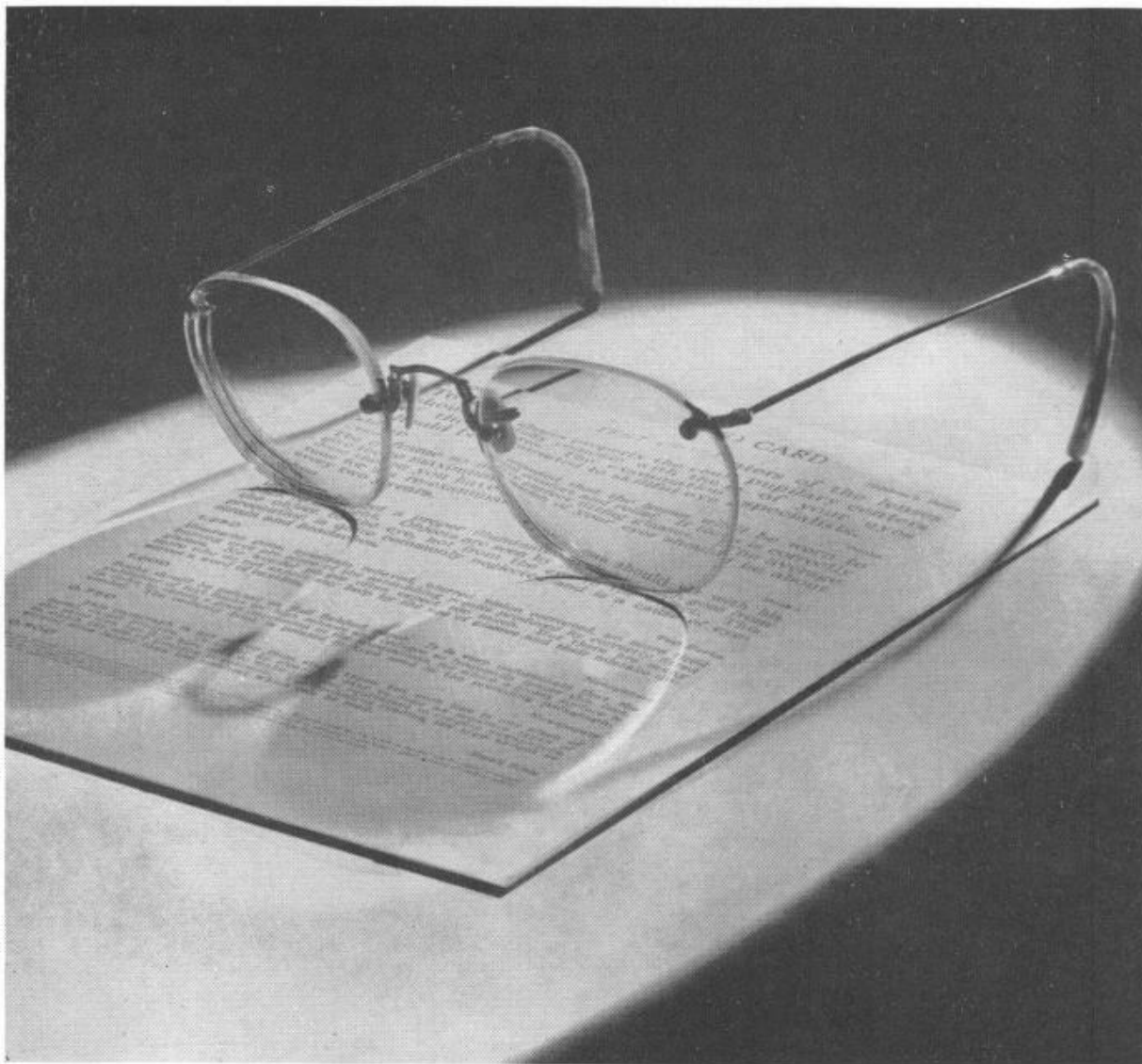
Methyl methacrylate is probably the nearest approach to organic glass thus far developed. Its optical properties make it suitable for spectacle lenses, camera lenses, magnifying glasses, and protective goggles. Spectacle lenses are now being made to prescription by molding. It is estimated that 900 molds will supply the requirements of about 98 percent of the prescriptions. The excellent light transmitting quality of methyl methacrylate permits its use in edge lighting, advertising displays, and instrument dials. It is also used in inspection windows in various types of machinery where curved sections are necessary and where glass might be broken.

A synthetic resin combining the properties mentioned, together with high tensile and impact strength, good dielectric properties, ultraviolet transmission, and resistance to water, oil, acids, and alkalis is an important contribution. The acrylates may be colored or have fillers added to give any desired translucency or opaqueness. They can be sawed, cut, blanked, turned, drilled, ground, polished, and sanded much the same as are nitrocellulose plastics.



AIRPLANE COCKPIT ENCLOSURES OF CAST ACRYLATE RESIN.

Source: Rohm & Haas Company, 222 W. Washington Square, Philadelphia, Pa.



SPECTACLE LENSES MOLDED TO OPTICAL PRESCRIPTION FROM ACRYLATE RESIN.

Source: Rohm & Haas Company, 222 W. Washington Square, Philadelphia, Pa.

A new and interesting application of the acrylate resins is as molded reflectors in a system of indirect highway lighting. The reflectors are pressed from colorless, transparent methyl methacrylate resin and are $1\frac{5}{8}$ inches in diameter. They are assembled in a pressed metal housing to form a double facing marker which is snap-locked to the top of an angle iron post. The posts are so located that the reflectors are accurately aligned 3 feet above the pavement edge. An installation has been made on U. S. Highway No. 16 between Detroit and Lansing, Mich., at a cost of about \$340 per mile. The motorist provides his own light from his headlights which strikes the reflectors and is returned as a narrow beam of brilliant illumination. The chief of the United States Bureau of Public Roads states that this is a definite contribution to the

safety and utility of the highways at night. The reflector is a group of tiny cube corners, over 300 in each disk. Each cube corner is a complete retrodirective optical system; a light ray entering the front surface is reflected from surface to surface of the cube and after the third reflection is directed back toward the headlight regardless of the entrance angle. If the cubes are made with a high degree of dimensional accuracy, the reflected light has a high candlepower, strong enough to be seen for a mile.

Other uses for these resins are in sound recording records, dentures, telephone and radio transmitter diaphragms, novelties, and lighting fixtures.

The monomer (unpolymerized methyl methacrylate) may be used to impregnate wood, cloth, wallboard, cork, paper, electrical coils, tile, or stone, and then polymerized to form the resin. Paper and cloth so treated have many uses, such as in the electrical and food-packaging industries. Laminated sheets find wide possibilities for use in the aircraft field, and for lamp shades. Wood may be impregnated with as much as 60 percent of the monomer. Solutions of these resins in organic solvents, such as ethylene dichloride, ethyl acetate, and toluol, are used in surface coatings, undercoats on difficult adhesion jobs, to impregnate paper and textiles, and in insulation. These coating solutions are marketed in the United States under the trade name Acryloids and in Europe under the trade names, Borron, Plexigum, and Acronol. They may be brushed, sprayed, dipped, and baked. Baking is recommended to give a higher gloss, better adhesion, and a harder film. The dried film has an elasticity of 1,000 percent at ordinary room temperature and the light transmission of clear films is intermediate between ordinary window glass and quartz.

Acrysol is an adhesive consisting of a dispersion of the resin in water and is recommended for use where adhesion is difficult, as on rubber or rubberized surfaces.

Production in the United States.

Commercial production of acrylate resins in the United States was started in 1931 by Rohm and Haas, Philadelphia, Pa., under United States Patents Nos. 1,388,016 of August 16, 1921, and 1,829,208 of October 27, 1931.

Commercial production of methyl methacrylate resins was started in 1937 by E. I. du Pont de Nemours & Co. This development is under United States Patent No. 1,980,483, issued in 1934. The liquid monomer is produced at Belle, W. Va., and shipped to Arlington, N. J., where it is polymerized by heat to the solid resin.

The output of acrylate resins was hardly more than experimental in 1935 but increased somewhat in 1936 and very appreciably in 1937. Although statistics of production are not publishable, it can be stated that in 1937 the output approached that of other synthetic resins made in commercial quantities. The properties of these resins indicate very large commercial production in the near future. Prices of the several types are still high as compared with other resins but should eventually be somewhat

lower than those of cellulose acetate and nitrocellulose plastics and slightly higher than those of cast phenolic resins.

Imports into and exports from the United States.

There have been no recorded imports of acrylate resins. The two domestic producers have agreements, licenses, or affiliations with the principal foreign makers of these products, one in England and one in Germany. Such arrangements would account for the absence of imports, except for sample or experimental lots, and might also limit export markets.

7. COUMARONE AND INDENE RESINS

Coumarone and indene are present in appreciable quantities in certain coal-tar fractions, especially in the solvent naphtha fractions distilling between 160° and 190° C. No attempt is made to isolate them from the solvent naphtha. Such a procedure would be difficult and expensive and, since polymerization readily takes place in dilute solutions, it is more economical to use fractions of solvent naphtha rich in these substances. The resins obtained are mixtures of polymerized coumarone and polymerized indene.

The solvent naphtha must be refined by fractional distillation and the polymerization very carefully controlled. The polymerizing agent is usually sulphuric acid although metallic salts, such as aluminum chloride, are sometimes used. The yield and color of the resin are affected by temperature and amount of acid used. Light colored resins are the most desirable. After polymerization the acid or metallic salt is removed, the product washed and neutralized and finally distilled. Several byproducts, such as naphtha, paracoumarone soap, and high boiling oils, are also obtained.

Description and uses.

Coumarone and indene resins are produced and marketed in the United States under the trade names Cumar and Neville. A number of grades are available, including the following:

Designation:	<i>Melting point</i>
Rubber grade, soft	50°-65° C.
Medium soft	65°-85° C.
Rubber grade, hard	85°-100° C.
Medium hard	100°-135° C.
Varnish grade	135°-160° C.

In addition to these, certain types are produced for special purposes.

The coumarone and indene resins are used to a large extent in varnishes for metal and wood. In this application they may be used to replace all or part of the higher priced natural resins and, to some extent, ester gum. Their application is somewhat limited by their rather short durability and elasticity. They are neutral, nonoxidizing and nonsaponifiable and impart to varnishes greater inertness and adhesion, fair dielectric strength, and shorter drying time than many of the natural resins. They cannot be used in nitrocellulose lacquer since they are not compatible with that plastic.

Another important use of these resins is as an ingredient in mastic floor tile, in the production of which a thermoplastic binder is used. Originally, asphalt was used, but demand for light colored tile necessitated some other binder, the requirements for which were met by the coumarone and indene resins.

The next largest application of these resins is in rubber compounding, their effect being to soften the rubber during milling and to facilitate its handling on rolls. They do not affect the aging qualities of rubber and are used as a softener for reclaimed as well as for new rubber.

Coumarone and indene resins are used, to some extent, in linoleum, for impregnating roofing felt, in electrical and friction tapes, paper and cloth sizing, printing inks, brake linings, adhesives, artificial leather, oil cloth, and shoe polishes. As a substitute for chicle as much as 10 percent may be incorporated in the chewing gum mixture. Their application in molded articles is very limited because of their brittleness and low tensile strength.

Production in the United States.

There are three domestic makers of these resins. Statistics of production and sales cannot be published without disclosing the operations of individual companies. The output, however, has increased appreciably in recent years and this type of synthetic resin is now among the most important produced.

Imports into and exports from the United States.

There have been no recorded imports of coumarone and indene resins in recent years. This is understandable because the duty alone would usually be more than the domestic price.^[4]

Official export statistics do not separately record these resins, although quantities are exported to nearby countries, including Canada.

8. PETROLEUM RESINS

Considerable research work has been done on the synthesis of resins from petroleum. It has long been known that cracked petroleum distillates, when stored for a time, have a tendency to form gums. This tendency is so pronounced that inhibitors are added to arrest such formation. These gums are of little value as resins, but it is possible to obtain good varnish resins by oxidation or controlled polymerization of certain distillates of petroleum cracking. By carefully controlling operations, resins of varied properties are obtained and several of them have become commercially important. The unsaturated compounds, largely olefins and diolefins, present in highly cracked petroleum distillates can be polymerized, with certain catalysts. The resin produced depends upon the types of unsaturated hydrocarbons present and upon the conditions of polymerization.

Properties and uses.

Several types of petroleum resins are on the market, one made from the “polymer slop” obtained in the high temperature, vapor-phase cracking operation, and the other prepared primarily for the production of resin. The former is marketed under the trade name Petropol and the latter as Santoresin.

The “Petropol” resins are marketed in two grades, No. 1158 and No. 2138. The specifications for these are as follows:

	Petropol No. 1158 liquid	Petropol No. 2138 liquid
Gravity	15.5-18.5	10-11 A. P. I.
Flash	175° F. minimum.	230° F. minimum.
Fire	215° F. minimum.	280° F. minimum.
Viscosity	200-225 at 212° F.	225-300 at 210° F.
Pour	0° F. maximum.	45° F. approximate.
Iodine No.	195 minimum.	200 minimum.
Molecular weight	300 approximate.	425 approximate.

Percent solids	60-65.	80-85.
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Miscible in all proportions with petroleum solvents.

Petropol No. 1158 is used by core oil makers to replace such vegetable oils as linseed, tung, and perilla. It is used also as a binder and waterproofing agent on rock wool insulation, replacing rosin and mineral oil. For spraying coal to minimize dusting, it has the advantage over calcium chloride of increasing the B. t. u. content of the fuel.

Petropol No. 2138 is a surface coating material which dries by polymerization. A low cost paint is obtained by combining a pigment and a plasticizer with the resin. Such paint dries in about one-fourth the time of linseed oil paints, adheres better to metal, and has greater resistance to water, acids, and alkalies. In varnishes and enamels it replaces 12 to 15 percent of tar-acid resin, minimizes skinning, and gives a higher luster and better flow. Another use of this Petropol is as a binder in brake linings, replacing certain tar-acid resins.

These two Petropol resins are among the lowest priced synthetics, selling at present (1938), in tank carlots, for 2 to 5 cents per pound.

The Santoresins are clear, hard, neutral products, melting at 100° C. They are soluble in drying oils, accelerate the gelatination of tung oil, are nonreactive with pigments, do not yellow on outdoor exposure, and are resistant to alkalies, acids, alcohol, and water. Applications are in protective coatings for wood, metal, paper, leather, cement, plaster, and other materials, in printing inks, plastic tile, linoleum, and fiber packages. Being odorless and tasteless they may be used to line food containers. Their high resiliency and purity recommend their use as a base for chewing gum. Other uses are as an agent for wetting and dispersing pigments in rubber and in surface coatings, to replace ester gum or modified tar-acid resins.

At present the Santoresins are offered at 15 cents per pound in lots of 20,000 pounds or more. Their approximate specifications are:

Appearance	A clear hard resin.
Melting point	110° to 120° C. A. S. T. M. (Ring & Ball).

Acid value	0 to 1.
Iodine value	125 to 135.
Specific gravity (at 20° C.)	1.02 or 8.5 lbs. per gallon.
Color (50 percent solution by weight in toluol)	13 to 15 Gardner Holt standard.

Odor slight when cold, sweet and aromatic when melted. Soluble in aromatic hydrocarbons, petroleum thinners, turpentine, and varnish oils. Insoluble in alcohols, esters, ketones, and not completely compatible with nitrocellulose.

Production.

In the United States two makers of petroleum resins are producing in commercial quantities and several others are carrying on extensive research. Production was small in 1935, but increased in 1936 and in 1937. The development and expansion of these resins over the past 2 years indicate that they will become important.

Imports into and exports from the United States.

There has been no importation of petroleum resins into the United States. Exports have been confined to samples and experimental quantities.

9. POLYSTYRENE RESINS

The polystyrene resins are thermoplastic products discovered about 100 years ago and are therefore the oldest synthetic resins known. Their practical application has been greatly retarded by the lack of inexpensive raw materials of high purity and by the difficulties experienced in their manufacture.

Ethylene, from petroleum or natural gas, is combined with benzene, from byproduct coke-oven operations, to form ethyl benzene, which is cracked to vinyl benzene or styrene. This monomer is polymerized by heat at 100°-150° C. The resin may be extremely tough or very brittle, depending on the conditions of polymerization. Products having different properties are obtained by (a) low temperature polymerization, (b) high temperature polymerization, and (c) catalytic polymerization.

The low-temperature polymers, sometimes designated as alpha-metastyrol, are produced by polymerizations of vinyl benzene at temperatures under 175° C. A transparent resin, colorless to light yellow, is produced which is remarkably tough, has excellent tensile strength, unusually good dielectric properties, and is resistant to most chemicals.

Polymerization at high temperatures (above 175° C.) yields a brittle resin designated as beta-metastyrol. This type is transparent but usually dark in color, has low tensile strength and shock resistance.

When vinyl benzene is polymerized in the presence of catalysts, the resulting resin is similar to resins obtained at high temperatures, except that it is lighter in color. It is sometimes designated as gamma-metastyrol. Oxidizing agents are usually the catalysts. Clear, colorless, vitreous resins are obtained by excluding air during polymerization.

Properties and uses.

Polystyrene resin is a clear, colorless, highly thermoplastic molding material with high insulating property, moisture resistance, inertness,

dimensional stability, and impact strength. It can be molded directly by heat and pressure, and the molded articles are remarkably resistant to discoloration by light. Polystyrene has a dielectric constant of 2.6, a power factor of 0.02 percent, and is equivalent to fused quartz as an electrical insulator of low dielectric loss. Films of 0.002 inch thickness have a dielectric strength of more than 2,000 volts per mil thickness, which is better than that of any other available synthetic resin and even better than that of shellac. The tensile strength of the resin is 5,500 to 7,000 pounds per square inch, and its impact resistance remains unchanged at temperatures as low as minus 70° C. It transmits all wave lengths of light down to 3,000 Angstrom units.

Polystyrene is adapted to large scale production of transparent, translucent, and opaque moldings in a wide variety of colors. It is easily molded by injection processes, softening at about 150° F. and is molded at 300° to 375° F., under 3,000 to 30,000 pounds pressure per square inch. As much as 40 percent filler may be used without seriously affecting the tensile strength, although the filler does affect the dielectric properties. Since the resin is thermoplastic there is no waste in the molding operation; scrap material may be reground and used again.

The unusual properties of these polystyrene resins should give them widespread applications when the cost is low enough to make them competitive with other materials. Potentially large volume outlets are in radio frequency insulation; in dentures because of the strength, low specific gravity, ease of coloring, and absence of odor and taste of the material; in electrical parts for submarine and aircraft storage battery cases and separators; and for the manufacture of glass eyes.

Other possible applications of polystyrene resins are in metal lacquers and in light colored enamels. Their toughness and light color, together with their solubility in cheap solvents, suggest their use for these purposes. Such lacquers are said to be quick-drying, resistant to water, and moderately so to acids and alkalies.

Production in the United States.

For a number of years, the Naugatuck Chemical Division of the United States Rubber Co. produced small quantities of polystyrene resins, which

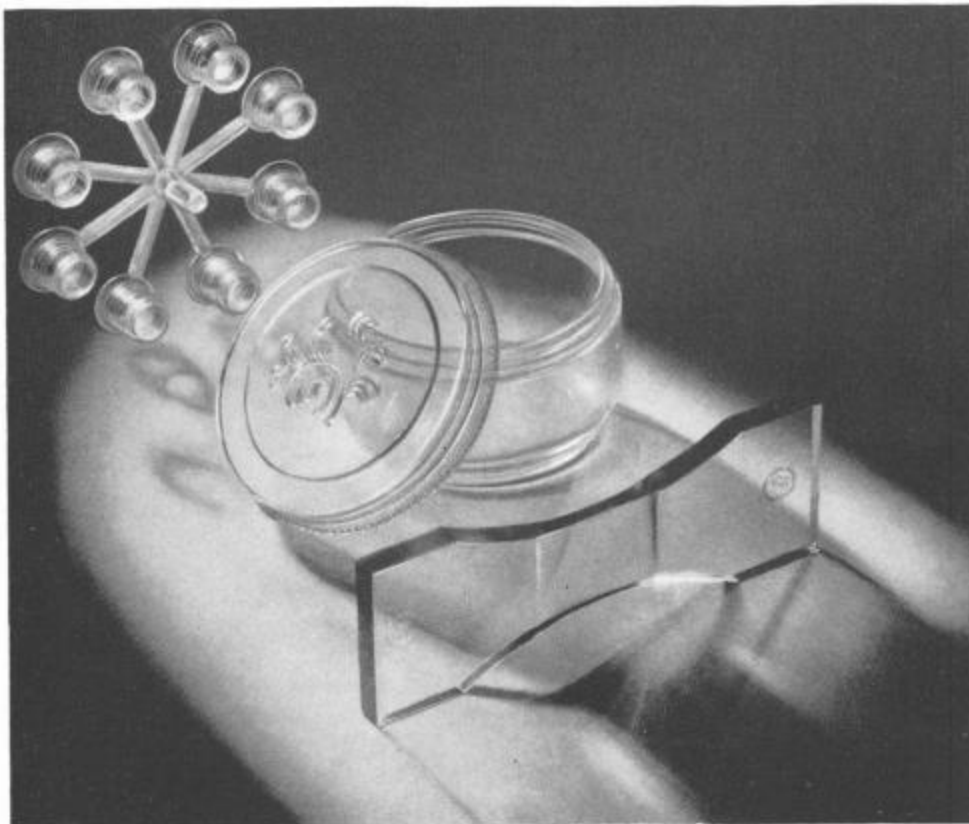
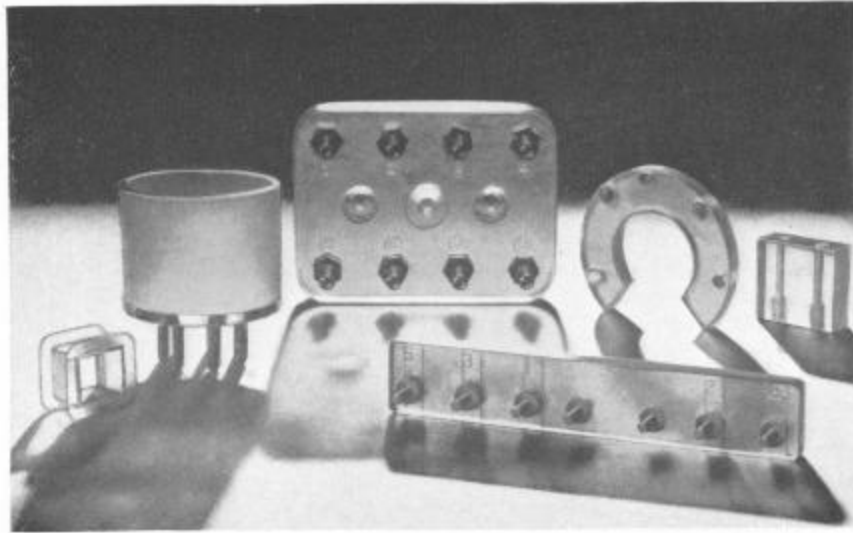
were marketed under the trade name Victron when for general purposes and under the trade name Marvelyn when for use in dentures. Little progress was made because of high costs and failure to produce a water-white product. The sales price was between \$1.50 and \$2 per pound. Early in 1937 the Naugatuck Chemical Division transferred its patents on polystyrene resins to the Carbide and Carbon Chemicals Corporation.

The Dow Chemical Co., Midland, Mich., late in 1937 announced commercial production of clear, colorless polystyrene in several forms. Styron is the trade name for the resin from this source. In January 1938, the Bakelite Corporation announced Bakelite Polystyrene. The plants manufacturing polystyrene have a capacity in excess of 2,000,000 pounds a year, and the resin is currently offered at 72 cents per pound.

At least one other domestic firm is doing research on the polystyrenes and expects to produce commercially in the near future.

Imports into and exports from the United States.

At least two commercial types of polystyrene resins are produced abroad. Both are made in Germany and marketed under the trade names Resoglas and Trolitul. Resoglas is a water-white, transparent thermoplastic resin softening at about 150° C. Its water absorption is low, it is nonoxidizing, and does not discolor on weathering and baking. Appreciable quantities are produced in Germany and the sales price there was reported to have been 40 cents per pound during 1936.



MOLDED POLYSTYRENE RESINS.

Source: Bakelite Corporation, 247 Park Avenue, New York, N. Y.

Small quantities of Resoglas and Trolitul have been imported from Germany in recent years. Table [9](#) shows the quantities imported in recent years.

TABLE 9.—*Resoglas and Trolitul: United States imports for consumption, 1933-37*

Year	Resoglas (polystyrol)			Trolitul		
	Quantity	Value	Unit value	Quantity	Value	Unit value
	<i>Pounds</i>			<i>Pounds</i>		
1933	771	(¹)		672	(¹)	
1934	991	(¹)		200	(¹)	
1935	110	\$97	\$0.88	4,608	\$3,782	\$0.82
1936	2,220	1,901	.86	4,671	3,641	.78
1937 ²	None	None		6,788	4,077	.60

¹ Not available.

² Preliminary.

Source: Analyses of invoices of paragraph 28, act of 1930—U. S. Tariff Commission.

With the more advanced development of polystyrol resins in Germany prior to 1938, evidenced by larger commercial production, by wider application, by the marketing of a water-white product at a considerably lower price, it might be expected that imports into the United States would have been in considerably larger amount than shown in table 9. That they were small was probably due to the high rate of duty which made them expensive as compared with other synthetic resins in the United States and thus limited their market to uses in which the others were less satisfactory. Resoglas was reported to have been selling for 40 cents per pound in Germany. The imported resin is assessed for duty under the provisions of paragraph 28 of the Tariff Act of 1930 at 45 percent ad valorem based on American selling price (as a competitive product) and 7 cents per pound. The American selling price of the resin made in the United States until late in 1937, as determined by the Bureau of Customs, Treasury Department, was \$1.85 per pound. The duty was therefore 90 cents per pound. Imports of Trolitul were valued at 75 cents per pound, giving a cost of \$1.75 per pound laid down, duty paid, in domestic markets. With the present American selling price of 72 cents per pound, the duty would be approximately 36 cents per pound.

10. VINYL RESINS

Vinyl acetate, vinyl chloride, and to a lesser extent vinyl chloroacetate, are the raw materials (monomers) for the several vinyl resins commercially produced in the United States, Canada, and Germany. These are all esters of the hypothetical vinyl alcohol and are made by the action of acetic and hydrochloric acids on acetylene.

The spontaneous polymerization of vinyl derivatives has been known for many years, although its significance and industrial application have been realized only recently. Vinyl acetate, probably the most important of the vinyl esters, was discovered in 1912 and first made in Canada in 1917.

Vinyl resins may be classified into (a) polyvinyl acetate, (b) copolymers of vinyl acetate and vinyl chloride, (c) polyvinyl chloride, and (d) polyvinyl chloroacetate.

Description and uses.

Polyvinyl acetate resins.—The several commercial types of vinyl acetate resins are marketed under the trade names Vinyloid A, Alvar, Gelva, Formvar, and Mowilith. The first of these is a product of Carbide and Carbon Chemicals Co., New York, the next three are products of Shawinigan Chemicals Limited, Shawinigan Falls, Canada, and the last is made by the Interessen Gemeinschaft Industrie A. G., Germany. Vinyloid A and Gelva represent the simplest series of vinyl acetate resins and are made by polymerizing the monomer. The softening point and viscosity of the polyvinyl acetate resins increase with higher polymerization. Such resins are colorless, tasteless, odorless, thermoplastic products. They are soluble in coal-tar solvents and are compatible with certain alkyd resins, tar-acid resins, and natural resins. Films of polyvinyl acetate resin are not discolored by exposure, and after irradiation they become opaque to ultraviolet light, are hard and tough, and have good adherence and endurance. Their dielectric strength is good and they do not show a carbon track after the

passage of an electric arc. Various grades having softening points from 80° to 200° C. are available.

Polyvinyl acetate resins are used in making transparent papers, paper to metal laminations, glassine papers for food packaging, as a substitute for chicle in chewing gum, and as a component of paints, varnishes, and lacquers. They have the desirable properties of compatibility, durability, resistance to abrasion, and rust inhibition in the surface-coating use. Having the same refractive index as pyrex glass, they leave no line of demarcation when used as a cement for that material. They have been used to stiffen toe-caps in shoes and articles made from paper pulp suspensions. Gelvas are not molded as such because of their tendency to cold flow. They are used, however, as a binder for ground mineral fillers in advertising signs and for wood flour in molded artificial wood carvings. In nitrocellulose lacquers they improve the adhesion, luster, and toughness.

Alvars are made by replacing part or all of the acetate groups in Gelva with acetaldehyde. Their viscosity varies with the degree of polymerization and their properties vary according to the extent of replacement of the acetate groups. The Alvar types do not cold flow when molded, are tougher, harder, and have better adhesion but are less resistant to weathering than the Gelva types. Other properties are about the same as those of the Gelvas. Alvars having 70 to 80 percent acetate group replacement are used chiefly in spirit type varnishes, lacquers, and enamels that must stand exposure to weather. Another Alvar type is used in injection and press molding. The high binding power of the resin permits the use of large percentages of filler without loss of desirable properties. Such moldings may be machined and polished, and take inserts, such as the wood core in shoe heels. Flexible phonograph and transcription records made from the Alvars have gained wide approval. An 85 percent (acetate replacement) type has better impact strength and is used in toilet articles. Sheets, rods, and tubes of this resin may be machined in much the same way as nitrocellulose plastic and used where noninflammability is an asset.

Formvars are made by replacing part or all of the acetate groups in Gelva with formaldehyde. These resins are colorless, odorless, tasteless, and thermoplastic. They have higher softening points and greater tensile and impact strength than the Alvars. They are resistant to alcohols, coal-tar solvents, fats, oils, or water. Moisture transmission rate through a film of

this resin is about one-tenth that through regenerated cellulose and one-fourth that through cellulose acetate.

The grades of the Formvars available are designated by the extent of replacement of the acetate group. The 75-percent replacement type has excellent mechanical strength and flexibility and is unaffected by sunlight. Formvars of 95 percent acetate displacement have a tensile strength as high as 10,000 pounds per square inch and offer possibilities in the manufacture of artificial silk and photographic film.

The vinyl resins have made possible a new type of safety glass superior to any heretofore marketed. By condensing butylaldehyde with vinyl acetate, a polymer is obtained which is used as the inner layer between two sheets of glass. Heat and pressure secure complete adhesion and yield a sheet with greater resistance to breakage at low temperatures than the types now in general use.

Although safety glass was invented in 1905, and many substitutes for the original nitrocellulose inner layer have been proposed, only two reached commercial importance before the development of the vinyl resins. These are cellulose acetate and the acrylate resins. Safety glass used in automobile windshields up to about 1930 discolored after a year or two of service. This discoloration was due to the action of the actinic rays of the sun on the nitrocellulose layer. Since 1930 this difficulty has been largely overcome by using an actinic ray filter glass (a special glass with a high iron content) in front of the nitrocellulose sheet, or by using cellulose acetate, which is not discolored to the same extent by light, as a substitute for nitrocellulose. Both cellulose nitrate and cellulose acetate, however, have a tendency to lose toughness and strength at low temperatures, to absorb moisture, and to separate from the glass around the edge unless sealed, and to lose their plasticizer and shrink.

Although a vast improvement over ordinary plate glass, laminated glass made with cellulose nitrate or acetate has the serious defect of being brittle at low temperatures, such as prevail in the winters of northern States. It is easily shattered at zero Fahrenheit, while at 60° F. and above it is quite strong. This shortcoming led to the development of the vinyl resin sheet for safety glass with a remarkable degree of toughness. At normal temperatures it has rubberlike toughness which, although decreased at low temperatures,

is not punctured by the impact of a half-pound steel ball falling from a 30-foot height at minus 10° F., whereas nitrocellulose or acetate laminated glass withstands the impact of a fall from not greater than one-tenth this height. A further advantage of the vinyl sheet is that it is water resistant, making the sealing of the edges of the glass unnecessary and thus reducing costs. Exposure to ultraviolet light in Florida sunlight for more than 2 years did not discolor it.

The many desirable properties of the vinyl resins, as outlined above, indicate their widespread use in laminated safety glass when it is available in sufficient quantities. It is estimated that our annual output of safety glass interlayer sheets exceeds 17,000,000 pounds, of which 25 to 30 percent are for windshields, and 70 to 75 percent for side and back windows of automobiles.

At least one of the series of Mowiliths made in Germany is polymerized vinyl acetate. It is recommended as an ingredient of water-white lacquers. It is compatible with nitrocellulose and is extremely durable and not disintegrated or discolored on exposure to weather.

Copolymers of vinyl acetate and vinyl chloride.—The simultaneous polymerization of mixtures of vinyl acetate and vinyl chloride yields resins with the desirable properties of the two reactants. The extent of plasticity is largely controlled by varying the ratio of the vinyl derivatives. Resins high in vinyl chloride content are better suited to molding, and those high in vinyl acetate are better lacquer ingredients. These resins are marketed as Vinylites by the Carbide and Carbon Chemicals Co., New York. They are thermoplastic, odorless, tasteless, and practically nonflammable. Their outstanding properties are resistance to water, soap, acids, alkalies, and alcohol, and their strength and good dielectric properties. Their stability to light is improved by the addition of ultraviolet absorbing compounds and their stability to heat by the addition of lead oleate, calcium stearate, or other bases. Water absorption and compatibility with other resins is increased as the chloride content increases.

The principal types of copolymers are:

Vinylite VYN, high molecular weight. This resin is used in dentures where good fatigue resistance, impact strength, and tensile strength are required. It contains 85 to 88 percent vinyl chloride.

Vynlite VYN, medium molecular weight. This resin is used in general molding and extending applications including sheets, rods, and tubes. Its vinyl chloride content averages 85 to 88 percent.

Vynlite VYN, low molecular weight. This resin is used in moldings, coated paper, lacquers, floor tile, phonograph records, and felt impregnation. It contains 85 to 88 percent vinyl chloride.

Vynlite VYC. This resin of low molecular weight is compatible with nitrocellulose and is used in lacquers and finishes for industrial applications. Lacquers from the Vynlites are called Vynloids.

The Vynlites for molding are thermoplastic and shrink very little, making them applicable to large moldings. They may be used in extension processes such as tooth-brush preforms, pipe lining, and wall trim. Fillers and pigments may be added, although pigments containing iron and zinc have harmful effects on the stability of the resin. The fillers used are wood flour, mica, talc, and alpha cellulose. Fillers reduce the mechanical strength of the resin and lessen its resistance to water. Plasticizers, such as dibutyl phthalate or tricresyl phosphate, give a softer, more flexible resin. Resins from the copolymers resemble the cellulose derivatives in their molding characteristics, mechanical strength, and appearance.

In lacquers the Vynlites offer high resistance to water, oils, and chemicals. The drying of such lacquers is by evaporation rather than by oxidation. They are suitable for lining food containers, coating concrete, coating paper for bottle cap liners, and as a stiffener for box toes of shoes. Their most successful application at present is as an inside coating for beer cans. Floor tile containing these resins mixed with slate flour or other filler has good possibilities.

Polyvinyl chloride resins.—Vinyl chloride may be polymerized to give nonflammable resins of varying solubilities. The completely polymerized resin is practically insoluble at ordinary temperatures and is used as a rubber substitute. It is marketed as Koroseal by B. F. Goodrich Rubber Co., Akron, O. Compared with natural rubber, it has greater resistance to acids, alkalis, oils, and alcohol, more flexing life, better resistance to sunlight, water, and oxidation. Solutions of this resin marketed as Korolac are used in special types of varnishes.

Polyvinyl chloroacetate resins.—These resins known as Mowiliths are made in Germany. Application is largely for surface coating. Practically no information on this type is available.

Divinyl acetylene and synthetic rubber.—Two products closely related to those described above but probably not synthetic resins as defined for this discussion are divinyl acetylene, a synthetic drying oil, and Neoprene, a synthetic rubber.

Acetylene, when passed into a solution of copper chloride and ammonium chloride, combines with itself. When two molecules of acetylene react monovinyl acetylene is formed, and when three molecules of acetylene react divinyl acetylene is formed. Monovinyl acetylene reacts with hydrochloric acid to give chloroprene, which is polymerized to synthetic rubber or Neoprene.

Divinyl acetylene is a colorless liquid which darkens on exposure to light and which has an onionlike odor. When polymerized liquids are formed, then as the reaction progresses viscous products and finally insoluble, infusible, inert resins. By arresting the reaction before the gel point is reached, an amber colored heavy liquid, soluble in aromatic hydrocarbons, is obtained. Since divinyl acetylene will continue to polymerize at ordinary temperatures, this property is taken advantage of in using it as a basis for paints, under the name “synthetic drying oil.” Clear, amber films are obtained from solutions of this oil in solvent naphtha. Divinyl acetylene is quick drying, is many times more impervious to moisture than linseed oil, and is thermosetting. It is not attacked by solvents but is attacked by strong oxidizing agents, and the gelled material may ignite spontaneously.

Although not classified as a resin, synthetic rubber is discussed here because of its close chemical relationship to the vinyl resins. It is made commercially by E. I. du Pont de Nemours & Co., Wilmington, Del., and is marketed as Neoprene. It is sold as a plastic polymer which is vulcanized and processed much the same as natural rubber except that sulphur is not essential to vulcanization. Synthetic rubber is higher in price than natural rubber, but it has certain properties which make it suitable for service conditions where natural rubber is unsatisfactory. Among these properties are its resistance to gasoline, oils, and greases, and to elevated temperatures. It does not check or crack on exposure to sunlight, nor does it oxidize as

rapidly as natural rubber. Its principal applications are in special gaskets, printing rolls, jackets for high tension cable, linings for gasoline or oil hose lines, balloon fabrics, diaphragms for regulators, and packing for compressors. Its existence acts as a limit to the increase in the price of natural rubber and assures a supply in emergencies.

Production in the United States.

Some of the products described are commercially produced in the United States; others in Canada or in Germany. Those made in the United States are usually not made by more than one firm, so that statistics of production and sales are not publishable. The vinyl acetate resins have been produced principally in Canada; the copolymers of vinyl chloride and vinyl acetate are domestic products. In 1935 the United States output of all vinyl resins exceeded 1,000,000 pounds, a figure that was increased in 1936 and 1937.

The Canadian output of Gelva and Alvar has reached commercial quantities; that of Formvar is still confined to experimental plant lots.

The acceptance of vinyl resin sheets for safety glass will greatly increase the output in 1938. The basic patent, known as the Morrison-Blake patent, United States No. 2,036,092 issued on March 31, 1936, is owned by Shawinigan Chemicals, Ltd., Montreal, Canada, who have licensed several domestic producers. The monomer (vinyl acetate) is now produced at Niagara Falls, N. Y., by the Niacet Chemicals Corp., which is jointly owned by this Canadian firm, Carbide and Carbon Chemicals Corporation, and E. I. du Pont de Nemours & Co. It is also produced by du Pont at Belle, W. Va. It is shipped, in tank cars, to polymerization and sheet-forming plants at Indian Orchard, Mass., Arlington, N. J., and Charleston, W. Va. The Indian Orchard plant, known as the Shawinigan Resin Products Co., and jointly owned by the Canadian firm and the Fiberloid Corporation, is now in operation. The plant of the du Pont Company at Arlington, N. J., began production in May 1938, and that of Carbide and Carbon Chemicals Corp., at Charleston, W. Va., is in production. These plants have a combined annual capacity of about 10 million pounds of vinyl resin sheets. According to present plans this new safety glass will be available for 1939 model automobiles. The resin sheet to be used is 0.0015 inch thick as compared with the 0.0025 inch thickness of the present cellulose acetate and

nitrocellulose sheet. Several trade names have been adopted for the vinyl resin sheets, among which are Vinylite X, and Butvar. The licenses granted to domestic makers under the Morrison-Blaike patent also permit them to make vinyl acetate resins for purposes other than safety-glass sheets. Considerable progress has been made in adapting these resins to injection molding operations for the production of tooth-brush handles, combs, closures, and other parts.

Imports into the United States.

The official statistics of imports of vinyl resins prior to 1936 are not satisfactory for purposes of comparison. Imports could be entered under either paragraph 2 or paragraph 11 and could be included either with the statistics of imports of vinyl acetate (see table [91](#), page [141](#)) or be thrown into a general group of non-coal-tar synthetic gums and resins, n. s. p. f., which in addition to vinyl resins would include the acrylates and ureas. Table [10](#) gives imports of synthetic resins under paragraph 11 of the Tariff Act of 1930.

TABLE 10.—*Synthetic resins classified under paragraph 11:*¹ *United States imports for consumption 1931-37*

Year	Quantity	Value	Unit value
	<i>Pounds</i>		
1931	453	\$173	\$0.38
1932	454	29	.06
1933	1,120	496	.44
1934	4,084	1,576	.39
1935	3,105	1,804	.58
1936	146	65	.45
1937 ²	1,963	439	.22

¹ Statistical classification 838.914, synthetic gums and resins, n. s. p. f. (not coal tar) 1931-35; 838.939 same, other than those in chief value of vinyl acetate, 1936 and 1937.

² Preliminary.

Source: Compiled by the U. S. Tariff Commission from official statistics of the U. S. Department of Commerce.

A better idea of the imports of vinyl resins prior to 1936 is obtained by an invoice analysis of imports through the Port of New York under paragraphs 2 and 11. Table [11](#) shows imports of vinyl acetate resins based on such an analysis for 1934 and 1935 and on official statistics for the years 1936 and 1937.

Similarly table [12](#) shows imports of Mowilith resins based upon import analysis for the period 1932-1935, and upon official statistics for 1936 and 1937.

TABLE 11.—*Vinyl acetate resins: United States imports for consumption, 1934-37*

Year	Quantity	Value	Unit value
	<i>Pounds</i>		
1934 ¹	42,000		
1935 ¹	240,000		
1936 ²	600,808	\$144,782	\$0.24
1937 ²³	652,730	201,213	.31

¹ Invoice analysis of imports entered through the New York customs district.

² Statistical classification 817.58 (par. 2), vinyl acetate, polymerized, and synthetic resins made in chief value from vinyl acetate, n. s. p. f. (excluding imports from Germany) and 838.938 (par. 11), synthetic resins made in chief value from vinyl acetate, n. e. s.

³ Preliminary.

Source: Compiled by the U. S. Tariff Commission from official statistics of the U. S. Department of Commerce.

TABLE 12.—*Mowilith resins: United States imports for consumption, 1932-37*

Year	Quantity	Value	Unit value
	<i>Pounds</i>		

1932 ¹	555	\$229	\$0.41
1933 ¹	741	247	.33
1934 ¹	2,950	1,668	.57
1935 ¹	3,372	3,175	.94
1936 ²	7,056	2,410	.34
1937 ²³	220	308	1.40

¹ Analysis of invoices of imports entered through the New York customs district.

² Imports from Germany under statistical classification 817.58 (par. 2), vinyl acetate, polymerized, and synthetic resins made in chief value of vinyl acetate.

³ Preliminary.

Source: Compiled by the U. S. Tariff Commission from official statistics of the U. S. Department of Commerce.

Prior to January 1, 1936, the rate of duty on imports of vinyl resins was 6 cents per pound and 30 percent ad valorem under paragraph 2, and 4 cents per pound and 30 percent ad valorem under paragraph 11 of the Tariff Act of 1930. Under the terms of the trade agreement with Canada, the duty under both paragraphs was reduced to 3 cents per pound and 15 percent ad valorem. This rate was generalized to the other countries from which we have received imports, with the exception of Germany.

Exports from the United States.

Exports of vinyl resins are not separately shown in official statistics.

11. OTHER SYNTHETIC RESINS

The synthetic resins already discussed are those in substantial commercial production but, by no means, the only ones known or produced. Several thousand new ones have been reported and the search continues in laboratories throughout the world. A successful new product must be one made from inexpensive raw materials or must possess some property or advantage that will permit its sale at a price level above that of other resins.

No attempt is here made to list the host of less important resins. Certain ones of unusual interest or possessing unique properties are described below. These include resins obtained from adipic acid, aniline, citric acid, diphenyl, furfural, lignin, sugar, and sulphonamide.

Adipic acid resins.

The resins from adipic acid are classed as alkyd resins. Those obtained by the condensation of adipic acid and glycerin are soft and rubbery and are used to some extent in surface coatings and in photographic films. In these the resin is formed in three stages as in other alkyd types: A soluble liquid, a viscous rubbery product, and a form insoluble in the usual solvents.

Commercial domestic production of these resins was reported for the first time in 1935 and the output has increased each year since then.

Aniline resins.

Resins obtained by condensing aniline and formaldehyde have been developed in recent years. Much of the research on this type of resin was done in Switzerland by the Ciba Co., which holds a number of patents on it. The Swiss product, called Cibanite, has excellent electrical and mechanical properties. At least one domestic manufacturer is licensed under the Swiss-owned patents.

Citric acid resins.

Considerable interest has recently been manifest in synthetic resins derived from citric acid. The sharp decline in the price of citric acid, as a result of large scale synthesis from sugar has placed it within the realm of possibility as a raw material for synthetic resins.

The citric acid resins, classed as alkyd resins, are obtained by condensing citric acid and glycerin. Commercial production is said to have started in Europe, but there is no known domestic production as yet.

Diphenyl resins.

A series of products known as Aroclors and made by chlorinating diphenyl are available in commercial quantities.

Diphenyl was commercially produced for the first time by Swann Research, Inc., at Anniston, Ala., about 1928. The demand for it as a heat-transfer medium resulted in large scale output. Later it was found that certain of the chlorinated compounds of diphenyl possess valuable resin properties.

The Aroclors range from a clear mobile oily liquid to an amber colored transparent solid. They are thermoplastic, do not polymerize or oxidize, and are therefore nondrying. They may be dissolved in varnish oils, such as tung oil and linseed oil, to give varnishes which are resistant to alkali and water. The diphenyl resins are good adhesives on metal and glass and give strong joints between such surfaces. They have a high dielectric constant, resistivity, and a low power factor. Their chief use is in wire insulation.

The domestic production of chlorinated diphenyls is, at present, solely by the Monsanto Chemical Company, St. Louis, Mo.

Furfural resins.

Large scale commercial production of furfural, an aldehyde obtained from oat hulls and other farm waste, has made it available for synthetic resin manufacture.

Tar-acid furfural resins possess certain outstanding properties, such as great dimensional accuracy, great reaction speed to the infusible solid stage, and unusual strength and toughness. They are available in dark shades only.

Printing plates as large as those of metropolitan daily papers are molded from them as are radio tube bases, all sorts of electrical parts, and machined parts requiring great dimensional accuracy. Other uses are in abrasive wheels, varnishes, and adhesives.

Probably the largest domestic maker of furfural resins is the Durite Plastics Division of Stokes and Smith Company, Philadelphia, Pa.

Resins from sugar.

Many attempts have been made to utilize sugar as a raw material for synthetic resins. United States Patent No. 1,949,831, dated March 6, 1934, claims a process for the manufacture of molding compounds by condensing saccharide with aldehydes and urea. Pure sucrose yields a clear, colorless, nonresilient resin, while molasses and cane sugar give dark-colored resins. The trade name Sakaloid is used to designate certain of these resins; there is, however, no known domestic production. Sucrolite is the trade name of a brand of resins from sugar produced in Europe.

Sulphonamide resins.

The sulphonamide resins were developed from para toluenesulphonamide, a byproduct obtained in the manufacture of saccharin (synthetic sweetening agent).

Para toluenesulphonamide, condensed with formaldehyde or other aldehyde, forms a viscous mass which, on heating, is converted to a hard colorless resin. Such resins are compatible with cellulose acetate or nitrocellulose in lacquers, the combination yielding clear, colorless lacquers of good gloss and adhesion. Other possible uses are as an adhesive in safety glass, in certain molding compositions, in insulating materials, and to deluster artificial silk.

Domestic production of sulphonamide resin is entirely by the Monsanto Chemical Co., St. Louis, Mo. It is marketed under the trade name Santolite.

12. ORGANIZATION OF THE SYNTHETIC RESIN INDUSTRY

The discussion of the various synthetic resins on pages [11 to 52](#) carries in each case, under the heading of production, a notation of the number of companies producing that particular resin; and the discussion on pages [86 to 141](#) of important raw materials for these resins describes briefly the conditions under which these materials are produced. We shall now consider the interrelationships between industries producing the several resins, and the relation of the resin industries to their raw materials and to some of the important resin-consuming industries.

No description of the organization of a rapidly expanding industry can be expected to remain accurate for long. But regardless of future changes that may be expected, the general pattern seems definite enough to make possible a few broad generalizations. At present the producers of synthetic resins may be classified in two groups: those making alkyd and tar-acid resins, and those making all other synthetic resins.

The alkyd resins and the tar-acid resins are produced in large volume, and for these resins the patent situation is such that there is nothing to exclude new producers. The result has been that new firms have entered the field and there has been a marked tendency for concerns using these resins on a large scale to produce them. This general situation may be expected to continue as long as the volume of consumption of these resins is rising. But when consumption levels off, it would not be surprising if increased competition for new business resulted in consolidations of some of the producing units.

Each of the other synthetic resins is produced by a small number of firms and this may be expected to continue as long as the production of a particular resin is small, or basic patents dominate the situation. When and if the situation in these respects changes for some of the other resins, they will probably develop the same tendencies as now exist in the production of the tar-acid and alkyd resins.

Horizontal relationships between resin producers.

Horizontal relationships between companies are those between different units in the same industry (say two tar-acid resin producers), or in different industries each operating at the same stage of industrial production (say a tar-acid resin producer and a producer of urea resin). As a rule, extensive horizontal relationships are not common in relatively young industries, and this is true of the production of synthetic resins. In general, it has not been necessary to absorb competitors to achieve a greater volume of sales, and efforts have been directed to exploiting the possibilities of expansion in a growing market. This necessitated solving technical problems concerning improvement of the product and its production on an ever larger scale; legal problems regarding patents (protection of those owned, and the policy to be adopted toward unadjudicated patents owned by others); and the marketing problem of convincing prospective customers of the worth of a new product. These and other problems incidental to successful competitive production and sale of a given type of synthetic resin have been sufficient to restrain the desire to produce more than one type.

The patent situation of most synthetic resins is extremely complicated. In the case of tar-acid molding resins the basic Baekeland patents have expired, but for other synthetic resins either the basic patent is still in force, or it is difficult to say which is the basic patent, because of lack of adjudication by the courts. In all cases dozens of supplementary patents are in force and sometimes hundreds. As a result the patent situation, though one of the bars against entering into a new field, frequently forces some relationship between producing units in the same synthetic resin field. Cast phenolic resins afford an example of patent-licensing of several corporations by another with the payment of royalties as compensation. In a number of other branches of the resin industry, such as the laminated tar-acid resins and the alkyd resins, the mutual desire of producers to avoid litigation has apparently resulted in “gentlemen’s agreements” not to sue.

Vertical relationships of resin producers.

A vertical relationship is one between producers operating at different stages of industrial production, such as a firm producing resin and a firm

producing a resin raw material or between the former and a firm that is a resin consumer. The incentive for a consuming industry operating on a large scale to make its own resins is naturally greater than for one using only small quantities. Therefore we may expect to find instances where a process consuming the resin in quantity and resin manufacture are both performed by the same company provided other conditions (such as the patent situation and knowledge of the art of manufacture) are favorable.

Tar-acid resins for molding.—The present practice of molding resins is favorable to large-scale production. The shaping of the mold is expensive, involving skilled labor upon hardened steel; but once the mold is made it may be used to produce tens or hundreds of thousands of units. Subsequent labor upon the molded product is usually limited to the simple task of smoothing the line where the flash is broken off, since the product comes from the mold in the color and with the surface and shape desired.

The usual arrangement at the present time is to have a battery of presses, grouped around central units which supply hydraulic pressure and steam for heat. A measured amount of molding powder or a pellet of compressed molding powder is applied to each cavity by the press operator, who controls by hand the time of application of heat and pressure and removes the article from the press. The cycle is a matter of minutes, and since each cycle produces a finished article if the molding is large, or a number of them if it is small, daily production per worker is high. The estimated average costs of the different elements in the process have been apportioned as follows: the cost of raw material is about one-third the cost of the finished product; and the combined cost of the mold allocated per unit, the labor cost per unit, and overhead the remaining two thirds.^[5] On small runs labor cost and particularly allocated mold cost would be much higher, so that molding is usually uneconomic where only small quantities of the finished product are desired.

In 1937 there were eight molders that produced their own tar-acid resins in whole or in part. One of these molders was the third largest producer of such resins. In the same year six producers of tar-acid resins for molding, including the first, second, fourth, and fifth largest, confined their activities to resin making. One producer of raw materials for tar-acid resins also made the resin on a moderate scale.

This picture of interstage relationship as it existed in 1937 may be somewhat modified by new developments in molding presses. There are now available self-contained presses which are not dependent upon other units for their supplies of heat and pressure and which are either semiautomatic or automatic. The semiautomatic press requires an operator for charging the cavity and removing the molded product, but once adjusted automatically applies the heat and pressure and controls the time of the pressing cycle. The automatic press, adapted as yet only to the simpler moldings, requires no attention whatever. These presses are more expensive, but may be set up anywhere and require less skilled labor. There is the possibility that they may be installed by some industrial users of molded articles, and thus take some business from the custom molder. If this occurs, such molders will presumably buy their resin from companies that are primarily resin makers, since their requirements of the material would not ordinarily be large enough to justify making their own.

Tar-acid resins for laminating.—The manufacture of laminated resin products is most economic when done on a large scale, in which case the impregnation of the paper or fabric becomes a continuous process, the material feeding from a roll through resin sirup and then through drying towers, where time and heat may be controlled. The impregnated material contains resin in the B-stage. The material is then cut up and the sheets piled together (the number depending on the thickness desired) and sent to huge presses which, with heat and pressure, compact and unite the layers and convert the resin to the C-stage. If it is desired to produce decorative panels with a smooth surface, the top sheet used is one colored or printed with a design (perhaps a photographic reproduction of the surface of a cabinet wood) and placed between polished chromium-plated metal sheets before going to the press. Rods and coil forms as well as flat sheets are commonly made from laminated material. Any of these forms may undergo subsequent fabrication; rods and coil forms cut to required length, thin sheets stamped to shape, gear blanks cut to final form on automatic gear machines, and decorative panels sawed to shape.

Many laminators purchase all their resin requirements, but a number of them make part or all of the tar-acid resin they use. In 1937 there were seven laminators who made tar-acid resins (including the second, third, and

fourth largest producers of such resins) and four producers of tar-acid resins for this use (including the largest) which did no laminating.

Cast phenolic resins.—The firms producing cast phenolic resins market them in sheets, rods, and tubes. The castings are made in molds of lead or glass, and the range of possible shapes is limited. The consumers of these products fabricate them into finished form by cutting, turning, and polishing, much as they might fabricate wood or soft metal. Since considerable labor is required per unit, fabrication is not particularly adapted to large-scale production. In 1937 there were nine producers of cast phenolic resins. One of the smaller producers was also a fabricator of cast resins, and another a producer of raw materials used in making the resin.

Tar-acid resins for coatings.—The use of tar-acid resins in surface coatings has been overshadowed by the more rapid development of alkyd resins. Nevertheless the volume of tar-acid resins used as raw materials by varnish and lacquer manufacturers is growing rapidly. They are used in marine varnishes unmodified by other synthetic resins, but to a greater extent in combination with other plastics, especially the alkyds and nitrocellulose. The coating industry includes many units producing on a large as well as a greater number producing on a smaller scale. In general, they are not producing their own tar-acid resins. In 1937 there were 11 producers of tar acid resins for coatings (including the three largest) who confined their activities to resin production. In addition there were eight manufacturers of varnishes and lacquers and one producer of resin raw materials, who also produced tar-acid resins for use in coatings.

Tar-acid resins for miscellaneous uses.—The chief uses for tar-acid resins other than for molding, casting, laminating, and in coatings are as a bonding material, and as an adhesive. These resins form a valuable bonding agent for asbestos in brake linings and chemical tanks, for abrasives and for ground cork in special uses. As an adhesive they are used in making moisture-resistant plywood.

In 1937 there were five producers of tar-acid resins for miscellaneous uses, including the largest, who confined their activities to the making of resins and two, including the second largest, who also made products in which these resins were consumed.

Alkyd resins made from phthalic anhydride.—The rapid increase in the production of alkyd resins for use in coatings is one of the most remarkable in the whole resin industry. They go into varnishes, lacquers, and enamels for spraying, brushing, and dipping. The coatings may be air-dried, with a wide range of drying time, or dried by oven baking. The volume of alkyd resins used by the coating industry has grown so large that a number of coating firms have gone into the production of alkyds and now make part or all of their own requirements. In 1937 there were 24 paint, varnish, and lacquer firms producing alkyd resins. Included in this number were the first and second largest producers of such resins. Eleven producers of these resins, including the third and fourth largest, made alkyd resins for sale only. Each of these groups included one firm which also made phthalic anhydride.

Alkyd resins made from maleic anhydride.—In 1937 there were seven producers of alkyd resins from maleic anhydride who produced for sale only. This group included the two largest producers and also one firm which produced maleic anhydride. In addition there were five paint, varnish, and lacquer firms producing part or all of their needs of resins of this type. The general conditions under which these resins are consumed are the same as for alkyd resins made from phthalic anhydride.

Urea resins for molding.—The conditions under which urea resins are molded are not greatly different from those already discussed for tar-acid resins. The molding cycle is somewhat longer and, because of the light colors used, special precautions must be taken to prevent discoloration of the molded product by dirt or flecks of molding powder from other operations, carried through the air or upon the person of the laborer. In 1937 there were four producers of urea resins for molding. Three of them, including the two largest, produced for sale only; the other consumed his own production.

Urea resins for other uses.—Until recently urea resins were thought of exclusively for molding, but they are now being used for laminating, for surface coatings, and also as an adhesive. Ordinarily the ureas are used only in impregnating the outside laminae of a laminated sheet where they are valuable for the light colors they make possible. The volume of urea resins used in surface coatings is small compared with the alkyd or tar-acid resins

used for this purpose, but is increasing. The use of urea resins in adhesives is still new but promises to become important.

In 1937 there were four producers of the ureas for uses other than molding, who produced for sale only; and two producers who consumed their own product.

Coumarone and indene resins.—Coumarone and indene resins are produced in connection with the production of solvent naphtha. There were three producers in 1937, all of whom sold their product. These resins go into varnishes, where they replace natural resins or ester gum.

Other resins.—In 1937 there were four producers of vinyl resins in the United States, and two of these also produced their raw materials. The vinyl resins were used chiefly in surface coatings, molding, and in safety glass. The polystyrene resins, used chiefly for molding and laminating, were offered by two producers for the first time in 1937. Two other producers offered acrylate resins, which are cast, molded, or used in surface coatings. In the same year petroleum resins were sold in good volume, their only producer obtaining them as a byproduct of the oil industry.

Relationship of the resin industry to other industries.

The term “synthetic resin industry” is a very broad one, referring in reality to a group of industries producing the varied synthetic resins—much as the term “steel industry” includes the manufacture of pig iron, structural steel, tin plate, and wire. But it is interesting to examine briefly the connection of the synthetic resin industry with some of the other large industrial groupings.

Relationship to the chemical industry.—Since the processes involved in the production of the synthetic resins are essentially of a chemical nature, the whole industry might be legitimately classed as a branch of the chemical industry. Historically, the synthetic resin industry in the United States developed outside of the chemical industry as it was constituted at the time, but with the passage of years and the development of a greater variety of resins the connections have multiplied. Chemical companies supply some of the important raw materials for synthetic resins; their skilled experts possess the technical training to develop new resin processes; their

research programs from time to time lead to the discovery of valuable facts regarding resin; and they possess, or can, more easily than a new company, obtain the capital necessary to exploit a process.

At present the interest of the large chemical corporations in synthetic resins ranges from active participation to apparent indifference; but the growing number of corporations thought of as chemical which are now engaged in experimental production would seem to indicate that in time they will be increasingly important in the production of synthetic resins. Some of the larger chemical companies that are important producers of synthetic resins in 1938 are:

American Cyanamid Co	Urea resins.
Carbide & Carbon Chemicals Corporation	Vinyl resins.
Dow Chemical Co	Polystyrene resins.
E. I. du Pont de Nemours & Co	Alkyd, acrylate, vinyl resins.
Monsanto Chemical Co	Petroleum resins.

Relationship to the surface coating industry.—The use of tar-acid, alkyd, urea, and vinyl resins as raw material for the surface coating industry has already been mentioned, and also the fact that the coating industry is manufacturing a substantial part of its consumption of alkyd resins.

At present the synthetic resins go chiefly into varnishes, lacquers, and enamels for inside use and into finishes for outside use on metal. Now that coatings incorporating synthetic resins are successfully adapted to outside finishes on wood, the incentive for the production of resins by the coating industry will presumably increase because of the large volume of house paints sold.

Relationship to the electric industry.—The electric industry offered one of the first large markets for synthetic resin products. Molded and laminated parts for appliances and fixtures gave good insulation at ordinary voltages, and frequently allowed a simplification of the design. This development, coming at a time of rapid expansion in the manufacture of electric equipment, was a distinct benefit to both the electrical and synthetic resin industries. The larger electrical manufacturing firms soon began to do their own molding and laminating and became important as custom molders.

Later the General Electric Co. and the Westinghouse Electric & Manufacturing Co. manufactured their own tar-acid resins.

Another important outlet for synthetic resins appeared with the development of the radio industry. Radio now offers a market for special synthetic resins possessing high dielectric constants at radio frequencies, and much larger volumes of tar-acid and urea resins are used in molding the smaller cabinets. As a rule the radio industry purchases its resin products already molded to order.

The relationship to the auto industry.—The automobile manufacturing industry and makers of automobile parts together furnish a substantial market for synthetic resins. In general, the automobile manufacturers purchase parts made of resin, already fabricated; parts makers usually purchase the resins they require. The Ford Motor Co. makes tar-acid resins for its own use. Working parts, such as timer heads and horn buttons, are usually of molding tar acid resin; the timing gear usually of laminated tar-acid resin. For decorative parts, such as dash instrument knobs and radiator ornaments, urea and cast phenolic resins have been used. Most of these parts are small, but altogether they have taken a substantial volume of synthetic resin. Safety glass for automobile windshields is now being made from vinyl resin.

The future possibilities are difficult to appraise. The automobile industry is constantly experimenting with new materials and methods, and its policy of bringing out models annually makes possible rapid adoption of new developments. Molded window frames have been tried, and such a use, or use for the complete instrument panel, would obviously consume synthetic resins in much larger volume. Even whole motor car bodies of laminated resin have been suggested.

13. THE UNITED STATES TARIFF AND INTERNATIONAL TRADE IN SYNTHETIC RESINS

Synthetic resins enter into the foreign trade of the United States only to a small extent. This becomes apparent if a comparison is made between the United States production of these resins and our imports and exports of them. Table [13](#) gives the imports and production of synthetic resins in the United States for 1934 through 1937. Exports are so small that they are not separately reported.

TABLE 13.—*Synthetic resins: United States production and imports, 1934-37*

[Pounds]				
	1934	1935	1936	1937
Production in the United States ¹	56,059,489	95,133,384	132,912,821	162,104,713
Imports into the United States	² 19,795	² 21,120	³ 626,608	³ 673,880

¹ Does not include coumarone and indene resins, sulfonamide resins.

² Does not include imports of vinyl acetate resins which were not shown separately until 1936.

³ Includes vinyl acetate resins and all other types imported.

The small size of the international trade in synthetic resins is also emphasized if we compare the imports of all synthetic resins with the imports or exports of some of the important raw materials used in their manufacture. Table [14](#) makes such a comparison.

TABLE 14.—*Comparison of international trade of the United States in synthetic resins and in certain raw materials for resins, 1934-37*

[1,000 pounds]

Imports into or exports from the United States	1934	1935	1936	1937 ¹
Imports:				
Resins	20	21	627	674
Crude cresylic acid ²	7,332	7,010	13,794	16,745
Crude naphthalene	47,995	48,455	39,806	52,664
Crude glycerin	15,081	8,220	11,149	13,441
Refined glycerin	2,214	69	3,447	7,535
Exports:				
Phenol	329	323	149	(³)
Formaldehyde	2,597	2,598	1,844	2,865

¹ Preliminary.

² Conversion factor 8.7 pounds per gallon.

³ Not available.

There are three factors that together largely account for the small size of our foreign trade in synthetic resins. As a result of the comparative youth of the resin industry, the complicated patent situation, and the substantial tariff rates upon imports of resins into the United States, domestic producers have experienced little competition from abroad. The first two of these forces plus the tariff barriers of other countries have caused them to pay little attention to export markets. But it should be observed that both of the first two forces will become less important with the passage of time. When home markets have been more fully exploited, problems of production have become less pressing, and most of the basic patents on resins have expired, international trade in synthetic resins may be expected to increase from its present low levels. If this occurs, the United States, with its large scale production for the home market and with its generally favorable position with regard to the raw materials and the technical skills necessary, is more likely to become a net exporter than a net importer of synthetic resins.

Rapid expansion of business in home markets.

Being young industries and having potentially large home markets awaiting development, the synthetic resin industries in the United States naturally began by concentrating first on their numerous production problems to meet a rapidly expanding domestic demand, improving their products and devising useful applications.

The tar-acid-formaldehyde resins for molding were the first to develop. The industry producing them may be said to have started around 1910, but did not become important until after the World War, when the drop in price of phenol made the resins available at lower prices. The alkyd resins and the urea-formaldehyde resins in the United States began to be important in 1929 and 1930, respectively. The others may be said to be still in their earliest stages of development as industries, however much research work may have been done as to their properties and production.

The effect of patents on international trade.

A second factor involved in limiting international trade in resins is that relating to patents. The basic patents on tar-acid resins have expired; but while they were in force, they prevented imports into the United States. In the United States a valid patent can be enforced at law not only against domestic products which infringe but also against imports. In addition to court action, the provisions of our tariff law prohibiting unfair competition in the import trade were invoked to prevent entry of synthetic phenolic (tar-acid) resin, form C, but when the basic patent for this material expired, the exclusion order no longer applied to single color material, except in the matter of certain marking requirements. ^[6]

The patent situation may militate against exports as well as imports. Where a company owns foreign patents it may set up a company to exploit them abroad, or it may license their use by others. Again, mutual interest may dictate an exchange (by cross-licensing) of certain patents. International licensing of patents is usually accompanied by divisions of international markets through formal or informal understanding. Such agreements may outlive the life of the patents, especially if bolstered with financial connections. But unless the original producers continue to

dominate their respective markets, any agreements between them are likely to diminish in importance, because after the patents expire new competitors would have a free hand in foreign as well as domestic markets.

The original United States producer of tar-acid resins set up or licensed companies to manufacture in a number of foreign countries. The urea-formaldehyde process was developed in Europe and the first American producer was a licensee of a British corporation. Similar arrangements exist with regard to most of the other resins.

The United States tariff on resins and resin products.

Synthetic resins.—Imports of tar-acid, alkyd, coumarone and indene, styrol, adipic, and aniline resins are dutiable under the provisions of paragraph 28 of the Tariff Act of 1930, which reads in part: “synthetic phenolic resin and all resinlike products prepared from phenol, cresol, phthalic anhydride, coumarone, indene, or from any other article or material provided for in paragraph 27 [coal-tar intermediates] or [paragraph] 1651 [coal-tar crudes], all these products whether in a solid, semisolid, or liquid condition; ... 45 per centum ad valorem [based on American selling price^[7] or United States value^[8]] and 7 cents per pound.” Where these resins are produced in the United States, imports are “competitive” and the dutiable value is based upon American selling price. If the American selling price is higher than the foreign value, the effect of this method of valuation is to increase the duty to which imports are subject. The duty of 45 per cent ad valorem and 7 cents per pound was equivalent to 54 per cent ad valorem on the American selling price of the small imports of coal-tar resins in 1937. If it could be calculated upon foreign value it would be much higher.

Synthetic resins of non-coal-tar origin, except vinyl resins, are dutiable under paragraph 11, which reads “synthetic gums and resins not specially provided for, 4 cents per pound and 30 per centum ad valorem” on foreign value. This rate was the equivalent of 48 per cent ad valorem upon the small amount of imports in 1937. The most important resins included are the urea and acrylate resins.

Between 1930 and 1936 there was some doubt whether vinyl resins were dutiable under paragraph 11 at the rate quoted or under paragraph 2 which

provided for “vinyl alcohol ... homologues and polymers of all the foregoing; ethers, esters, salts and nitrogenous compounds of any of the foregoing, whether polymerized or unpolymerized, ... not specially provided for, 6 cents per pound and 30 per centum ad valorem” on foreign value. But the Canadian trade agreement, effective January 1, 1936, reduced the rate on vinyl resins under either paragraph 2 or paragraph 11 to 3 cents per pound and 15 percent ad valorem.^[9] The reduced rate was equivalent to 25 percent ad valorem upon the imports in 1937.

Under these rates, imports of synthetic resins, other than vinyl resins, have been insignificant.^[10] After the reduction of duty, imports of vinyl resins in 1936 amounted to approximately 600,000 pounds, valued at \$145,000 and in 1937 to 650,000 pounds, valued at \$200,000. (See table [11](#).)

Articles made of synthetic resins.—Laminated products of which synthetic resin is the chief binding agent and manufactures of such products are dutiable under paragraph 1539 (b) at the following rates: 15 cents per pound and 25 percent on laminated sheets or plates^[11]; 50 cents per pound and 40 percent on laminated rods, tubes, blocks, strips, blanks, or other forms; and 50 cents per pound and 40 percent on manufactures of such laminated products. Paragraph 1539 (b) also provides a duty of 50 cents per pound and 40 percent on manufactures of any other product of which any synthetic resin is the chief binding agent. These are, for the most part, molded synthetic resin articles. Paragraph 1539 (b) does not cover articles made entirely of synthetic resin (cast synthetic resin articles). Such articles unless specifically provided for in the law are dutiable under paragraph 1558 as manufactured articles, not specially provided for, at 20 percent ad valorem.

A great many articles, which are made in whole or in part of synthetic resin, are not dutiable under either paragraph 1539 (b) or paragraph 1558. These are articles which are specifically mentioned in other paragraphs and subject to the duties provided therein. Table [15](#) lists a number of them.

TABLE 15.—*Tariff classification and rates of duty in Tariff Act of 1930 on certain articles made of synthetic resin*

Article	Tariff paragraph	Rate of duty
Beads	1503	75 percent ad valorem.
Buttons	1510	45 percent ad valorem.
Dice, dominoes, chessmen, and poker chips	1512	50 percent ad valorem.
Phonograph records	1542	30 percent ad valorem.
Cigar and cigarette holders	1552	5 cents each plus 60 percent ad valorem.
Ash trays, humidors, etc.	1552	60 percent ad valorem.
Umbrella handles	1554	75 percent ad valorem.

In general, the available statistics of imports do not segregate imports of the specified articles made of synthetic resin from those of the same articles made of other materials; and the same situation is true of imports of unspecified articles wholly of synthetic resin which enter under paragraph 1558. Imports of manufactured articles, n. s. p. f. in which synthetic resin is the chief binding agent under paragraph 1539 have been small. Figures for recent years are given in table [16](#).

TABLE 16.—*Manufactured articles n. s. p. f. in which synthetic resin is the chief binding agent: United States imports for consumption, 1931-37*

Type	1931	1932	1933	1934	1935	1936	1937 ¹
<i>Quantity (pounds)</i>							
Laminated products:							
Sheets and plates		10		13			
Rods, tubes, blocks, etc.	215	13			609	514	668

Manufactures, n. e. s.	203	453	787	783	1,703	3,260	10,397
Nonlaminated	17,623	8,511	5,352	5,729	8,423	8,069	8,759
Total	18,041	8,987	6,139	6,525	10,735	11,843	19,824
<i>Value (dollars)</i>							
Laminated products:							
Sheets and plates		9		16			
Rods, tubes, blocks, etc.	612	71			579	1,329	1,920
Manufactures, n. e. s.	1,001	883	2,133	2,299	3,778	9,468	39,232
Nonlaminated products	31,992	10,113	7,914	10,673	11,064	10,846	18,001
Total	33,605	11,076	10,047	12,988	15,421	21,643	59,153

¹ Preliminary.

Source: Compiled from Department of Commerce statistics.

14. SYNTHETIC RESIN PRICES, PROPERTIES, AND USES

Synthetic resins as substitutes.

Any new material will in the course of time be applied to the uses for which it has special advantages, displacing older materials which formerly served those purposes. The resulting product may sometimes be used in the same manner as before, or the properties of the substitute material may widen the usefulness of the finished product, or even make possible a product almost wholly new.

Before the development of molded synthetic resins, electrical plugs and sockets were usually made of porcelain or molded of marble dust and shellac. In this use substitution has been almost complete. Wall plates for electric switches and outlets were usually of brass. Today molded tar-acid or molded urea resins are substituted in part. In neither of these examples has the substituted material any important effect upon the use of the product.

An example of a substitute material widening the usefulness of the product is afforded by a new computing scale, where a molded urea resin casing (substituted for metal in the older model) has aided in decreasing the weight and has improved the appearance. Another example is the use of laminated synthetic resin coil forms in radio frequency transformers which, because of their better electrical properties at high frequencies, have aided in the design of more compact units.

Examples of synthetic resins making possible a wholly new product are more difficult to find, but the following will serve as illustrations: Cast acrylate sheets to form curved cockpit enclosures for airplanes; molded acrylate buttons for reflecting road markers; and new special coatings, which make possible the use of metal cans for preserving foods and beverages hitherto impossible to can without loss of flavor.

Motives for substitution.

One of the most important reasons why a manufacturer may decide to substitute a synthetic resin for another material is the resulting economy in the sense of economy in total costs. As a rule, the synthetic resin will be more expensive pound for pound than the material for which it is substituted; but frequently the manufacturing cost is enough lower to more than make good the difference in material cost, because the resin part will come from the mold almost in finished form, whereas the part made of wood or metal will require considerable fabrication. In some cases there may be a saving in marketing costs. For example, the shades for large office fixture lights are now made of synthetic resin as well as of opal glass. The resin shades are less expensive to ship because they are lighter and require less expensive packing.

Another incentive toward substitution is to give novelty, and hence sales appeal, to an old product. In many cases the use of synthetic resins fits in with the present tendency to redesign an old-style product so that it will be more compact, have more pleasing lines, and more color.

Still another incentive toward substitution is to give the product greater usefulness, or lower costs in use. The great expansion in the use of synthetic resins in surface coatings has come about because, with these materials, coatings can be developed to fit special purposes, and dry rapidly, which means an important saving to those who use them.

Materials displaced by synthetic resins.

The wide range of uses to which synthetic resins are now applied implies that the materials displaced are numerous. For example, cast or wrought iron or steel is displaced in timing gears and in many small machine parts, such as cradle-type telephones; nonferrous metals in small machine parts and novelties, such as inexpensive bracelets; glass in lamp shades and in cosmetic containers; natural resins in lacquers; plastics, such as cellulose acetate in safety glass or cellulose nitrate in colored lacquers; other adhesives in bonding plywood; and cork or metal in bottle closures.

In general, the quantity of material displaced is a very small part of that material's total market. Frequently, however, industries producing the finished product have had to make substantial changes in their equipment in

order to use synthetic resins. This has been true in the button industry, in the bottle closure industry, in the varnish and lacquer industry, and in the various electrical supply industries; and readjustment is now proceeding in the fancy container industry and in the safety glass industry.

Competition between synthetic resins.

Any particular synthetic resin must compete for its market with other synthetic resins, as well as with other materials. The basis of choice or substitution will be the same as that which has already been briefly discussed in connection with the displacement of other materials by resins. As between a number of resins with properties fitting them for a particular use, the total costs of using each will be compared and the choice will go to the least expensive; but where a resin has special advantages in a particular use it may win out over a less expensive resin.

It should be emphasized that this battle of materials for markets is a never-ending one. The fact that a specific synthetic resin has achieved a certain position is no guarantee that it may not lose it wholly or in part to some newer resin or other material. Thus cast phenolic resin was for a time the only resin available in light colors but urea resins became available in pastel shades and more recently water-clear polystyrene and acrylate resins have come on the market. Until recently tar-acid resins were without competition in laminating, but urea resins now are used to some extent for the surface laminae and the tar-acid resins now face a potential threat in a new product offered to laminators. If the use of this cellulose sheet, which looks much like blotting paper and which has lignin incorporated in it to act as a binder in the press, should materially decrease the cost of laminated sheets, it will mean serious new competition for the tar-acid laminating resins.

The general effect of the increase in number of types of synthetic resin has been to modify the market outlook of the producers of each type. They are now more inclined to view the market as being limited by the price at which they can supply their product and by the physical properties of each resin rather than attempt to exploit it as a universal resin for all purposes.

Resins classified by cost.

At present the resins produced in largest volume are the alkyd resins for use in surface coatings; the tar-acid resins for molding, laminating, and surface coatings; the urea resins, chiefly for moldings; and the cast phenolic resins. Roughly, the price per pound of pure resin material^[12] for these various resins may be compared as follows:

Type of resin:	<i>Average sales price of net resin, 1937 (per pound)</i>
Cast phenolic	\$0.41
Tar-acid:	
For molding	.18
For laminating	.13
For coatings	.17
Alkyd	.20
Urea	.45

Because the cost of the filler is less per pound than the cost of the resin, the cost of the tar-acid and urea molding powders will be less than the figures given for the pure resin. On the other hand, wholesale prices paid by consumers will include transportation and distribution costs not included in the figures of manufacturers' sales.

Vinyl resins, acrylate resins, and polystyrene resins are at present produced in much smaller volume than those just listed. If and when the volume of production is increased the price may be decreased. In 1937, the price per pound of pure resin^[12] was as follows:

Type of resin:	<i>Average sales price of net resin, 1937 (per pound)</i>
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Vinyl	\$0.69
Acrylate	1.66

Early in 1938, acrylate resins were being offered for sale at 85 cents per pound for molding powder and \$1.25 per pound for the cast material; polystyrene resins at 72 cents per pound.

Petroleum resins, in 1937, sold for an average of 2 cents per pound net resin content.^[12] This low price puts them beyond competition of the other synthetic resins in the uses in laminating and coating to which they are adapted.

The physical properties of a resin and its uses.

A more expensive resin will be used in preference to a cheaper one, only if the higher cost is more than offset by some physical property, such as color, which makes it more desirable in a particular use. The most common molding resin at present is the tar-acid type, but it is available only in the darker colors and therefore has been at a disadvantage, where a light color is desired, in competition with cellulose nitrate (celluloid) and cellulose acetate plastics or with urea and cast phenolic resins. In recent years the production of cellulose acetate molding compounds and of urea resins has increased rapidly, largely under this stimulus. The desire for color also promises well for the future of the acrylate and polystyrene resins which are produced in water-clear grades or colored with dyes or pigments.

TABLE 17.—*Synthetic resins and other plastics: Properties that affect appearance*

Type	Machining qualities	Clarity	Color possibilities	Burning rate	Effect of age	Effect of sun
Synthetic resins:						
Tar-acid—Formaldehyde:						
Molded, wood flour filler.	Fair to good.	Opaque	Limited	Very low	None	Little shade discoloration
Molded, mineral filler.	do	do	do	Nil	do	do
Molded, fabric filler.	do	do	do	Approximately nil	do	do
Laminated, paper base.	Fair to excellent.	do	do	Very low	Improves mechanical and electrical properties.	Low surface resistance
Laminated, fabric base.	do	do	do	do	do	do
Laminated, asbestos cloth base.	do	do	do	Approximately nil	do	do
Cast	Excellent	Transparent, translucent, opaque.	Unlimited	Very low	Hardens slightly	Color may change
Tar-acid—Furfural:						
Wood flour filler.	Fair to good	Opaque	Limited	do	do	Little shade discoloration
Mineral filler.	do	do	do	Nil	do	do
Fabric filler.	do	do	do	do	do	do
Urea—Formaldehyde.	Fair	Translucent, opaque	Unlimited pastel shades	Very low	do	No change
Vinyl, unfilled.	Good	Transparent, translucent,	Unlimited pastels to	Nil	Strength unaffected	Darkens

Vinyl, filled.	Excellent (organic filler).	opaque do	black do	Approximately nil	None	Disc
Acrylate	Very good	Transparent (95% light transmission).	Unlimited	Slow	do	No
Polystyrene	Poor to good	Transparent, translucent, opaque.	do	do	do	Yel
Other plastics: Shellac compound.	do	Opaque	Limited, pastels excluded	High (wood filler)		No
Cold molded: Nonrefractory.	Poor	do	Dark colors only	Nil		
Refractory.	do	do	Gray	do		
Rubber compounds: Chlorinated rubber.		Translucent, opaque	Unlimited	do	Slight embrittlement	Dar
Modified isomerized rubber.	Good	Transparent	do	Slow	None	Sli sur cra
Hard rubber.	Fair	Opaque	Limited	Medium	do	Disc sur resis decr
Casein	Good	Translucent, opaque	Unlimited	Very low	Hardens slightly	Co may
Cellulose compounds: Ethyl cellulose	do	Transparent, translucent, opaque	do	Slow	Slight	Sli
Cellulose acetate sheet	do	do	do	do	do	c
Cellulose acetate molding	do	do	do	do	do	c
Cellulose nitrate	do	do	do	Very high	Slight hardening	Disc a beco bri

¹ Specified refractive degree.

NOTE.—The values for the properties in this table are based upon maximum and minimum figures submitted to Modern Plastics by a number of manufacturers of each type of material. Differences in test procedures and sizes of test specimens may lead to erroneous conclusions in some cases if direct comparisons are attempted. Special grades of materials are often available which excel in one particular property.

Source: Modern Plastics, vol. 15, no. 2, opp. p. 120. October 1937.

TABLE 18.—*Synthetic resins and other plastics: Molding properties*

Type.	General molding qualities	Compression molding temperature	Compression molding pressure	Injection molding temperature	Injection molding pressure	Compression ratio
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		°F.	Pounds per square inch	°F.	Pounds per square inch	
Synthetic resins:						
Tar-acid—Formaldehyde:						
Molded, wood flour filler	Excellent	280-360	1,600-4,500	275-375	2,000-10,000	2.5-3.0
Molded, mineral filler	Excellent to fair	270-350	1,600-6,000	275-350	2,000-15,000	2.2-7.1
Molded, fabric filler	Good to fair	270-330	3,000-8,000			2.5-11.0
Laminated, paper base		250-365	1,000-3,000			1.5-3.0
Laminated, fabric base		250-365	1,000-3,000			1.5-3.0
Laminated, asbestos cloth base		250-325	1,000-3,000			
Cast						
Tar-acid—Furfural:						
Wood flour filler	Excellent	330-400	1,000-3,000	250-290	300-5,000	2.5-3.0
Mineral filler	do	330-360	1,000-3,000	250-290	300-5,000	2.5-6.0
Fabric filler	Good to fair	300-360	1,000-3,000	250-290	300-50,000	4.0-15.0
Urea—Formaldehyde (alpha cellulose filler)	Excellent	290-325	1,500-6,000			3.0
Vinyl, unfilled	Good	240-275	1,500-2,000			2.0
Vinyl, filled	Excellent	250-300	2,000-2,500			1.5-3.5
Acrylate	do	285-315	1,500-5,000	325-475	3,000-30,000	2.0
Polystyrene	Good	280-325	300-2,000	300-375	3,000-30,000	2.5
Other plastics:						
Shellac compound	do	240	1,000-1,200			
Cold molded:						
Nonrefractory	Fair		4,000-12,000			2.5
Refractory	do		4,000-12,000			3.5
Rubber compounds:						
Chlorinated rubber	do	200-225	2,000-5,000			2.0-3.0
Modified isomerized rubber	Good	260-300	1,200-4,000			3.0
Hard rubber	Fair	285-350	1,200-1,800	180-220	2,000-5,000	4.0-6.0
Casein	Poor	200-225	2,000-2,500			
Cellulose compounds:						
Ethyl cellulose	Excellent	212-300	1,000-5,000			2.2-2.9
Cellulose acetate sheet	do	210-320	500-5,000			
Cellulose acetate molding	do	250-350	500-5,000	300-440	3,000-30,000	2.0-2.8
Cellulose nitrate	Good	185-250	2,000-5,000			

¹ Positive and injection 0.002-0.003; semipositive 0.005-0.007; flash 0.008-0.009.

NOTE.—The values for the properties in this table are based upon maximum and minimum figures submitted to Modern Plastics by a number of manufacturers of each type of material. Differences in test procedures and

sizes of test specimens may lead to erroneous conclusions in some cases if direct comparisons are attempted. Special grades of materials are often available which excel in one particular property.

Source: Modern Plastics, vol. 15, No. 2, opp. p. 120. October 1937.

Table 17 lists the properties which affect appearance and gives in addition to the color range, the clarity, material, the burning rate, the effect of age and sunlight, the refractive index, and the machining quality of each synthetic resin.

Table 18 lists molding properties of synthetic resins. Of special interest are the possibilities of using a resin in injection molding. The thermoplastic resins and plastics (see softening point in table 20) are generally preferred to the thermosetting materials for injection molding because they permit the reuse of material otherwise wasted.

Table 19 lists the strength properties of the synthetic resins; table 20 the heat properties; table 21 the electrical properties; and table 22 the resistance to acids, alkalies, and solvents. All of these qualities are important in some uses and each quality may be paramount in a few. Each material has its limitations and its special advantages and the consuming industry must choose the one best suited to its purposes. The tie-up between specific properties and particular uses is exemplified by vinyl resins, which because of their great elasticity at low temperatures, are used in safety glass, and by the polystyrene resins, which because of their electrical properties at high frequencies, are used in laminated electrical parts. As production of the various resins increases new uses will probably be found for most of them.

TABLE 19.—*Synthetic resins and other plastics: Strength properties*

Type	Tensile strength	Elongation	Modulus of elasticity	Compressive strength	Flexural strength	Impact strength ¹ (foot pounds)
	<i>Pounds per square inch</i>	<i>Percent</i>	<i>Pounds per square inch × 10³</i>	<i>Pounds per square inch</i>	<i>Pounds per square inch</i>	
Synthetic resins:						
Tar-acid—Formaldehyde:						
Molded, wood flour filler	6,000-11,000		10-15	16,000-36,000	8,000-15,000	0.10-0.28; I, N
Molded, mineral filler	5,000-10,000		10-45	18,000-36,000	8,000-20,000	0.11-0.36; I, N
Molded, fabric filler	6,500-8,000		7-12	20,000-32,000	10,000-13,000	0.4-2.4; I, N
Laminated, paper base	6,000-13,000		5-20	20,060-40,000	13,000-20,000	0.4-1.2; I, N
Laminated, fabric base	8,000-12,000		5-15	20,000-44,000	13,000-20,000	0.8-5.2; I, N
Laminated, asbestos cloth base	9,000			18,000-40,000	17,000	
Cast	5,000-12,000		5-15	15,000-30,000		0.1-1.5; I, N
Tar-acid—Furfural:						
Wood flour filler	5,000-12,000		10-25	28,000-36,000	10,000-16,000	0.08-0.52; C, N
Mineral filler	4,000-12,000		10-45	24,000-36,000	8,000-14,000	0.08-0.48; C, N
Fabric filler	5,000-10,000		7-12	26,000-30,000	10,000-16,000	1.6-3.1; C, N
Urea—Formaldehyde	8,000-13,000		16	24,000-35,000	13,000-15,000	0.7-1.5; C, U
Vinyl, unfilled	8,000-10,000		3.5-4.1		10,000-13,000	0.3-0.6; I, N

Vinyl, filled	6,000-12,000		3.5-8.5			0.1-0.7; I, N
Acrylate	7,000-9,000	1.0	6	8,000	15,000-17,000	0.25-0.5; C, N
Polystyrene	5,500-7,500	1.0	4.6-5.1	13,000-13,500	6,500-8,000	0.16-0.25; I, N
Other plastics:						
Shellac compound	900-2,000					
Cold molded				6,000-15,000	5,300-7,500	0.4; C
Nonrefractory	}					
Refractory				16,000	6,000	0.4; C
Rubber compounds:						
Chlorinated rubber						3.0+; C, U
Modified isomerized rubber	4,300	0.013	4.7	8,500-11,000	7,000-9,000	2.6-6.2; I, N
Hard rubber	4,000-10,000	8-15	5.3	8,000-12,000		0.5; I
Casein	7,600		5.1-5.7			1.0; I
Cellulose compounds:						
Ethyl cellulose	2,000-7,000		2.8			1-4; I, N (per in. sq.)
Cellulose acetate sheet	6,000-11,000	20-55	1-3	4,000-16,000		2-7; C, N (per in. sq.)
Cellulose acetate molding	3,500-10,000	10-48	2-4	11,000-16,000	5,200-8,800	3-12; C, N (per in. sq.)
Cellulose nitrate	5,000-10,000	10-40	2-4			3-12; C, N (per in. sq.)

¹ ASTM D256-34T. C = Charpy; I = izod; N = notched; U = unnotched.

² 2.5 mm ball; 25 kg. load unless otherwise noted.

³ 50 kg. load.

⁴ 10 mm. ball; 500 kg. load.

⁵ Shore.

⁶ 10 kg. load.

NOTE.—The values for the properties in this table are based upon maximum and minimum figures submitted to Modern Plastics by a number of manufacturers of each type of material. Differences in test procedures and sizes of test specimens may lead to erroneous conclusions in some cases if direct comparisons are attempted. Special grades of materials are often available which excel in one particular property.

Source: Modern Plastics, vol. 15, No. 2, opp. p. 120; October 1937.

TABLE 20.—*Synthetic resins and other plastics: Heat properties*

Type	Thermal conductivity	Specific heat	Thermal expansion	Resistance to continuous heat	Softening point	Distortion under heat	Tende to co flow
Synthetic resins:	10 ⁻⁴ calories per second per square centimeter per 1°C. per centimeter	Calories per °C. per gram	10 ⁻⁶ per °C.	°F.	°F.	°F.	

Tar-acid—Formaldehyde:							
Molded, wood flour filler	4-12.2	0.35-0.36	3.7-7.5	350	None	240-285	Non
Molded, mineral filler	8-20	0.25-0.35	2.5-4	450	do.		Do
Molded, fabric filler	3-5	0.30-0.35	2-6	250-350	do.		Do
Laminated, paper base	5-8	0.3 -0.4	2	212-300	do.	320	Do
Laminated, fabric base	5-8	0.3 -0.4	3	212-350	do.		Do
Laminated, asbestos cloth base			2	400-500	do.		Do
Cast	3-5	0.3-0.4	2.8	160			
Tar-acid—Furfural:							
Wood flour filler	3.5-5	0.3-0.4	3	280-400	Chars 450	268-288	Do
Mineral filler	10-20	0.3-0.4	2	350-500	Chars 550	277-297	Do
Fabric filler	5-8	0.3-0.4	4.5	280-350	Chars 400		Do
Urea—Formaldehyde	7.13		1.5	160	None	260	Do
Vinyl, unfilled	4	0.244	6.9		130-160	140-150	Slight
Vinyl, filled	Varies	Varies	Varies		130-160	140-158	Do
Acrylate	4.3-6.8	0.45	8.5		170-235	158	Do
Styrol	1.9	0.324	10.2		110-200	185	Do
Other plastics:							
Shellac compound				150-190	150		Do
Cold molded:							
Nonrefractory				500			
Refractory				1,300			
Rubber compounds:							
Chlorinated rubber					175-230	140	Do
Modified isomerized rubber	2.6-2.9		7-8		165-220	167-221	Do
Hard rubber	3.2	0.33	8.0		150-190		Do
Casein			8			200	
Cellulose compounds:							
Ethyl cellulose						210-266	
Cellulose acetate sheet	5.4-8.7	0.3-0.4	14-16	140-180	140-230	122-212	Do
Cellulose acetate molding	5.4-8.7	0.3-0.45	14-16	140-180	145-260	122-212	Do
Cellulose nitrate	3.1-5.1	0.34-0.38	12-16	ca. 140	160-195		

NOTE.—The values for the properties in this table are based upon maximum and minimum figures submitted to Modern Plastics by a number of manufacturers of each type of material. Differences in test procedures and sizes of test specimens may lead to erroneous conclusions in some cases if direct comparisons are attempted. Special grades of material are often available which excel in one particular property.

Source: Modern Plastics, vol. 15, No. 2, opp. p. 120. October 1937.

TABLE 21.—*Synthetic resins and other plastics: Electrical properties*

Type	Volume resistivity (50 percent relative humidity) (ohm = cms)	Breakdown voltage, 60 cycles (volts per mil (instantaneous))	Dielectric constant			Power factor		
			60 cycles	10 ³ cycles	10 ⁶ cycles	60 cycles	10 ³ cycles	10 ⁶ cycles

Synthetic resins:								
Tar-acid—Formaldehyde:								
Molded, wood flour filler	10 ¹⁰ -10 ¹²	300-500	5-12	4-8	4.5-8	0.04-0.30	0.04-0.15	0.035-0.1.
Molded, mineral filler	10 ⁹ -10 ¹¹	250-400	5-20	4.5-20	4.5-20	0.10-0.30	0.10-0.15	0.005-0.1.
Molded, fabric filler	10 ⁹ -10 ¹¹	300-450	5-10	4.5-6	4.5-6	0.08-0.30	0.08-0.20	0.04-0.1.
Laminated, paper base	10 ¹⁰ -10 ¹³	400-1,300			4-6			0.02-0.05.
Laminated, fabric base	10 ¹⁰ -10 ¹²	150-600			4.5-7			0.02-0.08.
Laminated, asbestos cloth base		90						
Cast	10 ⁹ -10 ¹⁴	300-450	5-10		5-7	0.025-0.20	0.005-0.08	0.01-0.045.
Tar-acid—Furfural:								
Wood flour filler	10 ¹⁰ -10 ¹²	400-600		4-8	6-7.5		0.04-0.15	0.035-0.1.
Mineral filler	10 ⁹ -10 ¹¹	200-500		4.5-20	5-18		0.1-0.15	0.04-0.1.
Fabric filler	0.4 × 10 ¹¹	200-500		4.5-6	5-7.5		0.08-0.20	0.035-0.1.
Urea—Formaldehyde	(2-2.8) × 10 ¹³	650-720	6.6		6	0.034		0.01-0.03.
Vinyl, unfilled	10 ¹⁴	400-500			4		0.0143	0.0175.
Vinyl, filled	10 ¹¹	350-400		4.7	4		0.02-0.15	0.02-0.065.
Acrylate	10 ¹⁵	480	4-6		2.8	0.06-0.08		0.02.
Polystyrene	10 ¹⁷ -10 ¹⁸	500-700	2.6	2.65	2.7	0.0003	0.0001	0.0001.
Other plastics:								
Shellac compound		100-400						
Cold molded:								
Nonrefractory	1.3 × 10 ¹²	85	15		6	0.20		0.07.
Refractory								
Rubber compounds:								
Chlorinated rubber		2,300	ca. 3			0.003		
Modified isomerized rubber	(5-7) × 10 ¹⁶		2.7	2.68		0.006		0.0016.
Hard rubber	10 ¹² -10 ¹⁵	250-900	2.8		3			0.003-0.008.
Casein		400-700			6.15-6.8			0.052.
Cellulose compounds:								
Ethyl cellulose		1,500		3.72			0.011	
Cellulose acetate sheet	(5-30) × 10 ¹²	800-2,500	5.1-7.5		4.2-5.3	0.025-0.07		0.038-0.091.
Cellulose acetate molding	(4.2-6.2) × 10 ¹²	800-850	5.8-6.0		4.4-4.6	0.042-0.058		0.038-0.042.
Cellulose nitrate	(2-30) × 10 ¹⁰	600-1,200	6.7-7.3		6.15	0.062-0.144		0.074-0.097.

NOTE.—The values for the properties in this table are based upon maximum and minimum figures submitted to Modern Plastics by a number of manufacturers of each type of material. Differences in test procedures and

sizes of test specimens may lead to erroneous conclusions in some cases if direct comparisons are attempted.
Special grades of materials are often available which excel in one particular property.

Source: Modern Plastics, vol. 15, No. 2, opp. p. 120. October 1937.

TABLE 22.—*Synthetic resins and other plastics: Specific gravity, specific volume, and resistance to other substances*

Type	Specific gravity	Specific volume <i>Cubic inches per pound</i>	Water absorption, immersion 24 hours ¹	Effect of weak acids	Effect of strong acids	Effect of weak alkalis	
Synthetic resins:							
Tar-acid—Formaldehyde:							
Molded, wood flour filler	1.34-1.52	20.7-18.2	0.2-0.6	None to slight.	Varies ²	Slight to marked	De
Molded, mineral filler	1.70-2.09	16.4-13.3	0.01-0.3	do.	do. ²	do.	
Molded, fabric filler	1.37-1.40	20.2-19.8	1.0-1.3	do.	do. ²	do.	
Laminated, paper base	1.34-1.55	20.7-17.8	0.5-9.0	do.	do. ²	do.	
Laminated, fabric base	1.34-1.55	20.7-17.8	0.5-9.0	do.	do. ²	do.	
Laminated, asbestos cloth base	1.6-1.65	17.3-16.8	0.5	do.	do. ²	do.	
Cast	1.27-1.32	21.8-20.0	0.01-0.5	do.		do.	
Tar-acid—Furfural:							
Wood flour filler	1.3-1.4	21.3-19.8	0.2-0.6	do.	do. ²	do.	
Mineral filler	1.6-2.0	17.3-13.9	0.01-0.15	do.	do. ²	do.	
Fabric filler	1.3-1.4	21.3-19.8	0.8-1.4	do.	do. ²	do.	
Urea—Formaldehyde	1.48-1.50	18.7-16.5	1-2	do.	Decomposed or surface attacked.	do.	
Vinyl, unfilled	1.34-1.36	20.7-20.4	0.05-0.15	Resistant	Resistant	Resistant	I
Vinyl, filled	1.35-2.5	20.5-11.1	0.2-4.0	Dependent on filler.	Dependent on filler.	Dependent on filler.	D
Acrylate	1.18	23.3	0.3	None	Oxidizing acids attack surface.	None	(
Polystyrene	1.05-1.07	26.3-25.8	0	do.	None	do.	
Other plastics:							
Shellac compound	1.1-2.7	25.2-10.3		Deteriorates	Deteriorates	Deteriorates	De
Cold molded:							
Nonrefractory	1.98-2.0	14.0-13.9	1.5	Slight	Decomposes	Decomposes	De

Refractory	2.2	12.6	0.5-15	Decomposes	do.	None	
Rubber compounds:							
Chlorinated rubber	1.5	18.5	0.1-0.3	Resistant	Resistant	Resistant	I
Modified isomerized rubber	1.06	26.1	0.02	do.	do.	do.	
Hard rubber	1.12-1.8	24.7-15.4	0.02	do.	Attacked by oxidizing acids.	do.	
Casein	1.35	20.5	3-7	do.	Decomposes	Softens	De
Cellulose compounds:							
Ethyl cellulose	1.14	24.3	⁶ 1.25	Slight	Decomposes	None	
Cellulose acetate sheet	1.27-1.37	21.8-20.2	1.5-3.0	do	do	Slight	De
Cellulose acetate molding	1.27-1.63	21.8-17.0	1.4-2.8	do	do	do	
Cellulose nitrate	1.35-1.60	20.5-17.3	1.0-3.0	do	do	do	

¹ ASTM D48-33.

² Decomposed by oxidizing acids; reducing and organic acids no effect.

³ On bleed-proof materials.

⁴ Resists alcohols, aliphatic hydrocarbons, and oils. Soluble in ketones and esters; swells in aromatic hydrocarbons.

⁵ Soluble in ketones, esters, and aromatic hydrocarbons.

⁶ 48 hours.

⁷ Soluble in ketones and esters; softened by alcohols; little affected by hydrocarbons.

NOTE.—The values for the properties in this table are based upon maximum and minimum figures submitted to Modern Plastics by a number of manufacturers of each type of material. Differences in test procedure and sizes of test specimens may lead to erroneous conclusions in some cases if direct comparisons are attempted. Special grades of materials are often available which excel in one particular property.

Source: Modern Plastics, vol. 15, No. 2, opp. p. 120. October 1937.

15. SYNTHETIC RESINS IN OTHER COUNTRIES

Large-scale production of synthetic resins is confined principally to the United States, Germany, and Great Britain. There is small production in many other countries, of which the most important are France, Italy, Czechoslovakia, Canada, and Japan.

In 1934 the world output was estimated at 135 million pounds, of which the United States produced about 44 percent, Germany 26 percent, and Great Britain 24 percent. In 1937 world output was estimated at 360 million pounds, the United States' share of the total being almost 50 percent, followed by 27 percent for Germany, 20 percent for Great Britain, and the remaining 3 percent scattered.

European estimates indicate that about 40 percent of the output goes into surface coatings and that 60 percent of the surface-coating resins are tar-acid and 40 percent alkyds. The Tariff Commission found that in 1937 50 percent of the United States production of all synthetic resins went into surface coatings, 27 percent into molded articles, and the remaining 23 percent into laminating and miscellaneous uses. Approximately three-fourths of the surface-coating resins were alkyds and one-fourth tar-acid resins.

GERMANY

Production.

In recent years Germany's production of synthetic resins has increased rapidly, each succeeding year registering the attainment of a new record. In 1933 production totaled 17,500,000 pounds and by 1935 had increased to 55,000,000 pounds. A further expansion of about 30 percent to 70,000,000 pounds in 1936 and present production trends indicate a gain of about 40 percent more in 1937, to an estimated total of 100,000,000 pounds.

Although tar-acid resins comprise the bulk of the German output, considerable gains are shown for other types, notably injection molding resins of the polystyrene and vinyl types. The development of completely automatic injection molding machinery has given an impetus to these types. While technical progress, including improvement of molding equipment, has contributed to the expanded production, the use of synthetic resins in Germany has had a strong stimulus because they are made almost wholly of domestic materials. Under the "Four-Year Plan" for the greatest possible national economic independence, synthetic resins are replacing imported

materials, such as the heavier nonferrous metals, iron, hardwoods, cork, and natural gums and resins in many uses. This displacement of materials has also affected such domestic products as glass and porcelain, which caused the Government to intervene and impose restrictions upon the use of resins for purposes adequately served by other materials of German origin.

Germany's expanding production of synthetic resin has also been aided by a sharp increase in exports, which have increased well over 100 percent since 1932.

Tar-acid resins.—German output of tar-acid resins has been estimated at 35 million pounds in 1934, at 49 million pounds in 1935, and at 63 million pounds in 1936. Such resins comprise the bulk of the German production of molding resins.

There are at least seven producers of tar-acid resins in Germany and nine producers of molding powders and pellets. Tar-acid resins for surface coatings are produced by a number of these concerns. Among the important makers in Germany are The Bakelite Gesellschaft (organized in 1910 to operate under the Baekeland patents); the explosives and munitions firm of Dynamit A.G.; Dr. Kurt Albert G.m.b.H.; the I.G. Farbenindustrie; Beckacite Kunstharzfabrik G.m.b.H.; and Rohm & Haas A.G. The Beckacite firm has associates in the United States and in the United Kingdom, and Rohm & Haas, an associate in the United States.

Alkyd resins.—The manufacture of alkyd resins has developed in Germany in the past few years. Demand for these resins has been given a marked impetus by the development of a new standardized substitute for linseed-oil varnish known as El Varnish, the use of which is required by the Control Board for Industrial Fats for certain interior and exterior painting.

There are five makers of resins for paints, varnishes, and lacquers. The output of alkyd resins has increased sharply since 1934, probably reaching 10 million pounds in 1936.

Urea resins.—The output of urea resins in Germany is relatively small; two of the more important types are known as Locron and Pollopas.

Polystyrene and vinyl resins.—In 1936 Germany's production of thermoplastic resins exceeded 1 million pounds, principally of the polystyrene and vinyl types. Among the vinyl resins are Acronal and Mowilith, both of which are manufactured by the I.G. Farbenindustrie. This combine also produces several types of polystyrene resins known as Mollit and Metastyrol. Dynamit A.G. produces a polystyrene resin known as Trolitul.

Uses of synthetic resins.

The original and most important use of synthetic resins in Germany was for electrical insulation. This use was so extensive that the industry was organized in 1924 into an association known as non-rubber insulation materials industry. Materials were standardized and classified into 14 types, of which 5 were tar-acid resins and 1 was a urea resin. Every type must meet certain specifications in order to be recognized by the Reich Testing Institute. More than 100 firms produce insulating materials meeting the institute's specifications.

Radio panels of the popular sets sponsored by the Government are made of synthetic resins. Consumption in the automobile industry is increasing for such parts as instrument panels, electrical equipment, steering wheels, gear-shift knobs, and numerous others. The latest airplanes show increased use of synthetic resins, where they contribute light weight, great strength, and resistance to corrosion.

In cameras and moving-picture equipment, wood and metal have been in part replaced by synthetic resins. Other applications of resins in Germany include bearings for rolling mills, goggles and spectacles (including the lens), and perfume and medicine bottles.

Resins for surface coatings are undergoing rapid development in Germany, owing to the shortage of linseed oil. Alkyd resins in coatings are being promoted by the Government, which prohibits or limits the use of the older oil-type coatings for certain uses so as to decrease the use of linseed oil and other paint oils that must be imported and hence require outlays of foreign exchange. Penalties have been imposed for violating the regulations.^[13]

Organization.

The synthetic-resin industry in Germany is a unit within the national industrial organization. It is a subdivision of the industrial chemical group, called Fachgruppe Kunststoffe, or Group 13 of the 19 trade groups in the chemical division. This subdivision controls casein and cellulose plastics as well as synthetic resins, and is further divided as follows: (1) Casein plastics, (2) cast phenolic resins, (3) molding compositions, (4) resins for lacquers, (5) celluloid and zellon, (6) transparent sheeting, (7) linoleum, and (8) miscellaneous (such as vulcanized fiber, bottle caps, and die-casting resins).

There are two cartels distinct from the national organization, which expressly excludes all functions and activities of cartels. One cartel represents the firms interested in molding compositions and the other those interested in synthetic resins for other purposes. Some of the producers are members of both cartels.

Foreign trade.

Imports of synthetic resins are negligible, although the duty of 4.6 cents per pound (25 marks per 100 kilograms) on imports into Germany is not prohibitive. Exports have increased practically every year since 1930, when they were first recorded separately.

Table 23 shows the quantity and value of exports in recent years.

TABLE 23.—*Synthetic resins: German exports, 1930-37*

Year	Hardening resins			Nonhardening resins		
	1,000 pounds	Value		1,000 pounds	Value	
		1,000 marks	1,000 dollars		1,000 marks	1,000 dollars
1930	2,549	1,973	472			
1931	3,775	2,757	651			
1932	3,162	2,112	501			
1933	4,009	2,625	801	6,628	3,566	1,088
1934	4,924	3,162	1,246	7,076	3,415	1,346
1935	4,948	2,993	1,206	6,921	3,445	1,388
1936	6,392	3,501	1,411	7,764	3,820	1,539
1937 ¹	8,706	4,402	1,770	10,866	5,389	2,117

¹ Preliminary.

Source: Consular reports.

German exports of synthetic resins are, for the most part, destined to European countries, most of which have increased their purchases considerably in recent years. Exports to Latin American countries have increased recently, especially to Brazil. Table 24 shows the distribution of exports in recent years.

TABLE 24.—*Synthetic resins: German exports, by countries, 1934-37*

[Thousands of marks]				
Destination	1934	1935	1936	1937 ¹
Austria	259	352	446	593
Belgium	215	259	297	420
Czechoslovakia	347	345	604	825
Denmark	316	391	473	540

France	626	651	680	734
Great Britain	1,247	563	596	844
Hungary	240	135	182	(²)
Italy	252	359	523	615
Netherlands	530	572	645	1,031
Spain	225	302	178	57
Sweden	415	457	463	691
Switzerland	721	705	714	749
Other European countries	370	618	706	(²)
Argentina	250	207	194	(²)
Brazil	46	77	109	(²)
Other Latin American countries	17	18	75	(²)
All other countries	501	427	436	2,692
Total	6,577	6,438	7,321	9,791

¹ Preliminary.

² Included in all other countries.

Source: Official German statistics.

GREAT BRITAIN^[14]

As in most other countries, the history of the synthetic-resin industry in Great Britain begins with the acquisition of rights by a British concern to manufacture under the original Bakelite patents. The Damard Lacquer Co., Ltd. was probably the pioneer maker of phenolic resins in England. The principal product was a baking lacquer sold under the trade name Damarda, marketed for and used principally as a coating to prevent corrosion on brass. The outbreak of the World War created such an urgent demand for laminated materials that this firm started production of them for the British Government. In 1926 this concern was merged with Mouldesite, Ltd. and Redmanol, Ltd., under the name of Bakelite, Ltd.

Production.

Statistics of production of synthetic resins in Great Britain are available only for 1934 and 1935. They are given in table [25](#).

TABLE 25.—*Synthetic resins: Production in Great Britain, 1934 and 1935*

Type	1934	1935
	<i>Pounds</i>	<i>Pounds</i>
Solid, liquid, cured, uncured, and hardened	25,558,400	13,283,200
Molding powder, 50 percent or more resin		25,872,000
Laminated sheets, rods, blocks, tubes	1,164,800	1,646,400
Total	26,723,200	40,801,600

Source: Great Britain. Board of Trade, Census of Production.

Capital invested in the British industry is reported as 15,000,000 pounds sterling and direct employment as 20,000 people.

Tar-acid resins.—Many large moldings are made in England, such as large radio cases, desk files, trays, and drain boards. Cast phenolic resin production has just been started in England.

Among the novelties recently produced in England is a toy railway molded of tar-acid resin. The trains and track spacers are of nonconducting resin; the molded rails are made conductive by a thin covering of metal which is pressed in and secured at the ends. Two trains may be run on the same set of rails at different speeds, or one can go forward and another backward, since the two outer rails are separate conductors, the third rail acting as a common return.

Molded piano parts are being tried in an attempt to solve the troubles hitherto encountered with wood, owing to variations in humidity. Resins have long been used in facing the keys, but the production of piano action parts has presented many technical difficulties. The secret of success with molded resin parts lies in molding the joints in position when the main body is molded. There are 88 sections in each piano.

Urea resins.—British Cyanides, Ltd., well-known makers of synthetic resins in England, acquired the Pollopas patents for the manufacture of urea resins in the United Kingdom, in certain continental European countries, and in the British Empire except Canada. The agreement called for a full exchange of patents and other information with the other licensees of the Pollopas patents. These arrangements were made for the purpose of consolidating the patent position and for the pooling of technical data already existing on manufacture, with the object of improving quality.

Acrylate resins.—An outstanding development in Great Britain has been the production of the thermoplastic resins known as Diakon and Perspex. These are made from methyl methacrylate and are developments of the Imperial Chemical Industries, Ltd. Diakon is for molding powders and Perspex is in the form of cast sheets, rods, tubes, and optical forms.

These new commercial resins are considered the best combination thus far obtained of strength, transparency, and light weight. Applications in England include fittings for aircraft, transparent inspection covers for machinery, medical equipment, instrument windows, lenses and prisms in optical systems, and aircraft windscreens. They are used in subways for lenses for deflecting and diffusing light and in battery cases and coil forms.

The general properties of the acrylate resins include transparency to both visible and ultraviolet light, almost unlimited color range, resistance to acids and alkalis, and superior electrical properties.

Aniline resin.—Panilax is an aniline-formaldehyde condensation product made in England. It has high electrical and thermal insulating properties, great mechanical strength, is odorless and odor repelling, and practically unaffected by water, oil, and alkalis.

Organization.

Most of the British producers of synthetic resins are members of the British Plastics Federation, Ltd.

Several years ago a 10-year contract was made between the Imperial Chemical Industries, Ltd. and the Toledo Synthetic Products Co. (now Plaskon Co.) of Toledo, Ohio. This agreement provides for an exchange of all technical and commercial information on urea-resin products and processes and the granting of free licenses under present or future patents.

Agreements probably also exist between the British Bakelite Co. and the American firm on tar-acid resins; between Nobel Chemical Finishes, Ltd. and E. I. du Pont de Nemours & Co. on alkyd resins; between British Thompson Houston Co., Ltd., and the General Electric Co. on alkyd resins; between Imperial Chemical Industries, Ltd. and du Pont on acrylate resins; and between Beetle Products Co. and American Cyanamid Co. on urea resins.

Foreign trade in resins.

British imports of synthetic resins, by principal sources, are shown in table [26](#).

TABLE 26.—*Synthetic resins: Imports into the United Kingdom, in selected years, 1930-36*

[1,000 pounds]

Source	1930	1931	1933	1934	1935	1936
British countries.	1	(¹)	5	2	19	24
Germany	508	1,621	2,267	2,259	1,476	914
Netherlands	679	667	151	114	(²)	(²)
UNITED STATES	119	229	656	902	986	1,056
All other countries	65	281	246	257	323	435
Total	1,372	2,798	3,470	3,534	2,804	2,429

¹ Less than 500.

² Included in "All other countries."

Source: Official statistics of the United Kingdom.

British exports of synthetic resins to principal countries are shown in table [27](#).

TABLE 27.—*Synthetic resins: Exports from the United Kingdom, in selected years, 1930-36*

[1,000 pounds]						
Source	1930	1931	1933	1934	1935	1936
British countries	138	170	992	1,350	1,788	2,732
Sweden	40	69	242	452	558	650
Denmark	(¹)	(¹)	99	140	159	150
Belgium	(¹)	(¹)	104	205	237	203
Italy	(¹)	(¹)	49	95	(¹)	(¹)
Argentina	(¹)	(¹)	28	198	156	238
All other countries	104	171	366	505	735	1,084
Total	282	410	1,880	2,945	3,633	5,057

¹ Not available; included in "All other countries."

Source: Official statistics of the United Kingdom.

FRANCE[\[15\]](#)

Producers.

Statistics of French production and sales of synthetic resin are not available. Larousse Commercial Illustré describes the French synthetic resin industry as not

important and estimates the output in

1930 at 2,000,000 pounds. The *Revue Général des Matières Plastiques*, most important technical review in France, estimates the production in 1931 as about 3,500,000 pounds.

The comparatively few French companies producing synthetic resins are, for the most part, under British or German control. The types of synthetic resin made in France, the trade names, and the names of the manufacturers, follow:

Bakelite.—Tar-acid molding compounds and laminating materials; cast phenolic resins; Cie La Bakelite, Bezous, Seine.

Plastose and Ferodo.—Tar-acid molding compounds; Société Ferodo-Plastose, Saint Ouen, Seine.

Pollopas.—Urea molding compounds and laminating materials; Établissements Kuhlmann, Paris.

Foreign trade.

French imports of synthetic resins are classified under tariff item No. 0376 bis: Synthetic resins (solid or resinous products of the Bakelite, Albertol, Plastose types, etc.) derived from the condensation of aldehydes with phenols, amines, and amides. Several subclassifications are shown: (a) Soluble in oil and not polymerizable, (b) which may be rendered insoluble and infusible, and (c) infusible. Imports in recent years, from principal sources, are shown in table [28](#).

TABLE 28.—*Synthetic resins: French imports, by types and by countries, 1931 and 1933-37*

[Pounds]						
Source	1931	1933	1934	1935	1936	1937 ¹
	Soluble in oil					
Germany	563,860	1,003,860	1,359,600	1,164,470	1,085,766	(²)
UNITED STATES	174,900	126,280	185,680	284,458	162,699	(²)
United Kingdom	184,800	131,120	80,520	109,789	18,960	(²)
Austria		35,640	162,580	193,564	575,180	(²)
Netherlands		49,720		16,755	(²)	(²)
All other countries	4,620	5,720	3,080	11,023	33,069	(²)
Total	928,180	1,352,340	1,791,460	1,744,059	1,875,894	1,794,985

	Molding compounds					
United Kingdom	21,780	71,060	10,340	11,243	23,589	(²)
Germany	248,600	49,060	20,460	68,563	39,242	(²)
Switzerland		13,200	31,900	11,464	(²)	(²)
UNITED STATES	11,220	18,920	22,660	20,062	66,799	(²)
Belgium		31,240	49,500	7,716	(²)	(²)
All other countries	3,080		4,840	6,173	5,732	(²)
Total	284,680	183,480	139,700	125,221	135,362	105,380
	Molded, cast, and laminated articles					
Germany	12,980	7,700	4,840	9,039	17,857	(²)
Netherlands				220		(²)
Austria	4,840	440	220			(²)
United Kingdom			220			(²)
UNITED STATES		220		220		(²)
All other countries	1,320				1,984	(²)
Total	19,140	8,360	5,280	9,479	19,841	8,377

¹ Preliminary.

² Not separately reported.

Source: Consular reports.

Exports of synthetic resins from France, by principal markets, are shown in table [29](#).

TABLE 29.—*Synthetic resins: French exports 1931 and 1933-37*

[Pounds]						
Destination	1931	1933	1934	1935	1936	1937
Belgium	203,060	224,180	186,780	113,757	165,565	(¹)
Argentina		69,080	91,300	(¹)	(¹)	(¹)
Switzerland			16,940	12,787	37,258	(¹)
Italy	12,980			(¹)	(¹)	(¹)
All other countries	4,840	29,260	15,180	54,895	36,376	(¹)
Total	220,880	322,520	310,200	181,439	239,199	417,772

¹ Not separately reported.

Source: Consular reports.

CZECHOSLOVAKIA

Production of phenolic resins in Czechoslovakia has increased rapidly in recent years and is ample to supply domestic requirements. Most of the raw materials are imported from Germany, Great Britain, and France, but formaldehyde is produced locally in sufficient quantities.

The principal makers of synthetic resins in Czechoslovakia are:

- | | | |
|-----|-------------------------------|-------------|
| (1) | Bratislavská továrna na kable | Bratislava. |
| (2) | Schreiber & Co. | Lipník |
| (3) | Ing. Alex Reiber | Sereď |
| (4) | J. Elias | Prague |
| (5) | Mathias Oechsler & Sohn | Riegersdorf |
| (6) | J. Batistello, Jr. | Gablonz |

Resin products are widely used by the electrical industries for wall plates, plugs, switches, fuse boxes, etc. Other articles made of synthetic resins are: handles and knobs for furniture and kitchen equipment, bottle caps, fountain pens and pencils, clock and radio housings, tableware, cutlery handles, trays, buttons, toilet ware and toys.

Imports of synthetic resins in 1934 totaled 1,270,500 pounds; Germany supplied 46 percent and Great Britain 22 percent of this total. Exports of synthetic resins during the same year amounted to 166,540 pounds and went principally to Poland, Yugoslavia, Germany, and Argentina.

ITALY

The Società Italiana Resine, an affiliate of the important chemical firm, Chimiche Forestali, is a leading maker of tar-acid resins in Italy. A new and modern plant is located at Milan in close proximity to the electrical and textile industries, both important markets for resins.

In 1936 the Ministry of Corporations granted Montecatini Società Generale per l'Industria Mineraria, Milan, a permit to develop a factory for alkyd resins; and also Società Italiana Ebonite and Sostituti, Milan, one to produce tar-acid resins. In 1937 a permit was granted to Montecatini S.A. for a plant to manufacture acrylic acid resins at the Villadossola works of the Soc. Elettochimica del Toce.

JAPAN^[16]

The history of the synthetic resin industry in Japan goes back to 1913 when Dr. Jokichi Takamine, discoverer of adrenalin and takadiastase, acquired the right to manufacture and sell tar-acid resin Products in Japan. The business was financed by the Sankyo Co., Ltd., and a factory was built at Shinagawa, near Tokyo. In 1923 a subsidiary company known as the Japan Bakelite Co., Ltd., was formed with a paid-in capital of 1,200,000 yen. This firm considers itself an affiliate of the Bakelite Corporation of the United States and, according to an existing agreement, cannot export to the United States. Its territory includes the Japanese Empire and Manchukuo. China is considered an open market.

The original plant at Shinagawa was partially destroyed by fire in 1919, and the following year was moved to Mukojima, Tokyo. The firm makes tar-acid resins, and a full line of products covered by the patents of the American concern. Included are laminated sheets, molding compounds, molded articles, surface coating resins, laminated resin gears and spindles for rayon mills. An interesting development is the adaption of tar-acid resin lacquers to the production of Japanese lacquer ware such as bowls, vases, etc.

Since the establishment of the Japan Bakelite Co., several other firms have started the production of synthetic resins. The Tokyo Electric Co., an affiliate of the General Electric Co., makes tar-acid resins under the trade name Tecolite. Products are used principally for insulation, although molding compositions and molded articles such as are used by the electrical trade are commercially produced.

The Matsushita Electrical Works at Osaka are producers of tar-acid resins and articles made therefrom. The output is used largely for radio and electrical equipment. The Nissholite Manufacturing Co., Ltd., with a factory at Yasui-cho, Uzumasa, Kyoto specializes in decorative laminated material sold under the trade name Nissholite. The Japan Nitrogenous Fertilizer Co. (Nippon Chisso Hirijo Kabushiki Kaisha) is an important maker of tar-acid resins, marketing them under the trade names Chissolite, Safeloid, and Minaloid. The Yokahama Resin Co., a relatively small company, produces tar-acid resins and markets them in the form of molding powders. The firms listed account for practically all of the Japanese production of synthetic resins and for about 50 percent of the molded articles made from them. The remaining 50 percent of the output of molded articles is made by a large number of small firms, the majority being household industries. It is reported that there are about 2,000 of these so-called plants already engaged in this relatively new industry.

Production.

The Japanese production of manufactures of tar-acid resin reported by the Department of Commerce and Industry is shown in table 30. These data include the output of plants employing more than five operators and apparently account for only half of the total.

TABLE 30.—*Manufactures of tar-acid resins: Production in Japan, 1929-35*

Year	Quantity	Value		
		Of quantity reported	Additional ¹	Total
	<i>Pounds</i>			
1929	28,681	\$46,594	\$125,404	\$171,998
1930	607,800	52,409	442,583	494,992
1931	744,119	99,907	268,594	368,501
1932	286,422	36,584	367,220	403,804
1933	229,854	26,747	516,903	543,650
1934	1,435,977	193,857	926,951	1,120,808
1935	3,176,441	477,526	923,546	1,401,072

¹ Quantity not reported.

Source: Factory statistics of Department of Commerce and Industry, Japan.

Estimates from other sources of Japanese productions of tar-acid resins indicate an output of 2,600,000 pounds of resin and 3,600,000 pounds of molded resin articles in 1933, and of 4,900,000 pounds of resin and 7,500,000 pounds of resin articles in 1935.

It was recently announced that the Gosei Chemical Co. will manufacture vinyl resins in Japan. This firm's principal interest is in acetate fiber and rayon manufacture. Later in 1936 the Showa Fertilizer Co. announced the successful development of a process for making urea. Urea resins are in commercial production by the Toyo Gosei Kagaku Kogyo K.K., an affiliate of Chugoku Toyo K.K.

The resin industry in Japan is expected to undergo considerable development in the near future. Raw materials are available in sufficient quantities and the art of molding is fairly well developed.

CANADA

The producers of synthetic resins in Canada are:

Bakelite Corporation of Canada, Ltd.	Toronto.
Shawinigan Chemicals, Ltd.	Shawinigan Falls.
Canadian General Electric Co.	Toronto.
Canadian Industries, Ltd.	Toronto.

The Bakelite Corporation of Canada, Ltd., an affiliate of the firm of the same name in the United States, was formed in 1925. This plant makes molding materials, laminating materials, and an extensive line of technical varnishes. Molded parts were made at this factory until 1932.

Shawinigan Chemicals, Ltd. is the pioneer organic chemical maker in Canada. A modern plant at Shawinigan Falls, Quebec, produces synthetic acetic acid, acetaldehyde, vinyl acetate, vinyl resins, and other chemicals. The vinyl resins manufactured by this firm have already been described (see p. [44](#)). Appreciable quantities of these resins have been exported to the United States in the past but the construction of a factory (jointly owned by Shawinigan Chemicals, Ltd., and the Fiberloid Corporation) at Indian Orchard, Mass., for the manufacture of vinyl resins will probably result in a decrease of exports from Canada to the United States.

The Canadian General Electric Co. makes alkyd resins for use in surface coatings. Phthalic anhydride and other raw materials are imported from the United States. Canadian Industries, Ltd., produces alkyd resins at a plant in Toronto, Ontario.

There are about 14 molders of synthetic resins in Canada, of which all but 3 are in Ontario. These firms make a general line of molded articles including electrical articles, closures, costume jewelry, and smokers' accessories. Appreciable quantities of molded articles are imported from the United States and smaller quantities from Germany.

UNION OF SOVIET SOCIALIST REPUBLICS

The synthetic resin industry in the Union of Soviet Socialist Republics is concentrated in two public departments, known as Public Commissariates: (a) Public Department for Heavy Industry and (b) Public Department for Light Industry.

The Department for Heavy Industry, known as Soyuzchemplastmass, controls the following plants:

1. Karbolit-pawod in Ljubatschani, producing tar-acid resin laminated fabric known as Textolite.
2. Karbolit-stroj in Ljubatschani, making cast phenolic resins.
3. Karbolitni-pawer in Dubrowka, making tar-acid resin molding compounds. This plant has at least 350 molding presses producing electrical parts and automotive

parts. The number of presses was to have been increased to 1,000 in 1937.

4. Komsomolskaja prawda in Leningrad manufactures articles, including phone sets, from cast phenolic resins.

5. Ochtenski Chimkombinat in Ochta. This plant makes nitrocellulose plastics. No information could be obtained by our Chargé d’Affaires at Moscow concerning its production of synthetic resins, although it is believed to be considerable.

The Department for Light Industry has a resin section known as Mosplastmass producing casein plastics only.

THE NETHERLANDS

There has been no production of synthetic resins in the Netherlands; but a plant is under construction (October 1937) at Groningen for the manufacture of alkyd resins. The manufacture of surface coating and electrical parts from imported resins is carried on, chiefly by N. V. Philips’ Gloeilampenfabrieken, Afdeeling Inkoop, Eindhoven, manufacturers of radios, filament lamps, and electrical appliances. Efforts are being made to employ resins for other purposes, such as the bonding of plywood and the manufacture of closures and novelties, but little has been accomplished thus far. The relatively high cost of the resins is the principal difficulty. Molding compounds and laminated sheets, rods, and tubes are imported from Germany, Great Britain, Austria, and the United States.

The paint, varnish, and lacquer industry in the Netherlands has been experimenting with synthetic resins for several years. Alkyd resins of the glycerol phthalate type are being used by Dutch paint makers, imported principally from Germany and Austria. In spite of high cost, they have been found to have many advantages, especially better and more uniform quality. The prices of gums and resins in the Netherlands during the latter part of 1936 are shown in table [31](#).

TABLE 31.—*Prices of gums and resins in the Netherlands, 1936*

Type	Florins per 100 kilos
Damar	37.
Congo copal (various qualities)	12 to 45.
Indian copal (various qualities)	20 to 35.
Kauri (various qualities)	25 to 200.
Shellac (various qualities)	37 to 52.
Pine resin (rosin) (various qualities)	13 to 14.

The Dutch aviation industry is using tar-acid resins to bond plywood for wing surfacing on Fokker-type wooden planes. The advantages obtained are excellent adhesiveness and resistance to moisture and temperature changes. In this application they have replaced casein.

Germany supplies more than 85 percent of the Netherland imports of synthetic resins, as shown in table [32](#).

TABLE 32.—*Synthetic resins: Netherland imports by countries, 1931 and 1933-37*

[Pounds]						
Source	1931	1933	1934	1935	1936	1937 ¹
Germany	1,203,393	1,257,568	1,207,857	1,351,581	1,490,310	2,449,311
United Kingdom	8,520	47,843	64,458	94,565	335,099	1,223,553
Austria	63,758	7,297	30,886	63,642	(²)	132,276
UNITED STATES	3,168	24,193	27,434	50,888	(²)	(²)
Belgium	2,640	3,923		1,514	(²)	(²)
France		3,120	4,129	616	(²)	(²)
Czechoslovakia	3,326	4,948			(²)	(²)
Switzerland	1,789	4,193			(²)	(²)
Other countries	1,450	1,027	2,629	1,573	216,051	207,232
Total	1,288,044	1,354,112	1,337,393	1,564,379	2,041,460	4,012,372

¹ Preliminary.

² Not separately reported.

Source: Consular reports.

DENMARK

The annual output of synthetic resins in Denmark is about 500,000 pounds, almost entirely of the tar-acid type.

Bakelite is produced by the Nordiske Kabel and Traadfabrikker A. S. Fabrikvej at Copenhagen. Other brands made in Denmark are Nokait, Helomit, and Etronit. There are 14 manufacturers of finished products, making electrical equipment principally.

POLAND

Production of synthetic resins in Poland in 1936 totaled 660,000 pounds, entirely of the tar-acid type.

16. RAW MATERIALS FOR ALKYD RESINS

The alkyd resins are made chiefly from phthalic anhydride and glycerin. Phthalic anhydride in turn is made from naphthalene. Polybasic acids such as maleic, succinic, etc., may also be used with glycerin to form alkyd resins. Naphthalene, phthalic anhydride, maleic and other polybasic acids, and glycerin are discussed in the order named.

NAPHTHALENE

The discovery of naphthalene in coal tar was made simultaneously by Garden and Brande in 1819, and its composition was determined by Faraday in 1826 and later by Laurent in 1832. Naphthalene is almost invariably a constituent of the products obtained when organic matter is heated to comparatively high temperatures. For example, it is formed in small quantities when acetylene, alcohol, acetic acid, benzene, or toluene are heated to high temperatures. Together with certain aromatic hydrocarbons it is formed in the cracking of petroleum and in the hydrogenation of petroleum fractions. Naphthalene is a constituent of the principal varieties of tar produced from coal in the manufacture of gas and coke under ordinary conditions, but not of low-temperature tar. It is present in coal gas although its presence must be kept as low as possible to avoid blocking service pipes in cold weather. The proportion in gas tar varies with the kinds of coal used and is greater the higher the temperature used during carbonization; it usually amounts to 4 to 6 percent but is sometimes as much as 10 percent. In tars obtained from byproduct coke ovens the proportion of naphthalene and other aromatic hydrocarbons depends on the type of oven used. Byproduct coke-oven tar averages 10 to 11 percent naphthalene; blast-furnace tar contains only very small amounts.

Processes to synthesize naphthalene were described as early as 1873 by Aronheim, in 1876 by Wroden and Znutowicz, and in 1884 by Baeyer and Perkin. English Patent No. 26,061 of 1898 claims that it may be obtained by heating barium carbide with barium hydroxide to a high temperature. None of these processes has become of commercial importance.

Recovery of naphthalene.

Naphthalene is recovered in the distillation of coal tar, in the fraction boiling at 180° to 250° C., in the creosote oil fraction boiling at 240° to 270° C. and most abundantly in the carbolate or middle oil fraction boiling at 200° to 250° C. When these fractions are allowed to cool most of the naphthalene crystallizes out and is separated by draining and hot-pressing. This crude material is partially purified by washing with hot caustic soda solution to remove tar acids and then with mineral acid to remove basic substances. Refined naphthalene is obtained by subliming, or preferably by distilling the crude product.

Description and uses.

The Tariff Act of 1930 defines crude naphthalene as naphthalene solidifying under 79° C. after the removal of all water present; and refined naphthalene as that having a solidifying point at or above 79° C. after the removal of all water present.

Crude grades, melting between 70° and 78.5° C., are found in commerce as yellow, red, or brown crystalline solids. These grades are used in the manufacture of phthalic anhydride and other coal-tar intermediates; in the manufacture of lampblack; to enrich illuminating gas and sometimes motor fuel; in synthetic tanning materials; and in certain insecticides. Probably its most important outlets are as a raw material for phthalic anhydride (see p. [98](#)) and refined naphthalene.

Refined grades, melting above 79° C., are marketed as white, crystalline lumps or flakes. Their principal uses are in the manufacture of intermediates, dyes, medicinals, solvents, and textile assistants; as moth repellants; as a lubricant when mixed with rapeseed oil; to remove the "bloom" from lubricating oils; as a preservative for rubber goods and animal skins; and in explosives (trinitro naphthalene). In 1936 more than 75 coal-tar intermediates made from naphthalene were commercially produced in the United States. Of the 75 million pounds of these

intermediates produced in that year, 31 million pounds were phthalic anhydride, an important component of synthetic resins of the alkyd type.

United States production.

Crude naphthalene is produced in the United States by byproduct coke-oven operators, gas works that produce their own coal tar, and also by firms that purchase coal tar and distill it. Statistics of production by groups are shown in table 33.

TABLE 33.—*Crude naphthalene: United States production, 1918-37*

Year	By producers of tar			By purchasers of tar			Total production		
	Quantity	Value	Unit value	Quantity	Value	Unit value	Quantity	Value	Unit value
	<i>1,000 pounds</i>		<i>Per pound</i>	<i>1,000 pounds</i>		<i>Per pound</i>	<i>1,000 pounds</i>		<i>Per pound</i>
1918				40,138	\$1,281,440	\$0.032	40,138	\$1,281,440	\$0.032
1919				12,612	327,201	.030	12,612	327,201	.026
1923	11,872	\$201,824	\$0.017	41,453	652,148	.016	53,325	853,972	.016
1925	9,239	92,389	.010	34,135	519,773	.015	43,374	612,162	.014
1926	7,747	100,709	.013	45,166	494,986	.011	52,913	595,695	.011
1927	8,303	91,331	.011	45,298	470,806	.010	53,601	562,137	.010
1928	¹ 12,182	146,186	.012	35,180	395,059	.011	47,362	541,245	.011
1929	¹ 19,761	316,182	.016	19,502	366,491	.020	39,263	682,673	.017
1930	¹ 12,640	151,681	.012	18,617	304,574	.020	31,257	456,255	.015
1931	¹ 7,623	76,229	.010	13,311	199,665	.015	20,934	275,894	.013
1932	¹ 4,632	41,690	.09	8,961	125,453	.014	13,593	167,143	.012
1933	¹ 6,618	66,181	.010	24,003	360,040	.015	30,621	426,221	.014
1934	¹ 10,743	139,665	.013	27,179	489,222	.018	37,922	628,887	.016
1935	¹ 12,937	168,185	.013	34,716	624,890	.018	47,653	793,075	.017
1936	¹ 37,552	600,836	.016	51,984	1,195,632	.023	89,536	1,796,468	.020
1937	¹ 60,797	1,215,942	.020	55,182	1,545,100	.028	115,979	2,667,522	.023

¹ Crude and refined. Refined naphthalene included here is probably small so that the figures here and those for total production are substantially accurate.

Source: Bureau of Mines and U.S. Tariff Commission.

Refined naphthalene is obtained from domestic crude, imported crude, and recently from petroleum cracking and hydrogenation. Table 34 shows the annual production and sales of refined naphthalene since 1916. The difference between the figures for the quantity produced and that sold represents the amount used by refiners in the manufacture of other products.

TABLE 34.—*Refined naphthalene: United States production and sales, 1917-37*

Year	Production			Sales		
	Quantity	Value	Value	Quantity	Value	Value
	<i>1,000 pounds</i>	<i>1,000 dollars</i>	<i>Per pound</i>	<i>1,000 pounds</i>	<i>1,000 dollars</i>	<i>Per pound</i>
1917	35,343	2,334	\$0.07			
1918	28,112	2,163	.08			
1919	17,625	1,161	.07			
1920	30,231	2,309	.08			
1921	13,554			13,183	741	\$0.056

1922	17,420		14,060	794	.057
1923	28,184		21,871	1,271	.058
1924	15,324		11,961	603	.050
1925	17,581		12,508	610	.049
1926	18,072		12,456	576	.046
1927	21,233		(¹)		
1928	24,992		(¹)		
1929	31,144		21,120	1,027	.049
1930	31,956		20,171	949	.047
1931	34,959		21,260	829	.039
1932	25,825		18,877	783	.041
1933	42,708		28,658	1,065	.037
1934	38,730		21,257	1,100	.052
1935	46,564		28,761	1,212	.042
1936	52,694		30,499	1,841	.060
1937	52,194		29,657	1,893	.060

¹ Not publishable.

Source: Compiled from annual reports of the Tariff Commission on dyes and other synthetic organic chemicals in the United States.

Organization of the industry.—There are 10 domestic producers of crude naphthalene, operating 52 tar-distilling plants in the following States: Ohio (7), Pennsylvania (6), Illinois and New York (5 each), Alabama, Minnesota, and New Jersey (3 each), Missouri, Rhode Island, Wisconsin, Utah, West Virginia (2 each), and Michigan, Massachusetts, Maryland, Kentucky, Oregon, Connecticut, Tennessee, Indiana, Virginia, and Washington (1 each). Although these plants do not all recover naphthalene as such, they are equipped to recover a crude mixture of naphthalene and tar acids for shipment to a central extracting and refining plant. The principal producing plants are located in Pennsylvania (2), New Jersey (2), Illinois (1), Indiana (1), and West Virginia (1).

The purchasers of tar produced 77 percent of the total output of crude naphthalene in 1935 and 58 percent in 1936.

There are 8 producers of refined naphthalene located in the following States: New Jersey (3), Pennsylvania (2), California, Indiana, and Ohio (1 each).

Trend of production.—Although the United States is the largest producer of coal tar, the limited demand for the main products of tar distillation (creosote oil and pitch) has tended to restrict the amount distilled, thereby reducing the output of naphthalene and the tar acids to a point where the domestic output was not sufficient to meet our requirements. As a result, large quantities of these products have been imported. In 1936 we produced 560 million gallons of coal tar, containing from 400 million to 500 million pounds of naphthalene.^[17] In the same year we distilled about 300 million gallons of tar, containing 230 million to 270 million pounds of naphthalene,^[17] and our actual recovery of crude naphthalene was 89,536,000 pounds.

Early in 1935 the price of crude naphthalene was about 1.5 cents per pound or 15 cents per gallon, at which level there was little incentive to isolate it from the various fractions of tar distillation. Late in 1935 and in 1936 a serious shortage in naphthalene prevailed, largely because of increased demand by synthetic resin makers but also because of restrictions on exports from certain European countries. The price of the crude then advanced in domestic markets to from 2.5 to 3 cents per pound, with the demand greatly exceeding the supply. Between 1930 and 1936 the apparent consumption of crude naphthalene (production plus imports) increased from 59 million to 129 million pounds, or more than 100 percent. During the same period production increased from 31 million to 89 million pounds; and imports increased from 27 million pounds in 1930 to 48 million pounds in 1935 but declined to 40 million pounds in 1936.

Domestic producers of naphthalene are increasing their output, and they state that continued prices of 2.5 to 3 cents per pound for naphthalene solidifying at about 75° C. or slightly higher will stimulate production sufficiently to meet all present and near-future requirements. Estimates obtained in the summer of 1936 from the large tar distillers and a petroleum refiner indicate that production was appreciably greater in 1937 than in 1935. These

estimates included the potential output of two new tar distillation plants under construction, the topping of large quantities of tar (hitherto used as fuel without removing any of the products), the recovery of appreciable quantities of naphthalene by several petroleum refiners, and increased output by other producers.

Imports of crude naphthalene in 1937 amounted to 52,664,277 pounds valued at \$1,133,157, or 2.2 cents per pound.

World production.

The output of naphthalene in the principal producing countries, in 1933 and 1935, is shown in table 35. Most of these statistics were estimated from the output of tar or of other distillation products given in official reports of the countries or in consular reports.

The figures in table 35 indicate that the output in 1935 was an increase of about 100 million pounds over 1933 or 41 percent. Notwithstanding this sharp increase in world production, consumers had difficulty in obtaining their requirements. It is believed that the world output in 1937 substantially exceeded that in 1935.

TABLE 35.—*Naphthalene (all grades): World production, by countries, 1933 and 1935*

[In thousands of pounds]		
Country	1933	1935
Germany	109,148	145,530
Great Britain	45,750	¹ 55,000
UNITED STATES	30,620	47,653
France	30,000	¹ 33,000
Netherlands	¹ 15,000	¹ 15,000
Belgium	11,025	¹ 25,000
Czechoslovakia	6,835	10,805
U. S. S. R.	¹ 10,000	¹ 15,000
Poland	¹ 5,000	¹ 8,000
Spain	¹ 1,250	¹ 2,000
Italy	¹ 2,500	¹ 3,000
Canada	¹ 2,000	¹ 3,000
Total	269,128	362,988

¹ Estimated.

Source: Official statistics of the several countries and consular reports.

Germany.—Germany is the largest producer of naphthalene and the third largest producer of coal tar. With increased production of coal tar and intensive efforts to recover the maximum of naphthalene there has been a larger output of naphthalene in recent years, but increased consumption has created a scarcity in Germany as in all other important producing countries. As a result, greatly reduced quantities are available for export, a situation that is in marked contrast to earlier periods when superabundant production created a marketing problem. The manufacture of phthalic anhydride for alkyd resins is requiring increased quantities of naphthalene.

The demand for alkyd resins has been given a marked impetus by the development of a new standardized linseed oil varnish substitute known as El Varnish, the use of which is required by the Control Board for Industrial Fats in Germany for certain interior and exterior painting (see p. 77). Increased requirements for other important purposes such as intermediates, dyes, black pigments, and explosives have also contributed to the scarcity of naphthalene. In order to conserve domestic supplies, the Reich Government, from December 1935 until late in 1937 prohibited its export without special permit. The prospect of continued strong domestic demand apparently will curtail for an indefinite period the quantities available for export.

The international scarcity of naphthalene resulted in a sharp increase in its price in Germany as elsewhere. The export embargo augmented the domestic German supply, although a shortage still existed and large consumers found it difficult to secure adequate amounts. The shortage of foreign exchange greatly curtailed imports of naphthalene from nearby countries.

The German Government issued a decree requiring that beginning July 1, 1936, the entire national output of coal tar should be delivered to plants equipped for the recovery of tar products distilling up to 240° C. (naphthalene boils at 218° C.). This measure assured maximum recovery of benzol, toluol, xylol, solvent naphtha, phenol, cresol, xylenol, other tar acids, and naphthalene. The decree contemplated an official list of distillation units, and all tar producers were required to report to the official trade control board for mineral oil their monthly output, quantities distilled, and quantities delivered to other distillation plants.

German production, imports, exports, and apparent consumption of naphthalene are shown in table 36. Production increased from 108 million pounds in 1928 to 146 million pounds in 1935; imports decreased from 9 to 4 million pounds; exports decreased from 48 to 22 million pounds; and apparent consumption increased from 69 to 128 million pounds in the same years.

TABLE 36.—*Naphthalene: German production, imports, exports, and apparent consumption, 1928-37*

[In thousands of pounds]				
Year	Production	Imports	Exports	Apparent consumption ¹
1928	108,173	9,471	48,332	69,312
1929	124,362	8,032	39,739	92,655
1930	103,194	3,892	34,614	72,472
1931	92,169	2,403	39,077	55,495
1932	90,626	952	29,720	61,852
1933	109,148	7,483	31,842	84,783
1934	132,300	8,641	35,044	105,891
1935	145,530	4,246	22,169	127,603
1936	(²)	493	8,153	(²)
1937	(²)	33	24,966	(²)

¹ Production plus imports, minus exports.

² Not available.

Sources: Consular reports (production); Der auswärtige Handel (imports and exports).

Imports of naphthalene into Germany in past years have been supplied by nearby countries, notably the Saar (which became an integral part of Germany in February 1935), Belgium, Czechoslovakia, Poland, the Soviet Union, and others. The United States has been the most important foreign market for German naphthalene, taking from 50 to 75 percent of the total quantity exported. Other important buyers were Belgium, Italy, Japan, and France. Table 92 (see p. 144) shows the quantity and value of imports and exports by countries in recent years.

Great Britain.—The recovery and distillation of coal tar in Great Britain is highly developed. The annual output of tar, principally gas-works tar, is somewhat smaller than in the United States, although the quantities distilled for the recovery of separate components exceed the quantities distilled in the United States. In 1935 the tar distilled in England and Wales totaled 330 million gallons and in Scotland, 31 million gallons, or a combined total of 361 million gallons as compared with about 280 million distilled in this country.

Production of naphthalene in Great Britain is shown in table 37.

TABLE 37.—*Naphthalene: Production in Great Britain, in specified years*

Year	Production
------	------------

	<i>1,000 pounds</i>
1924	13,730
1930	41,400
1933	45,750
1935 ¹	55,000
1936 ¹	70,000

¹ Estimated.

Source: Consular reports.

Table 38 shows exports of naphthalene from Great Britain in recent years. The United States has been the best customer, in most recent years taking 50 percent or more of the total exported. Our imports from Great Britain have been entirely crude naphthalene, duty-free.

Imports of naphthalene into Great Britain are not shown separately in official statistics. It is known that the Netherlands exported small quantities to Great Britain in 1929 and 1933.

TABLE 38.—*Naphthalene: Exports from the United Kingdom*

Year	Quantity		Value			
	To all countries	To United States	To all countries		To United States	
			Pounds sterling	Dollars ¹	Pounds sterling	Dollars ¹
	<i>1,000 pounds</i>	<i>1,000 pounds</i>				
1928	5,132	(²)	20,607	100,278	(²)	(²)
1929	9,185	4,312	32,348	157,110	12,558	60,993
1932	11,132	7,514	26,869	94,205	14,274	50,046
1933	14,718	10,480	38,172	161,728	19,604	83,059
1934	11,955	6,492	35,226	177,514	13,025	65,637
1935	14,490	7,999	49,939	244,789	18,413	90,256
1936	26,332	13,412	120,372	598,357	46,158	229,447

¹ Conversion to dollars at annual average quotations of the Federal Reserve Board.

² Not available.

Source: The Trade of the United Kingdom, 1929 and 1936.

Belgium.—The distillation of coal tar is one of the oldest and most important branches of the Belgian chemical industry. Approximately 90 batteries of byproduct-coke ovens, with a total of 3,000 ovens are in operation. Practically all of the coal tar produced in these operations is distilled for the recovery of the several products. The output of naphthalene is shown in table 39.

TABLE 39.—*Naphthalene: Belgian production, 1928-35*

Year	Quantity
	<i>1,000 pounds</i>
1928	26,000
1929	26,500
1930	24,200
1931	22,000
1933	12,000
1935 ¹	25,000

¹ Estimated.

Source: Consular reports.

Belgian imports and exports of naphthalene, by countries, are shown in tables [93](#) and [94](#) (see pp. [146](#), [147](#)). Belgium is a net importer of crude naphthalene and a net exporter of refined naphthalene. In 1937, it imported 9 million pounds and exported 6.7 million of crude; it imported only 19 thousand pounds and exported 14 million pounds of refined.

Czechoslovakia.—The annual output of naphthalene in Czechoslovakia is shown in table [40](#).

TABLE 40.—*Naphthalene: Czechoslovak production, in specified years, 1928-35*

Year	Quantity
	<i>1,000 pounds</i>
1928	5,733
1930	6,174
1931	2,205
1932	1,543
1933	6,835
1934	9,040
1935	10,805

Source: Consular reports.

France.—Naphthalene is produced in France by a number of manufacturers, most of whom consume their production in their own factories. The French output is said to be insufficient to meet domestic requirements. Estimated production is given as approximately 30 million pounds annually. Appreciable quantities are imported from nearby countries. Imports from Belgium in recent years average between 1 million and 3 million pounds.

Poland.—Production of crude naphthalene in Poland is shown in table [41](#).

TABLE 41.—*Crude naphthalene: Polish production, 1928-36*

Year	Quantity
	<i>1,000 pounds</i>
1928	4,708
1929	5,257
1930	3,925
1931	3,486
1932	3,704
1933	4,795
1934	7,705
1935	5,021
1936	2,836

Source: Consular reports.

The Netherlands.—Statistics of production are not available. Exports in recent years, however, have averaged about 10 million pounds annually. It is believed that the production of crude naphthalene exceeds 15 million pounds a year.

Table [95](#) (see p. [148](#)) shows Netherland imports and exports of naphthalene by countries in recent years. Imports in 1937 amounted to 2 million pounds and exports to 15 million pounds.

Canada.—Statistics of production are not available. The annual output of crude naphthalene is estimated at 2 to 3 million pounds.

Imports of refined naphthalene are usually about 1 million pounds (see table 96, p. 150). Exports are probably small, although in 1929 and 1934 those to the United States alone were over 1 million pounds.

The Soviet Union.—Statistics of production of naphthalene in the Soviet Union are not available. The annual output has been estimated at 10 million pounds in 1933 and 15 million pounds in 1935. Exports have increased substantially in recent years, those to the United States from 1 million pounds in 1934 to 6 million pounds in 1935. Exports to Germany were 361 thousand pounds in 1933; 1 million pounds in 1934; and 531 thousand pounds in 1935.

Japan.—Japanese production of naphthalene has been small compared with the output of other tar products. The output of crude naphthalene in 1934 was reported to have been 381 thousand pounds. Expansion of the byproduct coking industry in Japan and Manchuria has increased the production of coal tar, byproduct ammonia, and benzol. Japan has imported large quantities of naphthalene in recent years, principally from Germany and Belgium. The increased consumption in Europe may so reduce supplies from these sources as to cause Japan to increase the recovery at home.

Japanese imports of naphthalene from principal sources, are shown in table 97 (see p. 150). In 1936, 12.6 million pounds were imported.

United States imports.

Rates of duty.—Prior to September 8, 1916, all grades of naphthalene were imported free of duty. Since that time crude naphthalene has remained free but refined naphthalene has been subject to the tariff treatment shown in table 42.

TABLE 42.—*Naphthalene: Rates of duty upon imports into the United States, 1916-38*

Period	Rate of duty		Authority
	Crude	Refined	
To Sept. 8, 1916	Free	Free	Free under par. 452 of the act of 1913 and under previous acts.
Sept. 9, 1916, to Sept. 8, 1921.	do.	15 percent ad valorem and 2½ cents per pound.	Revenue Act of 1916.
Sept. 9, 1921, to Sept. 21, 1922.	do.	15 percent ad valorem and 2 cents per pound.	Emergency Tariff Act of 1921. (Title V, prohibited imports for 3 months except when not obtainable in sufficient quantities or on reasonable terms as to quality, price, and terms of delivery).
Sept. 22, 1922, to Sept. 21, 1924.	do.	55 percent ad valorem and 7 cents per pound. ¹	Crude, free under par. 1549 and refined dutiable under par. 27 of the Tariff Act of 1922.
Sept. 22, 1924, to June 17, 1930.	do.	40 percent ad valorem and 7 cents per pound. ¹	Ad valorem rate on refined reduced as provided for in Tariff Act of 1922.
June 18, 1930, to Apr. 30, 1935.	do.	do.	Crude, free under par. 1651 and refined dutiable under par. 27 of Tariff Act of 1930.
May 1, 1935, to—	do.	20 percent ad valorem and 3½ cents per pound. ¹	Refined reduced under trade agreement with Belgium. ²

¹ Ad valorem based on American selling price or United States value.

² Generalized to all countries which do not discriminate against United States products.

Under the Tariff Act of 1930, crude naphthalene is on the free list^[18] and refined naphthalene is dutiable at 7 cents per pound and 40 percent ad valorem on the basis of American selling price, since it is competitive with refined naphthalene produced in this country.^[19] Under the trade agreement with Belgium, effective May 1, 1935, the duty on refined naphthalene was reduced to 20 percent^[20] ad valorem and 3½ cents per pound on imports from that country. Under the Trade Agreements Act this reduction applies also to imports from all other countries which do not discriminate against commerce of the United States. In July 1938 Germany was the only one not receiving the reduced rate, exports from that country being subject to the rates specified under the Tariff Act of 1930.

Import statistics.—Table 43 shows imports of crude naphthalene (solidifying at less than 79° C.) and table 44 of refined naphthalene (solidifying at or above 79° C.) The unit invoice values of imports of refined naphthalene in 1924, 1926, 1927, 1928, and 1935 indicate that the imported product was probably not naphthalene as recorded but one of its derivatives provided for elsewhere in paragraph 27.

TABLE 43.—*Crude naphthalene (solidifying at less than 79° C.): United States imports for consumption, in specified years, 1919-37*

Calendar year	Rate of duty	Quantity	Value	Value per pound
		<i>1,000 pounds</i>		
1919	Free	3,239	\$92,265	\$0.028
1920	do.	15,012	530,219	.035
1923	do.	20,992	575,702	.027
1924	do.	5,267	96,491	.018
1925	do.	1,980	26,593	.013
1926	do.	6,963	126,088	.018
1927	do.	6,576	131,436	.020
1928	do.	19,926	357,679	.018
1929	do.	35,007	598,718	.017
1930	do.	27,667	397,292	.014
1931	do.	30,971	318,578	.013
1932	do.	27,002	234,557	.009
1933	do.	42,786	451,019	.010
1934	do.	47,995	669,383	.014
1935	do.	48,455	643,249	.013
1936	do.	39,806	785,396	.020
1937 ¹	do.	52,664	1,133,157	.022

¹ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

TABLE 44.—*Refined naphthalene (solidifying at or above 79° C): United States imports for consumption, in specified years, 1919-37*

Calendar year	Rate of duty	Quantity	Value	Unit value	Computed ad valorem rate
		<i>Pounds</i>			<i>Percent</i>
1919	15 percent + 2½ cents per pound.	7,650	\$384	\$0.050	64.8
1920	do.	3,697,562	416,172	.112	37.2
1923	55 percent + 7 cents per pound	9,605	194	.020	401.6
1924	do.	4,549	1,147	.252	82.8
1925	do.				
1926	40 percent + 7 cents per pound	424	125	.295	63.7

1927	do.	18,668	3,077	.165	82.5
1928	do.	27	6	.222	71.5
1929	do.	}	None		
1930	do.				
Jan. 1-June 17	do.				
June 18-Dec. 31	do.				
1931	do.	}			
1932	do.				
1933	do.				
1934	do.				
1935	do. ¹	66	6	.091	116.7
1936	20 percent + 3½ cents per pound ²	99	31	.313	62.4
1937 ³	do. ²	30	20	.667	50.5
		5,055	1,085	.215	36.3

¹ From Germany. No imports under trade agreement rate.

² Belgo-Luxemburg trade agreement rate.

³ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

Table 45 shows the principal sources of our imports of crude naphthalene in recent years. Germany was the principal source until 1936; the United Kingdom, previously the next most important source, was first in 1936 and 1937. In the last three years appreciable quantities have been received from Poland, Czechoslovakia, and the Soviet Union, hitherto unimportant sources.

TABLE 45.—*Crude naphthalene (solidifying under 79° C.): United States imports for consumption from principal sources, in specified years*

Source	1929	1931	1933	1934	1935	1936	1937 ¹
Quantity in thousands of pounds							
Germany	21,931	17,444	20,797	22,219	15,742	2,712	12,129
Belgium	2,531	253	4,970	7,314	2,388	2,025	1,995
United Kingdom	8,096	11,339	15,704	6,968	10,689	16,301	17,594
Poland and Danzig				5,766	5,075	1,969	2,312
Canada	1,488	331	223	1,073	76	255	734
Netherlands	44	937	1,092	621	1,344	3,794	3,359
Czechoslovakia				2,984	6,960	6,595	6,414
U. S. S. R.				1,050	6,158	5,145	7,091
All other countries	918	667			22	1,010	1,038
Total	35,007	30,971	42,786	47,995	48,455	39,806	52,664
Value							
Germany	\$382,078	\$170,463	\$242,501	\$326,607	\$230,820	\$75,314	\$287,901
Belgium	48,508	2,506	57,243	90,424	31,375	55,503	51,227
United Kingdom	124,427	123,890	135,853	78,968	123,545	273,964	340,760
Poland and Danzig				89,002	63,992	35,439	55,184
Canada	23,344	3,808	2,729	18,703	1,169	4,093	7,941
Netherlands	614	11,837	12,693	8,739	19,724	105,404	93,045
Czechoslovakia				44,371	98,099	120,529	128,197
U. S. S. R.				12,569	74,354	97,815	146,331
All other countries	19,747	6,074			171	17,335	22,571
Total	598,718	318,578	451,019	669,383	643,249	785,396	1,133,157

	Value per pound						
Germany	\$0.017	\$0.010	\$0.012	\$0.015	\$0.015	\$0.028	\$0.024
Belgium	.019	.010	.012	.012	.013	.027	.026
United Kingdom	.015	.011	.009	.011	.012	.017	.019
Poland and Danzig				.015	.013	.018	.024
Canada	.016	.011	.012	.017	.015	.016	.011
Netherlands	.014	.013	.012	.014	.015	.028	.028
Czechoslovakia				.015	.014	.018	.020
U. S. S. R.				.012	.012	.019	.021
All other countries	.022	.009			.008	.017	.022
Average	.017	.010	.011	.014	.013	.020	.022
	Percent of total quantity						
Germany	62.7	56.3	48.6	46.3	32.5	6.8	23.0
Belgium	7.2	.8	11.6	15.3	14.9	5.1	3.8
United Kingdom	23.1	36.6	36.7	14.5	22.0	41.0	33.4
Poland and Danzig				12.0	10.5	5.0	4.4
Canada	4.3	1.1	.5	2.2	.2	.6	1.4
Netherlands	.1	3.0	2.6	1.3	2.8	9.5	6.4
Czechoslovakia				6.2	14.4	16.6	12.2
U. S. S. R.				2.2	12.7	12.9	13.4
All other countries	2.6	2.2				2.5	2.0
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0

¹ Preliminary.

Source: Compiled from official statistics of the United States Department of Commerce.

United States exports.

Exports are not shown separately; it is doubtful if any naphthalene is exported. Demand in the United States has exceeded domestic production.

Competitive conditions.

The commercial development and widespread application of surface coatings and finishes made from alkyd resins, in which phthalic anhydride and glycerin are the principal components, has resulted in a world-wide shortage of naphthalene, which is a raw material used in making phthalic anhydride. In recent years about one-half of domestic requirements of crude naphthalene have been imported (see table 46) from Europe, principally Germany and the United Kingdom. Increased demand for the same purposes, in these countries, has so reduced the quantities available for export as to create a serious shortage in the United States. Germany placed an embargo on exports late in 1935 and continued it until late in 1937.

TABLE 46.—*Crude naphthalene: United States production, imports, and apparent consumption in specified years*

Year	Production ¹	Imports ²	Apparent consumption ³	Percent supplied by imports
	<i>1,000 pounds</i>	<i>1,000 pounds</i>	<i>1,000 pounds</i>	
1923	53,325	20,992	74,317	28
1927	53,601	6,577	60,178	11
1929	39,263	35,007	74,270	47
1931	20,934	30,971	51,905	60

1932	13,593	27,002	40,595	66
1933	30,621	42,786	73,407	58
1934	37,922	47,995	85,917	56
1935	47,653	48,455	96,108	50
1936	89,536	39,806	129,342	31
1937 ⁴	115,979	52,664	168,643	31

¹ From table [33](#).

² From table [43](#).

³ Production plus imports.

⁴ Preliminary.

Vast quantities of naphthalene potentially available in this country were not recovered because of the low prices prevailing until 1936. Since then an increase in the price of crude naphthalene from 1.55 cents to 2.5 cents and 3 cents per pound has stimulated production and has led to additional recovery.

PHTHALIC ANHYDRIDE

Description and uses.

Phthalic anhydride is an aromatic polybasic organic acid anhydride made from naphthalene by vapor phase catalytic oxidation. It is marketed as white needle-shaped crystals or flakes having a melting point of 130° to 131° C. and boiling at 284° to 285° C. It is the cheapest and most widely used aromatic organic acid. Its most important use is in the manufacture of synthetic resins of the alkyd type. Other important uses are in dye intermediates; in phenolphthalein; in benzoic acid; in dyes such as indigo, phloxines, rhodamines, erythrosines; and in esters such as dibutyl phthalate (widely used as a plasticizer in nitrocellulose lacquers and films and of interest as a greaseless lubricant), diethyl phthalate (used as a perfume fixative and denaturant of alcohol), dimethyl phthalate (used as a plasticizer in cellulose acetate films), and diamyl phthalate (used as a plasticizer). Important new processes using phthalic anhydride as a raw material include the syntheses of anthraquinone, substituted anthraquinones, and benzoyl benzoic acid.

Before the World War phthalic anhydride was made by heating naphthalene with sulphuric acid in the presence of mercury; the sulphuric acid acted as an oxidizer, and sulphur dioxide and carbon dioxide were liberated. This process was used in Europe and in the United States to produce the small quantities of phthalic anhydride needed for the manufacture of certain dyes and intermediates. It proved highly unsatisfactory as to operation; the yield varying widely from batch to batch. The sales price of the phthalic anhydride produced at that time was as high as \$4.25 per pound, whereas it is 12 to 14 cents per pound today.

In September 1916, Gibbs and Conover, working in the Color Laboratory of the Bureau of Chemistry and Soils, United States Department of Agriculture, developed a process for the synthesis of phthalic anhydride by the direct vapor phase catalytic oxidation of naphthalene. This work was done under the United States Government's wartime investigation of the manufacture of intermediates and dyes. Gibbs and Conover were granted United States Patent No. 1,285,117 covering the basic process, and the invention was assigned to the people of the United States. This process revolutionized the manufacture of phthalic anhydride, causing the market price to drop to \$2.85 per pound in 1918, to 46 cents per pound in 1920, and to 13 cents in 1930. With each decline in price new outlets were found, and domestic production increased practically every year, rising from 227,000 pounds in 1918 to 23,500,000 pounds in 1935.

By a remarkable coincidence the same basic process was developed in Germany, by Alfred Wohl, in the laboratories of the Interessen Gemeinschaft Industrie A. G. (German I. G.), at almost the same time that Gibbs and Conover made their discovery. In 1920 Wohl applied for a United States patent covering this process, claiming invention in the summer of 1916. There was some doubt whether his discovery had been made 2 months earlier or 3 days later than that of Gibbs and Conover, but in July 1934 the United States Court of Customs and Patent Appeals rendered a decision in favor of the German inventor, allowing Wohl's claim filed with the German patent office on June 28, 1916. Therefore, Wohl's claim covering the air oxidization process was upheld and he was granted United States Patent No. 1,971,888, issued August 28, 1934 and assigned to the German I. G.

Several domestic firms began commercial production of phthalic anhydride about 1918 under the patent of Gibbs and Conover and have since operated the process continuously. Such manufacturers are presumably protected from possible patent litigation and the payment of royalties under the Wohl patent by section 3 of the so-called Nolan Act of 1921, which states: "No patent granted or validated ... shall affect the right of any citizen of the United States or his successor in business to continue the manufacture, use, or sale commenced before the passage of this Act, nor shall the continued manufacture, use, or sale by such citizen ... constitute an infringement."

United States production.

Table 47 shows the production and sales of phthalic anhydride from 1917 through 1937.

TABLE 47.—*Phthalic anhydride: United States production and sales, 1917-37*

Year	Production	Sales		Unit value
		Quantity	Value	
	<i>Pounds</i>	<i>Pounds</i>		
1917	138,857	138,857	\$587,240	\$4.23
1918	227,414	227,414	648,650	2.85
1919	290,677	290,677	290,037	.99
1920	796,210	796,210	362,431	.46
1921	202,471	202,471	79,162	.39
1922	1,629,182	1,317,625	461,944	.35
1923	2,343,802	2,091,100	596,508	.29
1924	2,787,308	2,277,073	556,265	.24
1925	3,900,332	3,560,429	701,840	.20
1926	4,379,108	3,446,175	604,949	.18
1927	4,549,820	4,064,476	686,946	.17
1928	6,030,854	5,445,432	888,156	.16
1929	9,168,946	7,450,037	1,147,953	.15
1930	6,693,001	5,614,012	724,909	.13
1931	(¹)			
1932	6,259,000	5,695,000	663,000	.12
1933	14,075,844	11,593,716	1,271,887	.11
1934	20,680,379	13,511,253	1,575,787	.12
1935	23,421,558	17,931,662	2,105,134	.12
1936	31,244,378	22,905,873	2,824,471	.12
1937	45,210,784	17,565,905	2,492,473	.14

¹ Not available.

Source: Compiled from annual reports of the Tariff Commission on dyes and other synthetic organic chemicals in the United States.

There are six domestic makers of phthalic anhydride, with producing units at Bridgeville, Pa., Buffalo, N. Y., Philadelphia, Pa., Deepwater Point, N. J., Saint Louis, Mo., and Detroit, Mich. Five of these firms have been producing in commercial quantities continuously for a number of years and it is therefore believed that these companies may continue to produce without the payment of royalties. New producers using this process, however, might be at a disadvantage unless licensed to operate without the payment of royalties by the owner of the patent.

The production of phthalic anhydride has increased remarkably since the discovery of the vapor phase catalytic process of manufacture. Until 1922 the only large outlet was the coal-tar dye industry. The development of new uses for phthalic esters, principally dibutyl phthalate, increased the demand during the period 1922-28. With the drop in price of phthalic anhydride, resins made from it and glycerin became of commercial interest and about 1929 their production began to increase sharply. Most of the increased output since that year is accounted for by its

use in alkyd resins. As previously stated, surface coatings made from these resins are now applied to practically all “indoor” surfaces, both wood and metal, and to “outdoor” use on metal. Largely as a result of the growing popularity of surface coatings of the alkyd type the domestic production of phthalic anhydride exceeded 45 million pounds in 1937 and may reach 50 million pounds in 1938. This estimate is based on the present trend of consumption of alkyd resins and current use therein of phthalic anhydride. Should other polybasic acids be used in greater proportion the estimate would have to be revised. Considerable research work is being done on certain polybasic acids, with very promising results in some instances. Maleic anhydride is being used commercially, as are adipic acid, malic acid, and succinic acid. Other possibilities include such acids as citric, tartaric, sebacic, fumaric, and oxalic.

Production in other countries.

Phthalic anhydride is manufactured in Germany, England, France, Italy, and Japan, but no statistics of foreign production or of international trade are available. The output in Germany is known to be increasing rapidly and is believed to be the principal reason for the German embargo on exports of naphthalene.

In England there are two makers: Imperial Chemical Industries, Ltd., and Monsanto Chemicals, Ltd. The latter is a branch of the American firm of the same name.

In Italy, production was started in 1928 by the A. C. N. A. at Cengio. Capacity is given as 600,000 pounds annually, and the process is essentially the same as in this country.

Japanese production is estimated at 6 million pounds a year. Nihon Seuryo's plant is the principal one, and the Nishijima mill, at Osaka, the next in importance.

United States foreign trade.

Imports of phthalic anhydride are dutiable under paragraph 27 at 7 cents per pound and 40 percent ad valorem based on American selling price. There were practically no imports since the World War until 1937, when 223,431 pounds were imported from England to relieve a temporary shortage.

Exports, if any, are not shown separately in official statistics.

Competitive conditions.

Phthalic anhydride is the cheapest polybasic organic acid and therefore the most widely used in the production of alkyd resins. The rapid rise in consumption of surface coatings and finishes made from these resins presages greater demand for phthalic anhydride (and glycerin) in the future, particularly if this type of outdoor finish for wood is successful.

The world-wide shortage of naphthalene, with attendant sharp increases in price, raises the question of whether there may not be partial or complete replacement of phthalic anhydride by other polybasic acids in certain types of alkyd resins. The probability of such replacement seems remote unless the use of other polybasic acids, at present much higher priced, so improves the properties of the resins as to give a superior product. Approximately 100 pounds of naphthalene are required to produce 109 pounds of phthalic anhydride. Naphthalene at 3 cents per pound gives phthalic anhydride a raw material cost of 2.75 cents per pound as compared with 1.45 cents per pound when naphthalene was 1.55 cents per pound. In other words, the increase of 1.5 cents per pound in naphthalene, meant an increase of only about $1\frac{1}{3}$ cents per pound in the raw material cost of phthalic anhydride, and only approximately $\frac{1}{4}$ cent per pound in the raw material cost of an alkyd resin surface coating containing about 20-percent phthalic anhydride.

POLYBASIC ACIDS OTHER THAN PHTHALIC ANHYDRIDE

Maleic acid and anhydride.

Maleic anhydride is obtained as a byproduct in the manufacture of phthalic anhydride and as a major product by the vapor phase catalytic oxidation of benzene. Domestic production, still small compared with phthalic anhydride,

has increased many fold during the past two or three years. In 1937 there were three producers of maleic anhydride, with an output totaling 2,114,176 pounds. The uses of maleic acid derivatives other than in making resins are minor.

Malic acid and malomalic acid.

Malic acid is widely distributed in the vegetable kingdom, occurring especially in unripe apples. Commercially it is obtained by synthesis. Domestic production was reported for the first time in 1935. Malomalic acid is formed by heating malic acid. United States Patent No. 1,091,627 covers a resin made from malic acid and glycerin which will increase the toughness of phthalate resins. United States Patent No. 1,667,198 suggests the use of malomalic acid to form resins of glass-like appearance.

Adipic acid.

Adipic acid is made by oxidation of cyclohexanol. When condensed with glycerin it yields an alkyd resin which is soft and rubbery and which does not harden when heated. Numerous patents have been granted on the preparation of adipic acid and its resins. Commercial production of adipic acid was first reported in 1935, and the output increased in 1936 and in 1937.

Succinic acid and anhydride.

Succinic acid is a white crystalline powder melting at 185° C. and boiling at 234° C., with decomposition to succinic anhydride. It may be obtained by the reduction of maleic acid. Condensation with glycerin gives a resin tougher and more flexible than is obtained with phthalic anhydride.

In 1937 there were two commercial producers of succinic acid. It is believed that small quantities are used in combination with phthalic anhydride in alkyd resins.

Fumaric acid.

Fumaric acid is a white crystalline powder obtained by the prolonged heating of or by the action of mineral acids on maleic acid. Fumaric acid and maleic acid are structurally identical and the former decomposes at about 280° C., forming the latter. In 1937 there was one domestic maker of fumaric acid.

GLYCERIN

Description and uses.

Glycerin (glycerol) is a clear, colorless or almost colorless, odorless, syrupy, hygroscopic liquid. It is obtained as a byproduct of the soap and fatty acid (oleic acid or red oil and stearic acid) industries. Other sources are insignificant; glycerin can be produced by the fermentation of carbohydrates such as molasses, but when glycerin prices are low this process is not profitable. The chief commercial grades of crude glycerin are "soap lye" glycerin, a byproduct of the soap industry, containing about 80 percent glycerin, and "saponification" grade, a byproduct of the fatty acid industry, containing about 88 percent glycerin. Chemically pure grades contain about 95 percent and dynamite grades about 98.5 percent glycerin. Other grades include "30° yellow distilled" containing about 96 percent glycerin.

The uses of glycerin are extremely varied, the most important being in the manufacture of alkyd resins and ester gums; in the manufacture of nitroglycerin and dynamite; as a moistening, antiseptic, and sweetening agent in tobacco; in pharmaceutical and medicinal preparations; and in certain soft drinks, soaps, and inks.

United States production.

The output of both the crude and the refined has increased in recent years, reaching new highs in 1937. Chemically pure glycerin constitutes about 60 percent of the total refined output and dynamite and other grades

about 40 percent. In the production statistics shown in table 48, grades such as yellow distilled are included with the dynamite grade. Since large soap makers refine their own crude glycerin, the sale of crude is only a small part of the total output.

Table 48 shows domestic production of glycerin by grades and table 49 production for sale.

TABLE 48.—*Glycerin: United States production by grades, in specified years, 1919-37*

[In thousands of pounds]				
Census year	Crude 80 percent basis	Refined		
		Chemically pure grade	Dynamite grade	Total
1919	61,793	36,693	25,655	62,348
1920	54,688	32,860	31,571	64,431
1923	99,579	47,992	52,369	100,361
1924	95,154	53,243	37,368	90,611
1925	103,407	55,448	52,658	108,106
1926	116,369	64,460	49,579	114,039
1927	128,209	59,126	49,266	108,392
1928	130,499	66,419	46,622	113,041
1929	140,080	66,791	58,981	125,772
1930	138,675	69,654	50,974	120,628
1931	140,002	70,528	43,366	113,894
1932	133,919	63,624	41,539	105,163
1933	119,812	58,585	45,534	104,119
1934	153,115	80,359	48,553	128,912
1935	141,185	74,705	48,685	123,390
1936	154,096	85,386	47,535	132,921
1937	167,882	92,889	51,794	144,683

Source: Bureau of the Census, U. S. Department of Commerce.

TABLE 49.—*Glycerin: United States production for sale, in specified years, 1919-35*

Year	Crude ¹			Refined		
	Quantity	Value	Value per pound	Quantity	Value	Value per pound
	<i>1,000 pounds</i>		<i>Cents</i>	<i>1,000 pounds</i>		<i>Cents</i>
1919	18,228	\$2,482,779	13.6	47,377	\$11,461,213	24.2
1923	27,444	3,124,470	11.4	74,105	12,214,012	16.5
1925	30,735	4,258,351	13.9	94,303	16,991,213	18.0
1927	27,000	3,942,991	14.6	89,585	19,184,806	21.4
1929	28,790	2,358,031	8.2	113,140	12,715,641	11.2
1931	27,530	1,673,733	6.1	102,510	10,316,347	10.1
1931 ²	25,964	1,551,573	6.0	101,615	10,222,850	10.1
1933	22,161	1,191,000	5.4	107,853	7,915,000	7.3
1935	24,042	2,366,481	9.8	121,262	12,984,684	10.7

¹ By chemical and soap manufacturing plants only.

² Adjusted for comparison with 1933.

Source: Bureau of the Census, U. S. Department of Commerce.

Crude glycerin is produced by about 200 soap makers and by about 12 producers of fatty acids. Soap factories are located in more than half the States, the principal ones being in Ohio, New York, Massachusetts, New Jersey, Illinois, California, and Pennsylvania; the fatty acid plants are located in five or six States, Ohio being of chief importance. Most of the smaller producers sell their output of crude glycerin. Refiners of glycerin are few in number compared to the producers of crude. The larger soap plants refine their own crude glycerin and in addition purchase crude from other plants for refining.

The process of recovering glycerin consists of chemically treating weak glycerin solutions separated from the soap or fatty acids, and then concentrating and distilling under reduced pressures. The average yield is less than 10 percent but varies from about 9 to 12 percent, depending upon the kinds of oils and fats used. When prices are high every effort is made to recover the maximum yield of glycerin; when prices are low, cost of chemical treatment and distillation makes it advisable to allow more glycerin to remain in the soap or to discard the weak solutions.

Production in other countries.

As in the United States, glycerin is produced in foreign countries, as a byproduct of the soap and fatty acid industries. The United Kingdom, Germany, and France, and recently the Soviet Union, are the leading producers. The output in each of these countries is estimated to be less than a third of the output in the United States. The British Census of 1930 reports the production of crude glycerin in the United Kingdom at 44 million pounds. Authentic statistics on production in other leading countries are not available, but most estimates show lower figures than for the United Kingdom. In some European countries the normal production of soap results in more glycerin than can be utilized.

International trade.

France is the principal net exporter of crude glycerin and the United Kingdom of refined glycerin. The Netherlands, Germany, and France are also net exporters of refined glycerin. The international trade of certain of the more important producing countries in crude and refined glycerin is shown in table 50.

TABLE 50.—*Glycerin: Imports and exports of principal countries, 1931 and 1933-37*

		[In thousands of pounds]									
		1931		1933		1934		1935		1936	
		Imports	Exports	Imports	Exports	Imports	Exports	Imports	Exports	Imports	Exports
Crude:											
UNITED STATES ¹		8,782	(²)	4,988	(³)	13,722	(³)	4,092	(²)	8,686	(²)
United Kingdom		1,702	2,662	4,778	2,951	472	2,825	1,119	2,365	2,322	3,070
Germany		4,120	2,313	5,232	2,939	4,746	1,599	4,091	326	8,247	122
France		1,269	7,962	426	3,488		(²)	862	(²)	1,176	(²)
Netherlands		5,133	859	3,027	2,300	2,605	3,796	5,441	4,912	7,185	5,644
Refined:											
UNITED STATES		1,966	328	2,776	(³)	2,214	(³)	69	3,354	3,448	1,146
United Kingdom		2,519	9,926	822	19,834	230	19,134	2	15,032		12,991
Germany		102	10,092	57	3,562	224	3,818	108	1,571	30	1,155
France		178	3,989	109	1,246	51	12,249	4	12,118	10	9,269
Netherlands		618	8,337	1,144	6,620	1,008	5,955	694	5,516	739	8,885
Belgium		534	758	1,193	429	1,206	2,360	998	1,945	188	1,981

¹ Imports from Cuba and the Philippines not included in the United States statistics. These imports, consisting of crude glycerin, averaged about 2,200,000 pounds annually for the period 1931-37.

² Included, if any, with refined.

³ Not separately reported.

⁴ Preliminary.

Source: Official statistics of each country.

United States imports.

Under the Tariff Act of 1922, paragraph 43, imports of crude glycerin were dutiable at 1 cent per pound and refined glycerin at 2 cents per pound. The Tariff Act of 1930, paragraph 42, carries the same rates. Imports of crude glycerin from Cuba enjoy a preferential rate; they were dutiable at 0.8 cent per pound up to September 3, 1934, and at 0.4 cent per pound thereafter. Under the trade agreement with the Netherlands, effective February 1, 1936, the rate on refined glycerin was reduced from 2 cents to 1½ cents per pound (¾ cent plus regular rate on crude, but not more than 1½ cents). Under the trade agreement with France, effective June 15, 1936, the rate on crude glycerin was reduced from 1 cent to 0.8 cent, which automatically further reduced the rate on refined glycerin to 1¼₃₀ (approximately 1.47) cents per pound. The rates under these last two trade agreements are generalized to all countries which do not discriminate against our commerce.

The amount of glycerin supplied by imports has greatly declined. Prior to the World War, imports of crude glycerin ranged from 30 million to 40 million pounds annually. After the war imports were less and after 1929 declined to comparatively small quantities, except in 1934. Imports of refined glycerin were relatively unimportant until 1924, except in 1920. They amounted to almost 11 million pounds in 1926, but declined thereafter. Some of the imports are reexported with benefit of drawback. In 1930, 1,006,164 pounds of imported crude glycerin and 396,792 pounds of imported refined were thus reexported, chiefly in the refined grades. Corresponding figures in 1932 and 1933 were 197,331 and 111,753 pounds of crude, and 40,011 and 10,056 pounds of refined.

Statistics of imports other than from Cuba and the Philippines are given in table 51. Table 52 shows imports of crude from Cuba and table 53 imports of crude from the Philippines.

TABLE 51.—*Glycerin: United States imports¹ for consumption 1919-20 and 1923-37*

Calendar year	Rate of duty	Quantity, 1,000 pounds	Value	Unit value	Computed ad valorem rate, percent
Crude					
1919	1 cent per pound	3,564	\$417,774	\$0.117	8.5
1920	do	22,272	2,912,430	.131	7.7
1923	do	14,120	1,382,249	.098	10.2
1924	do	13,659	1,413,593	.103	9.7
1925	do	18,624	2,161,413	.116	8.6
1926	do	26,729	3,849,222	.144	6.9
1927	do	13,666	2,026,175	.148	6.7
1928	do	3,889	282,615	.073	13.8
1929	do	13,681	786,598	.058	17.4
1930	do	10,022	577,406	.058	17.4
1931	do	8,782	446,897	.051	19.7
1932	do	3,952	145,329	.037	27.2
1933	do	4,988	176,080	.035	28.3
1934	do	13,722	932,389	.068	14.7
1935	do	4,092	353,925	.086	11.4
1936	Various ²	8,686	936,312	.108	7.7
1937 ³	do	10,171	1,716,351	.169	4.8
Refined					
1919	2 cents per pound	39	4,471	.114	17.5
1920	do	5,382	1,170,030	.217	9.2
1923	do	586	76,994	.131	15.2

1924	do	1,501	229,897	.153	13.1
1925	do	2,044	305,796	.150	13.4
1926	do	10,839	2,328,936	.215	9.3
1927	do	8,289	1,697,330	.205	9.8
1928	do	4,218	450,247	.107	18.7
1929	do	5,358	489,575	.091	21.9
1930	do	3,137	265,093	.085	23.7
1931	do	1,966	140,975	.072	27.9
1932	do	2,348	142,359	.061	33.0
1933	do	2,776	166,991	.060	33.2
1934	do	2,214	208,989	.094	21.2
1935	do	69	8,277	.121	16.6
1936	Various ²	3,447	594,036	.172	8.5
1937 ³	do	7,535	1,827,189	.242	6.2

¹ Does not include products of Cuba (duty less 20 percent) and the Philippine Islands (free).

² For changes in rates, see p. 105.

³ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

TABLE 52.—*Crude glycerin: United States imports (for consumption) from Cuba, in specified years, 1919-37*

Calendar year	Rate of duty	Quantity	Value	Unit value	Computed ad valorem rate
		<i>1,000 pounds</i>			<i>Percent</i>
1919	$\frac{8}{10}$ cent per pound	249	\$27,023	\$0.108	7.4
1920	do	139	21,941	.158	5.1
1923	do	429	47,438	.111	7.2
1924	do	768	85,971	.112	7.2
1925	do	624	73,538	.118	6.8
1926	do	835	134,893	.162	5.0
1927	do	1,119	170,723	.153	5.2
1928	do	690	48,963	.071	11.3
1929	do	921	60,158	.065	12.2
1930	do	843	53,905	.064	
1931	do	1,171	67,709	.058	13.8
1932	do	1,232	50,147	.041	19.7
1933	do	1,216	56,737	.047	17.2
1934	Various ¹	1,178	92,692	.079	
1935	$\frac{4}{10}$ cent per pound	2,551	228,011	.089	4.5
1936	do	2,160	230,340	.107	3.8
1937 ²	do	2,477	381,683	.154	

¹ Trade agreement of $\frac{4}{10}$ cent per pound, effective Sept. 3, 1934.

² Preliminary.

Source: Foreign Commerce and Navigation of the United States.

TABLE 53.—*Crude glycerin: United States imports (for consumption) from Philippine Islands 1925-37*

Calendar year	Rate of duty	Quantity	Value	Unit value
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		<i>1,000 pounds</i>		
1925	Free	16	\$1,418	\$0.089
1926	do	95	12,115	.128
1927	do	159	18,261	.115
1928	do	337	24,327	.072
1929	do	250	16,796	.067
1930	do	279	18,805	.067
1931	do	180	10,993	.061
1932	do	198	9,150	.046
1933	do	268	14,078	.052
1934	do	181	14,984	.083
1935	do	1,579	74,798	.047
1936	do	304	32,708	.108
1937 ¹	do	793	145,348	.183

¹ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

France has usually been the principal source of imports of crude glycerin, but since 1935 Cuba has ranked first. Cuban imports enter at a preferential rate of duty (0.8 cent per pound on crude until September 3, 1934, when it was reduced to 0.4 cent). Receipts from the Philippines are duty-free. Imports by countries for recent years are given in table 98, page 151.

The Netherlands has generally been the chief source of imports of refined glycerin, although the United Kingdom was first in 1934 and 1935 and France first in 1937. Imports by countries for recent years are given in table 99, page 152.

United States exports.

Exports of glycerin are insignificant compared with production and are small compared with imports. They go chiefly to Mexico and Canada, and, at times, also to Cuba, the Philippines, and Chile. Geographic propinquity is probably the principal factor accounting for these exports, although it is possible that some exports are destined to foreign branch factories of an American company for making dynamite.

Crude and refined grades were not separately distinguished in export statistics, but it is known that exports consist principally, if not entirely, of the refined. In 1933 and 1934 glycerin exports were not reported. Statistics of exports are given in table 54.

TABLE 54.—*Glycerin: United States exports, in specified years, 1919-37*

Year	Quantity	Value	Unit value
	<i>Pounds</i>		<i>Cents per pound</i>
1919	3,963,392	\$1,190,984	30.0
1920	1,742,708	429,116	24.6
1923	1,767,407	318,765	18.0
1924	1,415,882	237,639	16.8
1925	1,367,191	282,078	20.6
1926	767,698	192,220	25.0
1927	693,144	143,700	20.7
1928	2,051,937	259,100	12.6
1929	1,373,605	197,986	14.4
1930	607,690	102,892	16.9
1931	328,143	48,095	14.7
1932	260,339	28,609	11.0

1933	(¹)	(¹)	
1934	(¹)	(¹)	
1935	3,353,625	450,248	13.4
1936	1,146,026	182,592	15.9
1937 ²	1,375,036	338,148	24.6

¹ Not reported separately.

² Preliminary.

Source: Foreign Commerce and Navigation of the United States.

Competitive conditions.

Glycerin occurs in chemical combination in animal and vegetable oils and fats. Since it is obtained as a byproduct, the output is dependent primarily upon the output of the major products, soaps and fatty acids, and its production is largely independent of demand. At low prices, however, less glycerin is recovered.

The United States usually consumes more glycerin than it produces (see table 55), whereas leading European producing nations produce more than they consume. In recent years the domestic production has apparently been approaching domestic requirements. A factor tending to decrease demand is the increasing resort to substitutes in various uses, particularly ethylene glycol, ethyl alcohol, and methyl alcohol. On the other hand, the demand in the resin industry is expanding rapidly.

TABLE 55.—*Refined glycerin: United States production, imports, exports, and apparent consumption, in specified years*

Year	Production ¹	Imports ²	Exports ³	Apparent consumption ⁴
	<i>Pounds</i>	<i>Pounds</i>	<i>Pounds</i>	<i>Pounds</i>
1927	108,392	8,289	693	115,987
1929	125,772	5,358	1,374	129,757
1931	113,894	1,966	328	115,531
1932	105,163	2,348	260	107,250
1933	104,120	2,776	(⁵)	106,895
1934	128,912	2,214	(⁵)	131,126
1935	123,390	69	3,354	120,105
1936	132,922	3,447	1,146	135,223
1937 ⁶	(⁵)	7,535	1,375	(⁵)

¹ From table 48 (refined basis).

² From table 51 (refined).

³ From table 54 (grade not specified, but chiefly refined).

⁴ Production plus imports minus exports.

⁵ Not available.

⁶ Preliminary.

Up to 1924 (except 1920) imports consisted principally of crude glycerin, much of which was refined in the United States and included in United States production; thereafter imports of refined glycerin became important relative to the crude. Exports are insignificant compared to either production or imports.

17. RAW MATERIALS FOR TAR-ACID RESINS

The first tar-acid resins were made from phenol and formaldehyde. As a result this group of resins is frequently spoken of as phenolic resins. This is true despite the fact that coal-tar acids other than phenol, particularly the cresols and xylenols, are used in large volume today.

THE TAR ACIDS

The term coal-tar acids is applied to certain organic compounds either obtained from or known to be present in coal tar. Probably the best known is phenol or carbolic acid, produced in large quantities in the United States and abroad. Others of commercial importance are ortho, meta, and para cresol and the xylenols. All of these are definite chemical compounds available as such or in mixture with other tar acids. Cresylic acid is a term widely used in commerce for almost any mixture of tar acids. Formerly it was used to designate a mixture of ortho, meta, and para cresols in the proportions in which they are found in coal tar. The higher boiling tar acids (listed in table 56 below the xylenols) have little or no commercial importance at this time.

Table 56 lists the tar acids by commercial name, chemical name, boiling point of the pure compound, and average percentage present in coal tar. Boiling point is shown because the several tariff classifications covering tar acids under the acts of 1922 and 1930 (pars. 27 and 1651) depend upon distillation range (boiling points) for classification and assessment of duty (see pp. 119 and 124).

TABLE 56.—*Tar acids: Commercial and chemical names, boiling points and average percentage present in coal tar*

Commercial name	Chemical name	Boiling point ° C	Average percent in coal tar
Phenol	Phenol	181.5	0.7
Orthocresol	2-methyl phenol	190.8	.4
Metacresol	3-methyl phenol	202.8	.4
Paracresol	4-methyl phenol	201.8	.3
2-3 Xylenol	2-3-dimethyl phenol	218.0	.2
2-4 Xylenol	2-4-dimethyl phenol	211.5	
2-5 Xylenol	2-5-dimethyl phenol	211.5	
2-6 Xylenol	2-6-dimethyl phenol	212.0	
3-4 Xylenol	3-4-dimethyl phenol	225.0	
3-5 Xylenol	3-5-dimethyl phenol	220.0	
Ortho ethylphenol	2-ethyl phenol	206.5	.5
Meta ethylphenol	3-ethyl phenol	217.0	
Para ethylphenol	4-ethyl phenol	218.5	
s-methyl ethylphenol	3-methyl-5-ethyl phenol	232.5	
Iso pseudocumenol	2-3-5-trimethyl phenol	233.0	
Mesitol	2-4-6-trimethyl phenol	219.5	
Pseudocumenol	2-4-5-trimethyl phenol	234.0	

Since the quantities of tar acids present in coal tar are small (see table 56), it is usually uneconomical to distill coal tar completely unless the creosote oil and pitch can be marketed profitably. Beginning in 1936, production of tar acids in the United States was increased by the practice of topping. Topping is the recovery in tar distillation of the light fractions only, leaving a residual thin enough to flow through the pipe lines to supply fuel to open hearth and other type furnaces. These light fractions contain the naphthalene and tar acids. The practice permits recovery of these products from tar to be used as fuel, thus providing a new alternative intermediate between the two older practices of either complete distillation or using the undistilled tar as fuel.

In the United States, consumption of most of the tar acids greatly exceeds the quantities extracted from tar, necessitating large production of synthetic phenol and importation of large quantities of the cresols and xylenols. The calculated amount of these tar acids present in the tar produced in this country vastly exceeds present day requirements. Table 57 shows the approximate amounts of the several tar acids contained in the coal tar produced and distilled in 1936. These estimates are based on a 1936 production of coal tar of 560,385,578 gallons and a distillation of 292,140,249 gallons. The calculation is made by using the percentage of tar acids in tar shown in table 56 and converting the gallons to pounds in accordance with the specific weights of the pure tar acids. Actual production of all tar acids in the United States in 1936 was about 29 million pounds.

TABLE 57.—*Tar acids available in coal tar produced and distilled in 1936*

Tar acid	Available in tar produced in 1936 ¹	Available in tar distilled in 1936 ²
	<i>1,000 pounds</i>	<i>1,000 pounds</i>
Phenol	34,912	18,200
Orthocresol	19,277	10,050
Metacresol	19,277	10,050
Paracresol	14,290	7,450
Xylenols	10,200	5,316
Others	25,200	13,290

¹ 560,385,578 gallons.

² 292,140,249 gallons.

The several tar acids are discussed in detail under the following heads:

- (a) Phenol.
- (b) The cresols, xylenols, and cresylic acid.
- (c) Synthetic tar acids other than phenol.

PHENOL

Description and uses.

Phenol (commonly called carbolic acid) is a tar acid obtained from two sources: (a) From one of the fractions recovered in the distillation of coal tar, a byproduct resulting from the manufacture of coke in byproduct ovens, and from the manufacture of coal gas; (b) from benzol, by synthesis. The second source has been the more important since 1923. Phenol, when pure, is a colorless substance of interlaced or separate needle-shaped crystals with a characteristic aromatic odor. It is corrosive to the skin and to mucous membrane. When pure it is water white, melts at about 42° C., and boils at about 181.5° C. It was discovered in 1834 by Runge.

Phenol is used today chiefly as a component of tar-acid resins. It is also widely used as an antiseptic and disinfectant, in the manufacture of explosives (picric acid and ammonium picrate), and as an intermediate for certain dyes and medicinals. Salicylic acid and its derivatives—aspirin, salol, and methyl salicylate (artificial oil of wintergreen)—are important medicinals made from phenol. Another use is in the extraction of lubricating oils. The relative importance of these various uses in recent years is indicated by table 58, which gives the estimated domestic consumption of phenol by uses in 1936-37.

TABLE 58.—*Phenol: Estimated consumption by industries, 1936-37*

Use	Percent of total consumption
Synthetic resins	60-65
Extraction of lubricating oil	5
Insecticides and disinfectants	10
Dyes and intermediates	5
Other	15-20

United States production.

Prior to 1914 United States production of phenol averaged about a million pounds a year and was entirely the natural product obtained from distillates of coal tar. Increased demand during the World War was met by several synthetic phenol processes, which utilized in part the vast quantities of benzol available. Our output of phenol reached 64 million pounds in 1917 and 107 million pounds in 1918. When the armistice was signed stocks on hand in the United States totaled between 35 million and 40 million pounds, estimated at three times the annual consumption at that time for nonmilitary purposes. As a result the price dropped from about 45 cents to 6 cents a pound, and the synthetic plants were closed.

The limited quantities of phenol available to synthetic resin makers prior to and during the World War caused much concern to that industry and led to research work for substitutes, work resulting in the development of many new and modified types of resins in which tar acids other than phenol were used. But notwithstanding the use of these other tar acids the increased demand for synthetic resins used up the accumulated stocks of phenol sooner than was expected.

Of the phenol produced in the United States from 1919 through 1923 a large part was natural phenol but the rapid increase in demand and the improvement of processes for synthetic phenol had by 1923 resulted in four companies beginning production of the synthetic article. The rapid increase in output, from about 3 million pounds in 1923 to about 15 million pounds in 1925, was almost entirely in synthetic phenol. Since then a large part of the domestic production has continued to be synthetic, although the production of natural phenol since 1935 has been about four times that of 1929.

Adequate quantities of coal tar are usually available to produce sufficient natural phenol to meet a substantial part of our requirements, if it were all recovered, but the quantity actually produced is determined in part by the demand for other coal-tar products, and in part by the value of the tar as fuel. More than 50 percent of the tar produced has been burned as fuel, principally at the coke ovens or nearby steel mills.

The domestic production and sales of phenol, natural and synthetic combined, are shown in table 59.

TABLE 59.—*Phenol: United States production and sales, in specified years, 1918-37*

Census year	Production	Sales			Ratio of sales to total production
	Quantity	Quantity	Value	Unit value	
	<i>1,000 pounds</i>	<i>1,000 pounds</i>	<i>1,000 dollars</i>		<i>Percent</i>
1918	106,794	106,794	37,270	\$0.35	
1919	1,544	1,544	156	.10	
1923	3,311	2,180	590	.27	66
1925	14,734	8,524	1,771	.21	58
1926	8,691	5,480	988	.18	63
1927	8,041	4,595	684	.15	57
1928	10,227	7,746	912	.12	76
1929	24,178	19,939	2,248	.11	83
1930	21,147	17,715	1,976	.11	84
1931	17,981	14,002	1,446	.10	78
1932	13,965	12,181	1,269	.10	87
1933	33,220	27,923	2,881	.10	84
1934	44,935	36,241	3,887	.11	81
1935	43,419	34,575	3,431	.10	80
1936	48,724	40,942	4,235	.10	84
1937	65,690	57,176	6,153	.11	87

Source: Compiled from annual reports of the Tariff Commission on dyes and other synthetic organic chemicals in the United States.

Grades produced for resins.—Increased production of phenol in recent years is largely due to the demand from makers of synthetic resins. A number of grades are regularly produced for this purpose, though it is believed that the technical grade is the principal one used in resins. The several grades are as follows:

- (1) USP.—Either natural or synthetic. This grade contains not less than 98 percent phenol.
- (2) Technical.—Various grades containing from 80 to 95 percent phenol, of which the two most important are 82-84 percent and 90-92 percent.
- (3) Mixtures.—Containing from 30 percent to 80 percent phenol and the remainder of the isomeric cresols.

Producers.—Natural phenol is obtained in the distillation of coal tar and to a smaller extent in the purification of ammonia liquors in coke and gas plants. In 1937 there were four producers of natural phenol with plants located at Philadelphia, Pa., Follansbee, W. Va., Indianapolis, Ind., and Pittsburgh,

Pa. All these are tar distillers recovering creosote oil, pitch, cresylic acid, naphthalene, and other crudes from coal tar.

Synthetic phenol is made from benzene, either by sulfonation followed by alkaline fusion, or by chlorination and subsequent heating under pressure with caustic soda. It is produced in large quantities by two firms, one at Midland, Mich., and the other at St. Louis, Mo. A third producer is building a plant at North Tonawanda, N. Y., using a process recently developed in Germany. Operation of this unit will probably start late in 1938.

World production.

Natural phenol is recovered in practically all European countries and in Japan. Germany and the United Kingdom are the principal producers and have also been the leading exporters. Synthetic phenol was made in Germany as early as 1900, and during the World War. Plants for synthetic phenol recently installed are now in operation in Germany, Great Britain, Belgium, and Italy.

Table 60 shows the average annual world production of phenol in recent years by countries. Half of the total was produced in the United States.

TABLE 60.—*Phenol: Estimated average annual production, by countries, 1933-35*

Produced in—	Estimated annual production
	<i>1,000 pounds</i>
UNITED STATES	41,000
United Kingdom	18,000
Germany	10,000
Poland	3,000
Japan	3,000
Czechoslovakia	1,000
Belgium	2,000
France	2,000
Italy	1,500
Spain	700
Total	82,200

Source: Consular reports.

In the United Kingdom, where tar distillation is a well developed and highly organized industry, large quantities of gas-works tar, rich in phenol and other tar acids, are available. Prior to the World War the United Kingdom was the principal source of phenol, and of the other products of tar distillation. During the war several synthetic processes of commercial importance were developed, but they were discontinued after its close. A new synthetic unit has recently been installed in England and is now in operation. Increasing consumption of phenol in synthetic resins during the last decade, particularly in the last several years, has changed the United Kingdom from an exporter to an importer of phenol. Estimated consumption of phenol in the United Kingdom is given as 20 million pounds annually—principally in synthetic resins, and in lesser quantities in dyes, intermediates, antiseptics, and disinfectants.

In Germany the phenol recovered in 1936 amounted to about 20 million pounds, and recently three commercial units have been installed for the production of synthetic phenol, one with a reported daily output of 11,000 pounds.

Natural phenol is also recovered in Belgium, France, the Netherlands, Czechoslovakia, Poland, Italy, and Spain. Synthetic phenol has recently been produced for the first time in Belgium and Italy. The quantities normally produced in these countries are small and are supplemented by imports from Great Britain and Germany.

The production of phenol in Japan has increased rapidly and has been sufficient since 1930 to meet domestic requirements. The estimated output increased from 300,000 pounds in 1927 to more than 3 million pounds annually in recent years. The Miike Dyestuffs Works is reported to be producing synthetic phenol.

United States imports.

Rates of duty.—Prior to September 6, 1916, phenol was imported free of duty. Since that date it has been dutiable at the various rates shown in table 61. Under the act of 1930 the rate of duty is 3½ cents per pound and 20 percent ad valorem on the American selling price (the wholesale price of a similar competitive article manufactured in the United States).^[21]

TABLE 61.—*Phenol: Rates of duty upon imports into the United States, 1916-37*

Period	Rate of duty	Authority
To Sept. 8, 1916	Free	Free under Tariff Act of 1913 and previous acts.
Sept. 9, 1916, to Sept. 21, 1922.	2½ cents per pound plus 15 percent ad valorem on foreign value.	Under Revenue Act of 1916.
Sept. 22, 1922, to Sept. 21, 1924.	7 cents per pound plus 55 percent ad valorem on American selling price ¹ or United States value. ²	Under par. 27 of Tariff Act of 1922; special provision for first 2 years.
Sept. 22, 1924, to Nov. 29, 1927.	7 cents per pound plus 40 percent ad valorem on American selling price ¹ or United States value. ²	Under par. 27 of Tariff Act of 1922, rate provided for period after the first 2 years.
Nov. 30, 1927, to June 17, 1930.	3½ cents per pound plus 20 percent ad valorem on American selling price ¹ or United States value. ²	By Presidential proclamation following a cost of production investigation under sec. 315 of Tariff Act of 1922.
June 18, 1930	3½ cents per pound plus 20 percent ad valorem on American selling price ³ or United States value. ⁴	Under par. 27 (b) of Tariff Act of 1930.

¹ As defined in subdivision (f) of section 402, title IV, act of 1922.

² As defined in subdivision (d) of section 402, title IV, act of 1922.

³ As defined in subsection (g) of section 402, title IV, act of 1930.

⁴ As defined in subsection (e) of section 402, title IV, act of 1930.

Import statistics.—Imports for consumption are shown in tables 62 and 63. Table 62 shows imports of phenol or carbolic acid and table 63 imports of “all distillates of tars yielding below 190° C. an amount of tar acids equal to or more than 5 percent.” Imports under the latter classification prior to 1928 were probably chiefly phenol. Phenol imports consist entirely of the natural product.

TABLE 62.—*Phenol: United States imports for consumption, 1910-37*

Year	Rate of duty	Quantity	Value	Unit value	Computed ad valorem rate
		<i>Pounds</i>			<i>Percent</i>
1910 ¹	Free	4,507,693	\$275,600	\$0.061	
1911 ¹	do	4,371,014	265,780	.061	
1912 ¹	do	5,686,704	521,457	.092	
1913 ¹	do	8,345,631	688,771	.083	
1914 ¹	do	8,393,216	531,535	.063	
1915 ¹	do	3,106,445	179,685	.058	
1916 ¹	(²)	2,246,256	154,841	.069	
1917 ¹	15 percent + 2½ cents per pound	265,519	17,168	.065	53.7
1918	do	283,337	62,497	.221	26.3
1919	do	2,061	264	.128	34.5
1920	do	1,040	244	.235	25.8
1921	do	250	142	.568	19.4
1922	(³)	{ 280,224	30,414	.109	38.0
		{ 69,310	16,102	.230	85.1
1923	55 percent + 7 cents per pound ⁴	126,618	21,389	.169	96.4
1924	(⁵)	176,081	46,786	.266	81.4
1925	40 percent + 7 cents per pound ⁴	256,126	58,958	.230	70.4
1926	do	218,437	47,351	.217	72.3
1927	(⁶)	500	100	.200	75.0
1928	20 percent + 3½ cents per pound ⁴	1,653	298	.180	39.4
1929	do	433,385	44,226	.102	54.3
1930	do	500	115	.230	39.4
1931	do	2,365	639	.270	33.0
1932	do	None			
1933	do	3,440	641	.186	38.8
1934	do	None			
1935	do	2,605	211	.081	63.0
1936	do	71,429	8,302	.116	50.1
1937 ⁷	do	32,238	3,767	.117	50.0

¹ Fiscal year.

² 15 percent ad valorem and 2½ cents per pound effective Sept. 9, 1916.

³ 55 percent ad valorem and 7 cents per pound, effective Sept. 22, 1922.

⁴ Ad valorem based on American selling price or United States value under acts of 1922 and 1930.

⁵ Ad valorem reduced to 40 percent effective Sept. 22, 1924.

⁶ Duty reduced to 20 percent ad valorem and 3½ cents per pound effective Nov. 30, 1927.

⁷ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

TABLE 63.—*All distillates of tar yielding below 190° C. an amount of tar acids equal to or more than 5 percent: United States imports for consumption, 1918-37*

Calendar year	Rate of duty	Quantity	Value	Unit value	Computed ad valorem rate
		<i>Pounds</i>			<i>Percent</i>
1918	15 percent + 2½ cents per pound	1,550	\$2,008	\$1.30	16.93
1919	do	3,170	4,587	1.45	16.73
1920	do	85,474	36,041	.422	20.93
1921	do	16,240	11,811	.727	18.43
1922	(¹)	350,764	42,912	.122	46.27
1923	55 percent + 7 cents per pound ²	245,119	30,328	.124	111.58
1924	(³)	662,938	49,380	.074	134.43
1925	40 percent + 7 cents per pound ²	252,382	15,441	.061	154.41
1926	do	1,102	5,236	4.75	41.47
1927	(⁴)	2	16	8.00	40.88
1928-37		None			

¹ 55 percent ad valorem and 7 cents per pound, effective Sept. 22, 1922.

² Ad valorem based on American selling price or United States value under acts of 1922 and 1930.

³ Ad valorem reduced to 40 percent, effective Sept. 22, 1924.

⁴ Duty reduced to 20 percent ad valorem and 3½ cents per pound, effective Nov. 30, 1927.

Source: Foreign Commerce and Navigation of the United States.

United States exports.

Exports of phenol have not been separately shown in official statistics since 1924. In that year they went chiefly to Panama, Japan, Cuba, and Mexico. Table 64 shows exports from 1918 to 1924, inclusive, as furnished by the Department of Commerce.

TABLE 64.—*Phenol: United States exports, 1918-24*

Year	Quantity	Value	Unit value
	<i>Pounds</i>		
1918	6,477,841	\$2,666,634	\$0.412
1919	1,243,841	363,744	.292

1920	2,151,475	388,047	.180
1921	249,658	35,994	.144
1922	223,146	23,223	.104
1923	232,830	34,389	.148
1924	51,364	8,016	.156

Source: Commerce and Navigation of the United States.

Appreciable quantities of phenol have been exported in recent years to Japan and China, and to Great Britain and other European countries. Export statistics, collected by the U. S. Tariff Commission from the several domestic producers, show the following quantities exported in recent years.

TABLE 65.—*Phenol: United States exports, 1934-36*

Year	Quantity	Value	Unit value
	<i>Pounds</i>		
1934	2,622,900	\$329,269	\$0.126
1935	2,921,835	322,933	.111
1936	1,258,244	148,501	.118

Source: Data obtained by the U. S. Tariff Commission through questionnaires.

In 1934, the principal destinations in order of importance were China, Italy, and Canada; in 1935 Germany, China, Japan, and Belgium; and in 1936 China, Belgium, and the Netherlands.

Competitive conditions.

Before the World War our average annual consumption of phenol was 5 million pounds, of which about 80 percent was imported from Great Britain and Germany. These countries produced phenol in excess of their consumption, and phenol was on the free list in the United States.

In September 1916 phenol became dutiable. The demand was increasing rapidly because of the use of phenol in the manufacture of picric acid, an explosive. To meet the wartime demand at home and abroad large scale production of synthetic phenol sprang up in the United States. But the end of the war not only shut off the largest part of the demand but left the producers with large stocks on hand. The price dropped sharply and the production of synthetic phenol ceased.

Since 1922 there has been a gradually increasing demand for phenol in the United States, chiefly for use in the manufacture of synthetic resins, and production has increased to meet this demand as shown in table [66](#).

TABLE 66.—*Phenol: United States production, imports, exports, and apparent consumption in specified years, 1918-37*

[In thousands of pounds]				
Year	Production ¹	Imports ²	Exports ³	Apparent consumption ⁴

1918	106,794	285	6,478	(⁵)
1919	1,544	5	1,244	(⁵)
1923	3,311	372	233	3,450
1925	14,734	919	(⁶)	(⁷)
1926	8,691	220	(⁶)	(⁷)
1927	8,041	1	(⁶)	(⁷)
1928	10,227	2	(⁶)	(⁷)
1929	24,178	433	(⁶)	(⁷)
1930	21,147	1	(⁶)	(⁷)
1931	17,981	2	(⁶)	(⁷)
1932	13,965		(⁶)	(⁷)
1933	33,220	3	(⁶)	(⁷)
1934	44,935		2,623	42,312
1935	43,419	3	2,922	40,500
1936	48,724	71	1,258	47,537
1937	65,690	32	(⁶)	(⁷)

¹ From table [59](#).

² From tables [62](#) and [63](#).

³ From tables [64](#) and [65](#).

⁴ Production plus imports minus exports.

⁵ Not calculated because of importance of stocks on hand.

⁶ Not available.

⁷ Not available because of absence of export figures. Exports probably negligible up to 1929; substantial in 1933.

The manufacture of synthetic phenol was revived about 1923. Imports were quite small as compared with production, especially after 1927. At first this was probably due primarily to the protection given by the duty which had been increased in 1922.^[22] But with the increase in volume of production in the United States the price decreased and since 1933 the United States producers have enjoyed a substantial export business. It may therefore be doubted that in recent years there would have been any substantial imports even if phenol had been free of duty.

THE CRESOLS, XYLENOLS, AND CRESYLIC ACID

Reference to table [56](#), page [109](#), will show that, as distillation of coal tar proceeds and the temperature of distillation is increased, the phenol fraction is followed in order by the three cresols and then by the six xyleneols. Each of these tar acids is a definite chemical compound with definite physical properties. Consideration of them as raw materials for synthetic resins is complicated by the fact that they are generally used in mixtures and that the commercial term, cresylic acid, applied to many of these mixtures has no definite relationship to the precise chemical terminology. Yet since the term cresylic acid is so widely used in commerce, since the tariff provides for imports under that name, and since the statistics available are in part in terms of cresylic acid and in part in terms of cresols and xyleneols it is impossible to present the complete picture on the basis of the correct chemical terminology.

Description and uses.

The cresols.—The cresols are isomeric tar acids obtained from coal tar by fractional distillation. Their combined content averages about 1 percent of domestic coal tar. The total cresol content is divided in about the following proportions: 40 percent metacresol, 35 percent orthocresol, and 25 percent paracresol. The cresols are marketed in a number of types and grades including mixtures of ortho, meta, and para; mixtures of meta and para; separated ortho, meta, and para; and also in mixtures with phenol and the xyenols.

Metacresol (chemically, 3-methyl phenol) is a colorless to yellow liquid with a phenol-like odor. When pure, it melts at 11° C., boils at 202.8° C., and has a specific gravity of 1.03. It is used in the manufacture of synthetic resins, photographic developers, explosives, disinfectant soaps, paint and varnish removers, to remove ink from newsprint, to soften and reclaim rubber, and in intermediates for dyes and perfume materials.

Orthocresol (chemically, 2-methyl phenol) is a colorless, crystalline product with a phenol-like odor, melting at 30° C., boiling at 190.8° C., and having a specific gravity of 1.04. It is used in the manufacture of coumarin (flavor), antiseptics, disinfectants, and fumigants. It is not used to any extent in synthetic resins.

Paracresol (chemically, 4-methyl phenol) is a colorless, crystalline substance with a phenol-like odor, melting at 35° C., boiling at 201.8° C., and having a specific gravity of 1.03. It is used in the manufacture of intermediates, dyes, disinfectants, and fumigants, in medicine, and in mixture with metacresol in synthetic resins. Domestic production of synthetic paracresol was announced early in 1938 by Swann & Co., Birmingham, Ala.

Metaparacresol is a combination of approximately 60 percent meta and 40 percent para cresol obtained in the fractional distillation of mixed cresols. The ortho isomer is distilled off, leaving a residue of metaparacresol. It is widely used in the manufacture of synthetic resins.

Cresol.—The term cresol used without further qualification indicates a mixture of the three isomers in substantially the same proportions in which they are found in coal tar. The United States Pharmacopoeia describes cresol, a mixture of isomeric cresols obtained from coal tar, as a colorless or yellowish to brownish-yellow or a pinkish, highly refractive liquid, becoming darker with age and on exposure to light. It is widely used in synthetic resins, antiseptics and disinfectants, and in medicine.

The xylenols.—Shortly after the original patents on Bakelite resins expired extensive research was begun for raw materials that would give different properties to the resultant resins. This work led to a study of the high-boiling tar acids, and methods of recovery for some of them were commercially developed. Among those obtained from coal tar are the six isomeric xylenols, methylethyl phenol, and one of the trimethyl phenols (see table [56](#)). Coal tar contains about 0.2 percent xylenols and 0.5 percent other high-boiling tar acids.

The principal uses for these products have been in the preparation of high-phenol coefficient disinfectants, and recently in the replacement of phenol and cresols in synthetic resins. It was found, for example, that 3: 5 xyleneol reacts with formaldehyde faster than either metacresol or phenol. Numerous patents have been granted on the use of these high-boiling acids in the production of synthetic resins.

The xylenols, when pure, are colorless, crystalline substances boiling between 211° and 225° C. They are usually marketed in mixtures containing from 50 to 80 percent xylenols and 20 to 50 percent cresols. There is commercial production of at least one of the separated xylenols (3: 5). An appreciable part of our imports of crude cresylic acid and of our production of cresylic acid contains high percentages of the xylenols.

Other high-boiling tar acids.—The other high-boiling tar acids are ortho ethylphenol, meta ethylphenol, para ethylphenol, methyl ethylphenol, and the three isomeric trimethyl phenols. Several of these have been isolated from coal tar. All of them, when pure, are crystalline compounds with boiling points ranging between 206° and 235° C. There has been little, if any, commercial production of this group up to this time. They are known, however, to have very high phenol coefficients, a property which would make them suitable for use in disinfectants. Little is known as yet concerning their application in synthetic resins.

Cresylic acid.—Cresylic acid is a generic term now applied to mixtures of tar acids in widely varying proportions. As defined in the literature and as formerly used in commerce the term identified a mixture of ortho, meta, and para cresols in the proportions in which they occur in coal tar. This proportion is approximately 40 percent metacresol, 35 percent orthocresol, and 25 percent paracresol. But in recent years the designation cresylic acid has been applied to all sorts of mixtures of tar acids boiling above 190° C. Practically every maker of synthetic resins, antiseptics, and disinfectants has his own specifications for cresylic acid; it may be any mixture in almost any proportions of the three cresols, the six isomeric xylenols, and the higher boiling tar acids. Imports of crude cresylic acid are understood to be largely xyleneol mixtures containing low percentages of the cresols. This loose application of cresylic acid in recent years is due to the increased commercial application of the high-boiling tar acids, especially the xylenols.

Under the Tariff Act of 1930 refined cresylic acid, that having a purity of 75 percent or more, is dutiable under paragraph 27 at 3½ cents per pound and 20 percent ad valorem based on American selling price or United States value; while crude cresylic acid, that having a purity of less than 75 percent, is free under paragraph 1651. The provision in paragraph 27 reads, “cresylic acid which on being subjected to distillation yields in the portion distilling below two hundred and fifteen degrees centigrade, a quantity of tar acids equal to or more than 75 per centum of the original distillate.” Under this provision cresylic acid may include an endless number of combinations of tar acids and may or may not contain any of the isomeric cresols. Of the 17 or more tar acids known to exist in coal tar (see table 56), only 8 have boiling points above 215° C. It would seem to be more accurate and more in line with present day usage to have the tariff drop the designation cresylic acid in favor of more definite terms based on composition, such as cresols and cresol mixes, xyleneol and xyleneol mixes, etc.

About 60 percent of our consumption of cresylic acid is in synthetic resins and the remainder in the manufacture of insecticides, antiseptics, disinfectants, and other coal-tar products, such as intermediates for dyes, plasticizers for nitrocellulose, etc.

United States production.

The cresols.—There is large production of cresol, metaparacresol, and orthocresol in the United States. Commercial production of paracresol was reported for the first time in 1934, and of metacresol in 1935.

Statistics of domestic production and sales are publishable only for the year 1934 because of the small number of producers. The output in that year is shown in table 67. Production has increased appreciably since then.

TABLE 67.—*Meta, ortho, and para cresols: United States production and sales, 1934*

Type	Production	Sales		
		Quantity	Value	Unit value

	<i>Pounds</i>	<i>Pounds</i>		
Cresol	8,929,836	8,559,048	\$572,738	\$0.07
Metaparacresol	2,033,424	1,692,149	101,324	.06
Orthocresol	835,016	(¹)	(¹)	
Paracresol	(¹)	(¹)	(¹)	

¹ Not publishable; figures would reveal operations of individual firms.

Source: Dyes and Other Synthetic Organic Chemicals in the United States. U. S. Tariff Commission.

The trend of domestic production of the several cresols is upward. In 1937 the output of all grades and types of cresols was 13,745,271 pounds with sales of 13,251,345 pounds, valued at \$1,071,965. The practice of topping coal tar will greatly increase the output of the cresols as well as of other tar acids and naphthalene.

There are five domestic producers of cresol, three each of orthocresol and metaparacresol, and two of metacresol and paracresol. All except one of these makers recover natural phenol, cresylic acids, and other tar acids. Refining plants are located at Pittsburgh, Pa., Philadelphia, Pa., Indianapolis, Ind., and Follansbee, W. Va. Domestic production of synthetic paracresol was first announced in 1938.

The xyenols.—There has been a large domestic production of mixed xylenols in recent years. These mixtures, containing from 50 to 80 percent xylenols, are marketed as cresylic acid. Statistics of domestic production, and sales are therefore included in table 68. It is estimated that the output of xylenols and xylene mixtures in 1935 exceeded 750,000 pounds and exceeded 1,250,000 pounds in 1937. At least one of the separated xylenols (1: 3: 5) has been produced commercially in the United States since 1935, but statistics of its production are not publishable.

Other high-boiling tar acids.—There was no reported domestic production of the other high-boiling acids prior to 1935 and the data obtained for that year are probably incomplete. Estimated output was 200,000 pounds in 1935, 250,000 pounds in 1936, and 300,000 pounds in 1937. These estimates are based on production of mixtures of high-boiling acids.

Cresylic acid.—Domestic production and sales statistics for so-called crude cresylic acid are not publishable. It is known, however, that production of the crude is small compared with our output of refined cresylic acid. It is usually more economical for the producer to prepare the mixture of tar acids to the specifications of the purchaser, rather than to leave part of the refining operations to be performed by the latter. The fact that imports of cresylic acid are chiefly of crude is largely due to the different tariff treatment of crude and refined.

Domestic production of refined cresylic acid was confined to one or two firms until 1928, when there were four makers. Statistics of production and sales are not publishable for the years prior to 1929, though it may be stated that the annual domestic output increased each year to supply the increased demand. Table 68 shows production and sales from 1929 to 1934, inclusive. Data for later years are not publishable.

TABLE 68.—*Refined cresylic acid: United States production and sales, 1929-37*

Year	Production	Sales		
		Quantity	Value	Unit value
	<i>Pounds</i>	<i>Pounds</i>		<i>Per pound</i>
1929	14,601,534			\$0.10
1930	17,305,308	16,026,407	\$1,267,155	.08

1931	10,994,000	10,305,000	652,000	.06
1932	8,060,000	4,805,000	251,000	.05
1933	13,813,941	11,975,441	626,496	.05
1934	10,949,860	9,230,255	489,231	.05
1935	(¹)	(¹)	(¹)	
1936	(¹)	(¹)	(¹)	
1937	(¹)	(¹)	(¹)	

¹ Not publishable; figures would reveal operations of individual firms.

Source: Compiled from annual reports of the Tariff Commission on dyes and other synthetic organic chemicals in the United States.

As previously stated, the composition of cresylic acid has gradually been changed from a mixture of the isomeric cresols to mixtures of cresols, xyleneols, and high-boiling tar acids. The cresols, formerly included under cresylic acid statistics, are now shown separately. For this reason the data in table 68 do not fully reflect the increased output of these tar acids in recent years. Statistics for years prior to 1931 probably include all of the tar acids except phenol, while those for subsequent years do not include the separated cresols. In 1934 the production of refined cresylic acid was 10,949,860 pounds, and in addition recovery of the several cresols amounted to 11,798,276 pounds making a total of 22,748,136 pounds as compared with a total of 14,601,534 pounds in 1929 and 17,305,308 pounds in 1930.

During 1936 and the first part of 1937 a serious shortage of cresylic acid existed in the domestic market owing to increased demand by synthetic resin makers. The output in 1936 exceeded that in 1935 and the production in 1937 was appreciably higher than in 1936. These increases are due to the recovery of appreciable quantities at several new distillation plants, the topping of large amounts of tar hitherto not processed, and increased production by present recovery units.

There are many grades of cresylic acid, most of which are prepared by mixing or blending to individual specifications. Every large consumer apparently has his own specifications. In addition to these special mixtures there are the following standard blends:

- (1) Ninety-nine percent high-boiling, straw color.
- (2) Low-boiling, straw color.
- (3) Special resin grade, high-boiling.

There are four domestic producers of cresylic acid with recovery and refining units at Pittsburgh, Pa., Philadelphia, Pa., Indianapolis, Ind., and Follansbee, W. Va. The last three mentioned are refining plants operated in conjunction with a number of tar distillation units widely scattered throughout the country. These units usually recover crude tar-acid fractions in the distillation of tar and ship them to these refining plants for separation and refining. All in this group are purchasers of coal tar. The fourth producer operates a byproduct recovery unit in connection with the company's coke-oven operations. Part of the coal tar produced is distilled to recover the several products, including creosote oil, tar acids, and naphthalene, and the residual pitch is mixed with the remaining undistilled tar and used for fuel. The shortage of tar acids and naphthalene in 1936 caused this producer to begin the topping of tar.

Foreign production.

The cresols are produced in the United Kingdom, Germany, France, the Netherlands, Belgium, and other European countries. Coal tar recovered in the United Kingdom is principally gas tar, which is much richer in tar acids than coke-oven tar, the principal source in the United States. This is true because low temperature carbonization of coal yields greater quantities of tar acids than are obtained in the tar from byproduct coke ovens. Exports to the United States are chiefly mixtures which can enter as crude cresylic acid.

Production of cresol in Germany in recent years is shown in table [69](#).

TABLE 69.—*Cresol: German production, in specified years*

Year	1,000 pounds
1929	23,814
1931 ¹	15,435
1933	11,780
1934	10,476

¹ Includes 2,866,000 pounds of separated ortho, meta, and para cresols.

Source: Consular reports.

German imports and exports of cresol, in recent years, are shown in table [70](#).

TABLE 70.—*Cresol: German imports and exports, in specified years*

Year	Imports	Exports
	<i>1,000 pounds</i>	<i>1,000 pounds</i>
1929	2,037	8,494
1930	1,277	6,712
1931	1,874	7,980
1932	1,541	3,669
1933	1,832	4,970
1934	1,960	6,345
1935	2,117	8,528
1936	3,737	4,531
1937	1,787	4,423

Source: Consular reports (1929-33) and official German statistics (1934-37).

The output of cresols in Czechoslovakia in recent years is shown in table [71](#).

TABLE 71.—*Cresol: Production in Czechoslovakia, in specified years*

Year	Quantity
	<i>1,000 pounds</i>
1928	1,984
1931	1,477
1932	1,102

1933	1,599
1934	1,918

Source: Consular reports.

Cresylic acid is recovered in all the countries of Europe, Great Britain and Germany being the leading producers and the principal exporters. Increasing demand in these countries for synthetic resins made from cresylic acid has greatly reduced the quantities available for export in recent years.

Great Britain is probably the world's largest producer of cresylic acid, and for many years has been the principal exporter to the United States. This position is due to the large available supply of gas-house tar, and to an ample market for all the products of tar distillation. In 1935 the tar distilled in England, Wales, and Scotland totaled 360 million gallons, of which 55 percent was gas-house tar.

British production of all grades of cresylic acid averages between 35 million and 42 million pounds annually, of which from 12 million to 20 million pounds are exported. Many British producers market their tar products through pools and associations. There is a cresylic acid pool, a phenol pool, and at least two creosote oil export associations, a pitch marketing association, and a benzol association. One of the principal grades of cresylic acid produced in Great Britain is "American duty-free specification."

Table 72 shows British exports of cresylic acid, by countries, in recent years.

TABLE 72.—*Cresylic acid: British exports, by countries, 1933-37*

Destination	1933	1934	1935	1936	1937
	Quantity (in thousands of pounds)				
UNITED STATES	3,616	5,783	6,116	11,296	(¹)
Chile	1,381	2,958	2,814	662	(¹)
France	358	87	214	189	(¹)
Japan	632	203	685	2,002	(¹)
All other countries	5,464	6,965	7,333	9,250	(¹)
Total	11,451	15,997	17,162	23,399	26,697
	Value (in thousands of dollars)				
UNITED STATES	147	277	255	620	(¹)
Chile	28	89	87	31	(¹)
France	21	5	20	14	(¹)
Japan	38	20	43	118	(¹)
All other countries	189	307	330	537	(¹)
	422	698	734	1,321	2,262

¹ Not available.

Source: Official British statistics.

Imports into the United States.

Rates of duty.—Prior to September 8, 1916, the cresols were imported free of duty. Since that date they have been subject to the tariff treatment shown in table [73](#).

TABLE 73.—*The cresols: Rates of duty upon United States imports, 1916-37*

Period	Rate of duty			Authority
	Less than 75 percent pure	75 to 90 percent pure	90 percent or more pure	
To Sept. 8, 1916.	Free	Free	Free	Free under par. 452, Tariff Act of 1913, and under previous acts.
Sept. 9, 1916, to Sept. 8, 1921.	do	do	15 percent ad valorem and 2½ cents per pound.	Revenue Act of 1916.
Sept. 9, 1921, to Sept. 21, 1922.	do	do	15 percent ad valorem and 2 cents per pound.	Emergency Tariff Act of 1921. From May 28, 1921, to Sept. 21, 1922, imports prohibited except when not obtainable in sufficient quantities or on reasonable terms as to quality, price, and terms of delivery.
Sept. 22, 1922, to Sept. 21, 1924.	do	55 percent ad valorem and 7 cents per pound. ¹	55 percent ad valorem and 7 cents per pound. ¹	Free under par. 1549 and dutiable under par. 27 of Tariff Act of 1922.
Sept. 22, 1924, to June 17, 1930.	do	40 percent ad valorem and 7 cents per pound. ¹	40 percent ad valorem and 7 cents per pound. ¹	Same; ad valorem reduced to 40 percent under provisions of the Tariff Act of 1922.
June 18, 1930, to date.	do	do	20 percent and 3½ cents per pound. ¹	Free under par. 1651 and dutiable under par. 27 of the Tariff Act of 1930.

¹ Ad valorem based on American selling price or United States value.

Under the Tariff Act of 1930 metacresol, orthocresol, and paracresol as such or in mixture, if less than 75 percent pure, would be imported free under paragraph 1651.^[23] If from 75 to 90 percent, they are dutiable under paragraph 27 at 7 cents a pound and 40 percent based upon American selling price.

[24] And if 90 percent pure or more, they are dutiable under paragraph 27 at 3½ cents per pound and 20 percent, based upon American selling price. [25]

The duties on cresylic acid in recent years are shown in table 74. Under the Tariff Act of 1930, cresylic acid less than 75 percent pure is free under paragraph 1651. [23] If more than 75 percent pure it is dutiable under paragraph 27 (b) at 3½ cents per pound and 20 percent, based on American selling price. [26]

TABLE 74.—*Cresylic acid: Rates of duty upon United States imports, 1916-37*

Period	Rate of duty		Authority
	Less than 75 percent pure	75 percent pure or more	
To Sept. 8, 1916.	Free	Free	Free under par. 452 of Tariff Act of 1913 and previous acts.
Sept. 9, 1916, to Sept. 8, 1921.	do	15 percent ad valorem and 2½ cents per pound.	Revenue Act of 1916.
Sept. 9, 1921, to Sept. 21, 1922.	do	15 percent ad valorem and 2 cents per pound.	Emergency Tariff Act of 1921. From May 28, 1921, to Sept. 21, 1922, imports prohibited except when not obtainable in sufficient quantities or on reasonable terms as to quality, price, and terms of delivery.
Sept. 22, 1922, to Sept. 21, 1924.	do	55 percent ad valorem and 7 cents per pound. ¹	Free under par. 1549 and dutiable under par. 27 of the Tariff Act of 1922.
Sept. 22, 1924, to Aug. 18, 1927.	do	40 percent ad valorem and 7 cents per pound. ¹	Same; ad valorem reduced to 40 percent under provisions of the Tariff Act of 1922.
Aug. 19, 1927, to June 17, 1930.	do	20 percent ad valorem and 3½ cents per pound. ¹	Duty reduced by Presidential proclamation.
June 18, 1930, to date.	do	20 percent ad valorem and 3½ cents per pound. ¹	Free under par. 1651 and dutiable under par. 27 of the Tariff Act of 1930.

¹ Ad valorem based on American selling price or United States value.

Import statistics.—Imports of the separated and mixed cresols are combined in official statistics. Table 75 shows imports of the cresols “90 percent pure or more.” There have been no recorded imports of less pure grades.

Tables 76, 77, and 78 show, by principal sources, imports of metacresol, orthocresol, and paracresol, as obtained from invoice analyses by the United States Tariff Commission. The sum of the three tabulations does not equal the total shown for all cresols in table 75. The difference in 1934 of 38,744 pounds valued at \$12,906 is accounted for by mixed cresols. Undoubtedly the differences in other years may be similarly accounted for.

TABLE 75.—*Metacresol, orthocresol, and paracresol, 90 percent pure or more: United States imports for consumption, 1920, and 1923-37*

Calendar year	Rate of duty ¹	Quantity	Value	Unit value	Computed ad valorem rate
		<i>Pounds</i>			<i>Percent</i>
1920	2½ cents per pound + 15 percent	2,444	\$2,230	\$0.912	17.7
1923	7 cents per pound + 55 percent	8,754	5,410	.618	66.3
1924	{ do	15,326	1,995	.130	108.8
	{ 7 cents per pound + 40 percent	1,000	663	.663	50.6
1925	do	34,874	5,741	.165	82.5
1926	do	105,238	15,040	.143	89.0
1927	do	174,094	35,054	.201	74.8
1928	do	207,897	33,638	.162	83.3
1929	do	227,974	32,098	.141	89.8
1930:					
Jan. 1-June 17	do	131,134	14,973	.114	101.3
June 18-Dec. 31	3½ cents per pound + 20 percent	71,183	11,762	.165	41.2
Total, 1930		202,317	26,735	.132	74.9
1931	do	151,571	26,901	.177	39.7
1932	do	83,848	18,530	.221	35.8
1933	do	48,511	16,205	.334	30.5
1934	do	124,598	34,361	.276	32.7
1935	do	65,468	18,290	.279	32.5
1936	do	83,273	27,686	.332	30.5
1937 ²	do	167,278	36,227	.217	36.2

¹ Ad valorem rate based on American selling price or United States value under the Tariff Acts of 1922 and 1930.

² Preliminary.

Source: Foreign Commerce and Navigation of the United States.

TABLE 76.—*Metacresol: United States imports for consumption by principal sources, in specified years*

Imported from	1929	1931	1933	1934	1935	1936	1937 ¹
	Quantity (pounds)						
United Kingdom	113,057	916	15,769	21,054	6,500	6,800	40,878
Germany	3,235		4,432				

All other countries	² 1,106						³ 11,025
Total	117,398	916	20,201	21,054	6,500	6,800	51,903
Invoice value							
United Kingdom	(⁴)	(⁴)	\$5,264	\$8,400	\$2,645	\$2,589	\$6,951
Germany	(⁴)	(⁴)	1,548				
All other countries	(⁴)	(⁴)					³ 4,200
Total			6,812	8,400	2,645	2,589	11,151
Invoice unit value							
United Kingdom	(⁴)	(⁴)	\$0.334	\$0.399	\$0.407	\$0.380	\$0.170
Germany	(⁴)	(⁴)	.349				
All other countries	(⁴)	(⁴)					.381
Average			.337	.399	.407	.380	.215
Percent of total quantity							
United Kingdom	96.30	100.0	78.06	100.0	100.0	100.0	78.76
Germany	2.76		21.94				
All other countries	2.94						³ 21.24
Total	100.00	100.0	100.00	100.0	100.0	100.0	100.00

¹ Preliminary.

² Netherlands.

³ Switzerland.

⁴ Not available.

Source: Invoice analyses, compiled by U. S. Tariff Commission.

TABLE 77.—*Orthocresol: United States imports for consumption, by principal sources, in specified years*

Imported from—	1929	1931	1933	1934	1935	1936	1937 ¹
Quantity (pounds)							
United Kingdom	105,790	79,198	19,548	25,855	29,120	33,816	112,108
Germany	82,859	5,914		10	10		
France						4,480	
All other countries	30,600						
Total	219,249	85,112	19,548	25,865	29,130	38,296	112,108
Invoice value							
United Kingdom	(²)	(²)	\$1,591	\$2,707	\$2,529	\$3,178	\$14,940
Germany	(²)	(²)		4	4		
France						336	
Total			1,591	2,711	2,533	3,514	14,940
Invoice unit value							
United Kingdom	(²)	(²)	\$0.081	\$0.105	\$0.087	\$0.094	\$0.133

Germany	(²)	(²)		.400	.400		
France						.075	
Average			.081	.105	.087	.092	.133
Percent of total quantity							
United Kingdom	48.25	93.05	100.0	99.96	99.97	88.30	100.00
Germany	37.79	6.95		.04	.03		
France						11.70	
All other countries	13.96						
Total	100.00	100.00	100.0	100.00	100.00	100.00	100.00

¹ Preliminary.

² Not available.

Source: Invoice analyses, compiled by U. S. Tariff Commission.

TABLE 78.—*Paracresol: United States imports for consumption, by principal sources, in specified years*

Imported from—	1929	1931	1933	1934	1935	1936	1937 ¹
Quantity (pounds)							
United Kingdom	2,587	458	6,972	16,889	16,625	32,666	14,338
Netherlands		11,243					
Germany				8,818	11,023	6,076	
France				13,228	5	6	4
Total	2,587	11,701	6,972	38,935	27,653	38,748	14,342
Invoice value							
United Kingdom	(²)	(²)	\$2,652	\$4,797	\$4,485	\$10,739	\$5,415
Netherlands	(²)	(²)					
Germany	(²)	(²)		1,921	3,090	3,079	
France	(²)	(²)		3,626	7	7	3
Total			2,652	10,344	7,582	13,825	5,418
Invoice unit value							
United Kingdom	(²)	(²)	\$0.380	\$0.284	\$0.270	\$0.329	\$0.378
Netherlands	(²)	(²)					
Germany	(²)	(²)		.218	.280	.506	
France	(²)	(²)		.274	1.400	1.167	.750
Average			.380	.266	.274	.357	.378
Percent of total quantity							
United Kingdom	100.0	3.92	100.0	43.38	60.12	84.30	99.97
Netherlands		96.08					
Germany				22.65	39.86	15.68	
France				33.97	.02	.02	.03
Total	100.0	100.00	100.0	100.00	100.00	100.00	100.00

¹ Preliminary.

² Not available.

Source: Invoice analyses, compiled by U. S. Tariff Commission.

The processes for recovery of the cresols (fractional distillation) usually yield products more than 75 percent pure and most of the consumers of mixed or prepared cresols require products of high purity. This explains why there are no imports less than 75 percent pure, notwithstanding that they are duty-free under paragraph 1651.

Under the act of 1930 cresols of 90 percent or greater purity are assessed for duty at 20 percent ad valorem and 3½ cents per pound while cresols 75.1 to 89.9 percent pure are assessed for duty at 40 percent ad valorem and 7 cents per pound. Naturally, since the duty on imports below 90 percent pure is double that on imports over 90 percent pure there are no imports of the former.

Imports of crude cresylic acid are shown in table 79 and those of refined cresylic acid in table 80. Imports by principal sources are shown in tables 81 and 82, for crude and refined, respectively.

TABLE 79.—*Crude cresylic acid: United States imports for consumption, 1924-37*

Year	Quantity ¹	Value	Unit value
	<i>Pounds</i>		
1924	2,327,528	\$157,643	\$0.068
1925	2,163,557	122,742	.057
1926	5,702,740	331,550	.058
1927	9,136,516	567,802	.062
1928	10,687,109	678,177	.063
1929	17,856,765	952,110	.053
1930	9,009,674	501,418	.056
1931	4,937,078	244,631	.050
1932	4,077,700	164,379	.040
1933	5,523,733	178,824	.032
1934	7,163,511	284,051	.040
1935	6,849,113	265,485	.039
1936	13,476,427	722,575	.054
1937 ²	16,360,213	1,219,268	.075

¹ Conversion factor—8.5 pounds to gallon.

² Preliminary.

Source: Foreign Commerce and Navigation of the United States.

TABLE 80.—*Refined cresylic acid: United States imports for consumption, in specified years, 1919-37*

Calendar year	Rate of duty	Quantity	Value	Unit value	Computed ad valorem rate
		<i>Pounds</i>			<i>Percent</i>

[illegible]

United Kingdom	\$739,385	\$190,333	\$165,986	\$276,989	\$262,137	\$661,781	\$954,953
Germany	212,652	51,643	8,666	6,263	3,325	33,068	184,887
Netherlands	73	2,655	551			1,190	
All other countries			² 3,621	799	23	³ 26,536	79,428
Total	952,110	244,631	178,824	284,051	265,485	722,575	1,219,268
Unit value							
United Kingdom	\$0.0529	\$0.0500	\$0.0328	\$0.0400	\$0.0388	\$0.0536	\$0.0752
Germany	.0549	.0481	.0243	.0287	.0347	.0528	.0740
Netherlands	.0660	.0489	.0250			.0681	
All other countries			.0432	.0452	.0602	.0545	.0687
Average	.0533	.0495	.0324	.0396	.0388	.0536	.0745
Percent of total quantity							
United Kingdom	78.3	77.2	91.6	96.7	98.6	91.60	77.65
Germany	21.7	21.7	6.5	3.0	1.4	4.65	15.28
Netherlands		1.1	.4			.13	
All other countries			1.5	.3	(⁴)	3.62	7.07
Total	100.0	100.0	100.0	100.0	100.0	100.00	100.00

¹ Preliminary.

² Canada.

³ Canada and France.

⁴ Less than one-tenth of 1 percent.

Source: Compiled from official statistics of the United States Department of Commerce.

TABLE 82.—*Refined cresylic acid: United States imports for consumption, from principal sources, in specified years, 1929-37*

Imported from—	1929	1931	1932	1933	1934	1935	1936	1937 ¹
Quantity (pounds)								
Great Britain	1,855,844	604,404	456,783	121,634	23,464	16,602		46,379
Germany	212,918	102,701	185,028				512	
All other countries	274,767		88	500				100
Total	2,343,529	707,105	641,899	121,634	23,964	16,602	512	46,479
Value								
Great Britain	\$144,630	\$35,041	\$24,607	\$9,164	\$1,412	\$1,128		\$5,101
Germany	14,699	7,115	12,714				\$40	
All other countries	23,995		5		85			21
Total	183,324	42,156	37,326	9,164	1,497	1,128	40	5,122
Unit value								
Great Britain	\$0.078	\$0.058	\$0.054	\$0.075	\$0.060	\$0.068		\$0.110
Germany	.069	.069	.069				\$0.078	
All other countries	.087		.057		.170			.210

Average	0.78	.060	.058	.075	.062	.068	.078	.110
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¹ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

In 1931 practically all imports of refined cresylic acid were from the United Kingdom and consigned to one importer in New York. In 1932 about 73 percent of the total dutiable imports were consigned to the same firm. From these data and from a conference with representatives of the importer it would appear that the imports were not cresylic acid in its original meaning (a mixture of cresols in their natural proportions), nor in the broadened commercial meaning (including with the cresols, xylenols and higher boiling tar acids), but were chiefly a product consisting largely of a single cresol separated from its two isomers. Treasury Decision 46146, effective March 11, 1933, closed the classification of refined cresylic acid to products of this type and imports thereafter under this head have been much smaller. After 1927 substantial amounts of the imports were reexported with benefit of drawback.

The imports of crude cresylic acid are also not of the type which the domestic producer would sell by that name. Far from being a straight run mixture of the cresol and higher boiling tar acids, they are usually a mixture of fractions which have been separated, and then chosen and combined so that they will meet both the tariff requirement (i. e., less than 75 percent of the total product will distill over at 215° C.) and the specifications of the purchasers. Customer's specifications are so drawn that the product will fill his special needs or can easily be broken down by fractional distillation in this country into elements, one or more of which will be so usable. Thus although imported crude cresylic acid must keep within the limitations set by the tariff it approaches as nearly as possible the type of cresylic acid which, if produced in this country, would be termed refined, since it was produced to meet the specifications of the consumer.

United States exports.

Exports of the cresols and of cresylic acid are not shown in official statistics and exports of these products as such are probably negligible, but there are appreciable exports of antiseptics, insecticides, and disinfectants in which they are incorporated, as well as of products or parts of products molded of resins made from cresylic acid.

Competitive conditions.

British coal tar is principally of gas-house origin and contains a higher percentage of tar acids (cresylic acid and phenol) than coke-oven tar, the principal kind recovered in the United States. The recovery of these tar acids from either kind of tar is usually not practicable, unless the distiller can dispose of the major products, creosote oil and pitch. British distillers have in the past ordinarily had a market for all their products; exporting large quantities of creosote oil to the United States, pitch to continental Europe, and tar acids and naphthalene to the United States, Germany, and other countries. Domestic distillers have sold cresylic acid, creosote oil, naphthalene, etc., in local markets in competition with duty-free imports from the United Kingdom, the Netherlands, Belgium, and Germany, but have found it difficult to dispose of pitch. The domestic production of coal tar ordinarily exceeded 600 million gallons, approximately one-half of which has been burned as fuel. Since the profit in distilling depends upon the markets for all of the joint products of the distillation, the large amount remaining undistilled can be understood.

The domestic production of cresylic acid may be expected to increase substantially, for several reasons: (1) The principal foreign producing countries have decreased exports because of increased demand for some of the coal-tar distillation products at home; (2) increased world prices; and (3) the development of topping, which allows the production of tar acids and naphthalene from coal tar without complete distillation.

Imports of refined cresylic acid are unimportant because the duty on the refined is high relative to the duty-free condition of the crude. Most of the imports of refined are either reexported or used in the manufacture of one proprietary antiseptic. The principal domestic market for cresylic acid is as a raw material for synthetic resins, and most of the domestic refined and most of the duty-free imported crude is now consumed by this industry. It may therefore be said that practically all imports of cresylic acid are duty-free and that, while they are sometimes refined by the consumer, they compete directly with domestic production of refined grades.

Phenol, already discussed, is closely related to cresylic acid—in chemical composition, in production by distillation from coal tar, and in use as a raw material for synthetic resins. To a considerable extent the proportions of phenol and cresylic acid used in the manufacture of tar-acid resins can be altered to take advantage of the changing price differential between the two.

All of the separated or mixed cresols are produced in commercial quantities in this country. Consumption in the United States, especially of types and grades used in synthetic resins, has increased appreciably in recent years, and is supplied chiefly by domestic production. A comparison of the quantity and value of domestic production and of imports in 1934 is shown in table [83](#).

TABLE 83.—*The cresols: Comparison of production and imports, 1934*

Product	Production			Imports		
	Quantity	Value	Unit value	Quantity	Value	Unit value
	<i>Pounds</i>			<i>Pounds</i>		
Metacresol	(¹)	(¹)		21,054	\$8,400	\$0.399
Paracresol	(²)	(²)	\$0.350	38,935	10,344	.266
Metaparacresol	2,033,424	\$122,005	.060	(¹)	(¹)	
Orthocresol	835,016	66,801	.080	25,865	2,711	.105
Orthometapara cresol	8,929,836	625,088	.070	38,744	12,906	.333
Total	11,798,276	813,894		124,598	34,361	

¹ None, production reported for first time in 1935.

² Not publishable.

Sources: Production, Dyes and Other Synthetic Organic Chemicals in the United States; imports, invoice analyses, U. S. Tariff Commission.

SYNTHETIC TAR ACIDS OTHER THAN PHENOL

Certain synthetic tar acids other than synthetic phenol are used commercially in the manufacture of synthetic resins in the United States. Among these are para tertiary amyl phenol, para tertiary butyl phenol, ortho phenyl phenol, para phenyl phenol, and resorcinol.

Para tertiary amyl phenol.

Para tertiary amyl phenol is made by reacting amylene with phenol in the presence of sulphuric acid as a catalyst. At ordinary temperatures it is a solid, melting at about 88° C. and boiling between 250°-265° C. Its use is of increasing importance as a component in tar-acid resins, especially in oil-soluble varnish resins. Owing to its phenol coefficient of approximately 60, it is also used as a germicide, fumigant, and insecticide. Commercial production was reported for the first time in 1933. Since then the output has increased appreciably each year, accompanied by material reductions in sales prices.

According to United States Patent No. 1,800,295, dated April 14, 1931, a resin fast to light and soluble in oils is obtained by heating 82 parts of p-tertiary amyl phenol with 90 parts of formaldehyde, in the presence of sodium hydroxide. This substituted phenol resin passes slowly into the infusible state, thus permitting better control of the reaction.

Para tertiary butyl phenol.

Para tertiary butyl phenol is a white solid with an aromatic odor, melting at approximately 100° C. It is a new commercial product and is used in resins for paints and varnishes. It is the most important resin material in this group.

Phenyl phenols.

Both ortho and para phenyl phenol are commercially produced and are used to some extent in resins to replace phenol. The ortho isomer is a white solid boiling at 284° C. and melting at about 56° C. It is used chiefly as a germicide, though small quantities are used in resins.

Para phenyl phenol is a white solid melting at about 165° C. and boiling at 322° C. Commercial production was reported for the first time in 1933. The output has increased each year since and the selling price has gradually declined.

Resorcinol.

Resorcinol, usually obtained by fusing meta benzene disulphonic acid with caustic soda, is a colorless, crystalline substance with a peculiar odor. It melts at 119° C. and boils at 276° C. It is used in medicine, in the manufacture of intermediates and dyes, and to some extent in synthetic resins. Resorcinol condenses with formaldehyde at such a rapid rate that some means must be applied to slow up the reaction. It is used to increase the rate of condensation of tar-acid resins and to reduce the danger of sticking or undercure.

Domestic production of resorcinol has decreased in recent years. Its relatively high cost is probably an important factor in limiting its use in synthetic resins.

FORMALDEHYDE

Description and uses.

At ordinary temperature and pressure formaldehyde is a gas. It enters commerce as formalin, an aqueous solution containing 40 percent formaldehyde by volume (37 percent by weight) and from 6 to 14 percent methyl alcohol. It is generally made by the oxidation of methyl alcohol. Commercial formalin contains polymers which tend to precipitate in water solution; these are kept in solution by allowing from 6 to 14 percent methyl alcohol to remain in the solution.

The principal use of formaldehyde is in the manufacture of synthetic resins. Other uses are (in the order of their importance): In the manufacture of synthetic indigo; in the manufacture of hydrosulphite; as a disinfectant, deodorant, and preservative; as a fungicide; in embalming fluids; in tanning leather; and in the manufacture of coated paper and wallpaper.

United States production.

The domestic output of formaldehyde has increased with the increased demand by resin makers. Production and sales in 1937 were more than double those in 1930. There are three domestic makers, two of which produce methyl alcohol, the raw material. Their plants are located in New Jersey and Oklahoma.

Statistics of production and sales are shown in table [84](#).

TABLE 84.—*Formaldehyde: United States production and sales, in specified years*

Year	Production	Sales		
	Quantity	Quantity	Value	Value per lb.
	<i>1,000 pounds</i>	<i>1,000 pounds</i>		
1914	(¹)	8,426	\$655,174	\$0.078
1919	25,007	19,664	3,928,322	.200
1921	9,657	6,056	651,681	.108
1922	23,958	16,140	1,676,401	.104
1923	24,081	18,855	2,474,506	.131
1924	26,155	20,542	1,971,053	.096
1925	31,456	23,392	1,895,913	.081
1926	31,953	22,552	2,050,967	.091
1927	29,920	24,597	2,256,534	.092
1928	38,718	27,934	2,491,615	.089
1929	51,786	(²)	(²)	
1930	40,763	(²)	(²)	
1931	(¹)	(¹)	(¹)	
1932	(¹)	(¹)	(¹)	
1933	52,236	46,424	2,122,925	.046
1934-37	(²)	(²)	(²)	

¹ Not available.

² Not publishable; figures would disclose operations of individual firms.

Source: Compiled from annual reports of the Tariff Commission on dyes and other synthetic organic chemicals in the United States.

Production in other countries.

Formaldehyde is produced in England, Germany, France, Czechoslovakia, Italy, Sweden, the Soviet Union, Japan, and Canada. Production data are not available but Germany and England are probably the leading foreign producers. Estimated productive capacity in the Soviet Union is given as 10

million pounds annually; in Japan 6.5 million pounds; in France 4 million pounds; and in Italy 3 million pounds.

United States imports and exports.

Imports of formaldehyde have been negligible since 1920 when 428,444 pounds, valued at \$210,191, were imported. There were no imports from 1928 until 1935, when 375 pounds valued at \$72 were imported from Canada. In 1936 imports amounted to 20 pounds, valued at \$14, from Switzerland.

The United States exports about 5 percent of its production of formaldehyde. Canada is the principal destination of exports, and prior to 1934 Japan was an important market. Table 85 shows exports of formaldehyde to principal markets, in recent years.

TABLE 85.—*Formaldehyde: United States exports to principal markets, in specified years, 1929-37*

Exported to—	1929	1931	1933	1934	1935	1936	1937 ¹
Quantity (pounds)							
Canada	19,321	716,132	1,095,847	1,236,103	1,493,993	1,105,277	1,187,661
Japan	1,464,763	622,407	729,875	11,250			
Netherlands Indies	626,689	650,875		344,725	71,375	200	335,275
United Kingdom	303	345,896	305			2	
China	29,030	229,760	215,050	572,700	598,342	459,490	938,700
All other countries	448,063	339,777	332,101	433,068	334,100	279,289	403,236
Total	2,588,169	2,904,847	2,373,178	2,597,846	2,497,810	1,844,258	2,864,872
Value							
Canada	\$4,405	\$36,471	\$46,611	\$58,348	\$83,805	\$53,062	\$50,780
Japan	124,411	33,395	43,254	562			
Netherlands Indies	42,807	38,462		15,536	2,866	13	13,411
United Kingdom	510	18,629	192			1	
China	3,354	13,019	11,998	27,407	28,239	19,258	34,274
All other countries	49,742	22,709	18,966	23,444	18,676	16,520	20,498
Total	225,229	162,685	121,021	125,297	133,586	88,854	118,963
Unit value							
Canada	\$0.228	\$0.051	\$0.043	\$0.047	\$0.056	\$0.048	\$0.043
Japan	.085	.054	.059	.050			
Netherlands Indies	.068	.059		.045	.040	.065	.040
United Kingdom	1.683	.054	.630			.500	
China	.116	.057	.056	.048	.047	.042	.037
All other countries	.111	.067	.057	.054	.056	.059	.051
Average	.087	.056	.051	.048	.053	.048	.042
Percent of total quantity							
Canada	0.8	24.7	46.2	47.6	59.8	59.9	41.4
Japan	56.6	21.4	30.7	.4			
Netherlands Indies	24.2	22.4		13.3	2.8		11.7

United Kingdom		11.9					
China	1.1	7.9	9.1	22.0	24.0	24.9	32.8
All other countries	17.3	11.7	14.0	16.7	13.4	15.2	14.1
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0

¹ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

Competitive conditions.

The competitive situation with respect to formaldehyde is determined largely by the output and price of the raw material, methanol. The United States produces large quantities of synthetic and natural methanol and is a net exporter of that product. Both methanol and formaldehyde are produced in many foreign countries, and foreign production is expanding. Although methanol is now the main, if not the sole, raw material utilized in making formaldehyde, plants for making the latter direct from natural gas, or from petroleum gas hydrocarbons are contemplated or actually under construction. Should such processes develop to an appreciable extent, the competitive situation of the United States may change, but in such case it is unlikely that this country would be so affected as to change its position as a moderate exporter.

HEXAMETHYLENETETRAMINE

Description and uses.

Hexamethylenetetramine is a white crystalline powder made by the interaction of formaldehyde and ammonia. It is used in tar-acid resins to replace formaldehyde, though its higher cost has limited its use to small proportions as a finishing or hardening agent. Other uses are as an internal antiseptic in medicine (marketed under trade names such as Urotropin, Cystogen, Aminoform, Urisol, and Cystamin), as an accelerator in the vulcanization of rubber (a declining use), and in artificial cork. During the World War it was used in gas masks as an absorbent for phosgene.

United States production.

The domestic production of hexamethylenetetramine declined during the depression, but has been increasing in the last few years. Production in 1937, however, was still below that of 1929. Statistics of production are shown in table [86](#).

TABLE 86.—*Hexamethylenetetramine: United States production and sales, 1923, and 1925-37*

Year	Production (quantity)	Sales		
		Quantity	Value	Unit value
	<i>Pounds</i>	<i>Pounds</i>		
1923	1,381,073	1,155,083	\$974,877	\$0.84
1925	1,657,993	1,506,286	994,458	.66
1926	1,495,220	(¹)	(¹)	
1927	1,315,213	(¹)	(¹)	

1928	1,661,645	(¹)	(¹)
1929	2,368,020	(¹)	(¹)
1930	1,871,690	(¹)	(¹)
1931	(²)		(²)
1932	(²)		(²)
1933-37	(¹)		(¹)

¹ Not publishable; figures would disclose operations of individual firms.

² Not available.

Source: Compiled from annual reports of the Tariff Commission on dyes and other synthetic organic chemicals in the United States.

Hexamethylenetetramine is made by two firms in New Jersey and by one in West Virginia. The raw materials utilized are formaldehyde and liquid or anhydrous ammonia. One company makes its own requirements of both, and another makes its own formaldehyde. Most of the production of hexamethylenetetramine is sold, marketed in barrels, drums, kegs, and cans.

Production in other countries.

Hexamethylenetetramine is made in a number of foreign countries, with Germany probably the leading foreign producer. Exports from Germany declined from 445,000 pounds in 1931 to 182,000 pounds in 1934, the decline being due chiefly to the expansion of production in countries previously large importers, particularly the United Kingdom, Japan, Czechoslovakia, and France.

United States imports and exports.

Imports of hexamethylenetetramine to the United States are shown in table [87](#).

TABLE 87.—*Hexamethylenetetramine: United States imports for consumption, 1923-37*

Calendar year	Rate of duty	Quantity	Value	Unit value	Computed ad valorem rate
		<i>Pounds</i>			<i>Percent</i>
1923	25 percent	47,373	\$24,722	\$0.522	
1924	do	3,826	3,998	1.045	
1925	do	20,771	10,453	.503	
1926	do	23,481	10,237	.436	
1927	do	3,417	1,715	.502	
1928	do	5,898	1,643	.279	
1929	do	5,562	1,857	.334	
1930:					
Jan. 1-June 17	do				
June 18-Dec. 31	11 cents per pound				
1931	do				
1932	do	1,103	336	.305	36.1

1933	do	1,103	293	.266	41.4
1934	do	612	273	.446	24.7
1935	do				
1936	do	7,496	1,510	.201	54.6
1937 ¹	do	10,895	8,197	.200	54.9

¹ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

Imports of hexamethylenetetramine in 1928 came principally from the United Kingdom, the remainder from Germany. In 1929 they came wholly from Germany; in 1932 and 1933 from Belgium; and in 1934 principally from Canada, with the rest from the United Kingdom. In 1936 Belgium supplied 7,166 pounds valued at \$1,368 and Germany 330 pounds valued at \$142.

Exports of hexamethylenetetramine are not shown in official statistics. It is known, however, that some has been exported, and that in 1933, at least, exports exceeded imports.

Competitive conditions.

Hexamethylenetetramine is made from formaldehyde and ammonia, of which there are ample supplies in the United States. The market for hexamethylenetetramine is limited, and imports are small. It is made in numerous foreign countries, Germany being probably the principal potential competitor.

FURFURAL

Furfural is an aldehyde found in oat hulls, rice hulls, corn cobs, bran, and other farm waste products. Commercially it is obtained in the United States from oat hulls and in the Soviet Union from the husks of sunflower seeds. It is a colorless liquid, boiling at 158° to 162° C. and freezing at minus 38° C. Its principal use is in synthetic resins, of which tar acid-furfural is probably the most important. These resins are used in molding, for impregnating, and in coatings. Furfural is also used as a solvent for cellulose ethers and esters, natural gums and resins, and in the manufacture of derivatives useful as rubber chemicals.

Domestic production is entirely by one firm, located in Iowa. Production and sales statistics are not publishable, but the maker has stated that consumption is in "terms of millions of pounds per year."

18. RAW MATERIALS FOR UREA RESINS

The principal raw materials entering into the manufacture of urea resins are urea and formaldehyde. Formaldehyde has already been discussed (see pp. [133-135](#)) and urea and thiourea are discussed below.

UREA

Urea is a white crystalline material, made by condensing carbon dioxide and ammonia under heat and pressure. It is an excellent fertilizer because of its high nitrogen content (46.6 percent) but this use is limited by its relatively high cost. Urea is an important synthetic resin material, being a constituent of urea-formaldehyde resins, known commercially under the trade names Beetleware and Plaskon.

Production of urea in the United States was started in 1916, when the German supply was cut off. In 1920 the domestic output was estimated at more than 200,000 pounds of fertilizer grade. Production ceased in 1922. Urea in ammonia solution for use in the manufacture of mixed fertilizer was first produced in 1933.

Crystal urea production in the United States was begun in 1935 and was largely made possible by the larger volume of urea in ammonia solution manufactured for fertilizer use. Prior to that time our requirements of crystal urea were imported, principally from Germany. Consumers of resin grade urea report that the domestic product is as good or better than the imported from Europe. The domestic output of crystal urea in 1936 showed an appreciable increase over that in 1935.

Statistics of imports of urea are given in table [88](#), showing imports of all grades combined. Up to 1931, and again in 1936, the imports were probably all for fertilizer use. From 1931 through 1935 some portion of the imports went into the manufacture of resins, but even in this period most of the imports were probably used in fertilizer.

TABLE 88.—*Urea: United States imports for consumption, 1919-20 and 1923-37*

Year	Rate of duty	Quantity	Value	Unit value
		<i>Pounds</i>		
1919	25 percent	14,290	\$9,741	\$0.682
1920	do	23,693	14,085	.594
1923	35 percent	45,711	5,892	.129
1924	do	94,307	12,891	.137
1925	do	146,438	15,886	.108
1926	do	377,729	30,346	.080
1927	do	813,120	51,799	.064

1928	do	1,788,927	101,900	.057
1929	do	4,588,313	228,401	.050
1930:				
Jan. 1-June 17	do	2,459,140	120,263	.049
June 18-Dec. 31	Free	17,843,840	719,982	.040
1931	do	11,695,040	445,674	.038
1932	do	7,291,200	267,787	.037
1933	do	12,918,080	483,238	.037
1934	do	10,850,560	423,675	.039
1935	do	8,189,440	379,427	.046
1936	do	6,095,040	272,679	.045
1937 ¹	do	5,297,600	266,166	.050

¹ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

Table [89](#) shows the sources of imports of urea in recent years. Germany has supplied from 90 to 100 percent of the total quantity, and Canada and the Netherlands the greater part of the remainder. There have been occasional shipments from Belgium, France, and Japan.

TABLE 89.—*Urea: United States imports for consumption by countries, 1931 and 1933-37*

Imported from—	1931	1933	1934	1935	1936	1937 ¹
	Quantity (pounds)					
Germany	10,496,640	12,649,280	10,660,160	7,869,120	6,095,040	5,297,600
Netherlands	922,880	147,840	20,160			
Canada	185,920	120,960	168,000	320,320		
All other countries	89,600		2,240			
Total	11,695,040	12,918,080	10,850,560	8,189,440	6,095,040	5,297,600
	Value					
Germany	\$401,976	\$473,703	\$415,777	\$366,371	\$272,679	\$266,166
Netherlands	31,523	5,034	666			
Canada	9,107	4,501	7,032	13,056		
All other countries	3,068		200			
Total	445,674	483,238	423,675	379,427	272,679	266,166
	Unit value					
Germany	\$0.038	\$0.037	\$0.039	\$0.047	\$0.045	\$0.050
Netherlands	.034	.034	.033			
Canada	.049	.037	.042	.041		

All other countries	.034		.089			
Average	.038	.037	.039	.046	.045	.050

¹ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

Imports of urea enter the United States free of duty under paragraph 1793 of the Tariff Act of 1930. In spite of that fact a substantial production in the United States has been achieved. This is due, at least to a considerable extent, to the production of crude urea in ammonia solution, which is used in ammoniating superphosphates for fertilizer use. It is shipped by tank car but would be difficult to transport by ship. The volume market for this form of the product has aided in the production of crystal urea for both resin and fertilizer use.

THIOUREA

Thiourea (thiocarbamide) is a white crystalline solid, melting at 180° C. It is made commercially by treating a solution of calcium cyanamide with sulphur and ammonium sulphide or with hydrogen sulphide and ammonia. The principal uses of thiourea are in making intermediates and pharmaceuticals, as a photographic developer, as an insecticide, and in medicine. Because of the water resistance it imparts it was for some time widely used in urea resins. During the past few years, however, its use in resins has declined sharply owing to its deleterious action on ordinary molds and its slow rate of cure. In molding compounds, thiourea requires about 10 minutes curing time as compared with 3 minutes or less for urea resins and tar-acid resins. Since ways have been found to fabricate water-resistant urea resins without using thiourea, the consumption of thiourea in this use has declined.

There is no known commercial domestic production of thiourea.

Imports through the New York Customs District, according to invoice analyses made by the Tariff Commission, are shown in table 90. Thiourea is dutiable at 25 percent under paragraph 5 of the act of 1930.

TABLE 90.—*Thiourea: United States imports through the New York Customs District, 1931-37*

Year	Quantity (pounds)	Value	Unit value	Source
1931	81,560	\$24,254	\$0.297	Germany and Switzerland.
1932	19,347	4,760	.246	Germany.
1933				
1934	15,738	5,982	.380	Do.
1935	29,480	10,500	.356	Do.

1936	81,031	19,782	.244	Do.
1937	(¹)			

¹ Not available.

Source: Invoice analyses of paragraph 5, of Tariff Act of 1930. Compiled by U. S. Tariff Commission.

19. RAW MATERIALS FOR VINYL RESINS

Vinyl resins are made chiefly from vinyl acetate and vinyl chloride.

Description and uses.

Vinyl acetate is an unsaturated ester of the hypothetical vinyl alcohol. It is made from acetylene and acetic acid, and is a colorless liquid with a pleasant sweetish odor, boiling at 73° C. On account of its tendency to polymerize to polyvinyl acetate, a trace of copper salt is added for shipment. To render the vinyl acetate chemically active again, the copper salt is removed by distillation. At present the sole use of vinyl acetate is for the manufacture of synthetic resins. (See pp. [43-50](#).)

Vinyl chloride, a salt of vinyl alcohol, is obtained commercially from acetylene. It is a gas (boiling at about minus 14° C.) used in the manufacture of synthetic resins. Vinyl chloride mixed with vinyl acetate is polymerized to a synthetic resin.

United States production.

Until 1938 the one domestic maker of vinyl acetate produced only experimental lots, the bulk of our requirements being imported from Canada. In that year large units to manufacture vinyl acetate were built at Niagara Falls, N. Y., and at Belle, W. Va. The remarkable properties of safety glass made from vinyl resin sheets, together with several other new and important applications of these resins, indicate a demand for vinyl acetate sufficient to warrant these large manufacturing units. The United States patents covering the processes of manufacture of vinyl acetate are owned by the Canadian producer, who has licensed the domestic makers.

Domestic production of vinyl chloride has increased from experimental quantities in 1927 to large-scale commercial output, increasing substantially each year since 1933.

United States imports.

There has been no import of vinyl chloride. Imports of vinyl acetate (unpolymerized), entirely from Canada, are shown in table [91](#).

TABLE 91.—*Vinyl acetate, unpolymerized: United States imports for consumption, 1931-37*

Year	Quantity (pounds)	Value	Unit value
1931	77,269	\$11,489	\$0.149
1932	104,129	14,053	.135
1933	159,757	21,134	.132
1934	217,182	39,462	.182
1935	776,426	149,876	0.193
1936 ¹	449,905	58,499	.130
1937 ²	297,496	39,074	.131

¹ Duty reduced from 30 percent ad valorem and 6 cents per pound to 15 percent ad valorem and 3 cents per pound under Canadian trade agreement, effective Jan. 1, 1936.

² Preliminary.

Source: Foreign Commerce and Navigation of the United States.

Competitive conditions.

Domestic consumption of vinyl acetate and vinyl chloride has increased in recent years from experimental to commercial quantities. Many years of intensive research looking toward large outlets for the resins made from these compounds has apparently been successful. The largest single application indicated at this time is for safety glass sheets.

The large increase in domestic consumption expected in the immediate future will probably be supplied chiefly from expanding domestic production and imports from Canada will probably decline even under the reduced trade-agreement duty except to fill possible temporary shortages.

APPENDIXES

A. Statistical tables on foreign trade in raw materials for synthetic resins.

B. Trade names for synthetic resins made in the United States.

C. Trade names for synthetic resins made in Great Britain.

D. Trade names for synthetic resins made in Germany.

E. List of United States manufacturers of raw materials for synthetic resins.

F. Glossary.

APPENDIX A

STATISTICAL TABLES ON FOREIGN TRADE IN RAW MATERIALS FOR SYNTHETIC RESINS

TABLE 92.—*Naphthalene: German imports and exports, by countries, 1929 and 1932-37*

	1929	1932	1933	1934	1935	
Quantity (pounds)						
IMPORTS						
Total from all countries	8,032,019	952,167	7,482,633	8,640,930	4,245,839	
Czechoslovakia	1,688,283	(¹)	2,839,304	1,602,744		
Saar Basin	1,531,977	(¹)	1,833,125	3,641,338	1,129,858	
Netherlands	1,457,020	(¹)	1,164,029	26,014	18,078	
Poland	2,524,487	(¹)	832,457	1,060,633	33,730	
U. S. S. R	284,834	246,033	361,334	1,024,037	531,088	
Value						
	<i>Dollars</i>	<i>Dollars</i>	<i>1,000 reichsmarks</i>	<i>Dollars</i>	<i>1,000 reichsmarks</i>	<i>Dollars</i>
Total from all countries	110,948	12,112	271	82,704	273	107,494
Czechoslovakia	26,666	(¹)	114	34,790	51	20,081
Saar Basin	17,380	(¹)	56	17,090	78	30,712
Netherlands	14,999	(¹)	27	8,240	2	788
Poland	38,332	(¹)	31	9,461	29	11,419
U. S. S. R	4,762	4,750	22	6,714	48	18,900
Quantity (pounds)						
EXPORTS						
Total to all countries	39,738,576	29,720,213	31,842,140	35,043,660	22,169,458	
UNITED STATES	17,070,218	13,820,858	21,824,879	21,631,535	12,052,769	
Belgium-Luxemburg	8,835,596	7,399,960	3,958,800	5,685,663	2,010,816	
Japan	(¹)	(¹)	801,152	3,434,767	1,880,965	
Italy	4,500,691	2,068,797	1,163,589	695,992	492,728	
Netherlands	734,573	295,857	275,134	427,913	413,142	
Value						
	<i>Dollars</i>	<i>Dollars</i>	<i>1,000 reichsmarks</i>	<i>Dollars</i>	<i>1,000 reichsmarks</i>	<i>Dollars</i>
Total to all countries	774,256	288,790	1,392	424,809	1,504	592,200
UNITED STATES	329,273	132,758	921	281,070	879	346,106
Belgium-Luxemburg	98,330	42,749	95	28,992	130	51,188
Japan	(¹)	(¹)	50	15,259	203	79,931
Italy	78,330	23,324	50	15,259	30	11,812
Netherlands	19,761	4,987	21	6,409	29	11,419

¹ Not separately shown.

Source: Der Auswärtige Handel Deutschlands, 1929. Monatliche Nachweise über den auswärtigen Handel, Deutschlands, 1932-37.

TABLE 93.—*Crude naphthalene: Belgian imports and exports, 1932-37*

	1932	1933	1934	1935	1936	1937
Quantity (pounds)						
IMPORTS						
Total from all countries	14,114,070	10,935,698	15,328,363	12,114,718	17,102,405	9,178,4
Netherlands	8,187,443	7,583,163	9,363,598	8,786,874	10,315,764	5,983,0
Germany	5,800,744	3,122,816	5,290,599	1,297,628	178,573	
Value						
	<i>Dollars</i>	<i>1,000 francs</i>	<i>Dollars</i>	<i>1,000 francs</i>	<i>Dollars</i>	<i>1,000 francs</i>
Total from all countries	74,299	2,059	73,711	3,015	140,419	3,299
Netherlands	41,352	1,327	47,506	1,881	87,605	2,139
Germany	31,611	661	23,663	970	45,176	341
Quantity (pounds)						
EXPORTS						
Total to all countries	1,102,300	5,955,727	3,395,745	6,796,782	11,538,215	6,700,2
UNITED STATES		3,991,428	2,499,355	1,709,888	2,119,062	1,009,9
France	871,699	1,262,354	352,075	2,930,134	5,163,835	4,382,7
Value						
	<i>Dollars</i>	<i>1,000 francs</i>	<i>Dollars</i>	<i>1,000 francs</i>	<i>Dollars</i>	<i>1,000 francs</i>
Total to all countries	10,964	2,503	89,605	1,181	55,003	3,059
UNITED STATES		1,562	55,918	707	32,927	543
France	8,404	446	15,966	243	11,317	1,460

¹ Not separately reported.

Source: Bulletin Mensuel du Commerce.

TABLE 94.—*Refined naphthalene: Belgian imports and exports, 1932-37*

	1932	1933	1934	1935	1936	1937
Quantity (pounds)						
IMPORTS						
Total from all countries	7,055	1,323	7,275	112,214	2,866	
Value						
	<i>Dollars</i>	<i>1,000 francs</i>	<i>Dollars</i>	<i>1,000 francs</i>	<i>Dollars</i>	<i>1,000 francs</i>
Total from all countries	150	8	286	15	699	78
Quantity (pounds)						
EXPORTS						
Total to all countries	15,362,314	15,298,822	14,792,425	16,148,695	11,419,167	14,0
Japan	6,566,401	4,582,922	2,867,523	4,202,849	3,447,113	2,5
United Kingdom	695,772	1,297,407	2,168,004	315,919	455,029	1,5
Argentina	1,078,490	1,130,519	988,102	1,376,773	970,685	1,5
British India	792,995	996,920	1,153,226	705,252	243,477	

Canada	841,496	942,026	421,740	90,168	(¹)	
Netherlands	449,518	472,225	510,365	971,788	477,516	
Value						
	<i>Dollars</i>	<i>1,000 francs</i>	<i>Dollars</i>	<i>1,000 francs</i>	<i>Dollars</i>	<i>1,000 francs</i>
Total to all countries	326,026	11,310	404,889	10,032	467,224	11,219
Japan	130,650	3,498	125,226	1,966	91,563	2,949
United Kingdom	14,053	893	31,969	1,524	70,978	212
Argentina	24,154	811	29,033	719	33,486	976
British India	18,867	752	26,921	764	35,582	498
Canada	17,670	696	24,916	301	14,019	66
Netherlands	9,934	375	13,425	306	14,251	695

¹ Not separately reported.

Source: Bulletin Mensuel du Commerce.

TABLE 95.—*Crude and refined naphthalene: Netherland imports and exports by countries, 1929 and 1932-37*

	1929	1932	1933	1934	1935	1936	
Quantity (pounds)							
IMPORTS							
Total from all countries	896,908	775,759	908,597	1,186,361	1,177,256	1,666,678	
Belgium-Luxemburg	191,044	477,508	569,514	568,328	524,962	703,267	
Germany	705,820	270,262	330,921	522,993	418,409	(¹)	
Italy	(¹)	(¹)	2,315	(¹)	(¹)	(¹)	
United Kingdom		(¹)	97	(¹)	(¹)	604,060	
Value							
	<i>Dollars</i>	<i>Dollars</i>	<i>1,000 guilders</i>	<i>Dollars</i>	<i>1,000 guilders</i>	<i>Dollars</i>	<i>1,000 guilders</i>
Total from all countries	25,874	13,733	38	19,885	45	30,220	39
Belgium-Luxemburg	5,708	8,309	24	12,265	22	14,903	21
Germany	20,153	4,960	13	6,775	19	13,132	12
Italy	(¹)	(¹)	(¹)	681	(¹)	(¹)	(¹)
United Kingdom		(¹)	(¹)	23	(¹)	(¹)	21
Quantity (pounds)							
EXPORTS							
Total to all countries	4,816,436	6,807,060	10,770,410	9,764,389	10,851,416	14,422,493	
Belgium-Luxemburg	2,820,043	6,768,631	8,757,006	9,307,821	9,178,854	10,568,852	
Germany	1,511,897		1,174,745	(¹)	17,846	(¹)	
UNITED STATES	44,092		661,380	410,225	681,332	3,207,693	
United Kingdom	333,902	(¹)	100,001	(¹)	(¹)	(¹)	
Value							
	<i>Dollars</i>	<i>Dollars</i>	<i>1,000 guilders</i>	<i>Dollars</i>	<i>1,000 guilders</i>	<i>Dollars</i>	<i>1,000 guilders</i>
Total to all countries	49,058	40,784	158	81,490	132	89,135	170
Belgium-Luxemburg	25,503	39,886	124	63,928	122	82,146	129

Germany	13,877		17	8,756	(¹)	(¹)	(²)	257	(¹)	(¹)	(¹)
UNITED STATES	482		12	6,103	8	5,183	17	11,782	143	92,208	10
United Kingdom	5,556	(¹)	2	938	(¹)	(¹)	(¹)	(¹)	(¹)	(¹)	(¹)

¹ Not separately reported.

² Less than 500.

Source: Nederland-Jaarstatistiek (1929 and 1932-35) and Maandstatistiek (1936-37).

TABLE 96.—*Refined naphthalene: Canadian imports by countries, 1928-29 and 1932-37*

	1928	1929	1932	1933	1934	1935	1936	1937 ¹
	Quantity (pounds)							
Total from all countries	565,866	1,075,415	1,223,372	1,053,114	844,929	1,342,530	884,059	1,256,289
UNITED STATES	32,274	4,049	17,560	9,553	3,145	3,620	2,091	2,018
United Kingdom	26,000	8,600	32,400	148,144	484,868	1,321,310	879,548	1,254,271
Belgium	(²)	841,876	1,102,203	895,042	347,956	17,600	(²)	
	Value (United States dollars)							
Total from all countries	18,162	32,411	21,787	22,014	25,482	40,060	46,670	57,455
UNITED STATES	1,428	245 577	545	188 696	243 185	696	243	185
United Kingdom	1,014	363 670	3,779	13,603	38,865	38,865	46,229	57,270
Belgium	(²)	25,906	19,401	17,657	11,431	499	(²)	

¹ Preliminary.

² Not shown separately.

Source: Trade of Canada.

TABLE 97.—*Naphthalene: Japanese imports by countries, 1928-29 and 1932-36*

	1928	1929	1932	1933	1934	1935	1936
	Quantity (pounds)						
Total from all countries	2,773,320	2,902,686	6,765,572	7,876,566	7,364,557	8,979,696	12,640,000
Belgium	379,900	403,180	2,857,448	3,950,056	1,590,138	2,926,430	3,160,000
Germany	491,541	403,180	1,935,213	2,525,036	4,679,670	3,486,239	2,720,000
Kwangtung Province	1,790,766	864,298	1,272,372	169,976	241,146	1,103,322	6,760,000
	Value						
	<i>Dollars</i>	<i>Dollars</i>	<i>Dollars</i>	<i>1,000 yen</i>	<i>Dollars</i>	<i>1,000 yen</i>	<i>Dollars</i>
Total from all countries	62,653	94,504	97,265	625	160,286	560	166,404
Belgium	12,531	14,291	42,448	309	79,245	122	36,252
Germany	16,707	14,291	28,111	200	51,291	357	106,083
Kwangtung Province	29,238	21,667	13,774	13	3,334		12

Source: Annual Return of the Foreign Trade of Japan.

TABLE 98.—*Crude glycerin: United States imports for consumption,¹ by countries, 1929 and 1931-37*

Imported from—	1929 ²	1931	1932	1933	1934	1935	1936
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	Quantity (pounds)						
France	4,931,691	2,550,457	1,653,825	2,455,264	4,880,013	578,617	1,058
Cuba	1,074,271	1,170,667	1,232,219	1,216,395	1,178,238	2,550,617	2,158
United Kingdom	3,847,345	1,631,103	582,194	252,238	1,494,445	578,231	1,403
Belgium	759,448	739,892	310,855	440,862	2,358,479	257,290	404
Germany	1,072,173	674,109	260,005	242,901	1,539,919	198,767	77
Philippine Islands (free)	250,165	180,490	197,841	268,449	180,549	1,578,523	303
Argentina	494,638	458,068	154,525	288,832	680,443	100,902	1,154
Netherlands	262,299	425,796	125,733	226,994	1,393,072	267,366	1,037
Canada	1,304,220	1,161,085	80,440		629,880	1,946,450	671
Denmark	54,988	175,273		124,278	58,449		442
Union of Soviet Socialist Republics	132,334		64,969	889,618	146,695	14,883	2,017
All other countries	668,329	966,023	317,866	67,254	541,045	149,288	416
Total	14,851,901	10,132,963	4,980,472	6,473,085	15,081,227	8,220,934	11,148
	Value						
France	\$280,062	\$114,575	\$53,391	\$80,068	\$324,840	\$45,245	\$121
Cuba	69,668	67,709	50,147	56,737	92,692	228,011	230
United Kingdom	216,307	8 2,262	19,802	7,722	97,972	50,549	134
Belgium	49,568	46,275	12,362	17,627	160,301	19,217	46
Germany	65,446	40,596	12,240	10,745	103,401	22,699	8
Philippine Islands (free)	16,796	10,993	9,150	14,078	14,984	74,798	32
Argentina	29,758	23,532	6,516	7,947	45,251	7,972	115
Netherlands	18,963	23,301	5,349	10,664	92,754	23,208	127
Canada	67,821	59,495	4,246		51,716	172,426	70
Denmark	2,966	9,358		5,171	4,062		47
Union of Soviet Socialist Republics	9,113		2,738	34,060	10,137	1,463	222
All other countries	37,084	47,343	12,614	2,076	41,955	11,146	42
Total	863,552	525,599	188,555	246,895	1,040,065	656,734	1,198
	Percent of total imports by quantity						
France	33.2	25.2	33.2	37.9	32.4	7.1	
Cuba	7.2	11.6	24.7	18.8	7.8	31.0	
United Kingdom	25.9	16.1	11.7	3.9	9.9	7.0	
Belgium	5.1	7.3	6.3	6.8	15.6	3.1	
Germany	7.2	6.7	5.2	3.8	10.2	2.4	
Philippine Islands (free)	1.7	1.8	4.0	4.1	1.2	19.2	
Argentina	3.3	4.5	3.1	4.5	4.5	1.2	
Netherlands	1.8	4.2	2.5	3.5	9.2	3.3	
Canada	8.8	11.4	1.6		4.2	23.7	
Denmark	.4	1.7		1.9	.4		
Union of Soviet Socialist Republics	.9		1.3	13.7	1.0	.2	
All other countries	4.5	9.5	6.4	1.1	3.6	1.8	
Total	100.0	100.0	100.0	100.0	100.0	100.0	1

¹ Includes imports from Cuba and shipments from Philippine Islands.

² General imports.

³ Preliminary.

Source: Foreign Commerce and Navigation of the United States.

TABLE 99.—*Refined glycerin: United States imports for consumption, by countries, 1929 and 1931-37*

Imported from—	1929 ¹	1931	1932	1933	1934	1935	1936	1937 ²
Quantity (pounds)								
Netherlands	3,114,642	1,054,810	1,705,855	2,153,490	775,074	16,913	1,542,924	2,776,046
Germany	1,824,672	197,890	384,131	406,716	276,908	600	319	352,680
United Kingdom	165,770	38,561	153,289	50,421	951,196	28,176	572,919	373,416
France		592,979	44,905	125,889		584	413,977	2,967,528
Canada	105	81,295	14,520				765,676	19,782
Czechoslovakia							112,562	506,598
All other countries	388,282		44,808	39,171	210,764	22,293	39,110	539,070
Total	5,493,471	1,965,535	2,347,508	2,775,687	2,213,942	68,566	3,447,487	7,535,120
Value								
Netherlands	\$294,595	\$75,462	\$100,451	\$132,172	\$66,445	\$2,718	\$273,432	\$636,644
Germany	154,432	17,152	26,582	22,826	27,159	252	129	96,542
United Kingdom	18,301	2,850	9,666	3,111	93,938	2,760	99,204	76,001
France		40,005	2,278	7,210		97	79,242	751,816
Canada	19	5,506	1,358				114,523	4,922
Czechoslovakia							21,650	148,612
All other countries	33,383		2,024	1,672	21,447	2,450	5,856	112,652
Total	500,730	140,975	142,359	166,991	208,989	8,277	594,036	1,827,189
Percent of total imports by quantity								
Netherlands	56.7	53.7	72.7	77.6	35.0	24.7	44.7	36.8
Germany	33.2	10.1	16.4	14.7	12.5	.9		4.7
United Kingdom	3.0	1.9	6.5	1.8	43.0	41.1	16.6	5.0
France		30.2	1.9	4.5		.8	12.0	39.4
Canada		4.1	.6				22.2	.3
Czechoslovakia							3.3	6.7
All other countries	7.1		1.9	1.4	9.5	32.5	1.2	7.1
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

¹ General imports.

² Preliminary.

Source: Foreign Commerce and Navigation of the United States.

APPENDIX B

TRADE NAMES FOR SYNTHETIC RESINS MADE IN THE UNITED STATES

<i>Trade name</i>	<i>Maker</i>
Tar-acid resins:	
Amberol	Resinous Products & Chemical Co., Inc., Phila., Pa.
Artifex	Artifex Products Co., Camden, N. J.
Bakelite	Bakelite Corp., New York, N. Y.
Beckacite	Beck, Koller & Co., Inc., Detroit, Mich.
Beckasol	Beck, Koller & Co., Inc., Detroit, Mich.
Beckopol	Beck, Koller & Co., Inc., Detroit, Mich.
Catalin	Catalin Corp., New York, N. Y.
Celeron	Continental-Diamond Fibre Co., Newark, Del.
Coltrock	Colt's Patent Fire Arms Mfg. Co., Hartford, Conn.
Colasta	Colasta Co., Inc., Hoosick Falls, N. Y.
Dilecto	Continental-Diamond Fibre Co., Newark, Del.
Dura	Paramet Chemical Corp., Long Island City, N. Y.
Durez	General Plastics, Inc., N. Tonawanda, N. Y.
Durite	Durite Plastics, Phila., Pa.
Fabroil	General Electric Co., Schenectady, N. Y.
Fiberlon	Fiberloid Corp., Indian Orchard, Mass.
Fibroc	Fibroc Insulation Co., Valparaiso, Ind.
Gemstone	A. Knoedler Co., Lancaster, Pa.
Haveg	Haveg Corp., Newark, Del.

Herculite	Colasta Co., Inc., Hoosick Falls, N. Y.
Indur	Reilly Tar & Chemical Corp., Indianapolis, Ind.
Insurok	Richardson Co., Melrose Park, Ill.
Joanite	Joanite Corp., Long Island City, N. Y.
Kellite	Kellogg Switchboard & Supply Co., Chicago, Ill.
Lewisol	John D. Lewis, Mansfield, Mass.
Makalot	Makalot Corp., Boston, Mass.
Marblette	Marblette Corp., Long Island City, N. Y.
Micarta	Westinghouse Electric & Mfg. Co., Trafford, Pa.
Moldarta	Westinghouse Electric & Mfg. Co., Trafford, Pa.
Pandura	Paramet Chemical Corp., Long Island City, N. Y.
Panelyte	Panelyte Corp., Trenton, N. J.
Paranol	Paramet Chemical Corp., Long Island City, N. Y.
Phenac	American Cyanamid Co., New York, N. Y.
Phenalin	E. I. du Pont de Nemours & Co., Wilmington, Del.
Prystal	Catalin Corp., New York, N. Y.
Resinox	Resinox Corp., New York, N. Y.
Spauldite	Spaulding Fibre Co., Tonawanda, N. Y.
Syntex	Jones-Dabney Co., Inc., Louisville, Ky.
Synthane	Synthane Corp., Oaks, Pa.
Taylor	Taylor Fibre Co., Norristown, Pa.
Textolite	General Electric Co., Schenectady, N. Y.
Waterlite	Watertown Mfg. Co., Watertown, Conn.

Other makers of tar acid resins in the United States include:^[27]
 Aluminum Industries, Cincinnati, Ohio; California Flaxseed Products
 Co., Los Angeles, Calif.; Cook Paint & Varnish Co., Chicago, Ill.;

Ford Motor Co., Detroit, Mich.; Heresite & Chemical Co., Manitowoc, Wis.; Millergum Co., Chicago, Ill.; Nubian Paint & Varnish Co., Chicago, Ill.; Varcum Chemical Co., Niagara Falls, N. Y.; Vita Var Corp., Newark, N. J.

Alkyd resins:

Amberlac	Resinous Products & Chemical Co., Inc., Phila., Pa.
Aquaplex	Resinous Products & Chemical Co., Inc., Phila., Pa.
Beckol	Beck, Koller & Co., Inc., Detroit, Mich.
Beckosol	Beck, Koller & Co., Inc., Detroit, Mich.
Dulux	E. I. du Pont de Nemours & Co., Wilmington, Del.
Duraplex	Resinous Products & Chemical Co., Inc., Phila., Pa.
Esterol	Paramet Chemical Corp., Long Island City, N. Y.
Glyptal	General Electric Co., Schenectady, N. Y.
Lewisol	John D. Lewis, Mansfield, Mass.
Makalot	Makalot Corp., Boston, Mass.
Paraplex	Resinous Products & Chemical Co., Inc., Phila., Pa.
Rauzene	Robert Rauh, Inc., Newark, N. J.
Rezyl	American Cyanamid Co., New York, N. Y.
Syntex	Jones-Dabney Co., Inc., Louisville, Ky.
Teglac	American Cyanamid Co., New York, N. Y.

Other makers of alkyd resins in the United States include:^[27] Atlas Powder Co., Wilmington, Del.; Bakelite Corp., New York, N. Y.; Andrew Brown Co., Los Angeles, Calif.; California Flaxseed Products, Los Angeles, Calif.; Carboygen Chemical Co., Garwood, N. J.; General Paint Co., Tulsa, Okla.; Hercules Powder Co., Wilmington, Del.; Nubian Paint & Varnish, Chicago, Ill.; Pittsburgh

Plate Glass Co., Milwaukee, Wis.; Valentine & Co., Inc., New York, N. Y.

Urea resins:

Beetle	American Cyanamid Co., New York, N. Y.
Beckamine	Beck, Koller & Co., Detroit, Mich.
Plaskon	Plaskon Co., Inc., Toledo, Ohio.
RHoplex	Rohm & Haas, Philadelphia, Pa.
Syntex	Jones-Dabney Co., Inc., Louisville, Ky.
Uformite	Resinous Products & Chemical Co., Phila., Pa.

Other makers of urea resins in the United States include: E. I. du Pont de Nemours & Co., Wilmington, Del.; Bakelite Corp., New York, N. Y.

Acrylate resins:

Acryloid	Resinous Products & Chemical Co., Phila., Pa.
Acrysol	Resinous Products & Chemical Co., Phila., Pa.
Crystalite	Resinous Products & Chemical Co., Phila., Pa.
Lucite	E. I. du Pont de Nemours & Co., Wilmington, Del.
Plexiglas	Resinous Products & Chemical Co., Phila., Pa.
Plexigum	Resinous Products & Chemical Co., Phila., Pa.
Primal	Resinous Products & Chemical Co., Phila., Pa.

Coumarone-indene resins:

Cumar	Barrett Co., New York.
Neville	Neville Co., Pittsburgh, Pa.

Nevindene	Neville Co., Pittsburgh, Pa.
Petroleum resins:	
Santo-Resin	Monsanto Chemical Co., St. Louis, Mo.
Petropol	Pure Oil Co., Chicago, Ill.
Polystyrene resins:	
Bakelite polystyrene	Bakelite Corp., New York, N. Y. Carbide & Carbon Chemicals Corp., New York, N. Y.
Styron	Dow Chemical Co., Midland, Mich.
Vinyl resins:	
Butvar	Shawinigan Products, Inc., Indian Orchard, Mass.
Vinyloid	Carbide & Carbon Chemicals Corp., New York.
Vinylite	Carbide & Carbon Chemicals Corp., New York.
Vinal	Carbide & Carbon Chemicals Corp., New York.
Vinylseal	Carbide & Carbon Chemicals Corp., New York.
Flamenol	General Electric Co., Schenectady, N. Y.
Koroseal	B. F. Goodrich Co., Akron, Ohio. E. I. du Pont de Nemours & Co., Wilmington, Del.

APPENDIX C

TRADE NAMES FOR SYNTHETIC RESINS MADE IN GREAT BRITAIN

<i>Trade name</i>	<i>Maker</i>
Tar-acid resins for moulding and laminating:	
Bakelite	Bakelite, Ltd., London.
Britsulite	British Insulite Ltd., Manchester.
Elo	Birkby's Ltd., Liversedge, Yorkshire.
Fabrolite	British Thompson Houston Co., Ltd., Rugby.
Holite	E. S. Hole, London.
Indurite	Indurite Moulding Powders Ltd., Radcliffe, Lancashire.
Lorival	Lorival Mfg. Co., Ltd., Southall, Middlesex.
Mouldrite	Imperial Chemical Industries, Ltd., London.
Nestorite	James Ferguson & Sons, Ltd., London.
Permaplastic	Permastic Ltd., Weybridge, Surrey.
Rockite	F. A. Hughes & Co., Ltd., London.
Oil soluble tar-acid resins:	
Bakelaque	Attwater & Sons, Ltd., Preston, Lancashire.
Damard	Bakelite, Ltd., London.
Damarda	Bakelite, Ltd., London.
Epok	British Resin Products, Ltd., Kingston-on- Thames.
Erinite	Erinoid Ltd., Stroud, Gloucester.
Formapex	Ioco Rubber & Waterproofing Co., Ltd., Anniesland, Glasgow.
Keebush	Bushings, Ltd., Hebburn-on-Tyne.
Cast phenolic resin:	

Catalin	Catalin Ltd., Waltham Abbey, Essex.
Urea resins:	
Beetle	Beetle Products Co., Ltd., Oldbury, Worcestershire.
Mouldrite	Imperial Chemical Industries, Ltd., London.
Pollopas	Beetle Products Co., Ltd., Oldbury, Worcestershire.
Scarlat	Beetle Products Co., Ltd., Oldbury, Worcestershire.
Alkyd resins:	
Albertalates	Albert Products, Ltd., Erith, Kent.
Dulux	Nobel Chemical Finishes Ltd., London.
Glyptal	British Thompson Houston Co., Ltd., Rugby.
Micanite	Micanite and Insulators Co., Ltd., Walthamstow, London.
Paralac	Imperial Chemical Industries, Ltd., London.
Acrylate resins:	
Diakon and Perspex	Mouldrite Ltd., Division of Imperial Chemical Industries, Ltd., London.
Aniline resins:	
Panilax	Micanite and Insulators Co., Ltd., London.

APPENDIX D

TRADE NAMES FOR SYNTHETIC RESINS MADE IN GERMANY

<i>Trade name</i>	<i>Maker</i>
Tar-acid resins:	
Alberid	Dr. Kurt Albert G. m. b. H., Chemische Fabrik, Wiesbaden-Biebrich.
Albertol	Dr. Kurt Albert G. m. b. H., Chemische Fabrik, Wiesbaden-Biebrich.
Ambresit	Chemische Fabrik Ambra, Zittau i. Sachsen (in liquidation).
Ammoplaste	I. G. Farbenindustrie A. G., Frankfort-on-Main.
Backdura	Bakelite G. m. b. H., Berlin.
Bakelit	Bakelite G. m. b. H., Berlin.
Bakelit A (Resol)	Bakelite G. m. b. H., Berlin.
Bakelit C	Bakelite G. m. b. H., Berlin.
Bezetz Harz	Louis Blumen, Zwickau i. Sachsen.
Boschbakelite	Robert Bosch A. G., Stuttgart.
Celloresen	Louis Blumer, Zwickau i. Sachsen.
Dekorit	Dr. F. Raschig G. m. b. H., Ludwigshafen a. Rhein.
Durax	
Durophen	Dr. Kurt Albert G. m. b. H., Wiesbaden-Biebrich.
Elastolith	Herold A. G., Hamburg 33.
Ethyl cellulose	I. G. Farbenindustrie A. G., Frankfort-on-Main.
Faturan	Dr. Heinr. Traun & Sohne, Hamburg.
Ferrozell	Deutsche Ferrozell G. m. b. H., Augsburg.

Greif Faturan	Dr. H. Traun & Sohne, Hamburg.
Havegit	Saureschutz G. m. b. H., Berlin.
Herolith	Herold A. G., Hamburg.
Hornolith	Hornolith G. m b. H., Berlin S 59.
Ivorax	Herold A. G., Hamburg.
Koraton	Wedig & Reu.
Kunstharz 26 Z	I. G. Farbenindustrie A. G., Frankfort-on-Main.
Laccain	Louis Blumer, Zwickau i. Sachsen.
Leukorit	Dr. F. Raschig G. m. b. H., Ludwigshafen a. Rhein.
Lithocorn	A. Elhardt Sohne, Kempten, Bayern.
Marbolith	Herold A. G., Hamburg 33.
Metakalin	I. G. Farbenindustrie A. G., Frankfort-on-Main.
Neoresit	August Nowak A. G., Bautzen.
Novolack	Bakelite G. m. b. H., Berlin.
Novotext	Allgemeine Elektrizitats-Gesellschaft, Berlin.
Ornalith	Herold A. G., Hamburg
Phenolplaste	I. G. Farbenindustrie A. G., Frankfort-on-Main.
Redmanol (Bakelit A)	Bakelite G. m. b. H., Berlin.
Resenoplast	Louis Blumer, Zwickau i. Sachsen.
Resinit	Bakelite G. m. b. H., Berlin.
Resinole	Dr. F. Raschig G. m. b. H., Ludwigshafen a. Rhein.
Resit	Bakelite G. m. b. H., Berlin.
Resol	Bakelite G. m. b. H., Berlin.
Schellackersatz	I. G. Farbenindustrie A. G., Frankfort-on-Main. Dr. A. Wacker Gesellschaft für Elektrochemische Industrie G. m. b. H., München. Dr. Kurt Albert, Wiesbaden-Biebrich.
Sipilite	Bakelite G. m. b. H., Berlin.
Supraresen	Louis Blumer, Zwickau i. Sachsen.

Syntellac	Dr. A. Wacker G. m. b. H., München.
Tenazit	Allgemeine Elektrizitäts-Gesellschaft, Berlin.
Toplast	Louis Blumer, Zwickau i. Sachsen.
Trolon	Rheinisch-Westfälische Sprengstoff Fabriken, Troisdorf.
Turbax	Jaroslav's Erste Glimmerwarenfabrik, Berlin SO 36.
Vigorith	Dr. F. Raschig, Ludwigshafen a. Rhein.
Vinnapas	Dr. A. Wacker G. m. b. H., München.
Wackerschellack	Dr. A. Wacker G. m. b. H., München.
Wenjazit	Kunst-Rohstoff A. G., Hamburg-Einbeck.
Alkyd resins:	
Alftalate	Dr. Kurt Albert G. m. b. H., Chemische Fabrik, Wiesbaden-Biebrich.
Alkydal	I. G. Farbenindustrie A. G., Frankfurt-on- Main.
Beckacite	Beckacite Kunsthharzfabrik, G. m. b. H., Hamburg-Wandsbeck.
Beckosol	Beckacite Kunsthharzfabrik, G. m. b. H., Hamburg-Wandsbeck.
Duxol	Louis Blumer, Zwickau i. Sachsen.
Duxalid	Louis Blumer, Zwickau i. Sachsen.
Geaphthal	Allgemeine Elektrizitäts-Gesellschaft, Berlin.
Glyptal	Dr. Kurt Albert G. m. b. H., Wiesbaden- Biebrich.
Urea resins:	
Hares L	H. Rommler A. G., Berlin W 62.
Locron	I. G. Farbenindustrie, A. G., Frankfurt-on- Main.
Pluviusin	Kunsthharzfabrik Dr. F. Pollack, Vienna, German branch Berlin.
Pollopas	Rheinisch-Westfälische Sprengstoff Fabriken, Troisdorf, Bez. Köln.

Vinyl resins:

Acronal	I. G. Farbenindustrie, A. G., Frankfort-on-Main.
Mowilith	I. G. Farbenindustrie, A. G., Frankfort-on-Main.

Polystyrene resins:

Metastyrol	I. G. Farbenindustrie, A. G., Frankfort-on-Main.
Mollit	I. G. Farbenindustrie, A. G., Frankfort-on-Main.
Trolitul	Rheinisch-Westfalische Sprengstoff Fabriken, Troisdorf.

Acrylate resins:

Plexigum	Rohm & Haas, Darmstadt.
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Other resins:

Coumarone resins	Kokawerke & Chemische Fabriken A. G., Berlin N. W. 40.
Harz No. 30	Ciba A. G., Berlin-Wilmersdorf.
Harz No. 238	Ciba A. G., Berlin-Wilmersdorf.
Ultrasit	Chemische Fabrik Ambra, Zittau, Sachsen, (in liquidation).
Utilith	Deutsche Rohstoffindustrie, Augsburg.

APPENDIX E

LIST OF UNITED STATES MANUFACTURERS OF RAW MATERIALS FOR SYNTHETIC RESINS^[28]

Phenol:

Barrett Co., New York, N. Y.
Calco Chemical Co., Inc., Bound Brook, N. J.
Dow Chemical Co., Midland, Mich.
Koppers Co., Pittsburgh, Pa.
Monsanto Chemical Co., St. Louis, Mo.
Reilly Tar & Chemical Corp., Indianapolis, Ind.

Cresols:

Barrett Co., New York, N. Y.
Calco Chemical Co., Inc., Bound Brook, N. J.
Givaudan-Delawanna, Inc., Delawanna, N. J.
Koppers Co., Pittsburgh, Pa.
Reilly Tar & Chemical Corp., Indianapolis, Ind.
Swann & Co., Birmingham, Ala.

Xylenols:

Barrett Co., New York, N. Y.
Calco Chemical Co., Inc., Bound Brook, N. J.
Reilly Tar & Chemical Corp., Indianapolis, Ind.

Butyl phenol (p-tertiary), Dow Chemical Co., Midland, Mich.

Phenyl phenols, Dow Chemical Co., Midland, Mich.

Resorcinol, tech.:

E. I. du Pont de Nemours & Co., Wilmington, Del.

Pennsylvania Coal Products Co., Petrolia, Pa.

Naphthalene:

Barrett Co., New York, N. Y.

Calco Chemical Co., Inc., Bound Brook, N. J.

Coopers Creek Chemical Co., West Conshohocken, Pa.

E. I. du Pont de Nemours & Co., Wilmington, Del.

Reilly Tar & Chemical Corp., Indianapolis, Ind.

Shell Chemical Co., San Francisco, Calif.

Standard Naphthalene Products Corp., South Kearney, N. J.

White Tar Co. of N. J., Inc., Pittsburgh, Pa.

Phthalic acid and anhydride:

American Cyanamid Co., New York, N. Y.

Barrett Co., New York, N. Y.

E. I. du Pont de Nemours & Co., Wilmington, Del.

Monsanto Chemical Co., St. Louis, Mo.

National Aniline & Chemical Co., Inc., New York, N. Y.

Maleic acid and anhydride:

National Aniline & Chemical Co., Inc., New York, N. Y.

American Cyanamid Co., New York, N. Y.

Monsanto Chemical Co., St. Louis, Mo.

Malic acid, National Aniline & Chemical Co., Inc., New York, N. Y.

Adipic acid, E. I. du Pont de Nemours & Co., Wilmington, Del.

Succinic acid and anhydride:

American Cyanamid Co., New York, N. Y.

National Aniline & Chemical Co., Inc., New York, N. Y.

Urea, E. I. du Pont de Nemours & Co., Wilmington, Del.

Formaldehyde:

E. I. du Pont de Nemours & Co., Wilmington, Del.

Empire Oil & Refining Co., Bartlesville, Okla.

Heyden Chemical Corp., New York, N. Y.

Hexamethylenetetramine, tech.:

E. I. du Pont de Nemours & Co., Wilmington, Del.

Heyden Chemical Corp., New York, N. Y.

Monsanto Chemical Co., St. Louis, Mo.

Furfural, Quaker Oats Co., Chicago, Ill.

Vinyl acetate:

Carbide & Carbon Chemicals Corp., New York, N. Y.

E. I. du Pont de Nemours & Co., Wilmington, Del.

Niacet Chemicals Corp., Niagara Falls, N. Y.

Vinyl chloride, Carbide & Carbon Chemicals Corp., New York, N. Y.

APPENDIX F

GLOSSARY^[29]

Alkyd resin.—Any condensation product involving a polybasic acid and a polyhydric alcohol. Typical examples are phthalic glyceride and its modifications containing combined fatty acids or rosin. Representative examples are Rezyls and Glyptal.

Aminoplast.—General terms for synthetic resins from amino or amido compounds. A typical example is urea-formaldehyde.

Amorphous.—Devoid of crystalline structure. This condition is rare. Many substances which are apparently amorphous show microcrystallinity, particularly under X-ray examination.

A-stage resins.—Thermosetting resins reacted only to the initial stage where they are soluble and fusible. The normal stage of a resin used for impregnation.

Bonding strength.—The amount of adhesion between a binder and filler. More specifically, the measure of the extent to which the composite layers of a laminated product are bonded together.

Brittleness.—Liability to break, generally to a conchoidal fracture.

B-stage resins.—Thermosetting resins reacted to a stage where they soften when heated and swell in contact with liquids but do not entirely fuse or dissolve. This is the preferred stage for the resin in molding compositions.

Casting.—Forming a material into a shape by pouring it when liquid into a mold. The product from the mold is used as such or mechanically worked in various ways to the final articles, as by sewing, cutting, blanking, turning, drilling, forming, swaging, grinding, polishing, sanding, or routing.

Compressive strength.—Resistance to deformation under applied pressure.

Condensation.—A chemical reaction in which two or more molecules combine with a separation of water or some other simple substance. Applied to synthetic resins it means the formation of a resin by combination of a number of molecules with elimination of water, ammonia, hydrogen chloride, or other simple substance. Examples of condensation resins are alkyd, phenol-aldehyde, and urea-formaldehyde resins. The final products are also called condensation-polymers. (See Polymerization.)

Copolymerization.—The term applied when two or more substances polymerize at the same time to yield a product which is not a mixture of separate polymers but a complex having properties different from either polymer alone. For example, vinylite is produced by polymerization of a mixture of vinyl acetate and vinyl chloride.

C-stage resins.—Thermosetting resins in the final stage in which they are infusible and insoluble. The state of the resin in the final molded article.

Curing.—The change of a binder from the soluble-fusible condition to the substantially insoluble-infusible form by chemical action. The heat-setting of a resinoid. Action is analogous to vulcanization of rubber.

Dielectric strength.—Voltage gradient at which a continuous electrical discharge will take place between two electrodes when the material in question is placed between the electrodes and a potential difference is applied to them.

Elastic.—A substance which exhibits rubberlike properties or “high elasticity” over a wide range of applied forces.

Elastic deformation.—When a substance reverts to its original dimensions on release of an applied stress.

Elastic limit.—The point at which a body begins to yield under a stress; that is, when the stress is equal to or greater than the internal friction.

Elasticity.—The property by virtue of which a body reverts to its normal bulk or shape after deformation by an applied force.

Extrusion molding.—A molding procedure for extended shapes of uniform cross section, whereby a heat-softened substance is forced through an orifice of form coinciding with the cross section of the article.

Flexibility.—Capability of bending without breaking.

Gums.—Viscous vegetable secretions which harden but, unlike resins, are water soluble. The name is often applied, particularly in the varnish industry, to natural resins such as copals.

Hardness.—Property of substances determined by their ability to abrade or indent one another. Often measured by the extent or depth of indentation produced by a standard substance under a predetermined load.

Impact strength.—The measure of toughness of a material. Generally determined by the energy required to break a specimen in one blow.

Injection molding.—A molding procedure whereby a heat-softened plastic material is forced from a receptacle into a cavity which gives the article of desired shape. Used particularly for thermoplastics since the scrap can be reused. As soon as the composition in the mold cools sufficiently to be rigid, the mold is opened and the molded article removed. An analogy of injection molding in another field is shown by the linotype machine.

Inserts.—Parts of a finished molded article which are of different material from the molding composition but are set in place or positioned by the molding operation.

Laminated products.—Sheets of material united by a binder. For example, sheets of paper or wood coated and/or impregnated with a resinous composition and subjected to pressure, generally with heat.

Monomer.—The simplest repeating structural unit of a polymer. For addition polymers this represents the originally unpolymerized compound.

Phenoplast.—A general term for phenol-aldehyde resins. Synonymous with popular term “phenolics.”

Plastics.—All substances that can be molded. In general a plastic is a substance which behaves as a solid at stresses less than a certain amount known as the yield value and as a viscous liquid at stresses greater than this. The name is also applied to substances which originally but not ultimately fulfill this condition. For example, it is applied to thermoset compositions or resinoids in the final stages.

Plasticity.—Susceptibility to and the retention of deformation. Capacity of taking and retaining the form of a mold. The property of solids by virtue of which they hold their shape permanently under the action of small

shearing stresses but are readily deformed, worked, or molded under larger stresses.

Polymerization.—A chemical change resulting in the formation of a new compound whose molecular weight is a multiple of that of the original substances. The products of the reaction are called polymers. To distinguish from those resulting from condensation (q. v.), they are often designated addition polymers, since the reaction is that of successive addition of a large number of relatively small molecules (monomers) to form the final polymer.

Power factor.—In an insulating material, the ratio of total power loss (watts) in the material to the product of voltage and current in a capacitor in which that material is a dielectric.

Preforms.—Molding powders converted by pressure and without heat into a denser coherent form which approximates the shape of the final hot-pressed article. Molding material converted to preforms has about half the bulk factor of the original powder. Other forms of densified composition which do not necessarily approximate the shape of the final molding are tablets, briquettes, pellets, pills, and balls.

Resin.—A term generally referring to a physical condition at room temperature approximating the physical properties of natural resins. However, the temperature of reference should not be limited to room temperature and the term is here intended to embrace all substances which within a certain temperature range show these, properties. For example, many oil-modified alkyd resins are viscous liquids at room temperature but not at lower temperatures; polystyrene is a resin at room temperature but rubberlike when warmed.

Resinoids.—The class name applied to thermosetting resins. Temporary thermoplastics. The name is also often applied to the final cured resins.

Softening point.—Resins have no sharp melting point. Application of heat causes gradual change from a brittle or exceedingly thick and slow flowing material to a softer and less viscous liquid. The softening point is the temperature at which the material flows at a definite rate or to a definite distance.

Synthetic resin.—A complex, substantially amorphous, organic semisolid or solid material (usually a mixture of substances) built up by chemical reaction of comparatively simple compounds and, depending upon the temperature at which the examination is made, approximating the natural resins in various physical properties: namely, luster, fracture, comparative brittleness, insolubility in water, fusibility or plasticity when heated or exposed to heat and pressure, and, at a certain more or less narrow temperature range before fusion, showing a degree of rubberlike extensibility; but commonly deviating widely from natural resins in chemical constitution and behavior with reagents.

Synthetic rubber.—Caoutchouc synthesized in the laboratory. The term is a misnomer and most probably represents an impossibility.

Tensile strength.—The greatest internal force per unit of cross section which a material develops before failure under tension. [\[30\]](#)

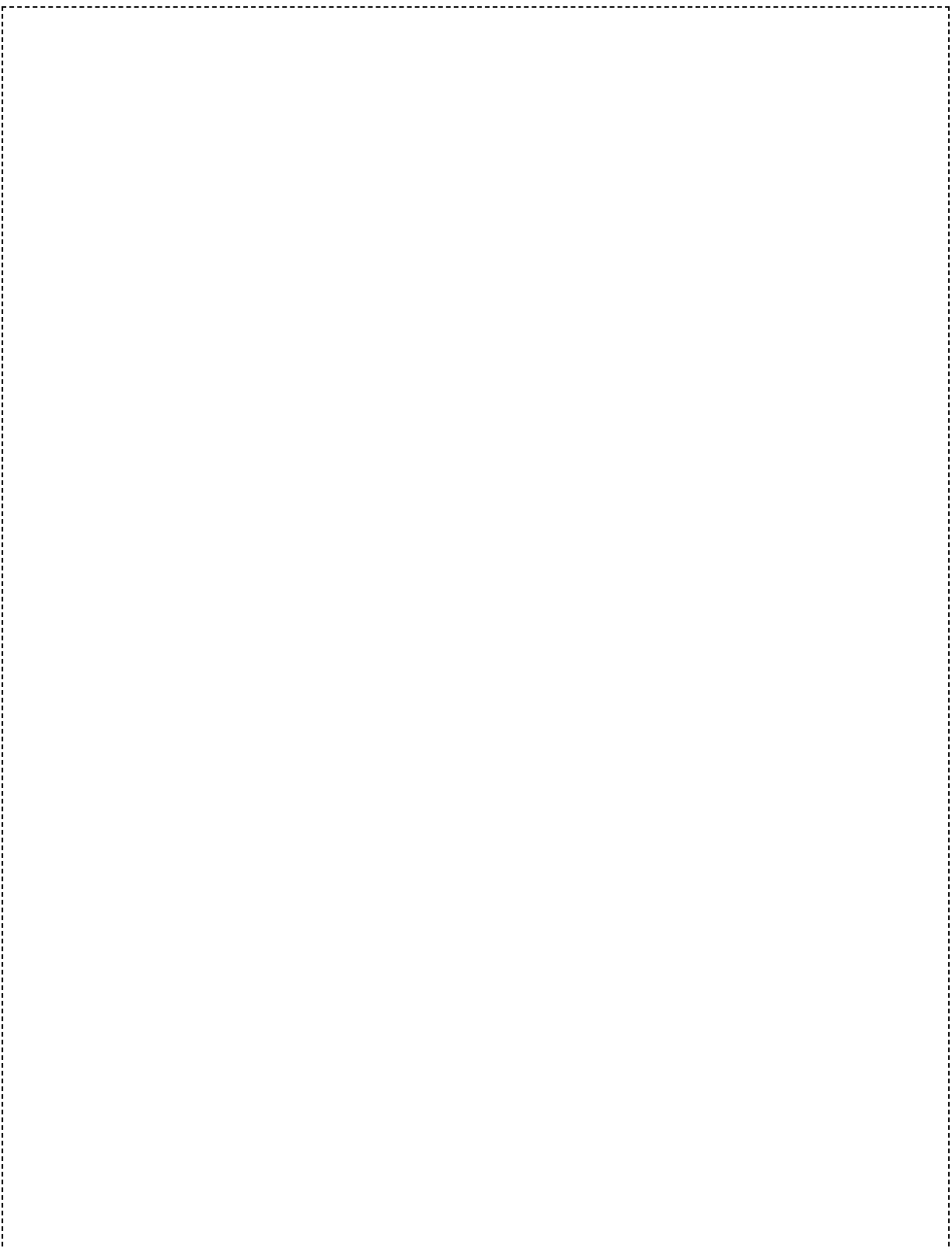
Thermoplastic.—The property of softening under heat. All molding materials are thermoplastic at the initial application of heat. One class (the so-called thermoplastics) remains soft permanently under heat; the other (thermosetting), after first softening, sets or cures more or less quickly to a more solid form. A practical distinction is that with the first class the mold must be cooled before the molded article is removed, but not with the second. A thermoplastic substance is adequately rigid at normal temperatures and under ordinary conditions of stress but is capable of deformation under heat and pressure.

Thermosetting.—The property of undergoing a chemical change when heated whereby a hardened product is obtained. Property most pronounced in phenol and urea formaldehyde resins and less so with alkyds. A thermosetting substance possesses initially the properties of a thermoplastic but under the influence of heat undergoes chemical change so that it is no longer thermoplastic but becomes permanently infusible.

Viscosity.—Internal friction or resistance to change of form of a liquid. The constant ratio of shearing stress to rate of shear.

Water-absorption.—Amount of water taken up when exposed to humid conditions or when immersed. Both rate of absorption and total absorption are important, also change in dimensions. A certain amount of absorbed

water may improve mechanical properties but usually weakens electrical characteristics.



FOOTNOTES

[1] A glossary of technical terms is included in [appendix F](#), p. 160 of this report. There are, however, certain fundamental terms which it would be advisable to have clearly in mind at this point.

[2] Shellac is a substance secreted by lac insects feeding on certain types of hardwood trees.

[3] Journal of the Society of Chemical industry, 1901. Vol. 20, p. 1075.

[4] Current sales prices in the United States average between 7 and 10 cents per pound and any imports would be dutiable under the provisions of par. 28 of the Tariff Act of 1930 at 45 percent ad valorem based on American selling price plus 7 cents per pound. Based on an American selling price of 7 cents per pound, the import duty would be slightly more than 10 cents per pound, while on an American selling price of 10 cents per pound the duty would be 11.5 cents per pound.

[5] Zelov, Victor I. Automatic Molding, Pt. 2, Advantages and limitations. Modern Plastics, v. 15, No. 2, p. 206; October 1937.

[6] For texts and interpretation of exclusion orders see Treasury Decisions 41512; 41895; 44411; 44491; 44776; and 44977.

[7] American selling price is defined in section 402, (g) as: "The American selling price of any article manufactured or produced in the United States shall be the price, including the cost of all containers and coverings of whatever nature and all other costs, charges, and expenses incident to placing the merchandise in condition packed ready for delivery, at which such article is freely offered for sale to all purchasers in the principal market of the United States, in the ordinary course of trade and in the usual wholesale quantities in such market, or the price that the manufacturer, producer, or owner would have received or was willing to receive for such merchandise when sold in the ordinary course of trade and in the usual wholesale quantities, at the time of exportation of the imported article."

[8] United States value is defined in section 402, (e) as: "The United States value of imported merchandise shall be the price at which such or similar imported merchandise is freely offered for sale, packed ready for delivery, in the principal market of the United States to all purchasers, at the time of exportation of the imported merchandise, in the usual wholesale quantities and in the ordinary course of trade, with allowance made for duty, cost of transportation and insurance, and other necessary expenses from the place of shipment to the place of delivery, a commission not exceeding 6 per centum, if any has been paid or contracted to be paid on goods secured otherwise than by purchase, or profits not to exceed 8 per centum and a reasonable allowance for general expenses, not to exceed 8 per centum on purchased goods."

[9] The reclassifications read:

PAR. 2. "Vinyl acetate, polymerized or unpolymerized, and synthetic resins made in chief value therefrom, not specially provided for."

PAR. 11. "Synthetic resins made in chief value from vinyl acetate, not specially provided for."

[10] See sections on import under each resin.

[11] Reduced May 23, 1934, from 25 cents per pound and 30 percent by Presidential proclamation under Section 336 of the Tariff Act of 1930.

[12] Based on the total sales in 1937 reported to the Tariff Commission; sales in dollars divided by quantity (net resin content).

[13] The new consumption-restriction regulations are—

Exterior use on plaster, brick, stone, and cement:

1. Surfaces already painted with oil paint may be repainted with oil paints, without restrictions;
2. Surfaces calcimined may be painted with paints containing not more than 15 percent oil;
3. Unpainted surfaces may be painted only with paints free of oil.

Exterior use on wood:

Linseed oil paint may be used for the first coat, and succeeding coats may contain up to 70 percent oil.

Exterior and interior use on metal:

Oil paints may be used without restriction.

Interior use on plaster, brick, stone, and cement:

1. Surfaces already painted with oil paint may only be repainted with paint containing not more than 15 percent oil.
2. Unpainted surfaces must be painted with paint free of oil.

Interior use on wood, to be cleaned with soap and soda:

El Varnish and oil varnish may be used for the first coat followed by paint containing up to 70 percent oil.

Interior use on wood, not cleaned with soap and soda:

As above, except that succeeding coats may contain not more than 40 percent oil.

[14] Acknowledgment: Most of the information about the industry in Great Britain was submitted by Norman Inwood of the staff of the American consulate general at London, England.

[15] Acknowledgment: Information obtained on the synthetic resin industry in France was furnished by Addison E. Southard, American consul general at Paris.

[16] Acknowledgment: Much of the information on the Japanese synthetic resin industry included herein was furnished by Carl H. Boehringer, Assistant Trade Commissioner at Tokyo at the request of the U.S. Tariff Commission.

[17] These figures are based on an average naphthalene content of coal tar of slightly less than 10 percent. The total amount contained would, of course, not be recovered even under ideal market conditions as to price and demand.

[18] Par. 1651. Coal-tar products: ... naphthalene which after the removal of all the water present has a solidifying point less than 79° C. ... (Free).

[19] Par. 27. Coal-tar products:

(a) (1), (5) ... naphthalene which after the removal of all water present has a solidifying point of 79° C. or above; all the foregoing products in this paragraph whether obtained, derived, or manufactured from coal tar or other sources; ... 40 percent ad valorem and 7 cents per pound.

(c) The ad valorem rates provided in this paragraph shall be based upon the American selling price (as defined in subdivision (g) of section 402, title IV), of any similar competitive article manufactured or produced in the United States. If there is no similar competitive article manufactured or produced in the United States then the ad valorem rate shall be based upon the United States value, as defined in subdivision (e) of section 402, title IV.

(d) For the purposes of this paragraph any coal-tar product provided for in this act shall be considered similar to or competitive with any imported coal-tar product which accomplishes results substantially equal to those accomplished by the domestic product when used in substantially the same manner.

[20] Upon American selling price or United States value.

[21] The relevant provisions of this act are as follows:

Par. 27 (b) ... phenol, carbolic acid which on being subjected to distillation yields in the portion distilling below one hundred and ninety degrees centigrade a quantity of tar acids equal to or more than 5 per centum of the original distillate, ..., and any mixture of any of the foregoing products with any of the products provided for in paragraph 1651, 20 per centum ad valorem and 3½ cents per pound.

(c) The ad valorem rates provided in this paragraph shall be based upon the American selling price (as defined in subdivision (g) of section 402, title IV), of any similar competitive article manufactured or produced in the United States. If there is no similar competitive article manufactured or produced in the United States then the ad valorem rate shall be based upon the United States value, as defined in subdivision (e) of section 402, title IV.

(d) For the purposes of this paragraph any coal-tar product provided for in this Act shall be considered similar to or competitive with any imported coal-tar product which accomplishes results substantially equal to those accomplished by the domestic product when used in substantially the same manner.

[22] In 1923 the unit value of domestic sales was 27 cents per pound and the duty on imports (computed specific rate per pound) was 16 cents; in 1925 the corresponding figures were 21 and 16 cents, respectively.

[23] Par. 1651. Coal-tar products: ..., all mixtures of any of these distillates and any of the foregoing pitches, and all other materials or products found naturally in coal tar, whether produced or obtained from coal tar or other source, and not specially provided for in pars. 27 or 28....

[24] Par. 27 (a) (2). Coal-tar products: All distillates (except those provided for in sub-paragraph (b)) of coal tar, blast-furnace tar, oil-gas tar, and water-gas tar, ..., which on being subjected to distillation yield in the portion distilling below two hundred and fifteen degrees centigrade a quantity of tar acids equal to or more than 75 per centum of the original distillate.

[25] Par. 27 (b). Metacresol having a purity of 90 per centum or more, orthocresol having a purity of 90 per centum or more, paracresol having a purity of 90 per centum or more, ... and any mixture of any of the foregoing products with any of the products provided for in paragraph 1651, ...

Par. 27 (c). The ad valorem rates provided in this paragraph shall be based upon the American selling price (as defined in subdivision (g) of section 402, title IV), of any similar competitive article manufactured or produced in the United States. If there is no similar competitive article manufactured or produced in the United States then the ad valorem rate shall be based upon the United States value, as defined in subdivision (e) of section 402, title IV.

Par. 27 (d). For the purposes of this paragraph any coal-tar product provided for in this Act shall be considered similar to or competitive with any imported coal-tar product which accomplishes results substantially equal to those accomplished by the domestic product when used in substantially the same manner.

[26] Par. 27 (b). ... cresylic acid which upon being subjected to distillation yields in the portion distilling below two hundred and fifteen degrees centigrade a quantity of tar acids equal to or more than 75 per centum of the original distillate....

[27] Some of the makers of these products do not care to be identified with their manufacture.

[28] Some of the makers of these products are not listed because they do not care to be so identified.

[29] Based on pp. 321-4, Modern Plastics, October 1937.

[30] Source: Peele's Mining Engineers' Handbook, Ed. 1, p. 2209.

*** END OF THE PROJECT GUTENBERG EBOOK SYNTHETIC
RESINS AND THEIR RAW MATERIALS ***

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